Introduction to Functional Analysis

MIT 18.102

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Course-note convention

Several theorems given in the homework are included in these course notes.

Definitions
Results
Examples
Notes

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1 Basic Banach Space

1.1 Normed spaces: an analogue for $\mathbb{R}^n$

Definition (recall)

A *vector space* V over a field $\mathbb{K}$ is a set of elements, which we call vectors, that comes with two operations:

$$+ : V \times V \longrightarrow V, \qquad (v_1, v_2) \longmapsto v_1 + v_2,$$

and

$$\cdot : \mathbb{K} \times V \longrightarrow V, \qquad (\alpha, v) \longmapsto \alpha v,$$

and satisfies the following axioms: commutativity, associativity, identity and inverse of addition, identity of scalar multiplication, and distributivity.

Examples

The spaces

$$\mathbb{R}, \mathbb{R}^2, \ldots, \mathbb{R}^n \qquad \text{and} \qquad \mathbb{C}, \mathbb{C}^2, \ldots, \mathbb{C}^n, \quad n \in \mathbb{N},$$

are vector spaces. Another example is

$$C([0,1]) = \{f : [0,1] \to \mathbb{C} \mid f \text{ is continuous}\}.$$

Recall

A function $f : \mathbb{C} \to \mathbb{C}$ is continuous at $z_0 \in \mathbb{C}$ if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for every $z \in \mathbb{C}$,

$$|z - z_0| < \delta \quad \Longrightarrow \quad |f(z) - f(z_0)| < \varepsilon.$$

Here $|\cdot|$ is the norm, or absolute value, of a complex number, and $\varepsilon, \delta \in \mathbb{R}$.

Definition

A vector space V is *finite-dimensional* if every linearly independent set is finite. In other words, if $E \subset V$ and, for every $v_1, v_2, \ldots, v_n \in E$,

$$\sum_{i=1}^{n} a_i v_i = 0 \quad \Longrightarrow \quad a_1 = a_2 = \cdots = a_n = 0,$$

then E must have finite cardinality. The space V is *infinite-dimensional* if it is not finite-dimensional.

Example

The space $C([0,1])$ is infinite-dimensional. Consider

$$E = \{f_n(x) = x^n \mid n \in \mathbb{N} \cup \{0\}\} \subset C([0,1]).$$

This set has infinite cardinality, yet it is linearly independent.

Proof

Suppose

$$\sum_{i=1}^{n} \alpha_i f_{a_i} = 0, \qquad \alpha_i \in \mathbb{R}, \quad a_i \in \mathbb{N}.$$

Here 0 is the zero map from $[0,1]$ to $\mathbb{C}$. We first show the claim for $f_{a_i} = x^i$. It then follows for $f_{a_i} = x^{a_i}$, because subsets of a linearly independent set are also linearly independent. In other words, we first work with $\{x, x^2, \ldots, x^n\}$ instead of $\{x^{a_1}, x^{a_2}, \ldots, x^{a_n}\}$.
We want to show that

$$\sum_{i=1}^{n} \alpha_i x^i = 0 \quad \Longrightarrow \quad \alpha_1 = \alpha_2 = \cdots = \alpha_n = 0 \qquad \text{for every } x.$$

Choose

$$x_1 = \frac{1}{2^0}, \quad x_2 = \frac{1}{2^1}, \quad \ldots, \quad x_n = \frac{1}{2^{n-1}}.$$

Then the coefficients $\alpha_1, \ldots, \alpha_n$ satisfy

$$\begin{cases} \alpha_1(2^{-0})^1 + \alpha_2(2^{-0})^2 + \cdots + \alpha_n(2^{-0})^n = 0, \\ \alpha_1(2^{-1})^1 + \alpha_2(2^{-1})^2 + \cdots + \alpha_n(2^{-1})^n = 0, \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \vdots \\ \alpha_1(2^{-(n-1)})^1 + \alpha_2(2^{-(n-1)})^2 + \cdots + \alpha_n(2^{-(n-1)})^n = 0. \end{cases}$$

The coefficient matrix of this homogeneous system is

$$A = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 2^{-1} & (2^{-1})^2 & \cdots & (2^{-1})^n \\ \vdots & \vdots & & \vdots \\ (2^{-(n-1)}) & (2^{-(n-1)})^2 & \cdots & (2^{-(n-1)})^n \end{pmatrix}.$$

This is a Vandermonde matrix of order n, and

$$|A| = \prod_{1 \le i < j \le n} \left(\frac{1}{2^j} - \frac{1}{2^i} \right) \neq 0.$$

Thus A is invertible, so the homogeneous system has only the trivial solution $\alpha_1 = \cdots = \alpha_n = 0$. Therefore $\{x, x^2, \ldots, x^n\}$ is linearly independent for every $n \in \mathbb{N}$, and hence E is linearly independent. Thus $C([0,1])$ is infinite-dimensional.

The Vandermonde determinant

For a Vandermonde matrix of order n,

$$\det\begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{pmatrix} = \det\begin{pmatrix} 1 & 1 & \cdots & 1 \\ 0 & x_2 - x_1 & \cdots & x_n - x_1 \\ \vdots & \vdots & & \vdots \\ 0 & x_2^{n-1} - x_2^{n-2}x_1 & \cdots & x_n^{n-1} - x_n^{n-2}x_1 \end{pmatrix}$$

$$= (x_2 - x_1)(x_3 - x_1)\cdots(x_n - x_1)\det\begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_2 & x_3 & \cdots & x_n \\ \vdots & \vdots & & \vdots \\ x_2^{n-2} & x_3^{n-2} & \cdots & x_n^{n-2} \end{pmatrix}.$$

The first equality uses the row operations $R_i \leftarrow R_i - x_1 R_{i-1}$ for $i = n, n-1, \ldots, 2$. We then factor each column after the expansion along the first column. Repeating these two steps gives

$$\det V(x_1, \ldots, x_n) = (x_2 - x_1)\cdots(x_n - x_1)(x_3 - x_2)\cdots(x_n - x_{n-1}) = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Finite- and infinite-dimensional spaces

When we deal with infinite-dimensional vector spaces, properties that hold in finite-dimensional spaces might no longer be valid. For example, the Heine–Borel theorem says that, in Euclidean space $\mathbb{R}^n$, a set $E \subset \mathbb{R}^n$ is closed and bounded if and only if it is compact. This is not always true in infinite-dimensional spaces.

Definition

A *norm* on a vector space V is a function

$$\|\cdot\| : V \longrightarrow [0, \infty)$$

satisfying, for every $v, v_1, v_2 \in V$ and every $\lambda \in \mathbb{K}$,

1) positive definiteness: $\|v\| = 0$ if and only if $v = 0$;
2) homogeneity: $\|\lambda v\| = |\lambda|\,\|v\|$;
3) the triangle inequality: $\|v_1 + v_2\| \le \|v_1\| + \|v_2\|$.

A *seminorm* is a function $\|\cdot\| : V \to [0, \infty)$ satisfying the second and third properties, but not necessarily the first. A vector space equipped with a norm is a *normed space*.

Definition

A *metric* on a set A is a function

$$d : A \times A \longrightarrow [0, \infty)$$

satisfying, for every $x, y, z \in A$,

1) identification: $d(x, y) = 0$ if and only if $x = y$;

2) symmetry: $d(x,y) = d(y,x)$;

3) the triangle inequality: $d(x,z) \le d(x,y) + d(y,z)$.

Theorem

If $\|\cdot\|$ is a norm on a vector space V, then

$$d(v,w) = \|v-w\|, \qquad v,w \in V,$$

defines a metric on V. It is called the metric induced by the norm.

Proof

The three metric properties follow from the norm properties:

$$\begin{aligned} d(v,w) = \|v-w\| = 0 \quad &\iff \quad v-w=0 \quad \iff \quad v=w, \\ \|v-w\| = \|(-1)(w-v)\| &= \|w-v\|, \\ d(x,z) = \|x-z\| \le \|x-y\| + \|y-z\| &= d(x,y) + d(y,z). \end{aligned}$$

Thus $d(x,y) = \|x-y\|$ is indeed a metric on V.

Example

On $\mathbb{R}^n$ or $\mathbb{C}^n$, the Euclidean norm is

$$\|x\|_2 = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2}.$$

Definition

The ℓ^∞-norm on $\mathbb{R}^n$ or $\mathbb{C}^n$ is

$$\|x\|_\infty = \max\{|x_1|, |x_2|, \ldots, |x_n|\}, \qquad x = (x_1, x_2, \ldots, x_n) \in \mathbb{K}^n.$$

More generally, for $1 \le p < \infty$, the ℓ^p-norm is

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}.$$

Remark

The expression $\|\cdot\|_p$ is a norm on $\mathbb{R}^n$ or $\mathbb{C}^n$; its triangle inequality is Minkowski's inequality, proved on page 27. The next proposition explains the notation $\|\cdot\|_\infty$.

Proposition

As $p \to \infty$, $\|x\|_p \to \|x\|_\infty$.

Proof

For every $x \in \mathbb{K}^n$ and every p,

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \leq n^{1/p} \max_i |x_i|,$$

while

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \geq \left[(\max_i |x_i|)^p\right]^{1/p} = \max_i |x_i|.$$

Therefore

$$\max_i |x_i| \leq \|x\|_p \leq n^{1/p} \max_i |x_i|.$$

The left and right sides both converge to $\max_i |x_i|$ as $p \to \infty$. The squeeze theorem gives $\|x\|_p \to \max_i |x_i| = \|x\|_\infty$.

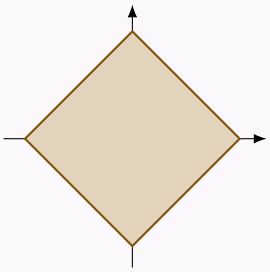
Recall: balls

In a metric space X, a ball of radius r centered at $x \in X$, denoted by $B(x, r)$, is

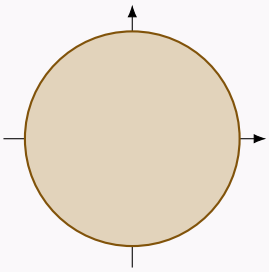
$$B(x, r) = \{y \in X \mid d(x, y) \leq r\}, \qquad r \in \mathbb{R}_{\geq 0}.$$

Unit balls for different ℓ^p-norms

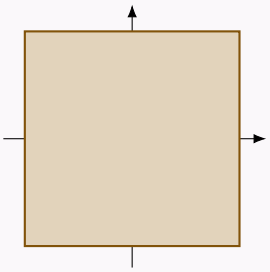
In $\mathbb{R}^2$, the unit balls for $p = 1, 2, \infty$ have the following shapes.



$p = 1$, $\|x\|_1 = |x_1| + |x_2|$



$p = 2$, $\|x\|_2 = (x_1^2 + x_2^2)^{1/2}$



$p = \infty$, $\|x\|_\infty = \max\{|x_1|, |x_2|\}$

More generally, these balls are nested:

$$B_1(0, 1) \subset B_2(0, 1) \subset B_p(0, 1) \subset B_\infty(0, 1).$$

This "sandwiching" eventually tells us that the p-norms are equivalent; we will return to this later in the course.

We now move to infinite-dimensional vector spaces.

Definition

Let X be a metric space. Define

$$C_\infty(X) = \{f : X \to \mathbb{C} \mid f \text{ is continuous and bounded}\}.$$

The subscript ∞ emphasizes that f is bounded, distinguishing this space from

$$C(X) = \{f : X \to \mathbb{C} \mid f \text{ is continuous}\}.$$

Indeed, $C_\infty(X)$ is a vector space.

Example

$$C_\infty([0,1]) = C([0,1]),$$

because continuous functions on a closed and bounded interval are bounded by the extreme value theorem.

Definition

Define $\|\cdot\|_\infty : C_\infty(X) \to \mathbb{R}_{\geq 0}$ by

$$\|u\|_\infty = \sup\{|u(x)| : x \in X\} = \sup_{x \in X} |u(x)|.$$

Here $|\cdot|$ means absolute value for a real number and the norm, or absolute value, for a complex number.

Theorem

The expression $\|\cdot\|_\infty$ defined above is a norm on $C_\infty(X)$.

Proof

For positive definiteness, suppose $\|u\|_\infty = 0$. Then, for every $x \in X$,

$$0 \leq |u(x)| \leq \sup_{x \in X} |u(x)| = 0,$$

so $|u(x)| = 0$ and hence $u(x) = 0$. Conversely, if $u(x) = 0$ for every $x \in X$, then $\|u\|_\infty = 0$. For homogeneity, if $\lambda \in \mathbb{K}$ and $u \in C_\infty(X)$, then

$$\|\lambda u\|_\infty = \sup_{x \in X} |\lambda u(x)| = |\lambda| \sup_{x \in X} |u(x)| = |\lambda| \, \|u\|_\infty.$$

For the triangle inequality, if $u, v \in C_\infty(X)$, then, for every $x \in X$,

$$|u(x) + v(x)| \leq |u(x)| + |v(x)| \leq \|u\|_\infty + \|v\|_\infty.$$

Thus $\|u\|_\infty + \|v\|_\infty$ is an upper bound for $|u(x) + v(x)|$. Taking the supremum gives

$$\|u + v\|_\infty \leq \|u\|_\infty + \|v\|_\infty.$$

A useful supremum technique

When we want to show

$$\sup_{x \in X} |u(x) + v(x)| \leq \varepsilon$$

or a similar inequality, it is enough to show first that ε is an upper bound and then take the

supremum.

Notation and convergence

We refer to $\|\cdot\|_\infty$ on $C_\infty(X)$ as the *uniform norm*, *supremum norm*, or *infinity norm*. We say $u_n \to u$ in $C_\infty(X)$ if and only if

$$\lim_{n\to\infty} \|u_n - u\|_\infty = 0.$$

Equivalently,

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N : \sup_{x\in X} |u_n(x) - u(x)| < \varepsilon,$$

which is equivalent to

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N \; \forall x \in X : |u_n(x) - u(x)| < \varepsilon.$$

Thus convergence in this metric is exactly uniform convergence on X.

Definition

For $1 \leq p \leq \infty$, define the ℓ^p-space by

$$\ell^p = \left\{ a = \{a_j\}_{j=1}^\infty \;\middle|\; \|a\|_p < \infty \right\},$$

where

$$\|a\|_p = \begin{cases} \left(\sum_{j=1}^\infty |a_j|^p \right)^{1/p}, & 1 \leq p < \infty, \\ \sup_{1\leq j<\infty} |a_j|, & p = \infty. \end{cases}$$

The finite-dimensional ℓ^p-norms above were defined only on $\mathbb{R}^n$ and $\mathbb{C}^n$. The space ℓ^p is a vector space, and this norm makes it a normed vector space.

Example

The sequence $\{1/j\}_{j=1}^\infty$ belongs to ℓ^p for $p > 1$, but does not belong to ℓ^1, because the harmonic series diverges:

$$\sum_{j=1}^\infty \frac{1}{j} = \infty.$$

1.2 Banach spaces: an analogue of the completeness of $\mathbb{R}$

Definition

A normed vector space is a *Banach space* if it is complete with respect to the metric induced by the norm; that is, every Cauchy sequence in it converges in it.

Example

The spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ are complete with respect to every ℓ^p-norm and are therefore Banach spaces.

Proof 1 (for $\mathbb{R}^n$)

Suppose $\mathbb{R}^n$ is equipped with an ℓ^p-norm, $1 \le p \le \infty$. We want to show that a sequence $\{v_n\}_{n=1}^{\infty} \subset \mathbb{R}^n$ is Cauchy if and only if it is convergent.
If $\{v_n\}$ is Cauchy, then it is bounded. By the Bolzano–Weierstrass theorem for Euclidean space $\mathbb{R}^n$, it has a convergent subsequence. A Cauchy sequence with a convergent subsequence converges to the same limit. Therefore a Cauchy sequence converges.
The converse follows directly from the triangle inequality for $\|\cdot\|_p$.

Proof 2 (for $\mathbb{C}$, $\mathbb{R}^n$, and $\mathbb{C}^n$)

First, $\mathbb{C}$ is complete with respect to the usual norm, the absolute value. Suppose $\{z_n\}_{n=1}^{\infty}$ is a Cauchy sequence in $\mathbb{C}$, with $z_n = x_n + iy_n$. The sequence $\{z_n\}$ is Cauchy if and only if both $\{x_n\}$ and $\{y_n\}$ are Cauchy sequences in $\mathbb{R}$.
Indeed, if $\{z_n\}$ is Cauchy, then for every $\varepsilon > 0$ there exists $M \in \mathbb{N}$ such that, for $n, \ell \ge M$,

$$|z_n - z_\ell| < \sqrt{\varepsilon}.$$

Since

$$|z_n - z_\ell|^2 = (x_n - x_\ell)^2 + (y_n - y_\ell)^2,$$

both $|x_n - x_\ell|$ and $|y_n - y_\ell|$ are less than $\sqrt{\varepsilon}$. Hence both real sequences are Cauchy. Conversely, suppose $\{x_n\}$ and $\{y_n\}$ are Cauchy. Given $\varepsilon > 0$, choose $M \in \mathbb{N}$ such that, for $n, \ell \ge M$,

$$|x_n - x_\ell| < \sqrt{\varepsilon/2}, \qquad |y_n - y_\ell| < \sqrt{\varepsilon/2}.$$

Then $|z_n - z_\ell| < \sqrt{\varepsilon}$, so $\{z_n\}$ is Cauchy.
By completeness of $\mathbb{R}$, there are $x, y \in \mathbb{R}$ such that $x_n \to x$ and $y_n \to y$. Put $z = x + iy$. Given $\varepsilon > 0$, for all sufficiently large n,

$$|x_n - x| < \frac{\varepsilon}{2}, \qquad |y_n - y| < \frac{\varepsilon}{2}.$$

Therefore

$$\begin{aligned} |z_n - z| &= |x_n + iy_n - (x + iy)| \\ &\le |x_n - x| + |i|\,|y_n - y| \\ &= |x_n - x| + |y_n - y| < \varepsilon. \end{aligned}$$

Thus every Cauchy sequence in $\mathbb{C}$ converges in $\mathbb{C}$, and $\mathbb{C}$ is complete with respect to the complex absolute-value norm.
Next, let $\mathbb{K}$ denote $\mathbb{R}$ or $\mathbb{C}$. We claim that $\mathbb{K}^n$ is complete with respect to the ℓ^p-norm. Let $\{Y_j\}_{j=1}^{\infty}$ be Cauchy in $\mathbb{K}^n$, where

$$Y_j = (z_{j1}, z_{j2}, \ldots, z_{jn}).$$

For s, ℓ sufficiently large,

$$\|Y_s - Y_\ell\|_p = \left(\sum_{i=1}^{n} |z_{si} - z_{\ell i}|^p \right)^{1/p} < \varepsilon.$$

It follows that, for $i = 1, 2, \ldots, n$,

$$|z_{si} - z_{\ell i}|^p < \varepsilon^p, \qquad \text{and hence} \qquad |z_{si} - z_{\ell i}| < \varepsilon.$$

Thus each component sequence $\{z_{ji}\}_{j=1}^{\infty}$ is Cauchy in $\mathbb{K}$. Since $\mathbb{R}$ and $\mathbb{C}$ are complete, for each i there is a $z_i \in \mathbb{K}$ with $z_{ji} \to z_i$.
Choose $N_0 \in \mathbb{N}$ such that, for $j \geq N_0$,

$$|z_{ji} - z_i| < n^{-1/p}\varepsilon \qquad \text{for } i = 1, 2, \ldots, n.$$

Let $Y = (z_1, z_2, \ldots, z_n) \in \mathbb{K}^n$. Then

$$\|Y_j - Y\|_p = \left(\sum_{i=1}^{n} |z_{ji} - z_i|^p\right)^{1/p} < \left(\sum_{i=1}^{n} (n^{-1/p}\varepsilon)^p\right)^{1/p} = \varepsilon.$$

Therefore every Cauchy sequence in $\mathbb{K}^n$ converges in $\mathbb{K}^n$. Thus $\mathbb{R}^n$ and $\mathbb{C}^n$ are complete.

Theorem

Let X be a metric space. The space of bounded continuous functions on X is complete. Thus $C_\infty(X)$ is a Banach space.

Proof

We want to show that every Cauchy sequence $\{u_n\}$ in $C_\infty(X)$ has a limit in $C_\infty(X)$. We first construct a candidate limit and then show that it lies in $C_\infty(X)$.

Step 1: the sequence is bounded. There is a $B > 0$ such that $\|u_n\|_\infty \leq B$ for every $n \in \mathbb{N}$. Because $\{u_n\}$ is Cauchy, there is $N_0 \in \mathbb{N}$ such that $\|u_n - u_m\|_\infty < 1$ whenever $n, m \geq N_0$. Hence, for $n \geq N_0$,

$$\|u_n\|_\infty \leq \|u_n - u_{N_0}\|_\infty + \|u_{N_0}\|_\infty < 1 + \|u_{N_0}\|_\infty.$$

For every $n \in \mathbb{N}$, therefore,

$$\|u_n\|_\infty \leq \sum_{i=1}^{N_0} \|u_i\|_\infty + 1 =: B. \tag{*}$$

The finiteness here follows because $\|u\|_\infty = \sup_{x \in X} |u(x)|$ and each u is continuous and bounded.
Step 2: a candidate limit. Fix $x \in X$. For every $n, m \in \mathbb{N}$,

$$|(u_n - u_m)(x)| \leq \sup_{x \in X} |(u_n - u_m)(x)| = \|u_n - u_m\|_\infty. \tag{**}$$

Since $\{u_n\}$ is Cauchy, $\{u_n(x)\}_{n=1}^{\infty}$ is Cauchy in $\mathbb{C}$ for every $x \in X$. Because $\mathbb{C}$ is complete, for each $x \in X$ there is a value $u(x) \in \mathbb{C}$ such that

$$u_n(x) \longrightarrow u(x).$$

In other words, this constructs a function $u : X \to \mathbb{C}$, our candidate for the limit. It remains to show that $u \in C_\infty(X)$ and that $u_n \to u$ in the uniform norm.

Step 3: the candidate is bounded. Since $u_n(x) \to u(x)$ for every $x \in X$,

$$|u(x)| = \lim_{n \to \infty} |u_n(x)|.$$

Moreover, $|u_n(x)| \le \|u_n\|_\infty$. Since limits preserve loose inequalities,

$$|u(x)| = \lim_{n\to\infty} |u_n(x)| \le B$$

by $(*)$. Thus u is bounded. **Step 4: uniform convergence and continuity.** It remains to show that u is continuous and that $u_n \to u$ under the uniform norm. Let $\varepsilon > 0$. Because $\{u_n\}$ is Cauchy, there is $N \in \mathbb{N}$ such that, whenever $n, m \ge N$,

$$\|u_n - u_m\|_\infty < \frac{\varepsilon}{2}.$$

Then, for every $x \in X$,

$$|u_n(x) - u_m(x)| \le \|u_n - u_m\|_\infty < \frac{\varepsilon}{2}.$$

Take the limit as $m \to \infty$. From the definition $u(x) = \lim_{m\to\infty} u_m(x)$, we obtain, for every $n \ge N$ and every $x \in X$,

$$|u_n(x) - u(x)| \le \frac{\varepsilon}{2}. \qquad (***)$$

Therefore

$$\|u_n - u\|_\infty = \sup_{x\in X} |u_n(x) - u(x)| \le \frac{\varepsilon}{2} < \varepsilon.$$

Hence $u_n \to u$ under the infinity norm on $C_\infty(X)$. This also follows from $(***)$ and the uniform-convergence criterion on page 6.
The uniform limit of a sequence of continuous functions is continuous, so u is continuous. Consequently $u \in C_\infty(X)$ and $u_n \to u$ under $\|\cdot\|_\infty$. Every Cauchy sequence in $C_\infty(X)$ therefore converges in $C_\infty(X)$, so $C_\infty(X)$ is complete and hence Banach.

Homework theorem

The space

$$c_0 = \left\{ a = \{a_j\}_{j=1}^\infty \in \ell^\infty \;\middle|\; \lim_{j\to\infty} a_j = 0 \right\}$$

is a Banach space.

Proof

We denote elements of ℓ^∞ by α instead of a, because we need to express a Cauchy sequence in ℓ^∞, and a_j is already used. Suppose $\{\alpha_i\}_{i=1}^\infty$ is a Cauchy sequence in c_0, where

$$\alpha_i = \{a_{ij}\}_{j=1}^\infty.$$

Then:

a) for every $\varepsilon > 0$, there is $M \in \mathbb{N}$ such that, for $n, m \ge M$, $\|\alpha_n - \alpha_m\|_\infty < \varepsilon$;

b) $\lim_{j\to\infty} a_{ij} = 0$, because $\alpha_i \in c_0$;

c) $\|\alpha_i\|_\infty < \infty$ for every i, because $\alpha_i \in c_0 \subset \ell^\infty$.

We want to find $\alpha \in c_0$ such that $\|\alpha_i - \alpha\|_\infty < \varepsilon$ for all sufficiently large i. From

$$\|\alpha_n - \alpha_m\|_\infty < \varepsilon \quad \Longrightarrow \quad \sup_j |a_{nj} - a_{mj}| < \varepsilon,$$

we obtain

$$|a_{nj} - a_{mj}| \leq \sup_j |a_{nj} - a_{mj}| < \varepsilon.$$

Thus, for each $j \in \mathbb{N}$, the sequence $\{a_{ij}\}_{i=1}^{\infty}$ is a Cauchy sequence in $\mathbb{C}$. By completeness of $\mathbb{C}$, there is an $a_j \in \mathbb{C}$ such that $a_{ij} \to a_j$ as $i \to \infty$. Combine these pointwise limits in the sequence

$$\alpha = \{a_j\}_{j=1}^{\infty} = (a_1, a_2, \ldots), \qquad a_j = \lim_{i\to\infty} a_{ij} \in \mathbb{C}.$$

This is our candidate for the limit of α_i. We now want to show that $\alpha \in c_0$. For this we need $\|\alpha\|_\infty < \infty$ and $\lim_{j\to\infty} a_j = 0$, and we must also show that $\|\alpha_i - \alpha\|_\infty < \varepsilon$.
Because $a_j = \lim_{i\to\infty} a_{ij}$,

$$|a_j| \leq \sup_i |a_{ij}| \leq \sup_i \left(\sup_j |a_{ij}| \right) = \sup_i \|\alpha_i\|_\infty, \qquad j \in \mathbb{N}.$$

By part (c), $\|\alpha_i\|_\infty < \infty$ for every i, so

$$|a_j| \leq \sup_i \|\alpha_i\|_\infty < \infty.$$

Therefore

$$\|\alpha\|_\infty = \sup_j |a_j| < \infty,$$

and hence $\alpha \in \ell^\infty$.
Since $\lim_{j\to\infty} a_{ij} = 0$ for every i, given $\varepsilon > 0$ and $i \in \mathbb{N}$, there is $M_i \in \mathbb{N}$ such that, for $j \geq M_i$,

$$|a_{ij}| < \frac{\varepsilon}{2}.$$

As $a_j = \lim_{i\to\infty} a_{ij}$, we obtain $|a_j| \leq \varepsilon/2 < \varepsilon$, and therefore $\lim_{j\to\infty} a_j = 0$. Thus $\alpha = \{a_j\}_{j=1}^{\infty} \in c_0 \subset \ell^\infty$.
Finally, we want to show that $\alpha_i \to \alpha$ as $i \to \infty$. We have

$$\|\alpha_i - \alpha\|_\infty = \sup_j |a_{ij} - a_j|.$$

Recall the supremum technique on page 6–6. Since $a_{ij} \to a_j$ as $i \to \infty$, for every j there exists $N_j \in \mathbb{N}$ such that, when $i \geq N_j$,

$$|a_{ij} - a_j| < \frac{\varepsilon}{2}.$$

Consequently

$$\|\alpha_i - \alpha\|_\infty = \sup_j |a_{ij} - a_j| \leq \frac{\varepsilon}{2} < \varepsilon$$

for sufficiently large i. Thus $\alpha_i \to \alpha$ under the uniform norm of ℓ^∞. Every Cauchy sequence in c_0 converges in c_0, so c_0 is complete and hence a Banach space.

Example

The space ℓ^p is a Banach space with respect to the ℓ^p-norm.

2 Bounded Linear Operators

Let V be a normed vector space.

Summable and absolutely summable series

A series

$$\sum_{n=1}^{\infty} v_n$$

in V is *summable* if its sequence of partial sums

$$\left\{\sum_{n=1}^{N} v_n\right\}_{N=1}^{\infty}$$

converges in V. It is *absolutely summable* if the scalar series

$$\sum_{n=1}^{\infty} \|v_n\|$$

converges.

Theorem

If $\sum_{n=1}^{\infty} v_n$ is absolutely summable, then its sequence of partial sums is a Cauchy sequence in V.

Remark

This is weaker than the corresponding statement for series in $\mathbb{R}$, since a general normed vector space need not be complete.

Proof

Since the scalar series $\sum_{n=1}^{\infty} \|v_n\|$ converges, its sequence of partial sums is Cauchy. Thus, given $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that, for $m > n \geq N$,

$$\sum_{j=n+1}^{m} \|v_j\| < \varepsilon.$$

By the triangle inequality,

$$\left\|\sum_{j=1}^{m} v_j - \sum_{j=1}^{n} v_j\right\| = \left\|\sum_{j=n+1}^{m} v_j\right\| \leq \sum_{j=n+1}^{m} \|v_j\| < \varepsilon.$$

Hence the sequence of partial sums is Cauchy in V.

An absolutely summable series that is not summable in $\mathbb{Q}$

Consider $\mathbb{Q}$ with its usual norm. The series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)} = 1 - 2\log 2$$

does not converge to an element of $\mathbb{Q}$. On the other hand,

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n(n+1)} \right| = 1.$$

Thus the series is absolutely summable in $\mathbb{Q}$ but is not summable there. The space $\mathbb{Q}$ is not a Banach space.

Theorem

A normed vector space V is a Banach space if and only if every absolutely summable series in V is summable.

Proof

Suppose first that V is a Banach space and $\sum_{n=1}^{\infty} v_n$ is absolutely summable. By the preceding theorem, the sequence of partial sums is Cauchy. Since V is complete, this sequence converges, so the series is summable.
Conversely, suppose that every absolutely summable series in V is summable, and let $\{v_n\}_{n=1}^{\infty}$ be a Cauchy sequence in V. For every $k \in \mathbb{N}$, choose $N_k \in \mathbb{N}$ such that

$$n, m \geq N_k \quad \Longrightarrow \quad \|v_n - v_m\| < 2^{-k}.$$

Set

$$n_k = N_1 + \cdots + N_k.$$

Then $n_{k+1}, n_k \geq N_k$, and therefore

$$\|v_{n_{k+1}} - v_{n_k}\| < 2^{-k}.$$

Consequently,

$$\sum_{k=1}^{\infty} \|v_{n_{k+1}} - v_{n_k}\| \leq \sum_{k=1}^{\infty} 2^{-k} < \infty.$$

The series $\sum_{k=1}^{\infty} (v_{n_{k+1}} - v_{n_k})$ is absolutely summable, so it is summable by assumption.
Its partial sums telescope:

$$\sum_{k=1}^{K} (v_{n_{k+1}} - v_{n_k}) = v_{n_{K+1}} - v_{n_1}.$$

It follows that the subsequence $\{v_{n_k}\}$ converges. A Cauchy sequence with a convergent subsequence converges to the same limit. Hence $\{v_n\}$ converges, and V is complete.

2.1 Operators and Functionals

Operators and functionals are the analogues of matrices and dot products.

An integral operator

Let $K \in C([0,1] \times [0,1])$, and define

$$Tf(x) = \int_0^1 K(x,y) f(y)\, dy.$$

Then

$$T : C([0,1]) \longrightarrow C([0,1]).$$

Such integral operators are inverse operators of differential operators.

For $f \in C([0,1])$, the function Tf is continuous. Moreover, if $\lambda_1, \lambda_2 \in \mathbb{K}$ and $f_1, f_2 \in C([0,1])$, then

$$\begin{aligned} T(\lambda_1 f_1 + \lambda_2 f_2)(x) &= \int_0^1 K(x,y)(\lambda_1 f_1(y) + \lambda_2 f_2(y))\, dy \\ &= \lambda_1 T f_1(x) + \lambda_2 T f_2(x). \end{aligned}$$

Linear transformation or linear operator

Let V and W be vector spaces over $\mathbb{K}$. A map $T : V \to W$ is *linear* if

$$T(\lambda_1 v_1 + \lambda_2 v_2) = \lambda_1 T v_1 + \lambda_2 T v_2$$

for all $v_1, v_2 \in V$ and all $\lambda_1, \lambda_2 \in \mathbb{K}$. Such a map is called a *linear transformation* or a *linear operator*.

We recall some multivariable analysis. Continuity for functions on subsets of $\mathbb{R}^n$ is defined in the same way as it is for functions on the real line.

Cluster point

Let $S \subset \mathbb{R}^n$. A point $x_0 \in \mathbb{R}^n$ is a *cluster point* of S if, for every $\delta > 0$, there exists $x \in S$ such that

$$0 < \|x - x_0\| < \delta.$$

Limit of a vector-valued function

Let $S \subset \mathbb{R}^n$, let x_0 be a cluster point of S, and let $f : S \to \mathbb{R}^k$. We say that

$$\lim_{x \to x_0} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in S, \quad 0 < \|x - x_0\| < \delta \quad \Longrightarrow \quad \|f(x) - L\| < \varepsilon.$$

Theorem

If $f = (f_1, \dots, f_k) : S \to \mathbb{R}^k$, then

$$\lim_{x \to x_0} f(x) = L = (L_1, \dots, L_k)$$

if and only if

$$\lim_{x \to x_0} f_j(x) = L_j \qquad (j = 1, \dots, k).$$

Continuity

Let $S \subset \mathbb{R}^n$ and $f : S \to \mathbb{R}^k$. The function f is *continuous at* $x_0 \in S$ if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in S, \quad \|x - x_0\| < \delta \quad \Longrightarrow \quad \|f(x) - f(x_0)\| < \varepsilon.$$

It is *continuous* if it is continuous at every point of S.

For an operator T, we usually write Tx instead of $T(x)$.

Recall the sequential criteria for limits and continuity. The same statements, with the same proofs, hold for maps between normed vector spaces V and W.

Theorem

Let V and W be normed vector spaces. A map $T : V \to W$ is continuous if and only if $T^{-1}(U)$ is open in V for every open set $U \subset W$.

Proof

Suppose that T is continuous and $U \subset W$ is open. Let $v \in T^{-1}(U)$. Since $Tv \in U$ and U is open, there exists $\varepsilon > 0$ such that

$$B_W(Tv, \varepsilon) \subset U.$$

By continuity of T at v, there exists $\delta > 0$ such that

$$\|x - v\|_V < \delta \quad \Longrightarrow \quad \|Tx - Tv\|_W < \varepsilon.$$

Thus $B_V(v, \delta) \subset T^{-1}(U)$, so $T^{-1}(U)$ is open.
Conversely, suppose that the inverse image of every open set in W is open in V. Let $v \in V$ and $\varepsilon > 0$. The set $T^{-1}(B_W(Tv, \varepsilon))$ is open and contains v. Therefore there exists $\delta > 0$ such that

$$B_V(v, \delta) \subset T^{-1}(B_W(Tv, \varepsilon)).$$

It follows that

$$\|x - v\|_V < \delta \quad \Longrightarrow \quad \|Tx - Tv\|_W < \varepsilon.$$

Hence T is continuous at v. Since v was arbitrary, T is continuous.

We use the notation

$$B(v, r) = \{x : \|x - v\| < r\}, \qquad \overline{B}(v, r) = \{x : \|x - v\| \le r\}$$

for open and closed balls, respectively.

Linearity gives a simpler characterization

For linear maps, continuity has a particularly simple characterization. Every linear map between finite-dimensional normed vector spaces is continuous. This need not be true for linear maps between infinite-dimensional Banach spaces.

For the next theorem, we temporarily distinguish the two norms by writing $\|\cdot\|_V$ and $\|\cdot\|_W$.

Theorem

Let V and W be normed vector spaces, and let $T : V \to W$ be linear. Then T is continuous if and only if there exists $C > 0$ such that

$$\|Tv\|_W \leq C\|v\|_V \qquad \text{for every } v \in V.$$

Bounded linear operator

The estimate above says that T maps bounded sets to bounded sets. In this case T is called a *bounded linear operator*. It does not mean that the entire image of T is a bounded set. Indeed, the only linear map whose entire image is bounded is the zero map. To see this, suppose $Tv \neq 0$ for some $v \in V$. Given $R > 0$, the Archimedean property gives $n \in \mathbb{N}$ such that

$$n\|Tv\|_W > R.$$

Then $T(nv) = nTv$, so $\|T(nv)\|_W > R$, and the image is not bounded.

Proof of the theorem

Suppose first that there is a $C > 0$ such that $\|Tv\|_W \leq C\|v\|_V$ for every $v \in V$. If $v_n \to v$, then

$$\|Tv_n - Tv\|_W = \|T(v_n - v)\|_W \leq C\|v_n - v\|_V \longrightarrow 0.$$

Thus $Tv_n \to Tv$, and the sequential criterion shows that T is continuous.
Conversely, suppose that T is continuous. Since the inverse image of an open set is open,

$$T^{-1}(B_W(0,1))$$

is an open set containing 0. Hence there is an $r > 0$ such that

$$B_V(0,r) \subset T^{-1}(B_W(0,1)).$$

If $v \neq 0$, then

$$\left\| \frac{r}{2\|v\|_V} v \right\|_V = \frac{r}{2} < r.$$

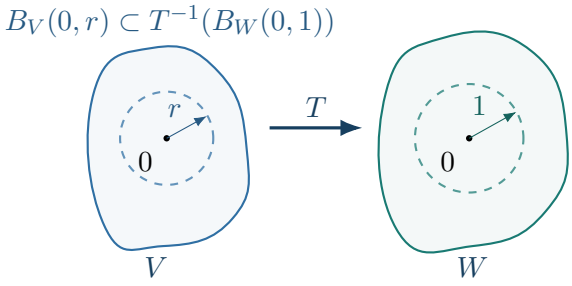
It follows that

$$\left\| T\left(\frac{r}{2\|v\|_V} v \right) \right\|_W < 1,$$

and therefore

$$\|Tv\|_W < \frac{2}{r}\|v\|_V.$$

The same estimate is immediate when $v = 0$. We may take $C = 2/r$.



From now on, we omit the subscripts on norms when it is clear which norm is being used.

The integral operator is bounded

Let $K \in C([0,1] \times [0,1])$, and define

$$Tf(x) = \int_0^1 K(x,y) f(y)\, dy.$$

Then $T : C([0,1]) \to C([0,1])$ is a bounded linear operator.

Since K is continuous on the compact set $[0,1] \times [0,1]$, its image is compact by the Heine–Borel theorem. In particular, K is bounded. For $f \in C([0,1])$,

$$\begin{aligned} |Tf(x)| &\le \int_0^1 |K(x,y)|\, |f(y)|\, dy \\ &\le \|K\|_\infty \|f\|_\infty. \end{aligned}$$

Taking the supremum over $x \in [0,1]$ gives

$$\|Tf\|_\infty \le \|K\|_\infty \|f\|_\infty.$$

Thus the preceding theorem applies with $C = \|K\|_\infty$. The function K is called the *kernel* of the integral operator.

Bounded sets, sequences, functions, and operators

Let V and W be normed vector spaces.

(i) A set $S \subset V$ is *bounded* if there is an $R > 0$ such that $S \subset B_V(0,R)$.

(ii) A sequence $\{v_n\}$ in V is *bounded* if its range $\{v_n : n \in \mathbb{N}\}$ is a bounded set.

(iii) A function $f : S \to W$ is *bounded* if $f(S)$ is a bounded set.

(iv) A linear operator is *bounded* if it maps bounded sets to bounded sets.

The space of bounded linear operators

For normed vector spaces V and W, let

$$\mathcal{B}(V,W) = \{T : V \to W : T \text{ is a bounded linear operator}\}.$$

For $T \in \mathcal{B}(V, W)$, define the *operator norm* by

$$\|T\| = \sup_{\|v\|=1} \|Tv\|.$$

Theorem

The operator norm is a norm on $\mathcal{B}(V, W)$.

Proof

If $\|T\| = 0$, then $\|Tv\| = 0$ for every v with $\|v\| = 1$. For $v \neq 0$, apply this to $v/\|v\|$ to obtain $Tv = 0$. Thus $T = 0$. The converse is immediate.
If $\lambda \in \mathbb{K}$, then

$$\|\lambda T\| = \sup_{\|v\|=1} \|\lambda T v\| = |\lambda| \sup_{\|v\|=1} \|Tv\| = |\lambda| \, \|T\|.$$

Finally,

$$\begin{aligned} \|T + S\| &= \sup_{\|v\|=1} \|(T+S)v\| \\ &\leq \sup_{\|v\|=1} (\|Tv\| + \|Sv\|) \\ &\leq \|T\| + \|S\|. \end{aligned}$$

Therefore the operator norm is a norm.

Two operator-norm estimates

For the integral operator,

$$\|Tf\|_\infty \leq \|K\|_\infty \|f\|_\infty,$$

so

$$\|T\| \leq \|K\|_\infty.$$

For every bounded linear operator T, the definition of the operator norm gives

$$\|Tv\| \leq \|T\| \, \|v\|.$$

The proof of the next theorem uses the following elementary fact.

Lemma

Let X and Y be metric spaces. If $f : X \to Y$ is continuous and $x_n \to x$ in X, then $f(x_n) \to f(x)$ in Y.

Corollary

The norm is a continuous function.

Proof of the corollary

The reverse triangle inequality gives

$$\big|\|x\| - \|y\|\big| \leq \|x - y\|.$$

Indeed, the triangle inequality implies

$$\|x\| \leq \|x - y\| + \|y\|,$$

so $\|x\| - \|y\| \leq \|x - y\|$. Interchanging x and y gives $\|y\| - \|x\| \leq \|x - y\|$, and the reverse triangle inequality follows. Hence the norm is Lipschitz continuous, and therefore continuous.

Theorem

Let V be a normed vector space and let W be a Banach space. Then $\mathcal{B}(V, W)$ is a Banach space.

Proof

We use the characterization of Banach spaces from page 12. Suppose that

$$\sum_{n=1}^{\infty} T_n, \qquad T_n \in \mathcal{B}(V, W),$$

is absolutely summable. Set

$$C = \sum_{n=1}^{\infty} \|T_n\| < \infty.$$

For each $v \in V$,

$$\sum_{n=1}^{\infty} \|T_n v\| \leq \sum_{n=1}^{\infty} \|T_n\| \, \|v\| = C\|v\| < \infty.$$

Thus $\sum_{n=1}^{\infty} T_n v$ is absolutely summable in W. Since W is complete, it is summable. Define

$$Tv = \sum_{n=1}^{\infty} T_n v.$$

The map $T : V \to W$ is linear, because for $\lambda_1, \lambda_2 \in \mathbb{K}$ and $v_1, v_2 \in V$,

$$\begin{aligned} T(\lambda_1 v_1 + \lambda_2 v_2) &= \sum_{n=1}^{\infty} T_n(\lambda_1 v_1 + \lambda_2 v_2) \\ &= \lambda_1 \sum_{n=1}^{\infty} T_n v_1 + \lambda_2 \sum_{n=1}^{\infty} T_n v_2 \\ &= \lambda_1 T v_1 + \lambda_2 T v_2. \end{aligned}$$

By the continuity of the norm,

$$\begin{aligned} \|Tv\| &= \left\| \sum_{n=1}^{\infty} T_n v \right\| \\ &= \lim_{N \to \infty} \left\| \sum_{n=1}^{N} T_n v \right\| \\ &\leq \sum_{n=1}^{\infty} \|T_n\| \, \|v\| \\ &= C\|v\|. \end{aligned}$$

Thus T is bounded. It remains to show that the partial sums converge to T in the operator norm. For $m \in \mathbb{N}$,

$$\begin{aligned}\left\| T - \sum_{n=1}^{m} T_n \right\| &= \sup_{\|v\|=1} \left\| Tv - \sum_{n=1}^{m} T_n v \right\| \\ &= \sup_{\|v\|=1} \left\| \sum_{n=m+1}^{\infty} T_n v \right\|.\end{aligned}$$

Therefore

$$\begin{aligned}\left\| T - \sum_{n=1}^{m} T_n \right\| &\leq \sup_{\|v\|=1} \sum_{n=m+1}^{\infty} \|T_n v\| \\ &\leq \sum_{n=m+1}^{\infty} \|T_n\| \longrightarrow 0.\end{aligned}$$

Hence $\sum_{n=1}^{\infty} T_n$ is summable in $\mathcal{B}(V, W)$. The characterization on page 12 now shows that $\mathcal{B}(V, W)$ is a Banach space.

Dual space and functionals

Let V be a normed vector space over $\mathbb{K}$. The *dual space* of V is

$$V' = \mathcal{B}(V, \mathbb{K}).$$

Since $\mathbb{K}$ is complete, V' is a Banach space. The elements of V' are called *functionals*.

Homework theorem: the dual of ℓ^p

Let $1 \leq p < \infty$, and let q be the Hölder conjugate of p, so that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then

$$(\ell^p)' \cong \ell^q.$$

In particular, ℓ^2 is self-dual. Also,

$$(\ell^1)' \cong \ell^\infty,$$

but the analogous statement for $p = \infty$ fails:

$$(\ell^\infty)' \ncong \ell^1.$$

The same phenomenon occurs for the L^p spaces: the dual space $(L^\infty)'$ is not L^1.

Proof of the homework theorem

Let $1 \leq p < \infty$, and let q be the Hölder conjugate of p, so that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

We prove the theorem in five claims.

Claim 1. If $A, B > 0$ and $t \in (0,1)$, then the function

$$f(x) = tx + (1-t)B - x^t B^{1-t}, \qquad x > 0,$$

has a minimum at $x = B$. Furthermore,

$$A^t B^{1-t} \leq tA + (1-t)B.$$

Claim 2 (Hölder's inequality). Let $n \in \mathbb{N}$, let $a_k, b_k \in \mathbb{C}$ for $1 \leq k \leq n$, and suppose $1 < p < \infty$ with $1/p + 1/q = 1$. Then

$$\sum_{k=1}^{n} |a_k b_k| \leq \left(\sum_{k=1}^{n} |a_k|^p \right)^{1/p} \left(\sum_{k=1}^{n} |b_k|^q \right)^{1/q}.$$

Claim 3. If $a = \{a_k\}_{k=1}^{\infty} \in \ell^p$ and $b = \{b_k\}_{k=1}^{\infty} \in \ell^q$, then

$$\sum_{k=1}^{\infty} |a_k b_k| \leq \|a\|_p \|b\|_q.$$

Claim 4. Let $b \in \ell^q$. Define $F_b : \ell^p \to \mathbb{C}$ by

$$F_b(a) = \sum_{k=1}^{\infty} a_k b_k, \qquad a \in \ell^p.$$

Then

$$F_b \in (\ell^p)' = \mathcal{B}(\ell^p, \mathbb{C}) \qquad \text{and} \qquad \|F_b\| = \|b\|_q.$$

Claim 5. The map

$$F : \ell^q \longrightarrow (\ell^p)', \qquad b \longmapsto F_b,$$

is a bounded bijective linear operator.
We now prove the claims.

Proof of Claim 1. We want to show

$$A^t B^{1-t} \leq tA + (1-t)B.$$

The function $f : \mathbb{R}_{>0} \to \mathbb{R}$ is differentiable, and

$$f'(x) = t - tB^{1-t}x^{t-1}, \qquad f''(x) = -t(t-1)B^{1-t}x^{t-2}.$$

If $f'(x) = 0$, then $x^{t-1} = B^{t-1}$, so $x = B$. Moreover, $f''(x) > 0$ for every $x > 0$. Thus f reaches a minimum at $x = B$. For every $A > 0$,

$$f(A) \geq f(B) = 0.$$

Therefore

$$tA + (1-t)B - A^t B^{1-t} \geq 0,$$

which proves

$$A^t B^{1-t} \leq tA + (1-t)B \qquad (A, B > 0,\ t \in (0,1)).$$

Proof of Claim 2. Apply Claim 1 with

$$A = |x|^p, \qquad B = |y|^q, \qquad t = \frac{1}{p}, \qquad 1 - t = \frac{1}{q}.$$

For every $x, y \in \mathbb{C}$, this gives

$$|x|\,|y| \le \frac{1}{p}|x|^p + \frac{1}{q}|y|^q. \tag{$*$}$$

Assume first that neither of the two finite norms below is zero. Applying $(*)$ term by term to the normalized quantities gives

$$\begin{aligned}
&\sum_{k=1}^{n} \frac{|a_k b_k|}{\left(\sum_{i=1}^{n} |a_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |b_i|^q\right)^{1/q}} \\
&\quad = \sum_{k=1}^{n} \frac{|a_k|}{\left(\sum_{i=1}^{n} |a_i|^p\right)^{1/p}} \frac{|b_k|}{\left(\sum_{i=1}^{n} |b_i|^q\right)^{1/q}} \\
&\quad \le \sum_{k=1}^{n} \left[\frac{1}{p} \frac{|a_k|^p}{\sum_{i=1}^{n} |a_i|^p} + \frac{1}{q} \frac{|b_k|^q}{\sum_{i=1}^{n} |b_i|^q}\right] \\
&\quad = \frac{1}{p} + \frac{1}{q} = 1.
\end{aligned}$$

Multiplying by the denominator, which is independent of k, yields

$$\sum_{k=1}^{n} |a_k b_k| \le \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |b_k|^q\right)^{1/q}.$$

The cases in which one of the two norms is zero are immediate.
The inequality $(*)$ is known as *Young's inequality.*

Proof of Claim 3. Recall that

$$\ell^p = \left\{a = \{a_j\}_{j=1}^{\infty} : \|a\|_p < \infty\right\}, \qquad \|a\|_p = \begin{cases} \left(\sum_{j=1}^{\infty} |a_j|^p\right)^{1/p}, & p < \infty, \\ \sup_{j \in \mathbb{N}} |a_j|, & p = \infty. \end{cases}$$

By Claim 2,

$$\sum_{k=1}^{n} |a_k b_k| \le \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |b_k|^q\right)^{1/q}.$$

Since

$$\sum_{k=1}^{n} |a_k|^p \le \sum_{k=1}^{\infty} |a_k|^p, \qquad \sum_{k=1}^{n} |b_k|^q \le \sum_{k=1}^{\infty} |b_k|^q,$$

and the series on the right exist because $a \in \ell^p$ and $b \in \ell^q$,

$$\sum_{k=1}^{n} |a_k b_k| \le \|a\|_p \|b\|_q.$$

The claim follows because limits in $\mathbb{R}$ preserve non-strict inequalities.

If $p = 1$, define $q = \infty$. Then

$$\sum_{k=1}^{\infty} |a_k b_k| \leq \sum_{k=1}^{\infty} |a_k| \sup_{k\in\mathbb{N}} |b_k| = \|a\|_1 \|b\|_\infty.$$

Proof of Claim 4. We first show that $F_b : \ell^p \to \mathbb{C}$ is linear and bounded. If $a_1 = \{a_{1k}\}_{k=1}^{\infty}$, $a_2 = \{a_{2k}\}_{k=1}^{\infty}$, and $\lambda_1, \lambda_2 \in \mathbb{C}$, then

$$\begin{aligned} F_b(\lambda_1 a_1 + \lambda_2 a_2) &= \sum_{k=1}^{\infty} (\lambda_1 a_{1k} + \lambda_2 a_{2k}) b_k \\ &= \lambda_1 \sum_{k=1}^{\infty} a_{1k} b_k + \lambda_2 \sum_{k=1}^{\infty} a_{2k} b_k \\ &= \lambda_1 F_b(a_1) + \lambda_2 F_b(a_2), \end{aligned}$$

by the linearity of series and limits. By Claim 3,

$$|F_b(a)| = \left| \lim_{n\to\infty} \sum_{k=1}^{n} a_k b_k \right| = \lim_{n\to\infty} \left| \sum_{k=1}^{n} a_k b_k \right| \leq \sum_{k=1}^{\infty} |a_k b_k| \leq \|a\|_p \|b\|_q.$$

Thus F_b is bounded, and $C = \|b\|_q$ works.
To prove that $\|F_b\| = \|b\|_q$, recall that

$$\|F_b\| = \sup_{\substack{a\in\ell^p \\ \|a\|_p = 1}} |F_b(a)|.$$

The estimate just proved shows that

$$\sup_{\|a\|_p = 1} |F_b(a)| \leq \|b\|_q.$$

It remains to find $a \in \ell^p$ such that

$$\|a\|_p = 1 \qquad \text{and} \qquad |F_b(a)| = \|b\|_q.$$

The idea is to choose a so that equality holds in the triangle inequality and

$$\sum_{k=1}^{\infty} a_k b_k = \left(\sum_{k=1}^{\infty} |b_k|^q \right)^{1/q}.$$

Assume first that $a_k b_k \geq 0$. We want

$$\sum_{k=1}^{\infty} a_k b_k = \left(\sum_{k=1}^{\infty} |b_k|^q \right)^{1-1/p}.$$

Equivalently,

$$\left(\sum_{k=1}^{\infty} a_k b_k \right) \left(\sum_{k=1}^{\infty} |b_k|^q \right)^{1/p} = \sum_{k=1}^{\infty} |b_k|^q.$$

This suggests

$$a_k b_k = \frac{|b_k|^q}{\|b\|_q^{q/p}}, \qquad a_k = |b_k|^{q-1}\|b\|_q^{-q/p}.$$

The condition $a_k b_k \geq 0$ is not guaranteed unless the terms are chosen carefully. Define

$$\operatorname{sgn}(b_k) = \begin{cases} \dfrac{b_k}{|b_k|}, & b_k \neq 0, \\ 0, & b_k = 0. \end{cases}$$

This definition is also valid for $b_k \in \mathbb{C}$. If $b_k \in \mathbb{C} \setminus \mathbb{R}$, however, one must be careful that

$$\operatorname{sgn}(b_k) = \frac{b_k}{|b_k|} \neq \frac{|b_k|}{b_k}.$$

For $b \neq 0$, consider

$$a_k = |b_k|^{q-1}\|b\|_q^{-q/p} \operatorname{sgn}(b_k)^{-1},$$

with $a_k = 0$ when $b_k = 0$. We claim that this choice works: namely, $a = \{a_k\}_{k=1}^{\infty} \in \ell^p$, $\|a\|_p = 1$, and

$$|F_b(a)| = \|b\|_q = \left(\sum_{k=1}^{\infty} |b_k|^q\right)^{1/q}.$$

Indeed, terms with $b_k = 0$ contribute zero, while for $b_k \neq 0$,

$$\begin{aligned} |F_b(a)| &= \left|\sum_{k=1}^{\infty} |b_k|^{q-1}\|b\|_q^{-q/p} \operatorname{sgn}(b_k)^{-1} b_k\right| \\ &= \left|\sum_{k=1}^{\infty} |b_k|^{q-1}\|b\|_q^{-q/p} \frac{|b_k|}{b_k} b_k\right| \\ &= \sum_{k=1}^{\infty} |b_k|^q \|b\|_q^{-q/p} \\ &= \left(\sum_{k=1}^{\infty} |b_k|^q\right)^{1-1/p} \\ &= \left(\sum_{k=1}^{\infty} |b_k|^q\right)^{1/q} = \|b\|_q. \end{aligned}$$

Now

$$\begin{aligned} \|a\|_p &= \left(\sum_{k=1}^{\infty} |a_k|^p\right)^{1/p} \\ &= \left(\sum_{k=1}^{\infty} |b_k|^{(q-1)p} \|b\|_q^{-q} \,|\operatorname{sgn}(b_k)|^p\right)^{1/p}. \end{aligned}$$

Since

$$\frac{1}{p} + \frac{1}{q} = 1 \quad \Longrightarrow \quad pq = p + q \quad \Longrightarrow \quad pq - p = q,$$

we have $(q-1)p = q$, and hence

$$\|a\|_p = \frac{(\sum_{k=1}^{\infty} |b_k|^q)^{1/p}}{(\|b\|_q^q)^{1/p}} = \frac{\|b\|_q^{q/p}}{\|b\|_q^{q/p}} = 1.$$

Therefore $a \in \ell^p$ and $\|a\|_p = 1$. We have shown both that

$$\sup_{\|a\|_p=1} |F_b(a)| \leq \|b\|_q$$

and that equality is attained. Thus $\|F_b\| = \|b\|_q$.

Proof of Claim 5. Recall that two vector spaces are isomorphic if there is a bijective map between them that preserves addition and scalar multiplication. By Claim 4,

$$\|F_b\| = \|b\|_q,$$

so $F : \ell^q \to (\ell^p)'$ is bounded, with constant $C = 1$.
To prove linearity, suppose $b_1 = \{b_{1k}\}_{k=1}^{\infty}$, $b_2 = \{b_{2k}\}_{k=1}^{\infty}$, and $\lambda_1, \lambda_2 \in \mathbb{C}$. For $a \in \ell^p$,

$$\begin{aligned} F_{\lambda_1 b_1 + \lambda_2 b_2}(a) &= \sum_{k=1}^{\infty} a_k(\lambda_1 b_{1k} + \lambda_2 b_{2k}) \\ &= \lambda_1 \sum_{k=1}^{\infty} a_k b_{1k} + \lambda_2 \sum_{k=1}^{\infty} a_k b_{2k} \\ &= \lambda_1 F_{b_1}(a) + \lambda_2 F_{b_2}(a). \end{aligned}$$

Thus F is linear.
For injectivity, suppose $F_{b_1} = F_{b_2}$. Equivalently, $F_b = 0$ for $b = b_1 - b_2$. We prove by contraposition that this implies $b = 0$. Suppose $b \neq 0$. As in Claim 4, take

$$a_k = |b_k|^{q-1} \operatorname{sgn}(b_k)^{-1} \|b\|_q^{-q/p}.$$

By Claim 4, $a \in \ell^p$. Since $b \neq 0$, there is an $s \in \mathbb{N}$ such that $b_s \neq 0$.
By the construction of a_k,

$$a_s b_s > 0, \qquad a_k b_k \geq 0 \quad \text{for every } k.$$

Therefore

$$F_b(a) = \sum_{k=1}^{\infty} a_k b_k \geq a_s b_s > 0.$$

Thus $F_b \neq 0$, proving injectivity.
It remains to prove surjectivity. Let

$$\varphi \in (\ell^p)' = \mathcal{B}(\ell^p, \mathbb{C}).$$

We want to find $b \in \ell^q$ such that $F_b = \varphi$. For each $k \in \mathbb{N}$, let

$$e^{(k)} = (0, 0, \ldots, 0, 1, 0, \ldots),$$

where the k-th entry is 1. Since $\|e^{(k)}\|_p = 1$, we have $e^{(k)} \in \ell^p$. Define

$$b = \{b_k\}_{k=1}^{\infty} = \{\varphi(e^{(k)})\}_{k=1}^{\infty}.$$

We claim that $b \in \ell^q$ and $F_b(a) = \varphi(a)$.
For $n \in \mathbb{N}$, construct $x^{(n)} = \{x_k^{(n)}\}_{k=1}^{\infty}$ by

$$x_k^{(n)} = \begin{cases} \dfrac{|b_k|^q}{b_k}, & b_k \neq 0 \text{ and } k \leq n, \\ 0, & b_k = 0 \text{ or } k > n. \end{cases}$$

Since $x^{(n)}$ has only finitely many nonzero terms, $x^{(n)} \in \ell^p$. By linearity and the definition of b_k,

$$\varphi(x^{(n)}) = \varphi\left(\sum_{k=1}^{\infty} x_k^{(n)} e^{(k)}\right) = \sum_{k=1}^{\infty} x_k^{(n)} b_k = \sum_{k=1}^{n} |b_k|^q.$$

Boundedness of φ now gives

$$\begin{aligned} \sum_{k=1}^{n} |b_k|^q &= |\varphi(x^{(n)})| \\ &\leq \|\varphi\| \, \|x^{(n)}\|_p \\ &= \|\varphi\| \left(\sum_{k=1}^{n} \left|\frac{|b_k|^q}{b_k}\right|^p\right)^{1/p}. \end{aligned}$$

Using $(q-1)p = q$, we obtain

$$\sum_{k=1}^{n} |b_k|^q \leq \|\varphi\| \left(\sum_{k=1}^{n} |b_k|^q\right)^{1/p}.$$

After dividing by the factor on the right, this becomes

$$\left(\sum_{k=1}^{n} |b_k|^q\right)^{1-1/p} = \left(\sum_{k=1}^{n} |b_k|^q\right)^{1/q} \leq \|\varphi\| < \infty$$

for every $n \in \mathbb{N}$. Letting $n \to \infty$ shows that $\|b\|_q < \infty$, so $b \in \ell^q$.
Finally, let $a \in \ell^p$ and define

$$a^{(n)} = (a_1, a_2, \ldots, a_n, 0, 0, \ldots).$$

By the choice $b_k = \varphi(e^{(k)})$,

$$\varphi(a^{(n)}) = \varphi\left(\sum_{k=1}^{n} a_k e^{(k)}\right) = \sum_{k=1}^{n} a_k \varphi(e^{(k)}) = \sum_{k=1}^{n} a_k b_k.$$

The sequence $\{a^{(n)}\}_{n=1}^{\infty}$ converges to a in ℓ^p. By the theorem on page 16, φ is continuous. Hence

$$\varphi(a) = \lim_{n\to\infty} \varphi(a^{(n)}) = \sum_{k=1}^{\infty} a_k b_k = F_b(a).$$

The series exists because $b \in \ell^q$, by Hölder's inequality in Claim 3. Thus $F_b = \varphi$, proving surjectivity and completing the proof.

The role of ℓ^2

The space ℓ^2 is the most important ℓ^p space because it is actually a Hilbert space.

Theorem

The dual space $(\ell^\infty)'$ is not isomorphic to ℓ^1. However,

$$(c_0)' \cong \ell^1,$$

where c_0 is the space introduced on page 10.

Homework theorem: Minkowski's inequality

For every $n \in \mathbb{N}$, all $a_k, b_k \in \mathbb{R}$ with $1 \leq k \leq n$, and $1 \leq p < \infty$,

$$\left(\sum_{k=1}^{n} |a_k + b_k|^p\right)^{1/p} \leq \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} + \left(\sum_{k=1}^{n} |b_k|^p\right)^{1/p}.$$

Remark

Minkowski's inequality is extremely important for showing later that L^p is a normed vector space.

Proof

By the triangle inequality,

$$|a_k + b_k| \leq |a_k| + |b_k|.$$

Therefore

$$|a_k + b_k|^p \leq |a_k|\,|a_k + b_k|^{p-1} + |b_k|\,|a_k + b_k|^{p-1},$$

and hence

$$\sum_{k=1}^{n} |a_k + b_k|^p \leq \sum_{k=1}^{n} |a_k|\,|a_k + b_k|^{p-1} + \sum_{k=1}^{n} |b_k|\,|a_k + b_k|^{p-1}.$$

Let q be the Hölder conjugate of p. By Hölder's inequality,

$$\begin{aligned}\sum_{k=1}^{n} |a_k|\,|a_k + b_k|^{p-1} &\leq \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |a_k + b_k|^{(p-1)q}\right)^{1/q} \\ &= \left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |a_k + b_k|^p\right)^{1-1/p}.\end{aligned}$$

Similarly,

$$\sum_{k=1}^{n} |b_k|\,|a_k + b_k|^{p-1} \leq \left(\sum_{k=1}^{n} |b_k|^p\right)^{1/p} \left(\sum_{k=1}^{n} |a_k + b_k|^p\right)^{1-1/p}.$$

It follows that

$$\sum_{k=1}^{n} |a_k + b_k|^p \leq \left[\left(\sum_{k=1}^{n} |a_k|^p\right)^{1/p} + \left(\sum_{k=1}^{n} |b_k|^p\right)^{1/p}\right] \left(\sum_{k=1}^{n} |a_k + b_k|^p\right)^{1-1/p}.$$

Dividing both sides by the final factor gives

$$\left(\sum_{k=1}^{n}|a_k+b_k|^p\right)^{1/p}\leq\left(\sum_{k=1}^{n}|a_k|^p\right)^{1/p}+\left(\sum_{k=1}^{n}|b_k|^p\right)^{1/p}.$$

3 Quotient Spaces and the Baire Category Theorem

3.1 Subspaces and Quotients

Subspace

Let V be a vector space. A subset $W\subset V$ is a *subspace* of V if, with the addition and scalar multiplication on V, it satisfies all ten axioms for a vector space.

Theorem: subspace criterion

Let V be a vector space and $W\subset V$. Then W is a subspace if and only if:

(1) $W\neq\varnothing$;

(2) $x,y\in W$ implies $x+y\in W$;

(3) $k\in\mathbb{K}$ and $x\in W$ imply $kx\in W$.

Theorem

Let W be a subspace of a Banach space V, equipped with the norm of V. Then W is a Banach space if and only if W is a closed subset of V with respect to the metric induced by the norm.

Proof

We first prove the following claim:

$$W \text{ is closed} \iff \text{every convergent sequence in } W \text{ converges to a point in } W.$$

Suppose W is closed and $\{x_n\}_{n=1}^{\infty}\subset W$ converges to $x\in V$. Then x is a limit point, or cluster point, of W, and closedness gives $x\in W$. Thus the sequence converges in W.
Conversely, suppose every convergent sequence in W converges in W. If x is a cluster point of W in V, we can construct a sequence $\{x_n\}\subset W$ such that $x_n\to x$. By assumption its limit belongs to W, so $x\in W$. Hence W contains all its cluster points and is closed.
With the claim proved, the theorem is immediate. Suppose first that W is a Banach space. Every convergent sequence in W is Cauchy, and every Cauchy sequence in W converges in W. By the claim, W is closed.
Conversely, suppose W is closed and let $\{x_n\}$ be a Cauchy sequence in W. Since V is a Banach space, $\{x_n\}$ converges in V. By the claim, it converges in W. Therefore W is a Banach space.

Closed versus compact

For comparison, recall that a set is *sequentially compact* if every sequence in it has a convergent subsequence whose limit belongs to the set. In a metric space, sequential compactness is equivalent to compactness. In $\mathbb{R}^n$, the Heine–Borel theorem further says that this is equivalent to being closed and bounded.

Quotient space

Let $W \subset V$ be a subspace. Define an equivalence relation on V by

$$v \sim v' \quad \Longleftrightarrow \quad v - v' \in W.$$

Let

$$[v] = \{v' \in V : v' \sim v\}$$

be the equivalence class of v. The *quotient space* is

$$V/W = \{[v] : v \in V\}.$$

We write

$$v + W = [v].$$

Equality of cosets

Two elements $v + W$ and $u + W$ of V/W are equal if and only if

$$v \sim u, \qquad \text{that is,} \qquad v - u \in W.$$

Recall that an equivalence relation is reflexive, symmetric, and transitive:

(1) $v \sim v$ for every $v \in V$;

(2) $v \sim v'$ if and only if $v' \sim v$;

(3) if $v \sim v'$ and $v' \sim v''$, then $v \sim v''$.

The relation defined above satisfies these properties.

Vector-space operations on the quotient

The quotient V/W is a vector space over $\mathbb{K}$, with

$$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$$

and

$$\lambda(v + W) = \lambda v + W.$$

The zero element is

$$W = 0 + W = w + W \qquad (w \in W).$$

Recall that a seminorm satisfies nonnegativity, homogeneity, and the triangle inequality, but may fail positive definiteness.

A seminorm that is not a norm

On the vector space of differentiable functions $f : \mathbb{R} \to \mathbb{R}$, define

$$\|f\| = \sup_x |f'(x)|.$$

This satisfies nonnegativity, homogeneity, and the triangle inequality, but not positive definiteness. For example, if $f = a$ is constant, then $\|f\| = 0$ although $f \neq 0$.

Theorem: a seminorm induces a quotient norm

Let $\|\cdot\|$ be a seminorm on V, and let

$$E = \{v \in V : \|v\| = 0\}.$$

Then E is a subspace of V, and the function on V/E defined by

$$\|v + E\|_{V/E} = \|v\|, \qquad v + E \in V/E,$$

is a norm.

Proof

First, E is a subspace. Let $\lambda_1, \lambda_2 \in \mathbb{K}$ and $v_1, v_2 \in E$. By homogeneity and the triangle inequality,

$$\|\lambda_1 v_1 + \lambda_2 v_2\| \leq |\lambda_1|\|v_1\| + |\lambda_2|\|v_2\| = 0.$$

Thus

$$0 \leq \|\lambda_1 v_1 + \lambda_2 v_2\| \leq 0,$$

so $\lambda_1 v_1 + \lambda_2 v_2 \in E$. The set E is nonempty because

$$\|0\| = \|0 \cdot 0\| = |0|\,\|0\| = 0.$$

We next check that the proposed norm is well-defined, or independent of the choice of representative. Suppose

$$v + E = v' + E.$$

Then $v - v' \in E$, so there is an $e \in E$ such that $v = v' + e$. By the triangle inequality for the seminorm,

$$\|v\| = \|v' + e\| \leq \|v'\| + \|e\| = \|v'\|.$$

Similarly, $\|v'\| \leq \|v\|$, and therefore

$$\|v + E\|_{V/E} = \|v' + E\|_{V/E}.$$

It remains to verify the norm axioms. For positive definiteness, suppose

$$\|v + E\|_{V/E} = 0.$$

Then $\|v\| = 0$, so $v \in E$, and hence $v + E = E$, the zero element in V/E. Conversely, if $v + E = E$, then $v \sim 0$, so $v \in E$ and $\|v + E\|_{V/E} = \|v\| = 0$.

For homogeneity, if $\lambda \in \mathbb{K}$, then

$$\|\lambda(v+E)\|_{V/E} = \|\lambda v + E\|_{V/E} = \|\lambda v\| = |\lambda|\,\|v\| = |\lambda|\,\|v+E\|_{V/E}.$$

For the triangle inequality,

$$\begin{aligned}\|(v_1+E)+(v_2+E)\|_{V/E} &= \|(v_1+v_2)+E\|_{V/E}\\ &= \|v_1+v_2\|\\ &\le \|v_1\| + \|v_2\|\\ &= \|v_1+E\|_{V/E} + \|v_2+E\|_{V/E}.\end{aligned}$$

Thus the function is a norm on V/E.

This gives one way to obtain a normed vector space from V. There is another.

Homework theorem: the quotient norm

Let V be a normed vector space and let $W \subsetneq V$ be a proper closed subspace. Define

$$\|\cdot\|_{V/W} : V/W \longrightarrow \mathbb{R}_{\ge 0}$$

by

$$\|v+W\|_{V/W} = \inf_{w\in W} \|v+w\|_V.$$

Then $\|\cdot\|_{V/W}$ is a norm on V/W.

Proof

The function is well-defined. Suppose $v+W = u+W$. Then

$$\{v+w : w\in W\} = \{u+w : w\in W\},$$

and therefore

$$\inf_{w\in W}\|v+w\| = \inf_{w\in W}\|u+w\|.$$

For positive definiteness, suppose

$$\|v+W\|_{V/W} = 0.$$

Then $\inf_{w\in W}\|v+w\| = 0$. For every $n\in\mathbb{N}$, choose $w_n \in W$ such that

$$0 \le \|v+w_n\| < \frac{1}{n}.$$

Thus $\|w_n - (-v)\| \to 0$, so $w_n \to -v$. Since W is closed, the closed-subspace criterion on page 28 gives $-v\in W$. Hence $v\in W$, so $v+W = W$.
Conversely, if $v+W = W$, then

$$0 \le \|v+W\|_{V/W} = \|W\|_{V/W} = \inf_{w\in W}\|w\| = 0.$$

Homogeneity follows from the homogeneity of the norm on V and the identity

$$\inf(\lambda A) = |\lambda|\inf(A).$$

For the triangle inequality, let $v_1, v_2 \in V$. Then

$$\begin{aligned}\|(v_1+W)+(v_2+W)\|_{V/W} &= \|(v_1+v_2)+W\|_{V/W}\\ &= \inf_{w\in W}\|(v_1+v_2)+w\|_V\\ &= \inf_{w\in W}\|(v_1+w)+(v_2+w)\|_V\\ &\le \inf_{w\in W}(\|v_1+w\|_V+\|v_2+w\|_V)\\ &= \inf_{w\in W}\|v_1+w\|_V+\inf_{w\in W}\|v_2+w\|_V\\ &= \|v_1+W\|_{V/W}+\|v_2+W\|_{V/W}.\end{aligned}$$

Therefore the quotient norm is a norm on V/W.

Homework theorem: quotients of Banach spaces

Let V be a Banach space, and let $W \subsetneq V$ be a closed subspace. Then V/W, equipped with the norm from the preceding theorem, is a Banach space.

Proof

We prove two claims.

Claim 1. For every $\varepsilon > 0$, there exists $v \in V$ such that

$$\|v\| = 1 \qquad \text{and} \qquad \|v+W\|_{V/W} \ge 1-\varepsilon.$$

Choose $u \in V \setminus W$. By positive definiteness of the quotient norm, $\|u+W\|_{V/W} > 0$. Since

$$\|u+W\|_{V/W} = \inf_{w\in W}\|u+w\|,$$

there is a $w \in W$ such that

$$\|u+W\|_{V/W} \le \|u+w\| \le \|u+W\|_{V/W} + \varepsilon\|u+W\|_{V/W}.$$

Set

$$v = \frac{u+w}{\|u+w\|}.$$

Then $\|v\| = 1$. Since $u+w \sim u$,

$$\begin{aligned}\|v+W\|_{V/W} &= \left\|\frac{u+w}{\|u+w\|}+W\right\|_{V/W}\\ &= \frac{1}{\|u+w\|}\|(u+w)+W\|_{V/W}\\ &= \frac{\|u+W\|_{V/W}}{\|u+w\|}\\ &\ge \frac{1}{1+\varepsilon} \ge 1-\varepsilon.\end{aligned}$$

Claim 2. Every absolutely summable series in V/W converges in V/W.

By the characterization on page 12, this claim proves that V/W is a Banach space. Suppose

$$\sum_{n=1}^{\infty}(v_n + W)$$

is absolutely summable in V/W. For every $n \in \mathbb{N}$, choose $w_n \in W$ such that

$$\|v_n + w_n\| \leq \|v_n + W\|_{V/W} + 2^{-n}.$$

The choice of 2^{-n}, rather than merely a sequence tending to zero, guarantees summability. Since

$$\sum_{n=1}^{\infty}\|v_n + W\|_{V/W} < \infty \qquad \text{and} \qquad \sum_{n=1}^{\infty} 2^{-n} < \infty,$$

the comparison test gives

$$\sum_{n=1}^{\infty}\|v_n + w_n\| < \infty.$$

Thus $\sum_{n=1}^{\infty}(v_n + w_n)$ is absolutely summable in V. Because V is a Banach space, the series converges in V. Hence there is an $\alpha \in V$ such that

$$\alpha = \sum_{n=1}^{\infty}(v_n + w_n).$$

The coset $\alpha + W$ is the candidate for the sum of $\sum_{n=1}^{\infty}(v_n + W)$.
We now verify convergence. Since $w_n \in W$,

$$\begin{aligned}
&\left\|\alpha + W - \sum_{n=1}^{N}(v_n + W)\right\|_{V/W} \\
&\quad = \left\|(\alpha + W) - \sum_{n=1}^{N}[(v_n + w_n) + W]\right\|_{V/W} \\
&\quad = \left\|\left[\alpha - \sum_{n=1}^{N}(v_n + w_n)\right] + W\right\|_{V/W} \\
&\quad = \inf_{w \in W}\left\|\alpha - \sum_{n=1}^{N}(v_n + w_n) + w\right\| \\
&\quad \leq \left\|\alpha - \sum_{n=1}^{N}(v_n + w_n)\right\|.
\end{aligned}$$

Therefore

$$0 \leq \left\|\alpha + W - \sum_{n=1}^{N}(v_n + W)\right\|_{V/W} \leq \left\|\alpha - \sum_{n=1}^{N}(v_n + w_n)\right\|.$$

The right-hand side tends to zero because

$$\sum_{n=1}^{N}(v_n + w_n) \longrightarrow \alpha.$$

By the squeeze theorem,

$$\left\| \alpha + W - \sum_{n=1}^{N} (v_n + W) \right\|_{V/W} \longrightarrow 0.$$

Thus

$$\sum_{n=1}^{N} (v_n + W) \longrightarrow \alpha + W.$$

Every absolutely summable series in V/W is summable, so the theorem on page 12 shows that V/W is a Banach space.

3.2 The Baire Category Theorem

Baire Category Theorem

Let M be a complete metric space. Let $\{C_n : n \in \mathbb{N}\}$ be a countable collection of closed subsets of M such that

$$M = \bigcup_{n \in \mathbb{N}} C_n.$$

Then at least one C_n contains an open ball. In other words, there are $x \in M$, $r > 0$, and $n \in \mathbb{N}$ such that

$$B(x, r) = \{y \in M : d(x, y) < r\} \subset C_n.$$

Equivalently, at least one C_n has an interior point.

Recall the elementary rules:

(1) an arbitrary union of open sets is open;

(2) a finite intersection of open sets is open;

(3) an arbitrary intersection of closed sets is closed;

(4) a finite union of closed sets is closed.

We now express these ideas in the language of topology.

Topology

Let X be a nonempty set. A *topology* τ on X is a collection of subsets of X such that:

(1) $\varnothing, X \in \tau$;

(2) τ is closed under arbitrary unions and finite intersections: if $S_\alpha \in \tau$ for every α in an index set A, then

$$\bigcup_{\alpha \in A} S_\alpha \in \tau,$$

and if $S_1, \ldots, S_N \in \tau$, then

$$\bigcap_{n=1}^{N} S_n \in \tau.$$

The pair (X, τ) is called a *topological space*, and the elements of τ are called *open sets*.

Closure

Let S be a subset of a topological space (X, τ). The *closure* of S, denoted by $\overline{S}$, is

$$\overline{S} = \{x \in X : \text{for every } V \in \tau \text{ with } x \in V,\ V \cap S \neq \varnothing\}.$$

Thus $x \in \overline{S}$ if and only if every open set containing x intersects S. The closure contains the limit points of S and the points of S itself.

Baire category for open sets

In applications, the sets C_n are often open rather than closed. The theorem still applies to their closures: if

$$M = \bigcup_{n \in \mathbb{N}} C_n,$$

then

$$M = \bigcup_{n \in \mathbb{N}} \overline{C_n},$$

and one of the closures $\overline{C_n}$ must contain an open ball.

Proof of the Baire Category Theorem

We argue by contradiction. Suppose there is a collection of closed subsets $\{C_n\}_{n \in \mathbb{N}}$ such that every C_n is nowhere dense, meaning that no C_n contains an open ball, and

$$\bigcup_{n \in \mathbb{N}} C_n = M.$$

We will construct a point that belongs to M but to none of the sets C_n, a contradiction. Because C_1 contains no open ball, $M \neq C_1$. Choose $p_1 \in M \setminus C_1$. Since C_1 is closed, its complement is open. Therefore there is an $\varepsilon_1 > 0$ such that

$$B(p_1, \varepsilon_1) \subset M \setminus C_1,$$

and hence

$$B(p_1, \varepsilon_1) \cap C_1 = \varnothing.$$

The ball $B(p_1, \varepsilon_1/3)$ is not contained in C_2, since C_2 contains no open ball. Thus there is a

$$p_2 \in B(p_1, \varepsilon_1/3) \qquad \text{with} \qquad p_2 \notin C_2.$$

Since C_2 is closed, its complement is open. We can choose $\varepsilon_2 < \varepsilon_1/3$ so that

$$B(p_2, \varepsilon_2) \subset M \setminus C_2,$$

or equivalently,

$$B(p_2, \varepsilon_2) \cap C_2 = \varnothing.$$

For the inductive step, suppose points and positive numbers

$$p_1, \ldots, p_k, \qquad \varepsilon_1, \ldots, \varepsilon_k$$

have been chosen so that

$$\varepsilon_j < \frac{\varepsilon_{j-1}}{3}, \qquad p_j \in B\left(p_{j-1}, \frac{\varepsilon_{j-1}}{3}\right), \qquad B(p_j, \varepsilon_j) \cap C_j = \varnothing.$$

Since $B(p_k, \varepsilon_k/3)$ is not contained in C_{k+1}, choose

$$p_{k+1} \in B(p_k, \varepsilon_k/3) \qquad \text{with} \qquad p_{k+1} \notin C_{k+1}.$$

The complement of the closed set C_{k+1} is open, so there is an $\varepsilon_{k+1} < \varepsilon_k/3$ such that

$$B(p_{k+1}, \varepsilon_{k+1}) \subset M \setminus C_{k+1}.$$

Therefore

$$B(p_{k+1}, \varepsilon_{k+1}) \cap C_{k+1} = \varnothing.$$

By induction, we obtain a sequence $\{p_k\}_{k=1}^{\infty}$ in M and a sequence $\{\varepsilon_k\} \subset (0, \varepsilon_1)$ such that

$$\cdots < 3^{k+1}\varepsilon_{k+1} < 3^k \varepsilon_k < \cdots < 3^2 \varepsilon_2 < 3\varepsilon_1,$$

$$p_j \in B\left(p_{j-1}, \frac{\varepsilon_{j-1}}{3}\right), \qquad B(p_j, \varepsilon_j) \cap C_j = \varnothing.$$

We show that $\{p_k\}$ is Cauchy. For $k, \ell \in \mathbb{N}$, repeated use of the triangle inequality gives

$$d(p_k, p_{k+\ell}) \le d(p_k, p_{k+1}) + \cdots + d(p_{k+\ell-1}, p_{k+\ell}).$$

Since

$$p_j \in B\left(p_{j-1}, \frac{\varepsilon_{j-1}}{3}\right),$$

we have

$$\begin{aligned} d(p_k, p_{k+\ell}) &< \frac{\varepsilon_k}{3} + \frac{\varepsilon_{k+1}}{3} + \cdots + \frac{\varepsilon_{k+\ell-1}}{3} \\ &< \frac{\varepsilon_1}{3^k} + \frac{\varepsilon_1}{3^{k+1}} + \cdots + \frac{\varepsilon_1}{3^{k+\ell}} \\ &< \varepsilon_1 \sum_{m=k}^{\infty} 3^{-m} = \frac{\varepsilon_1}{2}\, 3^{-k+1}. \end{aligned}$$

The final expression tends to zero as $k \to \infty$, so $\{p_k\}$ is Cauchy. Since M is complete, there is a $p \in M$ such that $p_k \to p$.

We finally show that $p \notin C_n$ for every $n \in \mathbb{N}$. For every $k, \ell \in \mathbb{N}$,

$$d(p_{k+1}, p_{k+1+\ell}) < \varepsilon_{k+1} \sum_{n=0}^{\infty} 3^{-n} = \frac{3}{2}\varepsilon_{k+1}.$$

Letting $\ell \to \infty$, we obtain

$$\begin{aligned} d(p_k, p) &\le d(p_k, p_{k+1}) + d(p_{k+1}, p) \\ &\le \frac{1}{3}\varepsilon_k + \frac{3}{2}\varepsilon_{k+1} \\ &< \frac{1}{3}\varepsilon_k + \frac{1}{2}\varepsilon_k < \varepsilon_k. \end{aligned}$$

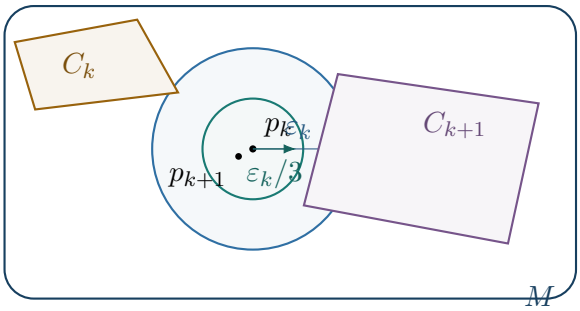
Thus $p \in B(p_k, \varepsilon_k)$ for every k. But

$$B(p_k, \varepsilon_k) \cap C_k = \varnothing,$$

so $p \notin C_k$ for every k. Therefore

$$p \notin \bigcup_{k \in \mathbb{N}} C_k = M,$$

even though $p \in M$, a contradiction.



$B(p_k, \varepsilon_k) \cap C_k = \varnothing$
$p_{k+1} \in B(p_k, \varepsilon_k/3)$
$p_{k+1} \notin C_{k+1}$

Consequences

The Baire Category Theorem can be used to prove that there exists a function that is continuous everywhere but differentiable nowhere. Another useful version says that, for a collection of open dense sets $\{G_n\}_{n\in\mathbb{N}}$, the intersection

$$\bigcap_{n=1}^{\infty} G_n$$

is also dense.

3.3 The Uniform Boundedness Theorem

Uniform Boundedness Theorem

Let B be a Banach space, let V be a normed vector space, and let

$$\{T_n\}_{n=1}^{\infty} \subset \mathcal{B}(B, V)$$

be a sequence of bounded linear operators. If

$$\sup_{n\in\mathbb{N}} \|T_n b\| < \infty \qquad \text{for every } b \in B,$$

then

$$\sup_{n\in\mathbb{N}} \|T_n\| < \infty.$$

Thus pointwise boundedness implies that the operator norms are uniformly bounded.

Pointwise boundedness

For a fixed $b \in B$, without taking a limit in n, each individual $\|T_n b\|$ is finite. The hypothesis is stronger: it says that the entire sequence $\{\|T_n b\|\}_{n=1}^{\infty}$ is bounded for each fixed b.

Proof

For each $k \in \mathbb{N}$, define

$$C_k = \left\{ b \in B : \|b\| \le 1,\ \sup_{n\in\mathbb{N}} \|T_n b\| \le k \right\}.$$

Each C_k is closed. Indeed, let $\{b_j\} \subset C_k$ and suppose $b_j \to b$. Continuity of the norm gives

$$\|b\| = \lim_{j\to\infty} \|b_j\| \leq 1.$$

For every $m \in \mathbb{N}$, the operator T_m is bounded and therefore continuous. Hence

$$\|T_m b_j\| \longrightarrow \|T_m b\|.$$

Since $\|T_m b_j\| \leq k$ for every j,

$$\|T_m b\| = \lim_{j\to\infty} \|T_m b_j\| \leq k.$$

This holds for every m, so

$$\sup_{m\in\mathbb{N}} \|T_m b\| \leq k.$$

Thus $b \in C_k$, proving that C_k is closed.
Moreover,

$$\{b \in B : \|b\| \leq 1\} = \bigcup_{k\in\mathbb{N}} C_k.$$

Indeed, for every $b \in B$, the pointwise-boundedness hypothesis gives a k such that $\sup_m \|T_m b\| \leq k$.
The closed unit ball

$$\{b \in B : \|b\| \leq 1\}$$

is a complete metric space because it is a closed subset of the Banach space B. By the Baire Category Theorem, one of the sets C_k contains an open ball $B(b_0, \delta_0)$.
For every $b \in B(0, \delta_0)$, we have

$$b_0 + b \in B(b_0, \delta_0) \subset C_k.$$

By the definition of C_k,

$$\sup_n \|T_n(b_0 + b)\| \leq k.$$

Since the T_n are bounded linear operators,

$$\begin{aligned}
\sup_n \|T_n b\| &= \sup_n \|T_n(b_0 + b) - T_n b_0\| \\
&\leq \sup_n (\|T_n(b_0 + b)\| + \|T_n b_0\|) \\
&\leq \sup_n \|T_n(b_0 + b)\| + \sup_n \|T_n b_0\| \\
&\leq k + k = 2k
\end{aligned}$$

for every $b \in B(0, \delta_0)$.
To obtain an operator-norm estimate, let $b \in B$ satisfy $\|b\| = 1$. Then

$$\left\| \frac{\delta_0}{2} b \right\| = \frac{\delta_0}{2} < \delta_0.$$

Therefore

$$\sup_n \left\| T_n \left(\frac{\delta_0}{2} b \right) \right\| \leq 2k,$$

so, for every $n \in \mathbb{N}$ and every b with $\|b\| = 1$,

$$\|T_n b\| \leq \frac{4k}{\delta_0}.$$

It follows that

$$\|T_n\| \leq \frac{4k}{\delta_0} \qquad \text{for every } n \in \mathbb{N},$$

and hence

$$\sup_{n\in\mathbb{N}} \|T_n\| \leq \frac{4k}{\delta_0} < \infty.$$

Why the conclusion is significant

The estimate found above uses a constant independent of b. Pointwise boundedness says that for every $b \in B$ there is a constant $C(b) > 0$, which may depend on b, such that

$$\sup_n \|T_n b\| < C(b).$$

Uniform boundedness produces one constant $C > 0$, independent of b, such that for every n and every $\|b\| = 1$,

$$\|T_n b\| < C.$$

4 The Open Mapping and Closed Graph Theorems

Recall the Baire Category Theorem: if M is a complete metric space, $\{C_n\}_{n\in\mathbb{N}}$ is a collection of closed sets, and

$$M = \bigcup_{n\in\mathbb{N}} C_n,$$

then at least one C_n has an interior point.

Open Mapping Theorem

Let B_1 and B_2 be Banach spaces, and let

$$T \in \mathcal{B}(B_1, B_2)$$

be a surjective linear operator. Then T is an *open map*: for every open subset $U \subset B_1$, the image $T(U)$ is open in B_2.

Proof

We first prove a claim.

Claim 1. If

$$B_1(0,1) = \{b \in B_1 : \|b\| < 1\},$$

then $T(B_1(0,1))$ contains an open ball in B_2 centered at 0.

Since T is surjective,

$$B_2 = \bigcup_{n\in\mathbb{N}} T(B_1(0,n)) = \bigcup_{n\in\mathbb{N}} \overline{T(B_1(0,n))}.$$

The Baire Category Theorem applies because B_2 is a Banach space. Therefore, for some $n_0 \in \mathbb{N}$, the set

$$\overline{T(B_1(0,n_0))}$$

contains an open ball.
Closures respect rescaling, and T is linear, so

$$n_0\,\overline{T(B_1(0,1))} = \overline{T(B_1(0,n_0))}.$$

Hence $\overline{T(B_1(0,1))}$ contains an open ball. Choose $v_0 \in B_2$ and $r > 0$ such that

$$B_2(v_0, 4r) \subset \overline{T(B_1(0,1))}.$$

By the definition of closure, there is a

$$v_1 \in T(B_1(0,1))$$

such that

$$\|v_0 - v_1\| < 2r.$$

Consequently,

$$B_2(v_1, 2r) \subset B_2(v_0, 4r) \subset \overline{T(B_1(0,1))}.$$

If $\|v\| < r$, then

$$2v + v_1 \in B_2(v_1, 2r)$$

because

$$\|2v + v_1 - v_1\| = \|2v\| < 2r.$$

It follows that

$$\frac{1}{2}(2v + v_1) \in \frac{1}{2}\,\overline{T(B_1(0,1))} = \overline{T(B_1(0,1/2))}.$$

Write $v_1 = T(u_1)$ for some $u_1 \in B_1(0,1)$. Then

$$\begin{aligned} v &= -T\left(\frac{u_1}{2}\right) + \frac{1}{2}(2v + v_1) \\ &\in -T\left(\frac{u_1}{2}\right) + \overline{T(B_1(0,1/2))}. \end{aligned}$$

This is the closure $\overline{T(B_1(0,1/2))}$, shifted by $-T(u_1/2)$. Since $u_1 \in B_1(0,1)$, we have $-u_1/2 \in B_1(0,1/2)$, and therefore

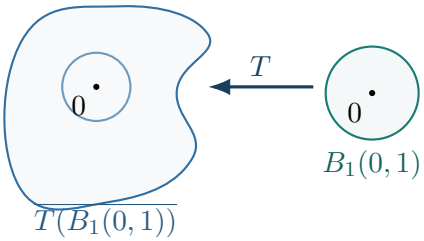
$$\begin{aligned} -T\left(\frac{u_1}{2}\right) + \overline{T(B_1(0,1/2))} &= \overline{T\left(-\frac{u_1}{2} + B_1(0,1/2)\right)} \\ &\subset \overline{T(B_1(0,1))}. \end{aligned}$$

We have shown that

$$B_2(0,r) \subset \overline{T(B_1(0,1))}.$$

By homogeneity of the norm, for every $n \in \mathbb{N}$,

$$B_2(0, 2^{-n}r) = 2^{-n}B_2(0,r) \subset 2^{-n}\overline{T(B_1(0,1))} = \overline{T(B_1(0,2^{-n}))}.$$



We now remove the closure after reducing the radius. We want to prove

$$B_2(0, r/4) \subset T(B_1(0,1)).$$

Let $\|v\| < r/4$. By what we have shown,

$$v \in \overline{T(B_1(0,1/4))}.$$

Thus there exists $b_1 \in B_1(0,1/4)$ such that

$$\|v - Tb_1\| < \frac{r}{4}.$$

The remainder belongs to

$$B_2(0, r/4) \subset \overline{T(B_1(0,1/4))}.$$

Repeating the process, choose $b_2 \in B_1(0,1/8)$ such that

$$\|v - Tb_1 - Tb_2\| < \frac{r}{8}.$$

Continuing inductively, we obtain a sequence $\{b_k\}$ in B_1 such that

$$\|b_k\| < 2^{-k-1}$$

and

$$\left\| v - \sum_{k=1}^{n} Tb_k \right\|_2 < 2^{-n-1} r.$$

The series $\sum_{k=1}^{\infty} b_k$ is absolutely summable, since

$$\sum_{k=1}^{\infty} \|b_k\| \leq \sum_{k=1}^{\infty} 2^{-k-1} < \infty.$$

Because B_1 is a Banach space, there is a $b \in B_1$ such that

$$b = \sum_{k=1}^{\infty} b_k.$$

Moreover, continuity of the norm and the triangle inequality give

$$\|b\| = \lim_{n\to\infty} \left\| \sum_{k=1}^{n} b_k \right\| \leq \sum_{k=1}^{\infty} 2^{-k-1} = \frac{1}{2} < 1.$$

Since T is continuous, limits interchange with T, and hence

$$Tb = \lim_{n\to\infty} T\left(\sum_{k=1}^{n} b_k\right) = \lim_{n\to\infty} \sum_{k=1}^{n} Tb_k = v.$$

Thus $b \in B_1(0,1)$ and $Tb = v$. Therefore

$$B_2(0, r/4) \subset T(B_1(0,1)).$$

This proves Claim 1.

Claim 2. The operator T is open.
Let $U \subset B_1$ be open, and let

$$b_2 = Tb_1 \in T(U).$$

Since U is open, there is an $\varepsilon > 0$ such that

$$b_1 + B_1(0,\varepsilon) = B_1(b_1,\varepsilon) \subset U.$$

By Claim 1, there is a $\delta > 0$ such that

$$B_2(0,\delta) \subset T(B_1(0,1)).$$

Therefore

$$\begin{aligned} B_2(b_2, \varepsilon\delta) &= b_2 + \varepsilon B_2(0,\delta) \\ &\subset b_2 + \varepsilon T(B_1(0,1)) \\ &= T(b_1) + T(B_1(0,\varepsilon)) \\ &= T(b_1 + B_1(0,\varepsilon)) \\ &= T(B_1(b_1,\varepsilon)) \subset T(U). \end{aligned}$$

Every point of $T(U)$ is therefore an interior point, so $T(U)$ is open.

Corollary

Let B_1 and B_2 be Banach spaces. If

$$T \in \mathcal{B}(B_1, B_2)$$

is bijective, then

$$T^{-1} \in \mathcal{B}(B_2, B_1).$$

Proof

By the continuity criterion from page 15, T^{-1} is continuous if and only if, for every open $U \subset B_1$, the inverse image

$$(T^{-1})^{-1}(U) = T(U)$$

is open in B_2. This is true by the Open Mapping Theorem.

Theorem: products of Banach spaces

If B_1 and B_2 are Banach spaces, then $B_1 \times B_2$, with the norm

$$\|(b_1, b_2)\| = \|b_1\| + \|b_2\|, \qquad (b_1, b_2) \in B_1 \times B_2,$$

is a Banach space.

We omit the proof. Both the norm axioms and completeness are proved in the same way as for $\mathbb{R}^n$.

Graph of an operator

For $T : B_1 \to B_2$, the *graph* of T is

$$\Gamma(T) = \{(u, Tu) : u \in B_1\} \subset B_1 \times B_2.$$

Closed Graph Theorem

Let B_1 and B_2 be Banach spaces, and let $T : B_1 \to B_2$ be a linear operator. Then

$$T \in \mathcal{B}(B_1, B_2) \iff \Gamma(T) \text{ is closed in } B_1 \times B_2.$$

Why this criterion is useful

For a linear operator, it is often easier to check directly that its graph is closed than to prove boundedness or continuity.

Proof

Suppose first that $T \in \mathcal{B}(B_1, B_2)$. Let

$$\{(u_n, Tu_n)\}_{n=1}^{\infty} \subset \Gamma(T)$$

be a sequence such that $u_n \to u$ and $Tu_n \to v$. Since T is continuous,

$$v = \lim_{n\to\infty} Tu_n = T\left(\lim_{n\to\infty} u_n\right) = Tu.$$

Thus $(u, v) = (u, Tu) \in \Gamma(T)$, so the graph is closed.
Conversely, suppose $\Gamma(T)$ is closed. Consider the projections

$$\pi_1 : \Gamma(T) \to B_1, \qquad \pi_2 : \Gamma(T) \to B_2,$$

defined by

$$\pi_1(u, Tu) = u, \qquad \pi_2(u, Tu) = Tu.$$

The graph $\Gamma(T)$ is a Banach space because it is a closed subspace of the Banach space $B_1 \times B_2$, by the theorem on page 28. Moreover,

$$\pi_1 \in \mathcal{B}(\Gamma(T), B_1), \qquad \pi_2 \in \mathcal{B}(\Gamma(T), B_2),$$

because

$$\|\pi_i(u, v)\| \le \|u\| + \|v\| = \|(u, v)\|.$$

The map π_1 is one-to-one and onto by the definition of $\Gamma(T)$. The corollary of the Open Mapping Theorem proved above shows that

$$\pi_1^{-1} \in \mathcal{B}(B_1, \Gamma(T)).$$

Finally,

$$T = \pi_2 \circ \pi_1^{-1}$$

is a composition of two bounded linear operators. Hence $T \in \mathcal{B}(B_1, B_2)$.

Logical relation

The proof above uses the Open Mapping Theorem to prove the Closed Graph Theorem. In fact, the two theorems are logically equivalent.

5 The Hahn–Banach Theorem

Given a general nontrivial normed vector space V, is its dual space

$$V' = \mathcal{B}(V, \mathbb{K})$$

always nontrivial?

For example, if $1 \leq p < \infty$, then

$$(\ell^p)' = \ell^{p'}, \qquad \frac{1}{p} + \frac{1}{p'} = 1.$$

Also,

$$(\ell^1)' = \ell^\infty, \qquad (\ell^\infty)' \ncong \ell^1, \qquad (c_0)' = \ell^1.$$

We first establish a few ideas from set theory.

Partial order

A *partial order* on a set E is a relation $\preceq$ on E with the following properties:

(1) $e \preceq e$ for every $e \in E$;

(2) if $e \preceq f$ and $f \preceq e$, then $e = f$;

(3) if $e \preceq f$ and $f \preceq g$, then $e \preceq g$.

Upper bound, maximal element, and minimal element

Let $D \subset E$ in a partially ordered set.

- An *upper bound* of D is an element $e \in E$ such that $d \preceq e$ for every $d \in D$.
- A *maximal element* of E is an element $e \in E$ such that, for every $f \in E$, $e \preceq f$ implies $e = f$.
- A *minimal element* of E is an element $e \in E$ such that, for every $f \in E$, $f \preceq e$ implies $e = f$.

Partial orders versus total orders

A maximal element does not necessarily sit above every other element: one first needs $e \preceq f$ before testing whether $e = f$. A total order has the additional property that, for every x, y, one has $x = y$, $x < y$, or $y < x$. In a partial order, two elements may have no relation at all. For example, two sets A and B may satisfy neither $A \subset B$ nor $B \subset A$.

Chain

Let $(E, \preceq)$ be a partially ordered set. A subset $C \subset E$ is a *chain* if, for every $e, f \in C$, either $e \preceq f$ or $f \preceq e$.

Zorn's Lemma

If every chain in a nonempty partially ordered set E has an upper bound, then E has a maximal element.

Hamel basis

Let V be a vector space. A *Hamel basis* $H \subset V$ is a linearly independent set such that every element of V is a finite linear combination of elements of H.

Hamel and Schauder bases

A Schauder basis allows countably infinite, or Schauder, linear combinations: it is a linearly independent set $S \subset V$ such that every element of V can be expressed as a Schauder combination of elements of S.

We take Zorn's Lemma as a given axiom from set theory. It can also be used to derive the Axiom of Choice.

5.1 Zorn's Lemma and the Hahn–Banach Theorem

We first use Zorn's Lemma to show that every vector space has a Hamel basis. The same pattern of argument will prove the Hahn–Banach Theorem.

Theorem: existence of a Hamel basis

Every vector space has a Hamel basis.

Proof

Let

$$E = \{e \subset V : e \text{ is linearly independent}\},$$

partially ordered by inclusion. Let $\mathcal{C} \subset E$ be a chain, and set

$$c = \bigcup_{e \in \mathcal{C}} e.$$

We claim that c is linearly independent. Indeed, take finitely many vectors $v_1, \dots, v_N \in c$. For each i, there exists $e_i \in \mathcal{C}$ such that $v_i \in e_i$. Since $\mathcal{C}$ is a chain, one of these sets contains all the others; hence all the vectors $v_1, \dots, v_N$ belong to a single linearly independent set. Therefore they are linearly independent, so $c \in E$. Moreover, c is an upper bound for $\mathcal{C}$.

By Zorn's Lemma, E has a maximal element H. We claim that H is a Hamel basis. If not, there would exist $v \in V$ which is not a finite linear combination of elements of H. Then $H \cup \{v\}$ would be linearly independent and would strictly contain H, contradicting the maximality of H. Thus H is a Hamel basis by definition.

Hahn–Banach Theorem

Let V be a normed vector space, let $M \subset V$ be a subspace, and let

$$u : M \longrightarrow \mathbb{C}$$

be linear and satisfy

$$|u(t)| \leq C\|t\| \qquad \text{for every } t \in M.$$

Then there exists

$$U \in \mathcal{B}(V, \mathbb{C})$$

such that

$$U\big|_M = u \qquad \text{and} \qquad |U(t)| \leq C\|t\| \quad \text{for every } t \in V.$$

Thus U is a continuous extension of u with the same bound C.

Lemma: extension by one dimension

Let V be a normed vector space, let $M \subset V$ be a subspace, and let $u : M \to \mathbb{C}$ be linear with

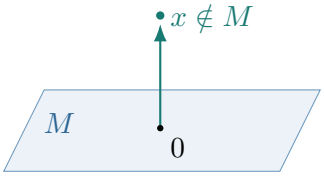
$$|u(t)| \leq C\|t\| \qquad \text{for every } t \in M.$$

If $x \notin M$, then there exists a linear functional

$$u' : M' = M + \mathbb{C}x = \{t + ax : t \in M,\ a \in \mathbb{C}\} \longrightarrow \mathbb{C}$$

such that

$$u'\big|_M = u \qquad \text{and} \qquad |u'(t')| \leq C\|t'\| \quad \text{for every } t' \in M'.$$



The strategy is to apply Zorn's Lemma to the collection of all continuous extensions of u. A maximal extension must be defined on all of V, just as the maximal linearly independent set in the preceding proof must span all of V. We first prove the Hahn–Banach Theorem using the lemma, and then prove the lemma.

Proof of the Hahn–Banach Theorem, assuming the lemma

Define

$$E = \{(v, N) : N \text{ is a subspace of } V,\ M \subset N,\ \text{and } v \text{ is a continuous extension of } u\},$$

where every extension is required to satisfy the same bound C. The set E is nonempty because

$(u, M) \in E$. Define a partial order on E by

$$(v_1, N_1) \preceq (v_2, N_2) \quad \Longleftrightarrow \quad N_1 \subset N_2 \text{ and } v_2\big|_{N_1} = v_1.$$

Let

$$\mathcal{C} = \{(v_i, N_i) : i \in I\}$$

be a chain in E, and put

$$N = \bigcup_{i \in I} N_i.$$

Then N is a subspace. Indeed, if $x_1 \in N_{i_1}$ and $x_2 \in N_{i_2}$, then, after interchanging the indices if necessary, $N_{i_1} \subset N_{i_2}$. Consequently, every linear combination of x_1 and x_2 belongs to $N_{i_2} \subset N$. Define $v : N \to \mathbb{C}$ by

$$v(t) = v_i(t) \qquad \text{if } t \in N_i.$$

This is well-defined. If $t \in N_{i_1} \cap N_{i_2}$, then, after interchanging the indices if necessary, $(v_{i_1}, N_{i_1}) \preceq (v_{i_2}, N_{i_2})$. Hence

$$v_{i_2}\big|_{N_{i_1}} = v_{i_1},$$

so $v_{i_1}(t) = v_{i_2}(t)$.
The same argument shows that v is linear. Namely, if $x_1 \in N_{i_1}$ and $x_2 \in N_{i_2}$, choose the larger of the two subspaces and compute there. The functional v has the same bound C as u and all the functionals v_i. Therefore (v, N) is an upper bound for the chain $\mathcal{C}$.
By Zorn's Lemma, E has a maximal element (U, B). We claim that $B = V$. If there were an $x \in V \setminus B$, the one-dimensional extension lemma would provide an extension φ on $B + \mathbb{C}x$ with the same bound, and hence

$$(U, B) \prec (\varphi, B + \mathbb{C}x),$$

contradicting maximality. Thus $B = V$, and U is an extension of u to all of V with the same bound C.

The proof of the one-dimensional extension lemma uses two elementary facts about linear functionals.

Terminology

We often use the term *functional* for a scalar-valued operator.

Lemma I

Every linear functional is either the zero functional or surjective.

Proof

Let $f \in \mathcal{B}(V, \mathbb{K})$, where V is a $\mathbb{K}$-Banach space. Suppose that $f \neq 0$. Then there exists $v \in V$ such that $f(v) = a \neq 0$. After rescaling,

$$f\left(\frac{1}{a}v\right) = 1.$$

For every $\lambda \in \mathbb{K}$, the vector $(\lambda/a)v$ belongs to V, and

$$f\left(\frac{\lambda}{a}v\right) = \lambda f\left(\frac{1}{a}v\right) = \lambda.$$

Thus a nonzero linear functional is surjective.

Lemma II

The complex extension of a real-linear functional is unique.

Proof

Let V be a complex vector space, and let

$$f : V \longrightarrow \mathbb{R}, \qquad f \in \mathcal{B}(V, \mathbb{R}),$$

be real-linear. Define $\widetilde{f} : V \to \mathbb{C}$ by

$$\widetilde{f}(x) = f(x) - if(ix).$$

We claim that $\widetilde{f}$ is a complex-linear extension of f, and that it is the unique such extension. Here "extension" means that

$$\Re \widetilde{f} = f.$$

Equivalently, complex linearity forces the imaginary part through the identity

$$\Im \widetilde{f}(x) = -f(ix).$$

Suppose that $g \in \mathcal{B}(V, \mathbb{C})$ also satisfies $\Re g = f$, and let

$$h = \widetilde{f} - g \in \mathcal{B}(V, \mathbb{C}).$$

Then $\Re h = 0$, so $h(V) \subset i\mathbb{R}$. By Lemma I, $h : V \to \mathbb{C}$ is either surjective or zero. Its range is contained in $i\mathbb{R}$, so it cannot be surjective onto $\mathbb{C}$. Therefore $h = 0$, and $\widetilde{f} = g$.

Important real-linearity warning

In the proof below, the real part of u is extended first, through auxiliary functionals w, w', and w'' that are only real-linear. In particular, real linearity alone does not imply

$$w''(it) = i\, w''(t).$$

The left-hand side is real-valued, whereas the right-hand side need not be.

Proof of the one-dimensional extension lemma

The set $M' = M + \mathbb{C}x$ is a subspace. Every $t' \in M'$ can be written uniquely as

$$t' = t + ax, \qquad t \in M, \quad a \in \mathbb{C}.$$

Indeed, if

$$t + ax = \widetilde{t} + \widetilde{a}x,$$

then

$$(a - \widetilde{a})x = \widetilde{t} - t \in M.$$

Since $x \notin M$, this forces $a = \widetilde{a}$, and then $t = \widetilde{t}$.

We need to choose $u' : M' \to \mathbb{C}$ of the form

$$u'(t + ax) = u(t) + a\,u'(x) = u(t) + \lambda a,$$

where $\lambda = u'(x)$.
If $C = 0$, then $u = 0$, and we take $u' = 0$. Suppose now that $C \neq 0$. It is enough to prove the result when $C = 1$: apply that case to $C^{-1}u$, then multiply the resulting extension by C. Thus assume

$$|u(t)| \leq \|t\|.$$

We seek $\lambda \in \mathbb{C}$ such that

$$|u(t) + a\lambda| \leq \|t + ax\| \qquad (t \in M,\ a \in \mathbb{C}). \tag{*}$$

The case $a = 0$ is immediate. When $a \neq 0$, rescaling by $-a$ shows that $(*)$ is equivalent to choosing $\lambda \in \mathbb{C}$ such that

$$|u(t) - \lambda| \leq \|t - x\| \qquad \text{for every } t \in M. \tag{**}$$

We begin with the real part. Define

$$w(t) = \Re u(t), \qquad t \in M.$$

We shall first extend this real-linear functional with the same bound. We seek $\alpha \in \mathbb{R}$ such that

$$|w(t) - \alpha| \leq \|t - x\| \qquad \text{for every } t \in M;$$

the number α will be the value of the extension at x.
For all $t_1, t_2 \in M$,

$$\begin{aligned} |w(t_1) - w(t_2)| &= |w(t_1 - t_2)| \\ &= |\Re u(t_1 - t_2)| \\ &\leq |u(t_1 - t_2)| \\ &\leq \|t_1 - t_2\| \\ &\leq \|t_1 - x\| + \|t_2 - x\|. \end{aligned}$$

Since w is real-valued,

$$w(t_1) - \|t_1 - x\| \leq w(t_2) + \|t_2 - x\|.$$

It follows that

$$\sup_{t \in M} \big(w(t) - \|t - x\|\big) \leq \inf_{t \in M} \big(w(t) + \|t - x\|\big).$$

Choose $\alpha \in \mathbb{R}$ between these two quantities. Then

$$|w(t) - \alpha| \leq \|t - x\| \qquad \text{for every } t \in M.$$

Consequently,

$$w' : M' = M + \mathbb{R}x \longrightarrow \mathbb{R}, \qquad w'(t + ax) = w(t) + a\alpha,$$

is a real-linear extension of w with the same bound.

We now repeat the argument to obtain a further continuous extension

$$w'' : M'' = M + \mathbb{C}x = (M + \mathbb{R}x) + \mathbb{R}(ix) \longrightarrow \mathbb{R}$$

of w'. We seek $\beta \in \mathbb{R}$ such that

$$|w''(t + b(ix))| = |w'(t) + b\beta| \leq \|t + b(ix)\|$$

for every $t \in M' = M + \mathbb{R}x$ and $b \in \mathbb{R}$. As above, it is enough to choose β such that

$$|w'(t) - \beta| \leq \|t - ix\| \qquad \text{for every } t \in M'.$$

The functional w' is bounded by the same constant, so

$$|w'(t_1) - w'(t_2)| \leq \|t_1 - t_2\| \leq \|t_1 - ix\| + \|t_2 - ix\|.$$

Because w' is real-valued,

$$w'(t_1) - \|t_1 - ix\| \leq w'(t_2) + \|t_2 - ix\|,$$

and hence

$$\sup_{t \in M'} \big(w'(t) - \|t - ix\|\big) \leq \inf_{t \in M'} \big(w'(t) + \|t - ix\|\big).$$

Choose $\beta \in \mathbb{R}$ between these two quantities. Then

$$|w'(t) - \beta| \leq \|t - ix\|.$$

As in the preceding step, this gives a real-linear functional

$$w'' : M'' = M' + \mathbb{R}(ix) \longrightarrow \mathbb{R}$$

with

$$w''\big(t + (a + bi)x\big) = w(t) + a\alpha + b\beta, \qquad t \in M, \quad a, b \in \mathbb{R},$$

which is a continuous extension of w with the same bound.
We are not done yet: we chose $w(t) = \Re u(t)$, which need not satisfy $w\big|_M = u$. Thus w'' is only a continuous extension of w, not of u. For $c = a + bi$, define

$$u' : M'' = M + \mathbb{C}x \longrightarrow \mathbb{C}$$

by

$$u'(t + cx) = w''\big(t + ax + b(ix)\big) - i\, w''\big(it - bx + a(ix)\big).$$

We claim that u' is a continuous extension of u.
By Lemma II, the functional $u' : M + \mathbb{C}x \to \mathbb{C}$ defined above is the unique complex-linear extension of w'', once we verify that it is indeed a linear functional. It is linear. Moreover, for $t \in M$,

$$\begin{aligned} u(t) &= w''(t) - iw''(it) \\ &= w(t) - iw(it) \\ &= \Re u(t) - i\Re u(it). \end{aligned}$$

Here only the real linearity of w'' is being used; we are not replacing $w''(it)$ by $iw''(t)$. In other words,

$$\Re u'(t) = w(t) = \Re u(t).$$

By Lemma II again, u' must agree with u on M, since the complex extension of $w : M \to \mathbb{R}$ is unique. Therefore

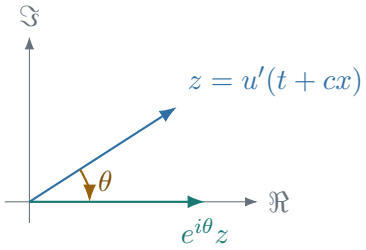
$$u'|_M = u.$$

It remains to show that the extension is continuous and does not increase the bound. For every $t \in M$ and $c \in \mathbb{C}$, choose the unique $\theta \in [0, 2\pi)$ such that

$$e^{i\theta} u'(t + cx) \in \mathbb{R}_{\geq 0}.$$

Then

$$\begin{aligned} |u'(t + cx)| &= e^{i\theta} u'(t + cx) \\ &= \Re u'(e^{i\theta} t + e^{i\theta} cx) \\ &= w''(e^{i\theta} t + e^{i\theta} cx) \\ &\leq \|e^{i\theta} t + e^{i\theta} cx\| \\ &= |\cos\theta + i \sin\theta| \, \|t + cx\|. \end{aligned}$$



We are choosing θ so that multiplication by $e^{i\theta}$ rotates $u'(t + cx)$ onto the nonnegative real axis. Finally,

$$|\cos\theta + i \sin\theta| \, \|t + cx\| = \sqrt{\cos^2\theta + \sin^2\theta} \, \|t + cx\| = \|t + cx\|.$$

Indeed, multiplication by $e^{i\theta}$ sends

$$r_0 e^{i\theta_0} = a_0 + b_0 i \quad \text{to} \quad r_0 e^{i(\theta_0 + \theta)},$$

so it rotates a complex number by θ while preserving its norm. Thus there exists a continuous extension u' of u without increasing the norm, exactly as required.

6 The Double Dual and the Outer Measure of a Subset

Recall the Hahn–Banach Theorem. If V is a normed vector space, $M \subset V$ is a subspace, and $u \in M' = \mathcal{B}(M, \mathbb{K})$ satisfies

$$|u(t)| \leq C\|t\| \qquad \text{for every } t \in M,$$

then there exists $U \in V'$ such that

$$U\big|_M = u \qquad \text{and} \qquad |U(t)| \leq C\|t\| \quad \text{for every } t \in V.$$

A consequence to be proved

We can use the Hahn–Banach Theorem to show that

$$(\ell^\infty)' \neq \ell^1,$$

and, in fact,

$$(c_0)' = \ell^1.$$

Theorem

If V is a normed vector space, then, for every $v \in V \setminus \{0\}$, there exists $f \in V' = \mathcal{B}(V, \mathbb{K})$ such that

$$\|f\| = 1 \qquad \text{and} \qquad f(v) = \|v\|.$$

Proof

Let $\mathbb{K}v = \operatorname{span}\{v\}$, and define the linear map

$$u : \mathbb{K}v \longrightarrow \mathbb{K}, \qquad u(\lambda v) = \lambda\|v\|.$$

The vector v is a basis of $\mathbb{K}v$, so every element has a unique expression of the form λv; therefore u is well-defined. By homogeneity of the norm,

$$|u(t)| = \|t\| \qquad \text{for every } t \in \mathbb{K}v,$$

and $u(v) = \|v\|$. By Hahn–Banach, there exists a continuous extension $f \in \mathcal{B}(V, \mathbb{K}) = V'$ which does not increase the bound. Thus

$$|f(t)| \leq \|t\| \qquad \text{for every } t \in V,$$

so $\|f\| \leq 1$. On the other hand,

$$\left| f\left(\frac{v}{\|v\|}\right) \right| = \frac{|f(v)|}{\|v\|} = 1.$$

Hence

$$\|f\| = \sup_{\|t\|=1} |f(t)| = 1.$$

6.1 The double dual

Double dual

The *double dual* of a normed vector space V, denoted by V'', is the dual of V':

$$V'' = (V')' = \mathcal{B}(V', \mathbb{K}).$$

Let $v \in V$. Define

$$T_v : V' \longrightarrow \mathbb{K}, \qquad T_v(v') = v'(v) \quad \text{for every } v' \in V'.$$

Then $T_v \in V''$; in other words, T_v is a bounded linear functional. Indeed, linearity follows from

$$T_v(\lambda_1 v_1' + \lambda_2 v_2') = (\lambda_1 v_1' + \lambda_2 v_2')(v) = \lambda_1 v_1'(v) + \lambda_2 v_2'(v).$$

Furthermore,

$$|T_v(v')| = |v'(v)| \leq \|v'\| \, \|v\|,$$

because $v' \in \mathcal{B}(V, \mathbb{K})$. Thus

$$\|T_v\| \leq \|v\|.$$

Isometric operator

Let V and W be normed spaces. An operator $T \in \mathcal{B}(V, W)$ is *isometric* if

$$\|Tv\| = \|v\| \qquad \text{for every } v \in V.$$

The canonical embedding into the double dual

For $v \in V$, define $T_v : V' \to \mathbb{K}$ by $T_v(v') = v'(v)$. Then the map

$$T : V \longrightarrow V'', \qquad T(v) = T_v,$$

is isometric.

Proof

We have already shown that $v \mapsto T_v$ belongs to $\mathcal{B}(V, V'')$ and that

$$\|T_v\| \leq \|v\|.$$

It remains to prove the reverse inequality. If $v = 0$, then $\|T_0\| = \|0\| = 0$. If $v \neq 0$, the theorem on page 51 gives an $f \in V'$ such that

$$\|f\| = 1 \qquad \text{and} \qquad f(v) = \|v\|.$$

Therefore

$$\|v\| = f(v) = |f(v)| = |T_v(f)| \leq \|T_v\| \, \|f\| = \|T_v\|.$$

Combining the two inequalities gives

$$\|T_v\| = \|v\|.$$

Thus T is isometric.

An isometry can map only 0 to 0, by positive definiteness of the norm. Hence the canonical map $T : V \to V''$ is injective. Is it also surjective?

Reflexive Banach space

A Banach space V is *reflexive* if

$$V \cong V'',$$

where the notation means that the canonical map

$$T : V \longrightarrow V'', \qquad v \longmapsto T_v,$$

is onto, and hence bijective.

Examples and nonexamples

If $1 < p < \infty$ and p' is the conjugate exponent, then ℓ^p is reflexive because

$$(\ell^p)'' = (\ell^{p'})' = \ell^p.$$

Here $\cong$ means isomorphism.
The space ℓ^1 is not reflexive because

$$(\ell^1)' = \ell^\infty, \qquad (\ell^1)'' = (\ell^\infty)' \neq \ell^1.$$

The space c_0 is not reflexive because

$$(c_0)' = \ell^1, \qquad (c_0)'' = (\ell^1)' = \ell^\infty \neq c_0.$$

Isomorphism of normed spaces

Let V_1 and V_2 be normed vector spaces. A bijective linear operator $S : V_1 \to V_2$ is an *isomorphism* if

$$S \in \mathcal{B}(V_1, V_2) \qquad \text{and} \qquad S^{-1} \in \mathcal{B}(V_2, V_1).$$

If an isomorphism exists between V_1 and V_2, we say that they are isomorphic.

Homework theorem: quotient by the kernel

Let V and W be Banach spaces, and let $T \in \mathcal{B}(V, W)$. Then

$$V/\operatorname{Ker}(T) \cong \operatorname{Ran}(T) \quad \Longleftrightarrow \quad \operatorname{Ran}(T) \text{ is closed in } W.$$

Proof

First, $\operatorname{Ker}(T)$ is closed in V. If x is a cluster point of $\operatorname{Ker}(T)$, choose $x_n \in \operatorname{Ker}(T)$ with $\|x_n - x\| < 1/n$. Since T is continuous,

$$Tx = T\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} Tx_n = 0,$$

so $x \in \operatorname{Ker}(T)$.
Suppose first that $V/\operatorname{Ker}(T) \cong \operatorname{Ran}(T)$. Then there is an isomorphism

$$S \in \mathcal{B}(V/\operatorname{Ker}(T), \operatorname{Ran}(T)), \qquad S^{-1} \in \mathcal{B}(\operatorname{Ran}(T), V/\operatorname{Ker}(T)).$$

Because $\operatorname{Ker}(T)$ is closed and V is Banach, the quotient theorem on page 32 shows that $V/\operatorname{Ker}(T)$ is Banach with norm

$$\|v + \operatorname{Ker}(T)\| = \inf_{w \in \operatorname{Ker}(T)} \|v + w\|.$$

We claim that the isomorphism transports completeness to $\operatorname{Ran}(T)$. Let (y_n) be Cauchy in $\operatorname{Ran}(T)$. Since S^{-1} is bounded,

$$\|S^{-1}(y_n - y_m)\| \leq C\|y_n - y_m\|,$$

so $(S^{-1}y_n)$ is Cauchy in $V/\operatorname{Ker}(T)$. By completeness, there is an $x \in V/\operatorname{Ker}(T)$ such that $S^{-1}y_n \to x$. Since S is bounded,

$$y_n = S(S^{-1}y_n) \longrightarrow Sx \in \operatorname{Ran}(T).$$

Thus every Cauchy sequence in $\operatorname{Ran}(T)$ converges in $\operatorname{Ran}(T)$, so $\operatorname{Ran}(T)$ is a Banach space and therefore is closed in W.
Conversely, suppose that $\operatorname{Ran}(T)$ is closed. Define

$$S : V/\operatorname{Ker}(T) \longrightarrow \operatorname{Ran}(T), \qquad S(v + \operatorname{Ker}(T)) = Tv.$$

We show that

$$S \in \mathcal{B}(V/\operatorname{Ker}(T), \operatorname{Ran}(T)) \qquad \text{and} \qquad S^{-1} \in \mathcal{B}(\operatorname{Ran}(T), V/\operatorname{Ker}(T)),$$

using the corollary to the Open Mapping Theorem on page 42.
The map S is well-defined. If

$$v_1 + \operatorname{Ker}(T) = v_2 + \operatorname{Ker}(T),$$

then $v_1 - v_2 \in \operatorname{Ker}(T)$, and hence

$$T(v_1 - v_2) = Tv_1 - Tv_2 = 0.$$

Therefore

$$S(v_1 + \operatorname{Ker}(T)) = Tv_1 = Tv_2 = S(v_2 + \operatorname{Ker}(T)).$$

Linearity is inherited from the linearity of T.
To prove boundedness, let $\varepsilon > 0$. By the definition of the quotient norm, there exists $w \in \operatorname{Ker}(T)$ such that

$$\|v + w\| < \|v + \operatorname{Ker}(T)\| + \varepsilon.$$

Then

$$\|S(v + \operatorname{Ker}(T))\| = \|Tv\| = \|T(v + w)\| \leq C\|v + w\| < C\big(\|v + \operatorname{Ker}(T)\| + \varepsilon\big).$$

Letting $\varepsilon \to 0$ gives

$$\|S(v + \operatorname{Ker}(T))\| \leq C\|v + \operatorname{Ker}(T)\|.$$

Thus S is bounded.
If $S(v + \operatorname{Ker}(T)) = 0$, then $Tv = 0$, so $v \in \operatorname{Ker}(T)$ and $v + \operatorname{Ker}(T) = 0$. Hence S is injective. It is surjective because every $w \in \operatorname{Ran}(T)$ has the form $w = Tv = S(v + \operatorname{Ker}(T))$.
By assumption, $\operatorname{Ran}(T)$ is a closed subspace of the Banach space W, so it is Banach. The quotient $V/\operatorname{Ker}(T)$ is also Banach. Since S is a bounded bijection between Banach spaces, the inverse corollary to the Open Mapping Theorem gives

$$S^{-1} \in \mathcal{B}(\operatorname{Ran}(T), V/\operatorname{Ker}(T)).$$

Thus

$$V/\operatorname{Ker}(T) \cong \operatorname{Ran}(T).$$

Counterexample: why the hypotheses cannot be dropped

This example shows that one cannot drop the hypotheses in the Open Mapping Theorem and the Closed Graph Theorem. Let

$$W = \left\{ a = (a_k)_{k=1}^{\infty} : \sum_{k=1}^{\infty} k|a_k| < \infty \right\},$$

equipped with the ℓ^1-norm. The space W is not complete because it is a proper dense subspace of ℓ^1. Indeed, if $b = (b_k) \in \ell^1$ and $\varepsilon > 0$, choose N so that the truncation

$$a = (b_1, \ldots, b_N, 0, \ldots) \in W$$

satisfies $\|a - b\|_1 < \varepsilon$.
Define

$$T : W \longrightarrow \ell^1, \qquad (Ta)_k = ka_k.$$

Then $\Gamma(T)$ is closed, yet T is not bounded. This is a counterexample to the Closed Graph Theorem without completeness of the domain. Moreover,

$$S = T^{-1} : \ell^1 \longrightarrow W$$

is bounded and surjective, yet it is not an open map. This is a counterexample to the Open Mapping Theorem without completeness of the codomain.

We now prove the assertion made at the beginning of this section that $(\ell^\infty)' \neq \ell^1$.

Homework theorem: a functional on ℓ^∞ not represented by ℓ^1

Let

$$M = \left\{ a = (a_k) \in \ell^\infty : \lim_{k\to\infty} a_k \text{ exists} \right\}$$

and define

$$u : M \longrightarrow \mathbb{C}, \qquad u(a) = \lim_{k\to\infty} a_k.$$

Then:

(1) u is a bounded linear functional on M.

(2) By the Hahn–Banach Theorem, there exists $v \in (\ell^\infty)'$ such that $v|_M = u$. There is no $b = (b_k) \in \ell^1$ such that, for every $a \in \ell^\infty$,

$$v(a) = \sum_{k=1}^{\infty} a_k b_k.$$

Proof

Linearity of u is clear. If $a \in M \subset \ell^\infty$, then

$$|u(a)| = \left| \lim_{k\to\infty} a_k \right| = \lim_{k\to\infty} |a_k| \leq \sup_k |a_k| = \|a\|_\infty.$$

Thus u is bounded.

For the second part, suppose, for a contradiction, that there exists $b = (b_k) \in \ell^1$ such that

$$v(a) = \sum_{k=1}^{\infty} a_k b_k \qquad \text{for every } a \in \ell^\infty.$$

Let $e_n = (\delta_{nk})_{k=1}^{\infty}$. Since $e_n \in M$, the way v was constructed gives

$$v(e_n) = u(e_n) = 0.$$

On the other hand, the assumed representation gives

$$v(e_n) = \sum_{k=1}^{\infty} \delta_{nk} b_k = b_n.$$

The functional v is not zero because it takes the value 1 on the constant sequence $(1, 1, \ldots)$. Hence $b \neq 0$, so at least one $b_N \neq 0$, a contradiction. Therefore

$$(\ell^\infty)' \neq \ell^1.$$

6.2 Lebesgue measure and integration

Consider

$$L^1_R([0,1]) = \{f : [0,1] \to \mathbb{C} : f \text{ is Riemann integrable on } [0,1]\}.$$

For a complex-valued function, the real and imaginary parts are integrated separately. If we define

$$\|f\|_1 = \int_0^1 |f(x)|\, dx,$$

there is a problem. For example, the function

$$f(x) = \begin{cases} 0, & x \in [0,1), \\ 1, & x = 1, \end{cases}$$

is nonzero, but $\int_0^1 |f| = 0$, so positive definiteness is not satisfied. For now, we treat this expression like a norm in order to see an even more serious issue arising with Riemann integration:

$$L^1_R([0,1]) \text{ is not a Banach space.}$$

One could obtain Lebesgue L^p-spaces by taking the completion, just as $\mathbb{R}$ is the completion of $\mathbb{Q}$, but we shall not do so here.

Lebesgue integration still calculates the area underneath a curve, but it is more general and more powerful than Riemann integration. In fact, the set of Riemann integrable functions is dense in the set of Lebesgue integrable functions.

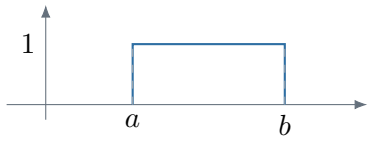
Given $E \subset \mathbb{R}$, how should we integrate its indicator function

$$\mathbf{1}_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \notin E? \end{cases}$$

If $E = [a, b]$, we would like

$$\int_{[a,b]} \mathbf{1}_E(x)\, dx = b - a = \ell([a, b]),$$

the length of the interval.



Similarly, for a general set E, we would like to define a "length" of E, called the *measure* of E and denoted by $m(E)$.

The task is to define the measure of subsets of $\mathbb{R}$ while retaining the following properties:

(1) We can measure every subset of $\mathbb{R}$. In other words, the measure is defined on the power set $\mathcal{P}(\mathbb{R})$.

(2) If $I \subset \mathbb{R}$ is an interval, then $m(I)$ is the length of the interval, whether it is open or closed.

(3) If (E_n) is a countable collection of pairwise disjoint sets and $E = \bigcup_n E_n$, then

$$m(E) = \sum_n m(E_n).$$

(4) If $E \subset \mathbb{R}$ and $x \in \mathbb{R}$, then

$$m(x + E) = m(E).$$

Unfortunately, it is impossible for a function

$$m : \mathcal{P}(\mathbb{R}) \longrightarrow [0, \infty]$$

to satisfy all four properties. It becomes possible if we drop the first requirement. We shall construct the Lebesgue measurable sets, which satisfy properties (2)–(4).

First we define an outer measure

$$m^* : \mathcal{P}(\mathbb{R}) \longrightarrow [0, \infty]$$

which satisfies (1), (2), (4), and almost satisfies (3). We then restrict m^* to appropriately well-behaved subsets of $\mathbb{R}$ in order to obtain (3).

6.3 Outer measure

If $I \subset \mathbb{R}$ is an interval, $\ell(I)$ denotes its length, regardless of whether it is open, closed, or half-closed.

Outer measure

For every subset $A \subset \mathbb{R}$, define the *outer measure* of A by

$$m^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \ell(I_n) : (I_n) \text{ is a countable collection of open intervals and } A \subset \bigcup_{n=1}^{\infty} I_n \right\}.$$

Directly from the definition,

$$m^*(A) \geq 0 \qquad \text{for every } A \subset \mathbb{R}.$$

A singleton has outer measure zero

For every $x_0 \in \mathbb{R}$,

$$m^*(\{x_0\}) = 0.$$

Indeed, for every $\varepsilon > 0$,

$$\{x_0\} \subset \left(x_0 - \frac{\varepsilon}{2}, x_0 + \frac{\varepsilon}{2}\right),$$

so

$$m^*(\{x_0\}) \leq \varepsilon.$$

Letting $\varepsilon \to 0$ proves the claim.

Countable sets have outer measure zero

If $A \subset \mathbb{R}$ is countable, then

$$m^*(A) = 0.$$

Proof

The assertion is immediate when A is finite. If A is countably infinite, enumerate it as

$$A = \{a_n : n \in \mathbb{N}\}.$$

Let $\varepsilon > 0$, and, for every $n \in \mathbb{N}$, set

$$I_n = \left(a_n - \frac{\varepsilon}{2^{n+1}}, a_n + \frac{\varepsilon}{2^{n+1}}\right).$$

Then

$$A \subset \bigcup_{n=1}^{\infty} I_n, \qquad \ell(I_n) = \frac{\varepsilon}{2^n},$$

and therefore

$$m^*(A) \leq \sum_{n=1}^{\infty} \ell(I_n) = \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon.$$

Letting $\varepsilon \to 0$ gives $m^*(A) = 0$.

We now see why outer measure "almost" satisfies countable additivity.

Monotonicity of outer measure

If $A \subset B$, then

$$m^*(A) \leq m^*(B).$$

Proof

Every countable open-interval covering of B is also a covering of A. Taking the infimum of the sums of lengths over the larger collection of covers gives $m^*(A) \leq m^*(B)$.

Countable subadditivity

Let (A_n) be a countable collection of subsets of $\mathbb{R}$, not necessarily disjoint. Then

$$m^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} m^*(A_n).$$

Proof

If one of the outer measures $m^*(A_n)$ is infinite, or if their sum diverges, the inequality is immediate. Suppose all the outer measures are finite and their sum converges. Let $\varepsilon > 0$. For every n, by the defining property of an infimum choose a countable collection of open intervals $(I_{nk})_{k=1}^{\infty}$ covering A_n such that

$$\sum_{k=1}^{\infty} \ell(I_{nk}) < m^*(A_n) + \frac{\varepsilon}{2^n}.$$

The collection $(I_{nk})_{n,k\in\mathbb{N}}$ is still countable, and

$$\bigcup_{n=1}^{\infty} A_n \subset \bigcup_{n,k\in\mathbb{N}} I_{nk}.$$

By monotonicity and the definition of outer measure,

$$\begin{aligned} m^*\left(\bigcup_{n=1}^{\infty} A_n\right) &\leq m^*\left(\bigcup_{n,k\in\mathbb{N}} I_{nk}\right) \\ &\leq \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \ell(I_{nk}) \\ &< \sum_{n=1}^{\infty} m^*(A_n) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \to 0$ completes the proof.

Homework theorem: translation invariance

Let $E \subset \mathbb{R}$ and $x \in \mathbb{R}$. If

$$E + x = \{y + x : y \in E\},$$

then

$$m^*(E + x) = m^*(E).$$

Thus outer measure is translation invariant.

Proof

Suppose (I_n) is a countable collection of open intervals covering E. Then $(I_n + x)$ is a countable collection of open intervals covering $E + x$, and

$$\ell(I_n + x) = \ell(I_n).$$

Taking infima gives $m^*(E+x) \leq m^*(E)$. Applying the same inequality to $E+x$ and the translation $-x$ gives the reverse inequality. Hence $m^*(E+x) = m^*(E)$.

Homework theorem: decomposition of open subsets of $\mathbb{R}$

Every open subset of $\mathbb{R}$ can be written as a countable union of open intervals.

Proof

Let $U \subset \mathbb{R}$ be a nonempty open set. For every $x \in U$, define

$$a_x = \inf\{a \in \mathbb{R} : (a, x] \subset U\}, \qquad b_x = \sup\{b \in \mathbb{R} : [x, b) \subset U\}.$$

Recall the definitions of supremum and infimum: $u \in \mathbb{R}$ is the supremum of $A \subset \mathbb{R}$ if

(1) $a \leq u$ for every $a \in A$;

(2) whenever $v \in \mathbb{R}$ is also an upper bound for A, one has $u \leq v$.

The definition of an infimum is the corresponding lower-bound statement.
Claim 1. For every $x \in U$,

$$(a_x, b_x) \subset U.$$

Indeed, let $z \in (a_x, b_x)$. Since a_x is an infimum and b_x is a supremum, there exist $a, b \in \mathbb{R}$ such that

$$a_x \leq a < z < b \leq b_x, \qquad (a, x] \subset U, \qquad [x, b) \subset U.$$

Thus $z \in (a, b) \subset U$.

Claim 2. If $y \in (a_x, b_x)$, then

$$(a_y, b_y) = (a_x, b_x).$$

By definition, $x > a$ implies $x > a_x$, and similarly $x < b_x$, so $x \in (a_x, b_x)$. If $y \in (a_x, b_x)$, the interval from a_x to y lies in U, so a_x is a lower bound for $\{a \in \mathbb{R} : (a, y] \subset U\}$. Hence $a_x \leq a_y$. Similarly, $b_y \leq b_x$. Thus

$$(a_y, b_y) \subset (a_x, b_x).$$

Interchanging x and y gives the reverse inclusion, so the intervals are equal. Intuitively, (a_x, b_x) is the largest open interval containing x and contained in U.
Claim 3. With indices written explicitly,

$$U = \bigcup_{q \in U \cap \mathbb{Q}} (a_q, b_q).$$

Since $U \cap \mathbb{Q} \subset \mathbb{Q}$ and $\mathbb{Q}$ is countable, this is a countable union. By Claim 1,

$$\bigcup_{q \in U \cap \mathbb{Q}} (a_q, b_q) \subset U.$$

For the reverse inclusion, let $x \in U$. Since U is open, there is an $\varepsilon_0 > 0$ such that

$$(x - \varepsilon_0, x + \varepsilon_0) \subset U.$$

By density of $\mathbb{Q}$ in $\mathbb{R}$, choose

$$q \in \mathbb{Q} \cap (x - \varepsilon_0, x + \varepsilon_0) \subset U.$$

The interval $(x - \varepsilon_0, x + \varepsilon_0)$ contains both q and x and is contained in U. By the definition of a_q, b_q,

$$x \in (a_q, b_q).$$

Therefore

$$U \subset \bigcup_{q \in U \cap \mathbb{Q}} (a_q, b_q),$$

which proves the theorem.

7 σ-Algebras

We first show that outer measure satisfies the interval-length property:

$$m^*(I) = \ell(I)$$

whenever $I \subset \mathbb{R}$ is an interval.

Recall that, for $A \subset \mathbb{R}$,

$$m^*(A) = \inf \left\{ \sum_n \ell(I_n) : (I_n) \text{ is a countable collection of open intervals and } A \subset \bigcup_n I_n \right\}.$$

We have already proved monotonicity,

$$A \subset B \quad \Longrightarrow \quad m^*(A) \leq m^*(B),$$

and countable subadditivity,

$$m^* \left(\bigcup_n E_n \right) \leq \sum_n m^*(E_n),$$

for countable collections that need not be disjoint.

Outer measure of an interval

If $I \subset \mathbb{R}$ is an interval, whether open, closed, or half-closed, then

$$m^*(I) = \ell(I).$$

Proof

First suppose that $I = [a, b]$ is closed and bounded. For every $\varepsilon > 0$,

$$[a, b] \subset (a - \varepsilon, b + \varepsilon),$$

so

$$m^*(I) \leq b - a + 2\varepsilon.$$

Letting $\varepsilon \to 0$ gives

$$m^*(I) \leq b - a.$$

For the reverse inequality, let $(I_n)_{n=1}^{\infty}$ be any countable collection of open intervals covering $[a, b]$. By the Heine–Borel Theorem, $[a, b]$ is compact, so the open cover has a finite subcover

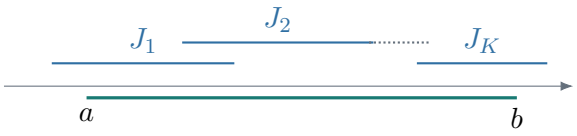
$$[a, b] \subset \bigcup_{k=1}^{N} J_k.$$

Write $J_k = (a_k, b_k)$. Since a lies in this finite union, after rearranging the intervals we may assume $a \in J_1$. If J_1 does not already cover through b, then $b_1 \in [a, b]$, so another interval of the subcover contains b_1. Call it J_2; then

$$a_2 < b_1 < b_2.$$

Repeating this process, we obtain some $K \leq N$ such that

$$a \in J_1, \qquad a_{k+1} < b_k < b_{k+1} \quad (1 \leq k < K), \qquad b < b_K.$$



Therefore

$$\begin{aligned}
\sum_{n=1}^{\infty} \ell(I_n) &\geq \sum_{k=1}^{K} \ell(J_k) \\
&= \sum_{k=1}^{K} (b_k - a_k) \\
&= b_K + (b_{K-1} - a_K) + \cdots + (b_1 - a_2) - a_1 \\
&\geq b_K - a_1 \\
&\geq b - a.
\end{aligned}$$

This is true for every countable open cover of I, so

$$b - a \leq m^*(I).$$

Combining the two bounds yields $m^*(I) = b - a = \ell(I)$.
Now let I be any bounded interval with endpoints $a < b$, including $[a, b)$, $(a, b]$, (a, b), and $[a, b]$. For every $\varepsilon > 0$,

$$[a + \varepsilon, b - \varepsilon] \subset I \subset [a - \varepsilon, b + \varepsilon].$$

By monotonicity and the result for closed bounded intervals,

$$b - a - 2\varepsilon \leq m^*(I) \leq b - a + 2\varepsilon.$$

The Squeeze Theorem, with $\varepsilon \to 0$, gives

$$m^*(I) = b - a.$$

Finally, suppose that I is an unbounded interval, such as

$$(-\infty, a), \quad (a, \infty), \quad (-\infty, a], \quad [a, \infty), \quad \text{or} \quad \mathbb{R}.$$

For instance, $[0, n] \subset [0, \infty)$ for every $n \in \mathbb{N}$, so

$$n = m^*([0, n]) \leq m^*([0, \infty)).$$

Since (n) is unbounded, $m^*([0, \infty)) = \infty$. The same argument applies to every unbounded interval. Hence $m^*(I) = \infty = \ell(I)$.

Approximation from outside by an open set

For every $A \subset \mathbb{R}$ and every $\varepsilon > 0$, there exists an open set O such that

$$A \subset O \qquad \text{and} \qquad m^*(A) \leq m^*(O) \leq m^*(A) + \varepsilon.$$

Proof

If $m^*(A) = \infty$, take $O = \mathbb{R}$. Suppose $m^*(A) < \infty$. By the definition of the infimum, choose a countable collection of open intervals (I_n) covering A such that

$$\sum_n \ell(I_n) \leq m^*(A) + \varepsilon.$$

Let

$$O = \bigcup_{n=1}^{\infty} I_n.$$

Then O is open, $A \subset O$, and, by monotonicity, countable subadditivity, and the interval theorem,

$$m^*(A) \leq m^*(O) \leq \sum_n m^*(I_n) = \sum_n \ell(I_n) \leq m^*(A) + \varepsilon.$$

This gives a useful approximation of A from outside by an open set.

7.1 Lebesgue measurable sets

Lebesgue measurable set

A set $E \subset \mathbb{R}$ is *Lebesgue measurable* if, for every $A \subset \mathbb{R}$,

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c).$$

By countable subadditivity,

$$m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c),$$

because

$$A = (A \cap E) \cup (A \cap E^c).$$

Thus, to prove measurability, it is enough to prove the reverse inequality

$$m^*(A \cap E) + m^*(A \cap E^c) \leq m^*(A) \qquad \text{for every } A \subset \mathbb{R}.$$

Elementary measurable sets and complements

The sets $\varnothing$ and $\mathbb{R}$ are measurable. Moreover, for every $E \subset \mathbb{R}$,

$$E \text{ is measurable} \iff E^c \text{ is measurable.}$$

Proof

For every $A \subset \mathbb{R}$,

$$A \cap \varnothing = \varnothing, \qquad A \cap \varnothing^c = A \cap \mathbb{R} = A.$$

Hence

$$m^*(A) = m^*(A \cap \varnothing) + m^*(A \cap \varnothing^c) = 0 + m^*(A),$$

so $\varnothing$ is measurable. The proof for $\mathbb{R}$ is the same.
If E is measurable, then

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c) = m^*(A \cap (E^c)^c) + m^*(A \cap E^c),$$

so E^c is measurable. The converse is identical.

Nonmeasurable sets

It is not easy to give a simple example of a nonmeasurable subset of $\mathbb{R}$. The construction relies on the Axiom of Choice. A classical example is the Vitali set, which is not measurable.

Sets of outer measure zero are measurable

If $E \subset \mathbb{R}$ and $m^*(E) = 0$, then E is Lebesgue measurable.

Proof

For every $A \subset \mathbb{R}$, monotonicity gives

$$m^*(A \cap E) \leq m^*(E) = 0,$$

and hence $m^*(A \cap E) = 0$. Therefore

$$m^*(A \cap E) + m^*(A \cap E^c) = m^*(A \cap E^c) \leq m^*(A),$$

again by monotonicity. Together with the reverse inequality from subadditivity, this proves that E is measurable.

In fact, every open set is measurable, and therefore every closed set is also measurable by taking its complement. Before proving that, we give another way to produce measurable sets.

Finite unions of measurable sets

If E_1 and E_2 are measurable, then $E_1 \cup E_2$ is measurable.

Proof

Let $A \subset \mathbb{R}$. Since E_2 is measurable, apply its defining identity to $A \cap E_1^c$:

$$m^*(A \cap E_1^c) = m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c).$$

By De Morgan's law,

$$(E_1 \cup E_2)^c = E_1^c \cap E_2^c,$$

and

$$A \cap (E_1 \cup E_2) = (A \cap E_1) \cup (A \cap E_2 \cap E_1^c).$$

Countable subadditivity gives

$$m^*(A \cap (E_1 \cup E_2)) \leq m^*(A \cap E_1) + m^*(A \cap E_2 \cap E_1^c).$$

Since E_1 is measurable,

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c) \\ &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c). \end{aligned}$$

Combining the last two displays yields

$$m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c) \leq m^*(A).$$

The reverse inequality follows from subadditivity, so $E_1 \cup E_2$ is measurable.

Corollary

If $E_1, \ldots, E_n$ are measurable, then

$$\bigcup_{k=1}^{n} E_k$$

is measurable.

Proof

This follows from the preceding theorem and induction.

Using this result, we can describe the structure of collections of measurable sets.

Algebra and σ-algebra

A nonempty collection $\mathcal{A} \subset \mathcal{P}(\mathbb{R})$ is an *algebra* if:

(1) $E \in \mathcal{A}$ implies $E^c \in \mathcal{A}$;

(2) if $E_1, \ldots, E_n \in \mathcal{A}$, then

$$\bigcup_{k=1}^{n} E_k \in \mathcal{A}.$$

An algebra $\mathcal{A}$ is a *σ-algebra* if, for every countable collection $(E_n)_{n=1}^{\infty} \subset \mathcal{A}$,

$$\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}.$$

Closure properties

Algebras are closed under complements and finite unions. σ-algebras are algebras that are also closed under countable unions. By De Morgan's laws,

$$\bigcap_{k=1}^{n} E_k = \left(\bigcup_{k=1}^{n} E_k^c \right)^c,$$

so algebras are also closed under finite intersections. In particular,

$$\varnothing = E \cap E^c \in \mathcal{A}, \qquad \mathbb{R} = \varnothing^c \in \mathcal{A}.$$

Similarly, σ-algebras are closed under countable intersections.

Examples of σ-algebras

The collections

$$\{\varnothing, \mathbb{R}\}, \qquad \mathcal{P}(\mathbb{R}),$$

are σ-algebras. So is the countable–cocountable collection

$$\mathcal{A} = \{E \subset \mathbb{R} : E \text{ is countable or } E^c \text{ is countable}\}.$$

To verify the last example, consider a countable collection $(E_n) \subset \mathcal{A}$. If every E_n is countable, then $\bigcup_n E_n$ is countable. Otherwise, there is an N_0 such that E_{N_0} is uncountable, in which case $E_{N_0}^c$ is countable. Then

$$\left(\bigcup_n E_n \right)^c = \bigcap_n E_n^c \subset E_{N_0}^c,$$

so the complement of $\bigcup_n E_n$ is countable. Thus $\bigcup_n E_n \in \mathcal{A}$.

We shall soon show that

$$\mathcal{M} = \{E \subset \mathbb{R} : E \text{ is Lebesgue measurable}\}$$

is a σ-algebra.

The Borel σ-algebra

Let

$$\Sigma = \{\mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra containing every open subset of } \mathbb{R}\}.$$

The collection

$$\mathcal{B} = \bigcap_{\mathcal{A} \in \Sigma} \mathcal{A}$$

is the smallest σ-algebra containing every open subset of $\mathbb{R}$. It is called the *Borel σ-algebra*.

The collection $\mathcal{B}$ is indeed a σ-algebra. If $E \in \mathcal{B}$, then $E \in \mathcal{A}$ for every $\mathcal{A} \in \Sigma$. Since every such $\mathcal{A}$ is a σ-algebra, $E^c \in \mathcal{A}$ for every $\mathcal{A} \in \Sigma$, and hence $E^c \in \mathcal{B}$. If $(E_n) \subset \mathcal{B}$, then $(E_n) \subset \mathcal{A}$ for every $\mathcal{A} \in \Sigma$, so $\bigcup_n E_n \in \mathcal{A}$ for every $\mathcal{A} \in \Sigma$. Thus $\bigcup_n E_n \in \mathcal{B}$.

The Borel σ-algebra contains more than open sets

The Borel σ-algebra is not merely the collection of all open subsets of $\mathbb{R}$. Since a σ-algebra is

closed under countable intersections, for $a < b$,

$$[a,b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n}\right) \in \mathcal{B}.$$

Also, $\varnothing, \mathbb{R} \in \mathcal{B}$.

Recall that we proved that every countable subset of $\mathbb{R}$ has outer measure zero and is therefore measurable. More generally, every set of outer measure zero is measurable. The converse is false: a measurable set of measure zero need not be countable.

The Cantor set

The Cantor set can be written as

$$C = [0,1] \setminus \bigcup_{n=0}^{\infty} \bigcup_{k=0}^{3^n-1} \left(\frac{3k+1}{3^{n+1}}, \frac{3k+2}{3^{n+1}}\right).$$

Its construction proceeds by repeatedly deleting the open middle third of every remaining interval:

$$[0,1] \longrightarrow \left[0,\frac{1}{3}\right] \cup \left[\frac{2}{3},1\right] \longrightarrow \left[0,\frac{1}{9}\right] \cup \left[\frac{2}{9},\frac{1}{3}\right] \cup \left[\frac{2}{3},\frac{7}{9}\right] \cup \left[\frac{8}{9},1\right] \longrightarrow \cdots.$$



The Cantor set is uncountable, yet it has outer measure zero: after n stages, the total length of the remaining intervals is $(2/3)^n$, which tends to zero.

8 Lebesgue Measurable Subsets and Measures

Recall that $E \subset \mathbb{R}$ is Lebesgue measurable if, for every $A \subset \mathbb{R}$,

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c),$$

and let

$$\mathcal{M} = \{E \subset \mathbb{R} : E \text{ is Lebesgue measurable}\}.$$

We showed on page 64 that $\mathcal{M}$ is an algebra. In this section we show that it is a σ-algebra containing the Borel σ-algebra $\mathcal{B}$, and we define Lebesgue measure on it. We begin with a general observation about algebras.

Disjointification lemma

Let $\mathcal{A}$ be an algebra, and let $(E_n)_{n=1}^{\infty} \subset \mathcal{A}$. Then there exists a sequence $(F_n)_{n=1}^{\infty} \subset \mathcal{A}$ of pairwise disjoint sets such that

$$\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} F_n.$$

Consequently, to check that an algebra is a σ-algebra, it is enough to check closure under countable unions of pairwise disjoint sets.

Proof

Let

$$G_n = \bigcup_{k=1}^{n} E_k.$$

Then $G_1 \subset G_2 \subset G_3 \subset \cdots$, and

$$\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} G_n.$$

Define

$$F_1 = G_1, \qquad F_{n+1} = G_{n+1} \setminus G_n \quad (n \geq 1).$$

Since $\mathcal{A}$ is an algebra, every $F_n \in \mathcal{A}$. The sets F_n are pairwise disjoint, and, for every n,

$$\bigcup_{k=1}^{n} F_k = G_n.$$

Therefore

$$\bigcup_{k=1}^{\infty} F_k = \bigcup_{n=1}^{\infty} G_n = \bigcup_{n=1}^{\infty} E_n.$$

We now return to measurable sets.

Finite additivity across disjoint measurable sets

Let $A \subset \mathbb{R}$, and let $E_1, \ldots, E_n$ be pairwise disjoint measurable sets. Then

$$m^*\left(A \cap \bigcup_{k=1}^{n} E_k\right) = \sum_{k=1}^{n} m^*(A \cap E_k).$$

The case of a set and its complement

For $E_1 = E$ and $E_2 = E^c$, the assertion is precisely the defining identity for the measurability of E.

Proof by induction

The case $n = 1$ is immediate. Suppose the equality holds for $n = m$, and let $E_1, \ldots, E_{m+1}$ be pairwise disjoint measurable sets. Since

$$\left(A \cap \bigcup_{k=1}^{m+1} E_k\right) \cap E_{m+1} = A \cap E_{m+1},$$

and

$$\left(A \cap \bigcup_{k=1}^{m+1} E_k\right) \cap E_{m+1}^c = A \cap \bigcup_{k=1}^{m} E_k,$$

the measurability of E_{m+1} gives

$$\begin{aligned} m^*\left(A \cap \bigcup_{k=1}^{m+1} E_k\right) &= m^*(A \cap E_{m+1}) + m^*\left(A \cap \bigcup_{k=1}^{m} E_k\right) \\ &= m^*(A \cap E_{m+1}) + \sum_{k=1}^{m} m^*(A \cap E_k) \\ &= \sum_{k=1}^{m+1} m^*(A \cap E_k), \end{aligned}$$

by the induction hypothesis.

The measurable sets form a σ-algebra

The collection $\mathcal{M}$ of Lebesgue measurable subsets of $\mathbb{R}$ is a σ-algebra.

Proof

We already proved on page 64 that $\mathcal{M}$ is an algebra: it is closed under complements and finite unions. By the disjointification lemma, it remains only to show closure under countable disjoint unions.
Let $(E_n)_{n=1}^{\infty}$ be a countable collection of pairwise disjoint measurable sets, and set

$$E = \bigcup_{n=1}^{\infty} E_n.$$

For an arbitrary $A \subset \mathbb{R}$, we prove

$$m^*(A \cap E) + m^*(A \cap E^c) \leq m^*(A).$$

For every $N \in \mathbb{N}$, the finite union $\bigcup_{n=1}^{N} E_n$ is measurable. Hence

$$m^*(A) = m^*\left(A \cap \bigcup_{n=1}^{N} E_n\right) + m^*\left(A \cap \left(\bigcup_{n=1}^{N} E_n\right)^c\right).$$

Since $\bigcup_{n=1}^{N} E_n \subset E$,

$$E^c \subset \left(\bigcup_{n=1}^{N} E_n\right)^c.$$

By monotonicity and the finite-additivity theorem,

$$m^*(A) \geq \sum_{n=1}^{N} m^*(A \cap E_n) + m^*(A \cap E^c).$$

Letting $N \to \infty$ gives

$$m^*(A) \geq \sum_{n=1}^{\infty} m^*(A \cap E_n) + m^*(A \cap E^c).$$

By countable subadditivity,

$$\sum_{n=1}^{\infty} m^*(A \cap E_n) \geq m^*\left(\bigcup_{n=1}^{\infty} (A \cap E_n)\right) = m^*(A \cap E).$$

Therefore

$$m^*(A) \geq m^*(A \cap E) + m^*(A \cap E^c),$$

so E is measurable. Thus $\mathcal{M}$ is a σ-algebra.

Why this σ-algebra matters

A fundamental property expected of a measure is countable additivity on pairwise disjoint sets. This is why the preceding lemmas and theorem are formulated in relation to σ-algebras. Once we prove that $\mathcal{M}$ contains every open subset of $\mathbb{R}$, the minimality of the Borel σ-algebra will give

$$\mathcal{B} \subset \mathcal{M}.$$

We first prove measurability for an easier class of open sets.

Open half-lines are measurable

For every $a \in \mathbb{R}$, the set

$$(a, \infty)$$

is Lebesgue measurable.

Proof

Let $A \subset \mathbb{R}$. As before, we use the fact that an inequality $X \leq Y + \varepsilon$ for every $\varepsilon > 0$ implies $X \leq Y$. Set

$$A_1 = A \cap (a, \infty), \qquad A_2 = A \cap (a, \infty)^c = A \cap (-\infty, a].$$

We show that

$$m^*(A_1) + m^*(A_2) \leq m^*(A).$$

Let $\varepsilon > 0$, and choose a countable collection of open intervals (I_n) covering A such that

$$\sum_n \ell(I_n) \leq m^*(A) + \varepsilon.$$

Define

$$J_n = I_n \cap (a, \infty), \qquad K_n = I_n \cap (-\infty, a].$$

Each J_n and K_n is either an interval or the empty set. Moreover,

$$A_1 \subset \bigcup_n J_n, \qquad A_2 \subset \bigcup_n K_n, \qquad \ell(I_n) = \ell(J_n) + \ell(K_n).$$

Therefore, by monotonicity, countable subadditivity, and the interval formula for outer measure,

$$\begin{aligned} m^*(A_1) + m^*(A_2) &\leq m^*\left(\bigcup_n J_n\right) + m^*\left(\bigcup_n K_n\right) \\ &\leq \sum_n m^*(J_n) + \sum_n m^*(K_n) \\ &= \sum_n \ell(J_n) + \sum_n \ell(K_n) \\ &= \sum_n \ell(I_n) \\ &\leq m^*(A) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \to 0$ proves the result.

Every open set is measurable

Every open subset of $\mathbb{R}$ is Lebesgue measurable. Consequently,

$$\mathcal{B} \subset \mathcal{M}.$$

Proof

We have shown that (a, ∞) is measurable for every $a \in \mathbb{R}$. Since $\mathcal{M}$ is closed under complements and countable unions,

$$(-\infty, b) = \bigcup_{n=1}^{\infty} (-\infty, b - 1/n] = \bigcup_{n=1}^{\infty} (b - 1/n, \infty)^c$$

is measurable. Because $\mathcal{M}$ is also closed under countable intersections, every bounded open interval is measurable:

$$(a, b) = (-\infty, b) \cap (a, \infty) \in \mathcal{M}.$$

Finally, every open subset of $\mathbb{R}$ is a countable union of open intervals by the theorem on page 61. Thus every open set is measurable. Since $\mathcal{B}$ is the smallest σ-algebra containing all open sets and $\mathcal{M}$ is such a σ-algebra, $\mathcal{B} \subset \mathcal{M}$.

We can now define a measure.

Lebesgue measure

If $E \in \mathcal{M}$ is Lebesgue measurable, its *Lebesgue measure* is

$$m(E) = m^*(E).$$

Basic properties of Lebesgue measure

If $A, B \in \mathcal{M}$ and $A \subset B$, then

$$m(A) \leq m(B).$$

Every interval $I \subset \mathbb{R}$ is measurable, and

$$m(I) = \ell(I).$$

Proof

Monotonicity and the interval-length formula are inherited directly from outer measure once the sets involved are measurable. It therefore remains only to check that every interval is measurable.
We already know that open intervals are measurable. Closed and half-closed bounded intervals can be written in terms of measurable half-lines; for example,

$$[a,b] = (b,\infty)^c \cap (-\infty,a)^c, \qquad [a,b) = (-\infty,b) \cap (-\infty,a)^c.$$

The other cases are similar. Thus every interval belongs to $\mathcal{M}$, and

$$m(I) = m^*(I) = \ell(I).$$

Why Lebesgue measurability is powerful

As mentioned on page 64, there is no simple nonmeasurable subset of $\mathbb{R}$. This is precisely why Lebesgue measure is so useful. For example, the Dirichlet function can now be integrated over $\mathbb{Q}$ or $\mathbb{R} \setminus \mathbb{Q}$.

The Vitali set and the Axiom of Choice

Define a relation on $\mathbb{R}$ by

$$x \sim y \quad \Longleftrightarrow \quad x - y \in \mathbb{Q}.$$

This is an equivalence relation on $\mathbb{R}$.
The Axiom of Choice states that, for any set X of nonempty sets, there exists a choice function that selects one element from each member of X. For a finite collection this appears elementary, but for infinite collections the Axiom of Choice is a fundamental principle of analysis, set theory, and modern mathematics.
Using the Axiom of Choice, select one representative from each equivalence class of $\sim$. The resulting set is a Vitali set. It is nonmeasurable: the translation classes force incompatible conclusions that its measure must behave both like zero and like a nonzero quantity.

We can now verify countable additivity for Lebesgue measure.

Countable additivity

Let $(E_n)_{n=1}^{\infty}$ be a countable collection of pairwise disjoint Lebesgue measurable sets. Then

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} m(E_n).$$

Proof

The union belongs to $\mathcal{M}$ because $\mathcal{M}$ is a σ-algebra. Countable subadditivity of outer measure gives

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} m(E_n).$$

For the reverse inequality, fix $N \in \mathbb{N}$. By finite additivity,

$$\sum_{n=1}^{N} m(E_n) = m\left(\bigcup_{n=1}^{N} E_n\right).$$

Since

$$\bigcup_{n=1}^{N} E_n \subset \bigcup_{n=1}^{\infty} E_n,$$

monotonicity gives

$$\sum_{n=1}^{N} m(E_n) \leq m\left(\bigcup_{n=1}^{\infty} E_n\right).$$

Letting $N \to \infty$ proves the reverse inequality and hence equality.

Outer measure is translation invariant, and therefore Lebesgue measure is translation invariant as well.

Continuity of measure from below

Let $(E_n)_{n=1}^{\infty}$ be a countable collection of measurable sets such that

$$E_1 \subset E_2 \subset \cdots.$$

Then

$$m\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n\to\infty} m\left(\bigcup_{k=1}^{n} E_k\right) = \lim_{n\to\infty} m(E_n).$$

Proof

The second equality follows from the inclusions, since

$$\bigcup_{k=1}^{n} E_k = E_n \qquad \text{for every } n.$$

It remains to prove the first equality. Rewrite the countable union as a disjoint union by setting

$$F_1 = E_1, \qquad F_k = E_k \setminus E_{k-1} \quad (k \geq 2).$$

Then the sets (F_k) are pairwise disjoint and measurable,

$$\bigcup_{k=1}^{n} F_k = E_n, \qquad \bigcup_{k=1}^{\infty} F_k = \bigcup_{n=1}^{\infty} E_n.$$

By countable additivity,

$$\begin{aligned} m\left(\bigcup_{n=1}^{\infty} E_n\right) &= m\left(\bigcup_{k=1}^{\infty} F_k\right) \\ &= \sum_{k=1}^{\infty} m(F_k) \end{aligned}$$

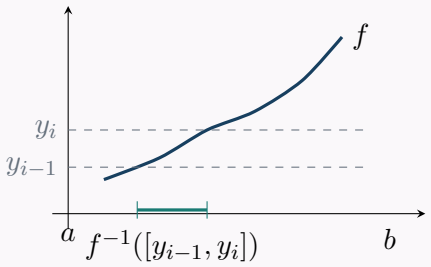
$$
\begin{aligned}
&= \lim_{N\to\infty} \sum_{k=1}^{N} m(F_k) \\
&= \lim_{N\to\infty} m\left(\bigcup_{k=1}^{N} F_k\right) \\
&= \lim_{N\to\infty} m(E_N).
\end{aligned}
$$

9 Lebesgue Measurable Functions

Motivation: slicing the range

Lebesgue's idea for constructing integrals is to slice the range instead of slicing the domain. Schematically,

$$\int_a^b f = \lim_{\text{partitions get finer}} \sum_{i=1}^{n} y_{i-1}\, \ell\Big(f^{-1}([y_{i-1}, y_i])\Big)\,.$$



Thus one uses inverse images in place of the "dx" of a Riemann integral. The problem is that $f^{-1}([y_{i-1}, y_i])$ might not be an interval. This naturally motivates considering functions f for which these inverse images are at least measurable.

Although we have already been using the following notation, we clarify it here. We consider the extended real line

$$[-\infty, \infty] = \mathbb{R} \cup \{-\infty, \infty\},$$

and we allow functions to take the values $\pm\infty$.

Recall that a sequence $(a_n)_{n=1}^{\infty} \subset \mathbb{R}$ converges to ∞ if, for every $N > 0$, there exists $M \in \mathbb{N}$ such that

$$n > M \quad \Longrightarrow \quad a_n > N.$$

We use the following conventions for algebraic operations involving $\pm\infty$:

- $x + (+\infty) = +\infty$ for every $x \in \mathbb{R}$;
- $0 \cdot (\pm\infty) = 0$. This is an algebraic convention, not a limiting procedure. For example,

$$0 \cdot (+\infty) \neq \lim_{x\to\infty} x \cdot \frac{1}{x};$$

- $$x \cdot (+\infty) = \begin{cases} +\infty, & x > 0, \\ -\infty, & x < 0. \end{cases}$$

Lebesgue measurable function

Let $E \subset \mathbb{R}$ be measurable, and let $f : E \to [-\infty, \infty]$. We say that f is *Lebesgue measurable* if, for every $\alpha \in \mathbb{R}$,

$$f^{-1}((\alpha, \infty]) \in \mathcal{M}.$$

An absolutely natural question is: why use the interval $(\alpha, \infty]$? Why not use one of $[-\infty, \alpha)$, $[-\infty, \alpha]$, or $[\alpha, \infty]$?

Equivalent tests for measurability

Let $E \subset \mathbb{R}$ be measurable, and let $f : E \to [-\infty, \infty]$. The following conditions are equivalent:

1. for every $\alpha \in \mathbb{R}$, $f^{-1}((\alpha, \infty]) \in \mathcal{M}$;
2. for every $\alpha \in \mathbb{R}$, $f^{-1}([\alpha, \infty]) \in \mathcal{M}$;
3. for every $\alpha \in \mathbb{R}$, $f^{-1}([-\infty, \alpha)) \in \mathcal{M}$;
4. for every $\alpha \in \mathbb{R}$, $f^{-1}([-\infty, \alpha]) \in \mathcal{M}$.

Proof

For $1 \implies 2$, note that

$$[\alpha, \infty] = \bigcap_{n=1}^{\infty} \left(\alpha - \frac{1}{n}, \infty\right].$$

Because inverse images respect set operations,

$$f^{-1}([\alpha, \infty]) = \bigcap_{n=1}^{\infty} f^{-1}\left(\left(\alpha - \frac{1}{n}, \infty\right]\right).$$

Every set on the right is measurable by assumption, so their countable intersection is measurable because $\mathcal{M}$ is a σ-algebra.
For $2 \implies 1$, similarly,

$$(\alpha, \infty] = \bigcup_{n=1}^{\infty} \left[\alpha + \frac{1}{n}, \infty\right],$$

and therefore

$$f^{-1}((\alpha, \infty]) = \bigcup_{n=1}^{\infty} f^{-1}\left(\left[\alpha + \frac{1}{n}, \infty\right]\right) \in \mathcal{M}.$$

For $3 \implies 4$, note that

$$[-\infty, \alpha] = \bigcap_{n=1}^{\infty} \left[-\infty, \alpha + \frac{1}{n}\right).$$

Hence

$$f^{-1}([-\infty, \alpha]) = \bigcap_{n=1}^{\infty} f^{-1}\left(\left[-\infty, \alpha + \frac{1}{n}\right)\right) \in \mathcal{M}.$$

For $4 \implies 3$, note that

$$[-\infty, \alpha) = \bigcup_{n=1}^{\infty} \left[-\infty, \alpha - \frac{1}{n}\right].$$

Thus

$$f^{-1}([-\infty, \alpha)) = \bigcup_{n=1}^{\infty} f^{-1}\left(\left[-\infty, \alpha - \frac{1}{n}\right]\right) \in \mathcal{M}.$$

Finally, $2 \iff 3$ because

$$[-\infty, \alpha) = [\alpha, \infty]^c,$$

and inverse images and σ-algebras respect complements.

Inverse images of Borel sets

Let $E \in \mathcal{M}$, and let $f : E \to \mathbb{R}$ be measurable. For every Borel set $F \in \mathcal{B}$, the inverse image $f^{-1}(F)$ is measurable.

Proof

Suppose that f is measurable. For every open interval $(a, b) \subset \mathbb{R}$,

$$f^{-1}((a, b)) = f^{-1}([-\infty, b)) \cap f^{-1}((a, \infty]).$$

Both sets on the right are measurable by the preceding theorem, so $f^{-1}((a, b))$ is measurable. Now let $S \subset \mathbb{R}$ be open. By the theorem on page 60, S is a countable union of open intervals. Hence $f^{-1}(S)$ is a countable union of measurable sets and is measurable.
Define

$$\mathcal{A} = \{U \in \mathcal{P}(\mathbb{R}) : f^{-1}(U) \text{ is measurable}\}.$$

Because inverse images respect complements and countable unions, $\mathcal{A}$ is a σ-algebra. We have shown that every open set belongs to $\mathcal{A}$. Since $\mathcal{B}$ is the smallest σ-algebra containing every open subset of $\mathbb{R}$,

$$\mathcal{B} \subset \mathcal{A}.$$

Therefore $f^{-1}(F)$ is measurable for every $F \in \mathcal{B}$. In words, the inverse images of Borel sets are measurable.

Inverse image versus inverse map

The notation $f^{-1}(Y)$ for an inverse image does not mean $\{f^{-1}(y) : y \in Y\}$. Given $f : X \to Y$, an inverse map exists if and only if f is bijective. In contrast, for a set A, the inverse image is always defined by

$$f^{-1}(A) = \{x \in X : f(x) \in A\}.$$

For example, if $f : [0, 1] \to [0, 1]$, then

$$f^{-1}([0, 1]) = f^{-1}(\mathbb{R}) = f^{-1}(\mathbb{C})$$

as inverse images, regardless of whether f is bijective. The inverse map f^{-1} need not exist, and expressions such as $f^{-1}(i)$ or $f^{-1}(100)$ need not exist as values of an inverse map. Equivalently,

$$f^{-1}(\mathbb{R} \setminus [0, 1]) = \varnothing, \qquad f^{-1}(\mathbb{C} \setminus [0, 1]) = \varnothing$$

as inverse images.

The infinite level sets are measurable

Let $f : E \to [-\infty, \infty]$ be measurable, where E is measurable. Then

$$f^{-1}(\{\infty\}) \quad \text{and} \quad f^{-1}(\{-\infty\})$$

are measurable.

Proof

Note that

$$f^{-1}(\{\infty\}) = f^{-1}\left(\bigcap_{n=1}^{\infty}(n, \infty]\right) = \bigcap_{n=1}^{\infty} f^{-1}((n, \infty]).$$

Each set $f^{-1}((n, \infty])$ is measurable because f is measurable, so $f^{-1}(\{\infty\})$ is measurable. Similarly, $f^{-1}(\{-\infty\})$ is measurable, also by the equivalent tests on page 76.

Thus, for every $F \in \mathcal{B} \cup \{\{-\infty\}, \{\infty\}\}$, the measurability of f implies that $f^{-1}(F)$ is measurable.

Continuous functions are measurable

If $f : \mathbb{R} \to \mathbb{R}$ is continuous, then f is measurable.

Proof

Recall that a function is continuous if and only if the inverse image of every open set is open. For every $\alpha \in \mathbb{R}$,

$$f^{-1}((\alpha, \infty]) = f^{-1}((\alpha, \infty)).$$

The set (α, ∞) is open in $\mathbb{R}$, so its inverse image is open. Every open set is measurable by the theorem on page 72. Therefore

$$f^{-1}((\alpha, \infty]) \in \mathcal{M} \qquad (\alpha \in \mathbb{R}),$$

and f is measurable.

Indicator functions

Let $E, F \subset \mathbb{R}$ be measurable. Define the indicator function $\chi_F : E \to \mathbb{R}$ by

$$\chi_F(x) = \begin{cases} 1, & x \in F, \\ 0, & x \notin F. \end{cases}$$

Then χ_F is measurable.
Indeed,

$$\chi_F^{-1}((\alpha, \infty]) = \begin{cases} \varnothing, & \alpha \geq 1, \\ E \cap F, & 0 \leq \alpha < 1, \\ E, & \alpha < 0. \end{cases}$$

All three cases are measurable, so χ_F is measurable.

Algebraic operations preserve measurability

Let $E \subset \mathbb{R}$ be measurable, let $f, g : E \to \mathbb{R}$ be measurable functions, and let $c \in \mathbb{R}$. Then

$$cf, \qquad f+g, \qquad fg$$

are measurable.

Proof: scalar multiplication

If $c = 0$, then $cf = 0$ is continuous and hence measurable by the preceding theorem.
If $c > 0$, then, for every $\alpha \in \mathbb{R}$,

$$cf(x) > \alpha \quad \iff \quad f(x) > \frac{\alpha}{c}.$$

Thus

$$(cf)^{-1}((\alpha, \infty]) = f^{-1}\left(\left(\frac{\alpha}{c}, \infty\right]\right),$$

which is measurable.
If $c < 0$, then similarly

$$(cf)^{-1}((\alpha, \infty]) = f^{-1}\left(\left[-\infty, \frac{\alpha}{c}\right)\right),$$

which is measurable by the equivalent tests on page 76. Therefore cf is measurable in every case.

Proof: addition

Let $\alpha \in \mathbb{R}$. Then

$$\begin{aligned} f(x) + g(x) > \alpha &\iff f(x) > \alpha - g(x) \\ &\iff f(x) > r > \alpha - g(x) \quad \text{for some } r \in \mathbb{Q}, \end{aligned}$$

by the density of $\mathbb{Q}$ in $\mathbb{R}$. Thus

$$x \in f^{-1}((r, \infty]) \cap g^{-1}((\alpha - r, \infty])$$

for some $r \in \mathbb{Q}$, and consequently

$$(f+g)^{-1}((\alpha, \infty]) = \bigcup_{r \in \mathbb{Q}} \left(f^{-1}((r, \infty]) \cap g^{-1}((\alpha - r, \infty])\right).$$

The right-hand side is a countable union of measurable sets, so it is measurable. Therefore $f + g$ is measurable.

Proof: multiplication

We first show that f^2 is measurable. If $\alpha < 0$, then

$$(f^2)^{-1}((\alpha, \infty]) = E,$$

which is measurable by assumption. If $\alpha \geq 0$, then

$$f^2 > \alpha \quad \Longleftrightarrow \quad f > \sqrt{\alpha} \ \text{ or } \ f < -\sqrt{\alpha},$$

and hence

$$(f^2)^{-1}((\alpha, \infty]) = f^{-1}((\sqrt{\alpha}, \infty]) \cup f^{-1}([-\infty, -\sqrt{\alpha})).$$

Both sets on the right are measurable by assumption and by the equivalent tests, so f^2 is measurable. Now use the identity

$$fg = \frac{1}{4}((f+g)^2 - (f-g)^2).$$

The functions $f+g$ and $f-g$ are measurable by what we have proved for scaling and addition. Since the square of a measurable function is measurable, it follows that fg is measurable.

Why extend the real line?

All of the preceding facts work perfectly well with $\mathbb{R}$. The new feature of the extended real line is closure under limiting processes.

Suprema, infima, and limit functions

Let $E \subset \mathbb{R}$ be measurable, and let $(f_n)_{n=1}^{\infty}$ be a sequence of measurable functions

$$f_n : E \to [-\infty, \infty].$$

Then the four functions

$$g_1(x) = \sup_n f_n(x), \qquad g_2(x) = \inf_n f_n(x),$$
$$g_3(x) = \limsup_n f_n(x), \qquad g_4(x) = \liminf_n f_n(x), \qquad x \in E,$$

are measurable.

Proof

First recall the identities

$$\limsup_n f_n(x) = \inf_{n \in \mathbb{N}} \left(\sup_{k \geq n} f_k(x) \right), \qquad \liminf_n f_n(x) = \sup_{n \in \mathbb{N}} \left(\inf_{k \geq n} f_k(x) \right).$$

Indeed, by definition,

$$\limsup_n f_n(x) = \lim_{n \to \infty} \left(\sup_{k \geq n} f_k(x) \right).$$

The sets of values satisfy

$$\{f_k(x) : k \geq n\} \subset \{f_k(x) : k \geq n-1\},$$

so

$$\sup_{k \geq n} f_k(x) \leq \sup_{k \geq n-1} f_k(x).$$

Thus $(\sup_{k\geq n} f_k(x))_{n=1}^{\infty}$ is monotonically decreasing. A bounded monotone real sequence converges, and a decreasing sequence converges to its greatest lower bound. Hence

$$\lim_{n\to\infty} \sup_{k\geq n} f_k(x) = \limsup_n f_n(x) = \inf_{n\in\mathbb{N}} \left(\sup_{k\geq n} f_k(x) \right).$$

Similarly, $(\inf_{k\geq n} f_k(x))_{n=1}^{\infty}$ is monotonically increasing and bounded above, so it converges to its least upper bound:

$$\lim_{n\to\infty} \inf_{k\geq n} f_k(x) = \liminf_n f_n(x) = \sup_{n\in\mathbb{N}} \left(\inf_{k\geq n} f_k(x) \right).$$

For the pointwise supremum g_1, observe that

$$\begin{aligned} x \in g_1^{-1}((\alpha,\infty]) &\iff \sup_n f_n(x) > \alpha \\ &\iff f_n(x) > \alpha \text{ for some } n \in \mathbb{N} \\ &\iff x \in \bigcup_{n=1}^{\infty} f_n^{-1}((\alpha,\infty]). \end{aligned}$$

Each $f_n^{-1}((\alpha,\infty])$ is measurable, so their countable union is measurable. Thus g_1 is measurable. Similarly,

$$x \in g_2^{-1}([\alpha,\infty]) \iff x \in \bigcap_{n=1}^{\infty} f_n^{-1}([\alpha,\infty]),$$

so g_2 is measurable.
Finally, the displayed identities for lim sup and lim inf express each as alternating pointwise infima and suprema. Since the pointwise supremum and infimum of a countable collection of measurable functions are measurable,

$$\limsup_n f_n = \inf_{n\in\mathbb{N}} \left(\sup_{k\geq n} f_k \right) \quad \text{and} \quad \liminf_n f_n = \sup_{n\in\mathbb{N}} \left(\inf_{k\geq n} f_k \right)$$

are measurable.

Pointwise limits preserve measurability

Let $E \subset \mathbb{R}$ be measurable, and let $f_n : E \to [-\infty,\infty]$ be measurable for every $n \in \mathbb{N}$. If $f_n \to f$ pointwise, then f is measurable.

Proof

If $f_n \to f$ pointwise, then, for every $x \in E$,

$$\lim_{n\to\infty} f_n(x) = f(x) = \limsup_n f_n(x) = \liminf_n f_n(x).$$

The preceding theorem therefore implies that f is measurable.

Almost everywhere

Let E be measurable. A statement $P(x)$ holds *almost everywhere* on E if

$$m(\{x \in E : P(x) \text{ does not hold}\}) = 0.$$

Outer measure formulation

The definition does not require us first to ask whether the exceptional set is measurable and has measure zero. We may equivalently use outer measure, which is defined on $\mathcal{P}(\mathbb{R})$:

$$m^*(S) = 0.$$

By the theorem on page 64, this implies that S is measurable and $m(S) = 0$.

Changing a function on a null set

Let $f, g : E \to [-\infty, \infty]$ satisfy

$$f = g \quad \text{almost everywhere on } E.$$

If f is measurable, then g is measurable. In words, varying a function on a set of measure zero without changing it elsewhere does not affect its measurability.

Proof

Let

$$N = \{x \in E : f(x) \neq g(x)\}.$$

By assumption, N has outer measure zero, and hence $m(N) = 0$. For every $\alpha \in \mathbb{R}$, define

$$N_\alpha = \{x \in N : g(x) > \alpha\} \subset N.$$

This set also has outer measure zero, because

$$0 \leq m^*(N_\alpha) \leq m^*(N) = 0.$$

Therefore

$$g^{-1}((\alpha, \infty]) = (f^{-1}((\alpha, \infty]) \cap N^c) \cup N_\alpha.$$

Indeed, if $g(x) \in (\alpha, \infty]$, then either $g(x) = f(x) > \alpha$, so $x \in f^{-1}((\alpha, \infty]) \cap N^c$, or $f(x) \neq g(x) > \alpha$, so $x \in N_\alpha$.
The set N is measurable, hence N^c is measurable. The set N_α is also measurable because $m^*(N_\alpha) = 0$. Therefore $g^{-1}((\alpha, \infty])$ is measurable for every $\alpha \in \mathbb{R}$, and so g is measurable.

10 Simple Functions

Recall that, if $E \subset \mathbb{R}$ is measurable, then a function $f : E \to [-\infty, \infty]$ is measurable if and only if, for every $\alpha \in \mathbb{R}$,

$$f^{-1}((\alpha, \infty])$$

is measurable.

Complex-valued measurable function

Let $E \subset \mathbb{R}$ be measurable, and let $f : E \to \mathbb{C}$. We say that f is measurable if the two real-valued functions

$$\operatorname{Re} f : E \to \mathbb{R}, \qquad \operatorname{Im} f : E \to \mathbb{R}$$

are both measurable. Thus

$$f = \operatorname{Re} f + i \operatorname{Im} f.$$

Operations on complex-valued measurable functions

If $f, g : E \to \mathbb{C}$ are measurable functions and $\alpha \in \mathbb{C}$, then

$$\alpha f + g, \qquad fg, \qquad \overline{f}, \qquad |f|$$

are measurable functions.

Proof

Writing $\alpha = \operatorname{Re}\alpha + i \operatorname{Im}\alpha$, we have

$$\begin{aligned} \alpha f + g = &((\operatorname{Re}\alpha)(\operatorname{Re} f) - (\operatorname{Im}\alpha)(\operatorname{Im} f) + \operatorname{Re} g) \\ &+ i((\operatorname{Re}\alpha)(\operatorname{Im} f) + (\operatorname{Im}\alpha)(\operatorname{Re} f) + \operatorname{Im} g), \end{aligned}$$

$$fg = ((\operatorname{Re} f)(\operatorname{Re} g) - (\operatorname{Im} f)(\operatorname{Im} g)) + i((\operatorname{Re} f)(\operatorname{Im} g) + (\operatorname{Im} f)(\operatorname{Re} g)),$$

and

$$\overline{f} = \operatorname{Re} f - i \operatorname{Im} f.$$

All real and imaginary parts in these formulas are measurable by the theorem on page 78. It remains to consider

$$|f| = \sqrt{(\operatorname{Re} f)^2 + (\operatorname{Im} f)^2}.$$

Define

$$h(x) = (\operatorname{Re} f(x))^2 + (\operatorname{Im} f(x))^2 = |f(x)|^2.$$

The function h is measurable. If $\alpha < 0$, then $(|f|)^{-1}((\alpha, \infty]) = E$. If $\alpha \geq 0$, then

$$x \in (|f|)^{-1}((\alpha, \infty]) \iff |f(x)| > \alpha \iff h(x) > \alpha^2.$$

Therefore

$$(|f|)^{-1}((\alpha, \infty]) = h^{-1}((\alpha^2, \infty])$$

is measurable. Hence $|f|$ is measurable.

Pointwise limits in the complex-valued case

Let $E \subset \mathbb{R}$ be measurable. Suppose that $f_n : E \to \mathbb{C}$ is measurable for every $n \in \mathbb{N}$, and

$$f_n(x) \longrightarrow f(x) \qquad \text{for every } x \in E.$$

Then f is measurable.

Proof

As shown when proving that $\mathbb{R}^n$ is a Banach space on page 7,

$$f_n(x) \to f(x) \iff \operatorname{Re} f_n(x) \to \operatorname{Re} f(x) \quad \text{and} \quad \operatorname{Im} f_n(x) \to \operatorname{Im} f(x).$$

The pointwise-limit corollary on page 81 implies that $\operatorname{Re} f$ and $\operatorname{Im} f$ are measurable. Therefore f is measurable.

Recall that, for measurable sets $E, F \subset \mathbb{R}$, the indicator function $\chi_F : E \to \mathbb{R}$ is defined by

$$\chi_F(x) = \begin{cases} 1, & x \in F, \\ 0, & x \notin F. \end{cases}$$

It takes only the two values 0 and 1. This motivates the next definition: every measurable function will be approximated by simple functions.

Simple function

A measurable function $\varphi : E \to \mathbb{C}$ is *simple* if its range is finite:

$$|\varphi(E)| < \infty.$$

Canonical form of a simple function

Every simple function is a complex linear combination of finitely many indicator functions.

Proof

Suppose

$$\varphi(E) = \{a_1, \dots, a_n\}.$$

For $i = 1, \dots, n$, let

$$A_i = \varphi^{-1}(\{a_i\}).$$

Each A_i is measurable. Indeed, since φ is measurable, its real and imaginary parts are measurable, and

$$(\operatorname{Re}\varphi)^{-1}(\{\operatorname{Re} a_i\}) = (\operatorname{Re}\varphi)^{-1}([\operatorname{Re} a_i, \infty)) \cap (\operatorname{Re}\varphi)^{-1}((\operatorname{Re} a_i, \infty))^c$$

is measurable; similarly,

$$(\operatorname{Im}\varphi)^{-1}(\{\operatorname{Im} a_i\})$$

is measurable. Hence

$$A_i = (\operatorname{Re}\varphi)^{-1}(\{\operatorname{Re} a_i\}) \cap (\operatorname{Im}\varphi)^{-1}(\{\operatorname{Im} a_i\})$$

is measurable.

Moreover, $A_i \cap A_j = \varnothing$ whenever $i \neq j$, because φ cannot map one point to two different values, and

$$\bigcup_{i=1}^{n} A_i = E.$$

Therefore, for every $x \in E$,

$$\varphi(x) = \sum_{i=1}^{n} a_i \chi_{A_i}(x).$$

Simple functions are closed under sums and products

Let $E \subset \mathbb{R}$ be measurable, let $\varphi_1, \varphi_2 : E \to \mathbb{C}$ be simple functions, and let $\lambda_1, \lambda_2 \in \mathbb{C}$. Then

$$\lambda_1\varphi_1 + \lambda_2\varphi_2 \qquad \text{and} \qquad \varphi_1\varphi_2$$

are simple functions.

Proof

By the theorem on page 82, the two displayed functions are measurable. Because $\varphi_1(E)$ and $\varphi_2(E)$ are finite, the sets

$$\{\lambda_1 a + \lambda_2 b : a \in \varphi_1(E),\ b \in \varphi_2(E)\}$$

and

$$\{ab : a \in \varphi_1(E),\ b \in \varphi_2(E)\}$$

are finite. They contain the respective ranges, so both functions are simple.

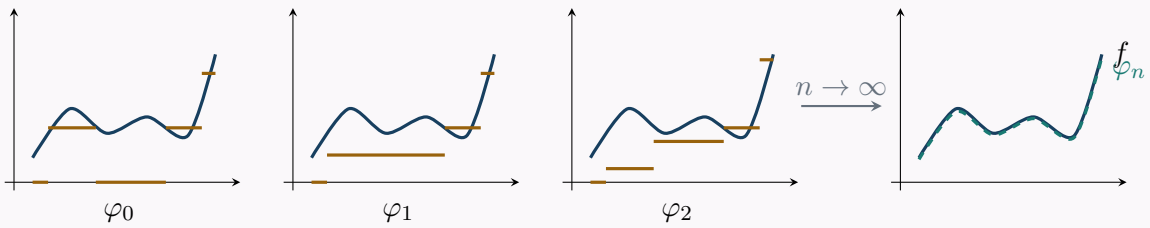
Approximation of a nonnegative measurable function

Let $f : E \to [0, \infty]$ be measurable. There exists a sequence of simple functions $(\varphi_n)_{n=0}^{\infty}$ such that:

1. for every $x \in E$,
$$0 \leq \varphi_0(x) \leq \varphi_1(x) \leq \varphi_2(x) \leq \cdots \leq f(x);$$
2. $\varphi_n(x) \to f(x)$ for every $x \in E$;
3. for every $B > 0$, $\varphi_n \to f$ uniformly on
$$\{x \in E : f(x) \leq B\}.$$

Idea of the construction

At stage n, divide the range interval $[0, 2^n]$ into subintervals of length 2^{-n}, and replace $f(x)$ by the left endpoint of the subinterval containing it. Values larger than 2^n are truncated to 2^n.



Proof

For $n = 0, 1, 2, \ldots$ and $0 \le k \le 2^{2n} - 1$, define

$$E_n^k = \{x \in E : k2^{-n} < f(x) \le (k+1)2^{-n}\}.$$

The sets E_n^k give the cuts of $[0, 2^n]$ along the range, each of length 2^{-n}. Also define

$$F_n = f^{-1}((2^n, \infty]).$$

Thus F_n consists of the points where $f(x) > 2^n$, the part not covered by the sets E_n^k.
Each E_n^k is measurable because

$$E_n^k = f^{-1}((k2^{-n}, \infty]) \cap f^{-1}([-\infty, (k+1)2^{-n}]),$$

and F_n is measurable because f is measurable. Define

$$\varphi_n = \sum_{k=0}^{2^{2n}-1} k2^{-n}\chi_{E_n^k} + 2^n\chi_{F_n}, \qquad n = 0, 1, 2, \ldots.$$

For example,

$$\varphi_1 = 0\chi_{f^{-1}((0,\frac{1}{2}])} + \frac{1}{2}\chi_{f^{-1}((\frac{1}{2},1])} + \chi_{f^{-1}((1,\frac{3}{2}])} + \frac{3}{2}\chi_{f^{-1}((\frac{3}{2},2])} + 2\chi_{f^{-1}((2,\infty])}.$$

Each φ_n is a simple function.
By construction, for every $x \in E$,

$$0 \le \varphi_n(x) \le f(x).$$

Indeed, if $x \in E_n^k$, then

$$k2^{-n} < f(x) \le (k+1)2^{-n}, \qquad \varphi_n(x) = k2^{-n} < f(x),$$

whereas, if $x \in F_n$, then

$$\varphi_n(x) = 2^n < f(x).$$

We next show

$$\varphi_0(x) \le \varphi_1(x) \le \varphi_2(x) \le \cdots.$$

Suppose first that $x \in F_n$. Then

$$\varphi_n(x) = 2^n > 2^{n-1} = \varphi_{n-1}(x).$$

If $x \in E_n^k$, then

$$k2^{-n} < f(x) \le (k+1)2^{-n}.$$

After refining the cut, $f(x)$ lies in one of the two intervals

$$((2k)2^{-(n+1)}, (2k+1)2^{-(n+1)}]$$

or

$$((2k+1)2^{-(n+1)}, (2k+2)2^{-(n+1)}].$$

Hence

$$x \in E_{n+1}^{2k} \cup E_{n+1}^{2k+1}.$$

If $x \in E_{n+1}^{2k}$, then

$$\varphi_n(x) = k2^{-n} = (2k)2^{-(n+1)} = \varphi_{n+1}(x).$$

If $x \in E_{n+1}^{2k+1}$, then

$$\varphi_n(x) = k2^{-n} = (2k)2^{-(n+1)} \leq (2k+1)2^{-(n+1)} = \varphi_{n+1}(x).$$

Since

$$E = \left(\bigcup_{k=0}^{2^{2n}-1} E_n^k \right) \cup F_n,$$

we have

$$\varphi_n(x) \leq \varphi_{n+1}(x) \qquad (x \in E,\ n \geq 0).$$

For pointwise convergence, fix $x \in E$. If $f(x) = \infty$, then $x \in F_n$ for every n, so

$$\varphi_n(x) = 2^n \quad \text{and} \quad \lim_{n\to\infty} \varphi_n(x) = f(x) = \infty.$$

Otherwise $f(x) \leq 2^n$ for some n. We claim that, whenever $f(x) \leq 2^n$,

$$0 \leq f(x) - \varphi_n(x) \leq 2^{-n}.$$

Indeed, the construction cuts $[0, 2^n]$ into intervals of resolution 2^{-n}, and

$$\bigcup_{k=0}^{2^{2n}-1} E_n^k = \{x \in E : f(x) \leq 2^n\}.$$

If $x \in E_n^k$, then

$$\varphi_n(x) = k2^{-n} < f(x) \leq (k+1)2^{-n} = \varphi_n(x) + 2^{-n}.$$

Thus

$$0 \leq f(x) - \varphi_n(x) \leq 2^{-n}$$

on $\{x \in E : f(x) \leq 2^n\}$. Letting $n \to \infty$ gives $\varphi_n(x) \to f(x)$.
Finally, let $B > 0$ and $\varepsilon > 0$. Choose $N_0 \in \mathbb{N}$ so that $B \leq 2^{N_0}$, and choose

$$N \geq \max\left\{ N_0, \log_2\left(\frac{1}{\varepsilon}\right) \right\}.$$

For every $n \geq N$ and every $x \in \{x \in E : f(x) \leq B\}$,

$$|f(x) - \varphi_n(x)| \leq 2^{-n} \leq 2^{-N} \leq \varepsilon.$$

Therefore $\varphi_n \to f$ uniformly on $\{x \in E : f(x) \leq B\}$.

We now extend the approximation theorem to functions taking values in $[-\infty, \infty]$ and in $\mathbb{C}$.

Positive and negative parts

Let $E \subset \mathbb{R}$ be measurable, and let $f : E \to [-\infty, \infty]$ be measurable. Define the positive and negative parts of f by

$$f^+(x) = \max\{f(x), 0\}, \qquad f^-(x) = \max\{-f(x), 0\}.$$

Properties of the positive and negative parts

For every $x \in E$,

$$f(x) = f^+(x) - f^-(x), \qquad |f(x)| = f^+(x) + f^-(x).$$

Also,

$$f^+(x) \geq 0, \qquad f^-(x) \geq 0,$$

even though f^- is called the negative part. Both f^+ and f^- are measurable. Indeed,

$$f^+ = \sup\{f, 0, 0, \ldots\}, \qquad f^- = \sup\{-f, 0, 0, \ldots\},$$

so their measurability follows from the theorem on page 80 and, for f^-, scalar multiplication.

Approximation of a complex-valued measurable function

Let $E \subset \mathbb{R}$ be measurable, and let $f : E \to \mathbb{C}$ be measurable. There exists a sequence of simple functions $(\varphi_n)_{n=0}^{\infty}$ such that:

1. for every $x \in E$,
$$0 \leq |\varphi_0(x)| \leq |\varphi_1(x)| \leq |\varphi_2(x)| \leq \cdots \leq |f(x)|;$$
2. $\varphi_n(x) \to f(x)$ for every $x \in E$;
3. for every $B > 0$, $\varphi_n \to f$ uniformly on
$$\{x \in E : |f(x)| \leq B\}.$$

Proof

We apply the preceding theorem to the four nonnegative measurable functions

$$(\operatorname{Re} f)^+, \qquad (\operatorname{Re} f)^-, \qquad (\operatorname{Im} f)^+, \qquad (\operatorname{Im} f)^-.$$

There are sequences of nonnegative simple functions

$$(\varphi_n^{(1)}), \quad (\varphi_n^{(2)}), \quad (\varphi_n^{(3)}), \quad (\varphi_n^{(4)})$$

such that each sequence is pointwise increasing, converges pointwise to the corresponding positive or negative part, and converges uniformly where that part is bounded. Define

$$\varphi_n(x) = (\varphi_n^{(1)}(x) - \varphi_n^{(2)}(x)) + i(\varphi_n^{(3)}(x) - \varphi_n^{(4)}(x)).$$

This is a simple function for every n.

We first prove the chain of inequalities in part 1. For any real-valued function u and any x, at least one of $u^+(x)$ and $u^-(x)$ equals zero. Indeed, if $u(x) > 0$, then

$$u^+(x) = u(x), \qquad u^-(x) = 0;$$

if $u(x) < 0$, then

$$u^+(x) = 0, \qquad u^-(x) = -u(x);$$

and, if $u(x) = 0$, then both parts are zero.
Fix $x \in E$. Suppose, without loss of generality, that $(\operatorname{Re} f)^+(x) = 0$. Since

$$0 \le \varphi_0^{(1)}(x) \le \varphi_1^{(1)}(x) \le \cdots \le (\operatorname{Re} f)^+(x) = 0,$$

we have

$$\varphi_n^{(1)}(x) = 0 \qquad (n = 0, 1, 2, \ldots).$$

Consequently,

$$\operatorname{Re} \varphi_n(x) = \varphi_n^{(1)}(x) - \varphi_n^{(2)}(x) = -\varphi_n^{(2)}(x).$$

Even though the inequalities for these real parts are reversed, their absolute values are increasing. If instead $(\operatorname{Re} f)^-(x) = 0$, the same conclusion follows with $\varphi_n^{(1)}$. Repeating the argument for the imaginary part gives the analogous conclusion.
For example, if $(\operatorname{Im} f)^+(x) = 0$, then

$$\varphi_n^{(3)}(x) - \varphi_n^{(4)}(x) = \operatorname{Im} \varphi_n(x) = -\varphi_n^{(4)}(x).$$

Thus, at every x,

$$|\operatorname{Re} \varphi_0(x)| \le |\operatorname{Re} \varphi_1(x)| \le \cdots \le |\operatorname{Re} f(x)|$$

and

$$|\operatorname{Im} \varphi_0(x)| \le |\operatorname{Im} \varphi_1(x)| \le \cdots \le |\operatorname{Im} f(x)|.$$

Squaring and adding gives

$$0 \le |\varphi_0(x)|^2 \le |\varphi_1(x)|^2 \le \cdots \le |f(x)|^2.$$

Taking square roots proves

$$0 \le |\varphi_0(x)| \le |\varphi_1(x)| \le \cdots \le |f(x)|.$$

For pointwise convergence, the linearity of limits in $\mathbb{C}$ gives

$$\begin{aligned} \lim_{n\to\infty} \varphi_n(x) &= \Big(\lim_{n\to\infty} \varphi_n^{(1)}(x) - \lim_{n\to\infty} \varphi_n^{(2)}(x)\Big) \\ &\quad + i\Big(\lim_{n\to\infty} \varphi_n^{(3)}(x) - \lim_{n\to\infty} \varphi_n^{(4)}(x)\Big) \\ &= \big((\operatorname{Re} f)^+(x) - (\operatorname{Re} f)^-(x)\big) \\ &\quad + i\big((\operatorname{Im} f)^+(x) - (\operatorname{Im} f)^-(x)\big) \\ &= \operatorname{Re} f(x) + i \operatorname{Im} f(x) = f(x). \end{aligned}$$

It remains to prove uniform convergence. Let $B > 0$ and $\varepsilon > 0$. By uniform convergence of the four approximating sequences on the corresponding bounded sets, there are $M_1, M_2, M_3, M_4 \in \mathbb{N}$

such that, for $n \geq M_j$, the error in the j-th approximation is less than $\varepsilon/8$ whenever the corresponding positive or negative part is at most B.
Choose

$$M = \max\{M_1, M_2, M_3, M_4\}$$

and set

$$E_B = \left\{x \in E : (\operatorname{Re} f)^+(x), (\operatorname{Re} f)^-(x), (\operatorname{Im} f)^+(x), (\operatorname{Im} f)^-(x) \leq B\right\}.$$

For $n \geq M$ and $x \in E_B$, the triangle inequality gives

$$\begin{aligned} |\varphi_n(x) - f(x)| &\leq |\varphi_n^{(1)}(x) - (\operatorname{Re} f)^+(x)| + |(\operatorname{Re} f)^-(x) - \varphi_n^{(2)}(x)| \\ &\quad + |\varphi_n^{(3)}(x) - (\operatorname{Im} f)^+(x)| + |(\operatorname{Im} f)^-(x) - \varphi_n^{(4)}(x)| \\ &< 4\frac{\varepsilon}{8} = \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$

Thus $\varphi_n \to f$ uniformly on E_B.
We now show

$$\{x \in E : |f(x)| \leq B\} \subset E_B.$$

If $|f(x)| \leq B$, then

$$|\operatorname{Re} f(x)| \leq B, \qquad |\operatorname{Im} f(x)| \leq B.$$

Using

$$|\operatorname{Re} f(x)| = (\operatorname{Re} f)^+(x) + (\operatorname{Re} f)^-(x)$$

and the nonnegativity of both terms gives

$$(\operatorname{Re} f)^+(x) \leq B, \qquad (\operatorname{Re} f)^-(x) \leq B.$$

Similarly,

$$(\operatorname{Im} f)^+(x) \leq B, \qquad (\operatorname{Im} f)^-(x) \leq B.$$

Therefore $x \in E_B$. Uniform convergence on E_B implies uniform convergence on $\{x \in E : |f(x)| \leq B\}$.

Interpretation

Every measurable function is, in this sense, almost a simple function.

10.1 Lebesgue Integral of a Nonnegative Function

If $E \subset \mathbb{R}$ is measurable, write

$$L^+(E) = \{f : E \to [0, \infty] : f \text{ is measurable}\}.$$

Integral of a nonnegative simple function

Suppose that $\varphi \in L^+(E)$ is simple and has the representation

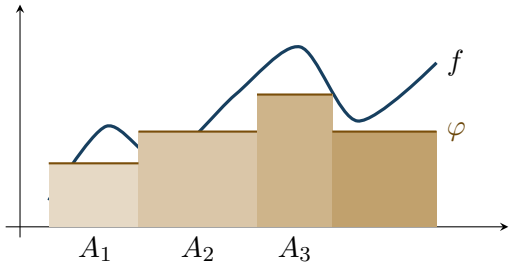
$$\varphi = \sum_{j=1}^{n} a_j \chi_{A_j},$$

where the measurable sets A_j are pairwise disjoint and

$$\bigcup_{j=1}^{n} A_j = E.$$

The Lebesgue integral of φ over E is

$$\int_E \varphi = \sum_{j=1}^{n} a_j m(A_j) \in [0, \infty].$$



The definition is independent of the chosen simple-function representation by the canonical-form lemma.

Basic properties of the simple-function integral

Suppose that $\varphi, \psi \in L^+(E)$ are simple functions. Then:

1. for every $c > 0$,
$$\int_E c\varphi = c \int_E \varphi;$$
2.
$$\int_E (\varphi + \psi) = \int_E \varphi + \int_E \psi;$$
3. if $\varphi \leq \psi$, then
$$\int_E \varphi \leq \int_E \psi;$$
4. if $F \subset E$ is measurable, then
$$\int_F \varphi = \int_E \chi_F \varphi \leq \int_E \varphi.$$

Proof of part 1

If

$$\varphi = \sum_{j=1}^{n} a_j \chi_{A_j},$$

then

$$c\varphi = \sum_{j=1}^{n} (ca_j) \chi_{A_j}.$$

Therefore

$$\int_E c\varphi = \sum_{j=1}^{n} (ca_j) m(A_j) = c \sum_{j=1}^{n} a_j m(A_j) = c \int_E \varphi.$$

Proof of part 2

Write

$$\varphi = \sum_{j=1}^{n} a_j \chi_{A_j}, \qquad \psi = \sum_{k=1}^{m} b_k \chi_{B_k},$$

where each family is a measurable partition of E. Then

$$A_j = \bigcup_{k=1}^{m} (A_j \cap B_k), \qquad B_k = \bigcup_{j=1}^{n} (A_j \cap B_k).$$

All sets $A_j \cap B_k$ are pairwise disjoint as j, k vary, and

$$\bigcup_{j=1}^{n} \bigcup_{k=1}^{m} (A_j \cap B_k) = E.$$

By additivity of Lebesgue measure,

$$\begin{aligned} \int_E \varphi + \int_E \psi &= \sum_{j=1}^{n} a_j m(A_j) + \sum_{k=1}^{m} b_k m(B_k) \\ &= \sum_{j=1}^{n} \sum_{k=1}^{m} a_j m(A_j \cap B_k) + \sum_{k=1}^{m} \sum_{j=1}^{n} b_k m(A_j \cap B_k) \\ &= \sum_{j=1}^{n} \sum_{k=1}^{m} (a_j + b_k) m(A_j \cap B_k). \end{aligned}$$

On the other hand,

$$\varphi + \psi = \sum_{j=1}^{n} \sum_{k=1}^{m} (a_j + b_k) \chi_{A_j \cap B_k}.$$

Hence

$$\int_E (\varphi + \psi) = \sum_{j=1}^{n} \sum_{k=1}^{m} (a_j + b_k) m(A_j \cap B_k) = \int_E \varphi + \int_E \psi.$$

Proof of part 3

Use the common refinement from the proof of part 2. If $\varphi(x) \le \psi(x)$ for every $x \in E$, then

$$a_j \le b_k \qquad \text{whenever } A_j \cap B_k \neq \varnothing.$$

Therefore

$$\int_E \varphi = \sum_{j=1}^{n} \sum_{k=1}^{m} a_j m(A_j \cap B_k)$$

$$\leq \sum_{j=1}^{n}\sum_{k=1}^{m} b_k m(A_j \cap B_k)$$
$$= \sum_{k=1}^{m} b_k m(B_k) = \int_E \psi.$$

If $\int_E \varphi = 0$, the conclusion is immediate because the integrals are nonnegative; the same displayed argument covers the nonzero case.

Proof of part 4

Since φ and χ_F are simple, their product $\chi_F\varphi$ is simple. If

$$\varphi = \sum_{j=1}^{n} a_j \chi_{A_j},$$

then the sets $F \cap A_j$ are pairwise disjoint and measurable, and

$$\bigcup_{j=1}^{n} (F \cap A_j) = F.$$

For $x \in E$,

$$\chi_F(x)\varphi(x) = \sum_{j=1}^{n} a_j \chi_{F\cap A_j}(x).$$

To view this as a partition of all of E, add the set $E \setminus F$ with coefficient 0. Thus, by the definition of the integral,

$$\int_E \chi_F\varphi = \sum_{j=1}^{n} a_j m(F \cap A_j) = \int_F \varphi.$$

Finally, since $\chi_F(x) \leq 1$ for every $x \in E$,

$$\chi_F\varphi \leq \varphi.$$

Part 3 therefore gives

$$\int_F \varphi = \int_E \chi_F\varphi \leq \int_E \varphi.$$

11 The Lebesgue Integral and Convergence Theorems

Recall that

$$L^+(E) = \{f : E \to [0, \infty] : f \text{ is measurable}\}.$$

If $\varphi \in L^+(E)$ is simple and has canonical form

$$\varphi = \sum_{j=1}^{n} a_j \chi_{A_j},$$

where the A_j are pairwise disjoint and $\bigcup_{j=1}^{n} A_j = E$, then

$$\int_E \varphi = \sum_{j=1}^{n} a_j m(A_j).$$

Every simple function has such a finite linear combination of indicator functions; from now on, this is its canonical form.

Lebesgue integral of a nonnegative measurable function

If $f \in L^+(E)$, define

$$\int_E f = \sup \left\{ \int_E \varphi : \varphi \in L^+(E) \text{ is simple and } \varphi \leq f \right\}.$$

A null domain has zero integral

Let $E \subset \mathbb{R}$ be measurable. If $m(E) = 0$, then

$$\int_E f = 0 \qquad \text{for every } f \in L^+(E).$$

Proof

Suppose that

$$\varphi = \sum_{j=1}^{n} a_j \chi_{A_j}$$

is a nonnegative simple function with $\varphi \leq f$. Because $\bigcup_j A_j = E$, every $A_j \subset E$, and hence

$$m(A_j) \leq m(E) = 0.$$

Therefore

$$\int_E \varphi = \sum_{j=1}^{n} a_j m(A_j) = 0.$$

Taking the supremum over all such φ gives $\int_E f = 0$.

Agreement with the simple-function integral

If $\varphi \in L^+(E)$ is simple, then the definition on this page agrees with the simple-function definition on page 90:

$$\int_E \varphi = \sup\left\{\int_E \psi : \psi \in L^+(E) \text{ is simple and } \psi \le \varphi\right\} = \sum_{j=1}^{n} a_j m(A_j),$$

where $\varphi = \sum_{j=1}^{n} a_j \chi_{A_j}$ is in canonical form.

Elementary properties of the integral

For $f, g \in L^+(E)$ and $c \in [0, \infty)$:

1. $$\int_E cf = c\int_E f;$$
2. if $f \le g$, then $$\int_E f \le \int_E g;$$
3. if $F \subset E$ is measurable, then $$\int_F f \le \int_E f.$$

Proof

Taking suprema preserves inequalities. For example, recall that if $A \subset [0, \infty]$, then

$$\sup(cA) = c \sup A \qquad (c \in [0, \infty)).$$

Since the simple-function integral satisfies $\int_E c\varphi = c\int_E \varphi$,

$$\begin{aligned}\int_E cf &= \sup_{\substack{\varphi \text{ simple} \\ 0 \le \varphi \le f}} \int_E c\varphi \\ &= \sup_{\substack{\varphi \text{ simple} \\ 0 \le \varphi \le f}} c\int_E \varphi = c\int_E f.\end{aligned}$$

The other two statements follow in the same way from monotonicity and restriction for simple functions.

Unions and intersections of integration domains

Let $A, B \subset E \subset \mathbb{R}$ be measurable, and let $f \in L^+(E)$. Then

$$m(A \cup B) + m(A \cap B) = m(A) + m(B)$$

and

$$\int_{A\cup B} f + \int_{A\cap B} f = \int_A f + \int_B f.$$

In particular, if $A \cap B = \varnothing$, then

$$\int_{A\cup B} f = \int_A f + \int_B f.$$

Proof strategy

The assertions are immediate for simple functions. The corresponding facts for arbitrary nonnegative measurable functions follow from the Monotone Convergence Theorem below. This theorem was not used in proving that convergence theorem, so there is no circular reasoning.

Almost-everywhere monotonicity

If $f, g \in L^+(E)$ and

$$f \le g \quad \text{almost everywhere on } E,$$

then

$$\int_E f \le \int_E g.$$

Proof

Let

$$F = \{x \in E : f(x) \le g(x)\}.$$

This set is measurable. One may separate the finite and infinite cases:

$$F = \{x \in E : f(x) < \infty \text{ and } f(x) \le g(x)\} \cup \big(\{x \in E : f(x) = \infty\} \cap \{x \in E : g(x) = \infty\}\big).$$

Each set is measurable by the level-set characterizations of measurable functions. Since $f \le g$ almost everywhere, $m(F^c) = 0$. Therefore

$$\int_E f = \int_F f + \int_{F^c} f = \int_F f \le \int_F g \le \int_F g + \int_{F^c} g = \int_E g.$$

The next goal

Additivity is the principal property still missing from the general Lebesgue integral. The convergence theorem below supplies it.

Monotone Convergence Theorem

Let $(f_n)_{n=1}^{\infty} \subset L^+(E)$ satisfy

$$f_1 \le f_2 \le f_3 \le \cdots$$

pointwise on E, and suppose that $f_n \to f$ pointwise on E. Then $f \in L^+(E)$ and

$$\lim_{n\to\infty} \int_E f_n = \int_E f.$$

Comparison with Riemann integration

Pointwise convergence is much weaker than the uniform convergence normally required to interchange limits and Riemann integrals.

Proof

Monotonicity gives

$$\int_E f_1 \leq \int_E f_2 \leq \cdots,$$

so $(\int_E f_n)$ is an increasing sequence in $[0,\infty]$ and has a limit. Since $f_n(x) \to f(x)$ increasingly, $f_n \leq f$, and therefore

$$\int_E f_n \leq \int_E f.$$

It follows that

$$\lim_{n\to\infty} \int_E f_n \leq \int_E f.$$

For the reverse inequality, let $\varepsilon \in (0,1)$, and let $\varphi \leq f$ be a nonnegative simple function with canonical form

$$\varphi = \sum_{j=1}^{m} a_j \chi_{A_j}.$$

Define

$$E_n = \{x \in E : f_n(x) \geq (1-\varepsilon)\varphi(x)\}.$$

Because $f_n(x) \to f(x)$ and $(1-\varepsilon)\varphi(x) < f(x)$ wherever $\varphi(x) > 0$, every point of E lies in some E_n; points where $\varphi = 0$ lie in every E_n. Hence

$$\bigcup_{n=1}^{\infty} E_n = E.$$

Moreover, $f_1 \leq f_2 \leq \cdots$ implies

$$E_1 \subset E_2 \subset E_3 \subset \cdots.$$

Thus

$$\int_E f_n \geq \int_{E_n} f_n \geq (1-\varepsilon) \int_{E_n} \varphi = (1-\varepsilon) \sum_{j=1}^{m} a_j m(A_j \cap E_n). \qquad (*)$$

The last equality follows by restricting the canonical form:

$$\varphi|_{E_n} = \sum_{j=1}^{m} a_j \chi_{A_j \cap E_n},$$

and the sets $A_j \cap E_n$ remain pairwise disjoint.
We claim that

$$m(A_j \cap E_n) \longrightarrow m(A_j).$$

Indeed,

$$A_j \cap E_1 \subset A_j \cap E_2 \subset \cdots$$

are measurable and

$$\bigcup_{n=1}^{\infty} (A_j \cap E_n) = A_j \cap \left(\bigcup_{n=1}^{\infty} E_n \right) = A_j.$$

Continuity from below of Lebesgue measure, proved on page 74, yields the claim.

Taking limits in $(*)$, we obtain

$$\lim_{n\to\infty}\int_E f_n \geq (1-\varepsilon)\sum_{j=1}^{m} a_j \lim_{n\to\infty} m(A_j\cap E_n)$$
$$= (1-\varepsilon)\sum_{j=1}^{m} a_j m(A_j) = (1-\varepsilon)\int_E \varphi.$$

Letting $\varepsilon \downarrow 0$ gives

$$\lim_{n\to\infty}\int_E f_n \geq \int_E \varphi$$

for every nonnegative simple $\varphi \leq f$. Taking the supremum over all such φ gives

$$\lim_{n\to\infty}\int_E f_n \geq \int_E f.$$

Together with the reverse inequality, this proves the theorem.

Computing an integral through approximation

The Monotone Convergence Theorem proves the earlier assertion that integrals are additive over disjoint measurable domains. It also lets us evaluate an integral using any increasing simple-function approximation, instead of directly taking the supremum over every simple minorant.

Integral from an increasing simple approximation

Let $E \subset \mathbb{R}$ be measurable, let $f \in L^+(E)$, and let $(\varphi_n)_{n=0}^{\infty}$ be simple functions satisfying

$$0 \leq \varphi_0 \leq \varphi_1 \leq \varphi_2 \leq \cdots \leq f, \qquad \varphi_n \to f \quad \text{pointwise.}$$

Then

$$\int_E f = \lim_{n\to\infty}\int_E \varphi_n.$$

Proof

This is the Monotone Convergence Theorem applied directly.

Additivity of the nonnegative Lebesgue integral

If $f, g \in L^+(E)$, then

$$\int_E (f+g) = \int_E f + \int_E g.$$

Proof

Choose increasing sequences of nonnegative simple functions (φ_n) and (ψ_n) such that

$$\varphi_n \to f, \qquad \psi_n \to g$$

pointwise. Then

$$0 \le \varphi_1 + \psi_1 \le \varphi_2 + \psi_2 \le \cdots \le f + g,$$

and $\varphi_n + \psi_n \to f + g$ pointwise. Each $\varphi_n + \psi_n$ is simple. By monotone convergence and additivity for simple functions,

$$\begin{aligned}\int_E (f+g) &= \lim_{n\to\infty} \int_E (\varphi_n + \psi_n) \\ &= \lim_{n\to\infty} \left(\int_E \varphi_n + \int_E \psi_n \right) \\ &= \int_E f + \int_E g.\end{aligned}$$

Interchanging a nonnegative series and an integral

If $(f_n)_{n=1}^{\infty} \subset L^+(E)$, then

$$\int_E \sum_{n=1}^{\infty} f_n = \sum_{n=1}^{\infty} \int_E f_n.$$

Proof

The pointwise sum belongs to $L^+(E)$ as the pointwise limit of its measurable partial sums. By finite additivity, for every $N \in \mathbb{N}$,

$$\int_E \sum_{n=1}^{N} f_n = \sum_{n=1}^{N} \int_E f_n.$$

The partial sums increase pointwise to $\sum_{n=1}^{\infty} f_n$. Therefore the Monotone Convergence Theorem gives

$$\int_E \sum_{n=1}^{\infty} f_n = \lim_{N\to\infty} \int_E \sum_{n=1}^{N} f_n = \lim_{N\to\infty} \sum_{n=1}^{N} \int_E f_n = \sum_{n=1}^{\infty} \int_E f_n.$$

Zero integral and vanishing almost everywhere

Let $f \in L^+(E)$. Then

$$\int_E f = 0 \quad \Longleftrightarrow \quad f = 0 \quad \text{almost everywhere on } E.$$

Proof

If $f = 0$ almost everywhere, then $f \le 0$ almost everywhere. Almost-everywhere monotonicity gives

$$0 \le \int_E f \le \int_E 0 = 0.$$

Conversely, suppose $\int_E f = 0$, and define

$$F_n = \left\{ x \in E : f(x) > \frac{1}{n} \right\}, \qquad F = \{x \in E : f(x) > 0\}.$$

Then

$$F = \bigcup_{n=1}^{\infty} F_n, \qquad F_1 \subset F_2 \subset F_3 \subset \cdots.$$

For every n,

$$0 \leq \frac{1}{n} m(F_n) = \int_{F_n} \frac{1}{n} \leq \int_{F_n} f \leq \int_E f = 0.$$

Hence $m(F_n) = 0$ for every n. Continuity from below gives

$$m(F) = m\left(\bigcup_{n=1}^{\infty} F_n\right) = \lim_{n\to\infty} m(F_n) = 0.$$

Therefore $f = 0$ almost everywhere.

Monotone Convergence Theorem: almost-everywhere version

Let $(f_n)_{n=1}^{\infty} \subset L^+(E)$. Suppose

$$f_1 \leq f_2 \leq f_3 \leq \cdots$$

almost everywhere on E, and suppose $f_n(x) \to f(x)$ for almost every $x \in E$. Then

$$\int_E f = \lim_{n\to\infty} \int_E f_n.$$

Proof

Let

$$F = \{x \in E : f_1(x) \leq f_2(x) \leq \cdots \text{ and } f_n(x) \to f(x)\}.$$

By assumption, $m(E \setminus F) = 0$. The functions $\chi_F f_n$ increase pointwise to $\chi_F f$. The ordinary Monotone Convergence Theorem gives

$$\int_E \chi_F f = \lim_{n\to\infty} \int_E \chi_F f_n.$$

Changing a nonnegative measurable function on a null set does not change its integral, by the preceding zero-integral theorem. Hence

$$\int_E f = \int_E \chi_F f, \qquad \int_E f_n = \int_E \chi_F f_n,$$

which proves the assertion. Thus sets of measure zero do not affect Lebesgue integrals.

Fatou's lemma

Let $(f_n)_{n=1}^{\infty} \subset L^+(E)$. Then

$$\int_E \liminf_{n\to\infty} f_n \leq \liminf_{n\to\infty} \int_E f_n.$$

Proof

Set

$$g_n = \inf_{k \geq n} f_k.$$

Then (g_n) is increasing and

$$g_n \longrightarrow \liminf_{n\to\infty} f_n.$$

By the Monotone Convergence Theorem,

$$\int_E \liminf_{n\to\infty} f_n = \lim_{n\to\infty} \int_E g_n. \tag{*}$$

For every $j \geq n$, $g_n \leq f_j$, so

$$\int_E g_n \leq \inf_{j \geq n} \int_E f_j.$$

Taking limits in the preceding inequality and using $(*)$ gives

$$\int_E \liminf_{n\to\infty} f_n = \lim_{n\to\infty} \int_E g_n \leq \lim_{n\to\infty} \inf_{j\geq n} \int_E f_j = \liminf_{n\to\infty} \int_E f_n.$$

Finite integral implies finiteness almost everywhere

If $f \in L^+(E)$ and

$$\int_E f < \infty,$$

then

$$\{x \in E : f(x) = \infty\}$$

has measure zero.

Proof

Let

$$F = \{x \in E : f(x) = \infty\}.$$

For every $n \in \mathbb{N}$,

$$n\chi_F \leq f\chi_F.$$

Integrating gives

$$n\, m(F) = \int_E n\chi_F \leq \int_E f\chi_F \leq \int_E f < \infty.$$

This holds for every n. Sending $n \to \infty$ forces $m(F) = 0$.

12 Lebesgue Integrable Functions and Dominated Convergence

Lebesgue integrable real-valued functions

Let $E \subset \mathbb{R}$ be measurable. A measurable function $f : E \to \mathbb{R}$ is *Lebesgue integrable* if

$$\int_E |f| < \infty.$$

Write

$$f^+(x) = \max\{f(x), 0\}, \qquad f^-(x) = \max\{-f(x), 0\}.$$

Then

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-,$$

and hence

$$\int_E |f| < \infty \quad \Longleftrightarrow \quad \int_E f^+ < \infty \text{ and } \int_E f^- < \infty.$$

For a Lebesgue integrable f, define

$$\int_E f = \int_E f^+ - \int_E f^-.$$

The hierarchy of Lebesgue theory



Linearity and additivity

Let $f, g : E \to \mathbb{R}$ be Lebesgue integrable.

1. For every $c \in \mathbb{R}$, cf is Lebesgue integrable and
$$\int_E cf = c \int_E f.$$
2. The function $f + g$ is Lebesgue integrable and
$$\int_E (f + g) = \int_E f + \int_E g.$$
3. If $A, B \subset E$ are disjoint measurable sets, then
$$\int_{A \cup B} f = \int_A f + \int_B f.$$

Proof

For the first assertion,

$$|cf| = |c|\,|f|,$$

so cf is integrable. If $c \geq 0$, then

$$(cf)^+ = cf^+, \qquad (cf)^- = cf^-,$$

and therefore

$$\int_E cf = c\int_E f^+ - c\int_E f^- = c\int_E f.$$

If $c < 0$, then

$$(cf)^+ = (-c)f^-, \qquad (cf)^- = (-c)f^+,$$

which gives the same conclusion.
For the second assertion,

$$|f+g| \leq |f| + |g|,$$

so $f+g$ is integrable. Pointwise,

$$f^+ - f^- + g^+ - g^- = (f+g)^+ - (f+g)^-,$$

and hence

$$f^+ + g^+ + (f+g)^- = f^- + g^- + (f+g)^+.$$

Every term is nonnegative and integrable, so additivity for nonnegative functions, proved on page 98, gives

$$\int_E f^+ + \int_E g^+ + \int_E (f+g)^- = \int_E f^- + \int_E g^- + \int_E (f+g)^+.$$

Rearranging the finite quantities yields

$$\int_E (f+g) = \int_E f + \int_E g.$$

For the third assertion, let $A, B \subset E$ be disjoint measurable sets, and apply additivity for nonnegative functions separately to f^+ and f^-:

$$\begin{aligned}\int_{A\cup B} f &= \int_{A\cup B} f^+ - \int_{A\cup B} f^- \\ &= \left(\int_A f^+ + \int_B f^+\right) - \left(\int_A f^- + \int_B f^-\right) \\ &= \int_A f + \int_B f.\end{aligned}$$

Basic inequalities and almost-everywhere properties

Let $f, g : E \to \mathbb{R}$ be Lebesgue integrable.

1. $$\left|\int_E f\right| \leq \int_E |f|.$$

2. If $f = g$ almost everywhere on E, then

$$\int_E f = \int_E g.$$

3. If $f \le g$ almost everywhere on E, then

$$\int_E f \le \int_E g.$$

Proof

Since $\int_E f^+$ and $\int_E f^-$ are nonnegative,

$$\left|\int_E f\right| = \left|\int_E f^+ - \int_E f^-\right| \le \int_E f^+ + \int_E f^- = \int_E |f|.$$

If $f = g$ almost everywhere, then $|f| = |g|$ almost everywhere. The almost-everywhere properties for nonnegative functions on page 95 show that

$$\int_E |f| = \int_E |g| < \infty$$

and

$$\int_E f^\pm = \int_E g^\pm.$$

Thus f is integrable exactly when g is, and their integrals are equal.
Finally, if $f \le g$ almost everywhere, then $g - f \ge 0$ almost everywhere. Consequently,

$$0 \le \int_E (g - f) = \int_E g - \int_E f,$$

which proves the third assertion.

Dominated Convergence Theorem

Let $g : E \to [0, \infty]$ be Lebesgue integrable, and let $(f_n)_{n=1}^\infty$ be a sequence of real-valued measurable functions on E. Suppose that

$$|f_n| \le g \quad \text{almost everywhere on } E$$

for every n, and that $f_n \to f$ almost everywhere on E. Then f is Lebesgue integrable and

$$\lim_{n\to\infty} \int_E f_n = \int_E f.$$

Why this is stronger than the Riemann theorem

Only almost-everywhere pointwise convergence is required; uniform convergence is not.

Proof

The pointwise almost-everywhere limit of measurable functions is measurable, by the theorems on page 81–page 81. Outside a null set,

$$|f| = \lim_{n\to\infty} |f_n| \le g,$$

so

$$\int_E |f| \le \int_E g < \infty.$$

Thus f is integrable. Modifying all functions on one null set does not change any integral, so we may assume throughout that

$$|f_n| \le g, \qquad f_n \to f$$

at every point of E.
The functions $g - f_n$ are nonnegative. By Fatou's Lemma (page 101),

$$\begin{aligned}\int_E (g - f) &= \int_E \liminf_{n\to\infty} (g - f_n) \\ &\le \liminf_{n\to\infty} \int_E (g - f_n) \\ &= \int_E g - \limsup_{n\to\infty} \int_E f_n.\end{aligned}$$

Therefore

$$\limsup_{n\to\infty} \int_E f_n \le \int_E f.$$

Apply Fatou's Lemma again to the nonnegative functions $g + f_n$:

$$\begin{aligned}\int_E (g + f) &\le \liminf_{n\to\infty} \int_E (g + f_n) \\ &= \int_E g + \liminf_{n\to\infty} \int_E f_n.\end{aligned}$$

Hence

$$\int_E f \le \liminf_{n\to\infty} \int_E f_n.$$

Combining the two estimates,

$$\limsup_{n\to\infty} \int_E f_n \le \int_E f \le \liminf_{n\to\infty} \int_E f_n.$$

Since the reverse inequality between the limit inferior and limit superior always holds, both are equal to $\int_E f$.

Agreement of the Lebesgue and Riemann integrals

If $f : [a, b] \to \mathbb{R}$ is continuous, then f is Lebesgue measurable and Lebesgue integrable, and

$$\int_{[a,b]} f \, dm = \int_a^b f(x)\, dx,$$

where the integral on the right is the Riemann integral.

Proof

A continuous function is measurable by page 78. Since f is bounded on the compact interval $[a, b]$,

$$\int_{[a,b]} |f| \leq \left(\max_{x \in [a,b]} |f(x)| \right) (b - a) < \infty.$$

By applying the result to f^+ and f^-, it suffices to assume that $f \geq 0$.
For each n, choose a partition

$$a = x_0^{(n)} < x_1^{(n)} < \cdots < x_n^{(n)} = b$$

whose mesh tends to zero. For every j, choose $\xi_j^{(n)} \in [x_{j-1}^{(n)}, x_j^{(n)}]$ such that

$$f(\xi_j^{(n)}) = \min_{x \in [x_{j-1}^{(n)}, x_j^{(n)}]} f(x).$$

Let N be the union of all partition endpoints. It is countable and therefore has measure zero. Define the simple function

$$f_n(x) = \sum_{j=1}^{n} f(\xi_j^{(n)}) \chi_{(x_{j-1}^{(n)}, x_j^{(n)})}(x),$$

with $f_n = 0$ on N. Then

$$0 \leq f_n \leq f \leq \max_{[a,b]} f$$

and

$$\int_{[a,b]} f_n = \sum_{j=1}^{n} f(\xi_j^{(n)})(x_j^{(n)} - x_{j-1}^{(n)}),$$

the lower Riemann sum for the partition.
Fix $x \in [a, b] \setminus N$. If $x \in (x_{j-1}^{(n)}, x_j^{(n)})$, then

$$|x - \xi_j^{(n)}| \leq x_j^{(n)} - x_{j-1}^{(n)} \longrightarrow 0.$$

Continuity gives $f_n(x) = f(\xi_j^{(n)}) \to f(x)$. Thus $f_n \to f$ almost everywhere. The constant $\max_{[a,b]} f$ is an integrable dominating function, so the Dominated Convergence Theorem gives

$$\int_{[a,b]} f = \lim_{n \to \infty} \int_{[a,b]} f_n.$$

The right-hand side is the limit of the lower Riemann sums and hence is the Riemann integral of f.

Lebesgue integrable complex-valued functions

A measurable function $f : E \to \mathbb{C}$ is *Lebesgue integrable* if

$$\int_E |f| < \infty.$$

In that case, define

$$\int_E f = \int_E \operatorname{Re} f + i \int_E \operatorname{Im} f.$$

Integral triangle inequality for complex functions

If $f : E \to \mathbb{C}$ is Lebesgue integrable, then

$$\left| \int_E f \right| \leq \int_E |f|.$$

Proof

If $\int_E f = 0$, the result is immediate. Otherwise, let

$$\alpha = \frac{\overline{\int_E f}}{\left| \int_E f \right|}, \qquad |\alpha| = 1.$$

Then

$$\left| \int_E f \right| = \alpha \int_E f = \operatorname{Re}\left(\alpha \int_E f \right) = \int_E \operatorname{Re}(\alpha f) \leq \int_E |\alpha f| = \int_E |f|.$$

The three convergence theorems

The logical progression is

$$\text{Monotone Convergence Theorem} \implies \text{Fatou's Lemma} \implies \text{Dominated Convergence Theorem}.$$

The Monotone Convergence Theorem also permits infinite integrals and does not require an integrable dominating function.

13 Littlewood's Principles and Approximation

Step function and symmetric difference

A *step function* on $[a, b]$ is a simple function whose level sets are finite unions of intervals. For sets $A, B \subset \mathbb{R}$, their symmetric difference is

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

Littlewood's First Principle

Let $E \subset \mathbb{R}$ have finite outer measure. Then E is Lebesgue measurable if and only if, for every $\varepsilon > 0$, there is a finite union U of open intervals such that

$$m^*(U \triangle E) < \varepsilon.$$

Proof: measurable implies approximable

Suppose that E is measurable and let $\varepsilon > 0$. By the open cover characterization of outer measure, there are open intervals $(I_j)_{j=1}^{\infty}$ such that

$$E \subset \bigcup_{j=1}^{\infty} I_j, \qquad \sum_{j=1}^{\infty} |I_j| < m(E) + \frac{\varepsilon}{2}.$$

Choose N such that

$$\sum_{j=N+1}^{\infty} |I_j| < \frac{\varepsilon}{2}$$

and set $U = \bigcup_{j=1}^{N} I_j$. Since

$$U \setminus E \subset \left(\bigcup_{j=1}^{\infty} I_j \right) \setminus E$$

and

$$E \setminus U \subset \bigcup_{j=N+1}^{\infty} I_j,$$

we obtain

$$m^*(U \setminus E) \leq \sum_{j=1}^{\infty} |I_j| - m(E) < \frac{\varepsilon}{2}$$

and

$$m^*(E \setminus U) \leq \sum_{j=N+1}^{\infty} |I_j| < \frac{\varepsilon}{2}.$$

Thus $m^*(U \triangle E) < \varepsilon$.

Proof: approximable implies measurable

Suppose that the approximation property holds. Let $A \subset \mathbb{R}$ and $\varepsilon > 0$. Choose a finite union U of open intervals such that

$$m^*(U \triangle E) < \varepsilon.$$

Since U is measurable,

$$m^*(A) = m^*(A \cap U) + m^*(A \cap U^c).$$

Moreover,

$$A \cap E \subset (A \cap U) \cup (E \setminus U),$$

and

$$A \cap E^c \subset (A \cap U^c) \cup (U \setminus E).$$

Consequently,

$$\begin{aligned} m^*(A \cap E) + m^*(A \cap E^c) &\leq m^*(A \cap U) + m^*(A \cap U^c) + m^*(E \setminus U) + m^*(U \setminus E) \\ &< m^*(A) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ proves the Carathéodory criterion and hence the measurability of E.

Continuous approximation of a bounded step function

Let $\psi : [a,b] \to \mathbb{R}$ be a step function with

$$\sup_{x\in[a,b]} |\psi(x)| \le B.$$

For every $\varepsilon > 0$, there are a continuous function $g : [a,b] \to \mathbb{R}$ and a measurable set $E \subset [a,b]$ such that

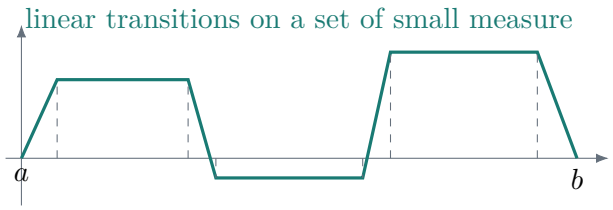
$$g(a) = g(b) = 0, \qquad \sup_{x\in[a,b]} |g(x)| \le B, \qquad m(E) < \varepsilon,$$

and

$$|\psi(x) - g(x)| < \varepsilon \qquad (x \in E^c).$$

Proof

The step function is constant on finitely many intervals. Around every endpoint and every jump, choose a small open interval; choose them so that the sum of their lengths is less than ε. Let E be their union. Define $g = \psi$ where no transition is required, join the neighboring constant values by line segments inside the chosen intervals, and similarly join the endpoint values to 0 at a and b. Then g is continuous, has the stated endpoint values and bound, and agrees with ψ on E^c.



Step approximation of a bounded simple function

Let $\varphi : [a,b] \to \mathbb{R}$ be a simple measurable function with

$$\sup_{x\in[a,b]} |\varphi(x)| \le B.$$

For every $\varepsilon > 0$, there are a step function $\psi : [a,b] \to \mathbb{R}$ and a measurable set $E \subset [a,b]$ such that

$$\sup_{x\in[a,b]} |\psi(x)| \le B, \qquad m(E) < \varepsilon,$$

and

$$|\varphi(x) - \psi(x)| < \varepsilon \qquad (x \in E^c).$$

Proof

Write the range of φ as $\{a_1, \dots, a_n\}$. For each i, Littlewood's First Principle gives a finite union U_i of open intervals such that

$$m(\varphi^{-1}(\{a_i\}) \triangle U_i) < \frac{\varepsilon}{n}.$$

The overlaps of the U_i have measure zero: if $i \neq j$, then the true level sets $\varphi^{-1}(\{a_i\})$ and $\varphi^{-1}(\{a_j\})$ are disjoint, so an overlap is contained in the corresponding symmetric-difference errors. Remove these overlaps successively, obtaining pairwise disjoint finite unions of intervals I_i, and define

$$\psi = \sum_{i=1}^{n} a_i \chi_{I_i}.$$

Then $\sup|\psi| \leq \sup|\varphi| \leq B$. The exceptional set

$$E = \bigcup_{i=1}^{n} \left(\varphi^{-1}(\{a_i\}) \triangle I_i\right)$$

is measurable and satisfies

$$m(E) \leq \sum_{i=1}^{n} m\left(\varphi^{-1}(\{a_i\}) \triangle I_i\right) < \varepsilon.$$

For $x \in E^c$, the functions have the same level value, so $\varphi(x) = \psi(x)$.

Measurable set operations

If $A, B \subset \mathbb{R}$ are Lebesgue measurable, then

$$A \setminus B, \qquad B \setminus A, \qquad A \triangle B$$

are Lebesgue measurable. Indeed,

$$A \triangle B = (A \cap B^c) \cup (B \cap A^c).$$

Littlewood's Second Principle

Let $a < b$, and let $f : [a, b] \to \mathbb{R}$ be measurable with

$$\sup_{x \in [a,b]} |f(x)| \leq B.$$

For every $\varepsilon > 0$, there are a continuous function $g : [a, b] \to \mathbb{R}$ and a measurable set $E \subset [a, b]$ such that

$$g(a) = g(b) = 0, \qquad \sup_{x \in [a,b]} |g(x)| \leq B, \qquad m(E) < \varepsilon,$$

and

$$|f(x) - g(x)| < \varepsilon \qquad (x \in E^c).$$

In other words, every measurable function is almost continuous.

Proof

By the simple-function approximation theorem on page 88, there are simple functions $(\varphi_n)_{n=0}^{\infty}$ such that

$$0 \leq |\varphi_0(x)| \leq |\varphi_1(x)| \leq \cdots \leq |f(x)|, \qquad \varphi_n(x) \to f(x)$$

pointwise, and the convergence is uniform on every set $\{x \in [a,b] : |f(x)| \le B\}$. Here this is all of $[a,b]$, so $\varphi_n \to f$ uniformly and $\sup|\varphi_n| \le B$.
Choose M so that

$$|\varphi_M(x) - f(x)| < \frac{\varepsilon}{3} \qquad (x \in [a,b]).$$

By the preceding step-approximation theorem, there are a step function ψ and a measurable E_1 such that

$$\sup|\psi| \le B, \qquad m(E_1) < \frac{\varepsilon}{3}, \qquad |\psi - \varphi_M| < \frac{\varepsilon}{3} \quad \text{on } E_1^c.$$

The continuous-approximation theorem for step functions gives a continuous g and a measurable E_2 such that

$$g(a) = g(b) = 0, \quad \sup|g| \le B, \quad m(E_2) < \frac{\varepsilon}{3}, \quad |g - \psi| < \frac{\varepsilon}{3} \quad \text{on } E_2^c.$$

By subadditivity,

$$m(E_1 \cup E_2) \le m(E_1) + m(E_2) < \varepsilon.$$

For $x \notin E_1 \cup E_2$,

$$|f(x) - g(x)| \le |f(x) - \varphi_M(x)| + |\varphi_M(x) - \psi(x)| + |\psi(x) - g(x)| < \varepsilon.$$

A union of null sets

If $(A_n)_{n=1}^{\infty}$ is a sequence of sets such that, for every $\varepsilon > 0$, $m(A_n) < \varepsilon$, then

$$m\left(\bigcup_{n=1}^{\infty} A_n\right) = 0.$$

Indeed, choose $m(A_n) < \varepsilon/2^{n+1}$; then

$$m\left(\bigcup_n A_n\right) \le \sum_n m(A_n) < \varepsilon.$$

Approximation of an integrable function

Let $f : [a,b] \to \mathbb{R}$ be Lebesgue integrable, and let $\varepsilon > 0$.

1. There is a bounded measurable function $h : [a,b] \to \mathbb{R}$ such that
$$\int_{[a,b]} |f - h| < \varepsilon.$$
2. There is a step function $\psi : [a,b] \to \mathbb{R}$ such that
$$\int_{[a,b]} |f - \psi| < \varepsilon.$$
3. There is a continuous function $g : [a,b] \to \mathbb{R}$, with $g(a) = g(b) = 0$, such that
$$\int_{[a,b]} |f - g| < \varepsilon.$$

Proof of part 1

For $n \in \mathbb{N}$, let

$$E_n = f^{-1}([-n, n]), \qquad h_n = f\chi_{E_n}.$$

Each h_n is measurable and bounded by n. Moreover, $h_n \to f$ pointwise and

$$|f - h_n| \leq |f| + |h_n| \leq 2|f|.$$

The function $2|f|$ is integrable, so the Dominated Convergence Theorem gives

$$\lim_{n\to\infty} \int_{[a,b]} |f - h_n| = \int_{[a,b]} \lim_{n\to\infty} |f - h_n| = 0.$$

Choose M for which the integral is less than ε, and take $h = h_M$.

Proof of part 2

Apply part 1 to obtain a bounded measurable h with

$$\int_{[a,b]} |f - h| < \frac{\varepsilon}{3}.$$

By the simple-function approximation theorem (page 88), there are simple functions $\varphi_n \to h$ pointwise with

$$|\varphi_n - h| \leq 2|h|.$$

Because a bounded measurable function on a measurable set of finite measure is integrable, the Dominated Convergence Theorem gives

$$\int_{[a,b]} |\varphi_n - h| \longrightarrow 0.$$

Choose M such that

$$\int_{[a,b]} |\varphi_M - h| < \frac{\varepsilon}{3}.$$

Let $B = \sup |h|$. The simple-to-step approximation on page 108 provides a step function ψ and a measurable set E such that

$$\sup |\psi| \leq B, \qquad m(E) < \frac{\varepsilon}{6B}, \qquad |\varphi_M - \psi| < \frac{\varepsilon}{3m([a,b])} \quad \text{on } E^c.$$

Therefore

$$\begin{aligned} \int_{[a,b]} |f - \psi| &\leq \int |f - h| + \int |h - \varphi_M| + \int |\varphi_M - \psi| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + 2B\, m(E) + m([a,b]) \frac{\varepsilon}{3m([a,b])} \\ &< \varepsilon. \end{aligned}$$

Second proof of part 2, using Littlewood's First Principle

As in the first proof, choose a bounded measurable h and then a simple function

$$\varphi_M = \sum_{i=1}^{N} c_i \chi_{A_i}$$

with pairwise disjoint measurable A_i, so that the first two L^1-errors are small. By Littlewood's First Principle, choose finite unions U_i of open intervals satisfying

$$m(A_i \triangle U_i) < \frac{\varepsilon}{N|c_i|}$$

whenever $c_i \neq 0$. Define the step function

$$\psi_M = \sum_{i=1}^{N} c_i \chi_{U_i}.$$

Then

$$|\varphi_M - \psi_M| \leq \sum_{i=1}^{N} |c_i| \chi_{A_i \triangle U_i},$$

and hence

$$\int_{[a,b]} |\varphi_M - \psi_M| \leq \sum_{i=1}^{N} |c_i| m(A_i \triangle U_i) < \varepsilon.$$

Choosing the three error tolerances to sum to the prescribed ε proves the assertion.

Proof of part 3

Choose the step function in part 2 so that

$$\int_{[a,b]} |f - \psi| < \frac{\varepsilon}{2}.$$

Apply Littlewood's Second Principle to ψ, with an exceptional set and pointwise error small enough that

$$\int_{[a,b]} |\psi - g| < \frac{\varepsilon}{2}.$$

Then g is continuous, $g(a) = g(b) = 0$, and

$$\int_{[a,b]} |f - g| \leq \int |f - \psi| + \int |\psi - g| < \varepsilon.$$

14 The Riemann–Lebesgue Lemma

Riemann–Lebesgue Lemma

Let $f : [-\pi, \pi] \to \mathbb{C}$ be Lebesgue integrable. Its Fourier coefficients are

$$\widehat{f}(n) = \frac{1}{2\pi} \int_{[-\pi,\pi]} f(x) e^{-inx}\, dx, \qquad n \in \mathbb{Z}.$$

Then

$$\lim_{|n|\to\infty} \widehat{f}(n) = 0.$$

Proof for step functions

First suppose that $f = \psi$ is a step function. Write

$$\psi = \sum_{j=1}^{N} a_j \chi_{U_j},$$

where the U_j are pairwise disjoint finite unions of open intervals whose union is $[-\pi, \pi]$, up to endpoints of measure zero. Therefore

$$\widehat{\psi}(n) = \sum_{j=1}^{N} \frac{a_j}{2\pi} \int_{U_j} e^{-inx}\, dx.$$

Decompose the U_j into a total of M open intervals (c_k, d_k), and absorb the appropriate coefficients into α_k. Then

$$\widehat{\psi}(n) = \sum_{k=1}^{M} \frac{\alpha_k}{2\pi} \int_{c_k}^{d_k} e^{-inx}\, dx.$$

For $n \neq 0$,

$$\int_{c_k}^{d_k} e^{-inx}\, dx = \frac{e^{-ind_k} - e^{-inc_k}}{-in}.$$

Since $|e^{-inx}| = 1$,

$$0 \leq |\widehat{\psi}(n)| \leq \sum_{k=1}^{M} \frac{|\alpha_k|}{2\pi|n|} \left| e^{-ind_k} - e^{-inc_k} \right| \leq \frac{1}{\pi|n|} \sum_{k=1}^{M} |\alpha_k|.$$

The right-hand side tends to zero as $|n| \to \infty$, so $\widehat{\psi}(n) \to 0$.

Proof for a general integrable function

Let $\varepsilon > 0$. By the approximation theorem on page 111, choose a step function $\psi : [-\pi, \pi] \to \mathbb{C}$ such that

$$\int_{[-\pi,\pi]} |f - \psi| < 2\pi\varepsilon.$$

For every $n \in \mathbb{Z}$,

$$\begin{aligned} |\widehat{f}(n) - \widehat{\psi}(n)| &= \frac{1}{2\pi}\left|\int_{[-\pi,\pi]} (f-\psi)e^{-inx}\,dx\right| \\ &\le \frac{1}{2\pi}\int_{[-\pi,\pi]} |f-\psi|\,|e^{-inx}|\,dx \\ &< \varepsilon. \end{aligned}$$

Since $\widehat{\psi}(n) \to 0$, it follows that $\widehat{f}(n) \to 0$.

15 The Lebesgue Criterion for Riemann Integrability

Lebesgue integrability criterion

Let $f : [a,b] \to \mathbb{R}$ be bounded. Then f is Riemann integrable if and only if its set D of points of discontinuity has measure zero; that is, if and only if f is continuous almost everywhere on $[a,b]$.

Oscillation

Let $f : S \to \mathbb{R}$ be bounded. Its oscillation on S is

$$\Omega_f(S) = \sup_{x,y\in S} |f(x) - f(y)|.$$

If $T \subset S$, then

$$\Omega_f(T) \le \Omega_f(S).$$

For $c \in S$, the oscillation of f at c is

$$\omega_f(c) = \lim_{r\downarrow 0} \Omega_f(S \cap N_r(c)), \qquad N_r(c) = \{x \in S : |x-c| < r\}.$$

Earlier Riemann integration

So far we have used the Riemann integral only for continuous functions, where uniform continuity shows that finer partitions have a common limit independent of the tags. We now give the general bounded-function definitions.

Lower and upper Riemann sums and integrals

Let $f : [a,b] \to \mathbb{R}$ be bounded, and let

$$\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}$$

be a partition. Put

$$m_k = \inf_{x_{k-1}\le x\le x_k} f(x), \qquad M_k = \sup_{x_{k-1}\le x\le x_k} f(x).$$

Define

$$L(f,\mathcal{P}) = \sum_{k=1}^{n} m_k(x_k - x_{k-1}), \qquad U(f,\mathcal{P}) = \sum_{k=1}^{n} M_k(x_k - x_{k-1}).$$

The lower and upper Riemann integrals are

$$\underline{\int_a^b} f = \sup_{\mathcal{P}} L(f,\mathcal{P}), \qquad \overline{\int_a^b} f = \inf_{\mathcal{P}} U(f,\mathcal{P}).$$

Integrability Criterion I

Let $f : [a,b] \to \mathbb{R}$ be bounded. Then f is Riemann integrable on $[a,b]$ if and only if

$$\underline{\int_a^b} f = \overline{\int_a^b} f.$$

If this holds, then

$$\int_a^b f = \underline{\int_a^b} f = \overline{\int_a^b} f.$$

Riemann Integrability Criterion

Let $f : [a,b] \to \mathbb{R}$ be bounded. Then f is Riemann integrable on $[a,b]$ if and only if, for every $\varepsilon > 0$, there is a partition $\mathcal{P}$ of $[a,b]$ such that

$$U(f,\mathcal{P}) - L(f,\mathcal{P}) < \varepsilon.$$

Geometric meaning

The difference $U(f,\mathcal{P}) - L(f,\mathcal{P})$ is the total area between the upper and lower rectangles. Requiring this difference to be small forces increasingly fine partitions near the varying parts of the graph.

Gauge and a fine tagged partition

Let $\delta : [a,b] \to \mathbb{R}_{>0}$. A tagged partition

$$\mathcal{P} = \{a = x_0 < x_1 < \cdots < x_n = b\}, \qquad \xi = \{\xi_1, \ldots, \xi_n\},$$

is δ-fine, with gauge δ, if

$$[x_{k-1}, x_k] \subset N_{\delta(\xi_k)}(\xi_k) \qquad (k = 1, \ldots, n).$$

Existence of a fine tagged partition

For every gauge $\delta : [a,b] \to \mathbb{R}_{>0}$, there is a δ-fine tagged partition of $[a,b]$.

Proof

The set $\delta([a,b]) \subset (0,\infty)$ is bounded below, so let

$$d = \inf_{x\in[a,b]} \delta(x).$$

Choose $M \in \mathbb{N}$ such that

$$M > \frac{b-a}{d},$$

and use a uniform partition with mesh $\Delta x = (b-a)/M$. Choose any tag $\xi_k \in [x_{k-1}, x_k]$. Then

$$x_k - \xi_k \le \Delta x < d \le \delta(\xi_k), \qquad \xi_k - x_{k-1} \le \Delta x < d \le \delta(\xi_k).$$

Thus

$$[x_{k-1}, x_k] \subset N_{\delta(\xi_k)}(\xi_k)$$

for every k.

Proof of the Lebesgue criterion: forward direction

Assume that f is Riemann integrable. For $i \in \mathbb{N}$, define

$$D_i = \left\{ x \in [a,b] : \omega_f(x) \ge \frac{1}{2^i} \right\}.$$

A bounded function is continuous at c if and only if $\omega_f(c) = 0$. Hence the discontinuity set is

$$D = \bigcup_{i=1}^{\infty} D_i.$$

Fix i and $\varepsilon > 0$. By the Riemann integrability criterion, there is a partition $\mathcal{P} = \{a = x_0 < \cdots < x_n = b\}$ such that

$$U(f,\mathcal{P}) - L(f,\mathcal{P}) = \sum_{k=1}^{n} (M_k - m_k)(x_k - x_{k-1}) < \frac{\varepsilon}{4^i}. \tag{$*$}$$

If $x \in D_i \cap (x_{k-1}, x_k)$, then for some $r > 0$, $N_r(x) \subset (x_{k-1}, x_k)$, and therefore

$$\frac{1}{2^i} \le \omega_f(x) \le \Omega_f(N_r(x)) \le M_k - m_k.$$

Let

$$T = \{k \in \{1,\dots,n\} : D_i \cap (x_{k-1}, x_k) \ne \varnothing\}.$$

Then, by $(*)$,

$$\frac{1}{2^i} \sum_{k\in T} (x_k - x_{k-1}) \le \sum_{k=1}^{n} (M_k - m_k)(x_k - x_{k-1}) < \frac{\varepsilon}{4^i},$$

so

$$\sum_{k\in T} (x_k - x_{k-1}) < \frac{\varepsilon}{2^i}.$$

The set D_i is covered by the intervals $[x_{k-1}, x_k]$, $k \in T$, together with the finitely many partition endpoints, and hence

$$m(D_i) < \frac{\varepsilon}{2^i}.$$

Thus $m(D_i) = 0$ for every i, and by countable subadditivity,

$$m(D) = m\left(\bigcup_{i=1}^{\infty} D_i\right) = 0.$$

Proof of the Lebesgue criterion: reverse direction

Assume that the discontinuity set

$$D = \{c \in [a, b] : f \text{ is discontinuous at } c\}$$

has measure zero. Choose $M > 0$ such that

$$|f(x)| \leq M \qquad (x \in [a, b]),$$

and let $\varepsilon > 0$. By the definition of measure, there are open intervals $(J_i)_{i=1}^{\infty}$ such that

$$D \subset \bigcup_{i=1}^{\infty} J_i, \qquad \sum_{i=1}^{\infty} |J_i| < \frac{\varepsilon}{4M}.$$

Define a gauge $\delta : [a, b] \to \mathbb{R}_{>0}$ as follows.

- If $t \in D$, choose i with $t \in J_i$. Since J_i is open, choose $r > 0$ such that $N_r(t) \subset J_i$, and set $\delta(t) = r$.
- If $t \in D^c$, continuity of f at t gives an $r > 0$ such that

$$x \in N_r(t) \quad \Longrightarrow \quad |f(x) - f(t)| < \frac{\varepsilon}{4(b-a)}.$$

  Set $\delta(t) = r$.

Choose a δ-fine tagged partition $(\mathcal{P}, \xi)$. Let

$$A = \{k : \xi_k \in D\}, \qquad B = \{k : \xi_k \in D^c\}.$$

Then

$$U(f, \mathcal{P}) - L(f, \mathcal{P}) = \sum_{k \in A} (M_k - m_k)\Delta x_k + \sum_{k \in B} (M_k - m_k)\Delta x_k.$$

For $k \in A$,

$$M_k - m_k \leq 2M.$$

For $k \in B$, δ-fineness and the definition of the gauge give

$$|f(x) - f(\xi_k)| < \frac{\varepsilon}{4(b-a)} \qquad (x \in [x_{k-1}, x_k]).$$

Hence, for x, y in that subinterval,

$$|f(x) - f(y)| \le |f(x) - f(\xi_k)| + |f(\xi_k) - f(y)| < \frac{\varepsilon}{2(b-a)},$$

so

$$M_k - m_k < \frac{\varepsilon}{2(b-a)}.$$

If $k \in A$, then the whole subinterval $[x_{k-1}, x_k]$ lies in one of the J_i. Therefore

$$\sum_{k \in A} \Delta x_k \le \sum_{i=1}^{\infty} |J_i| < \frac{\varepsilon}{4M}.$$

It follows that

$$\begin{aligned} U(f, \mathcal{P}) - L(f, \mathcal{P}) &\le 2M \sum_{k \in A} \Delta x_k + \frac{\varepsilon}{2(b-a)} \sum_{k \in B} \Delta x_k \\ &< 2M \frac{\varepsilon}{4M} + \frac{\varepsilon}{2(b-a)}(b-a) = \varepsilon. \end{aligned}$$

The Riemann integrability criterion on page 116 now shows that f is Riemann integrable.

Two comments on the gauge argument

- The proof can also be written without gauges by splitting the points into two classes.
- Gauges are not redundant. They lead to the Henstock–Kurzweil integral, a more general integral on the real line that integrates some functions which are not Lebesgue integrable.

16 L^p Space Theory

Where the theory now stands

The space of Lebesgue integrable functions contains the continuous functions. The Dominated Convergence Theorem showed, among other things, that the Lebesgue integral agrees with the Riemann integral for continuous functions. More generally, a bounded function on $[a, b]$ is Riemann integrable precisely when it is continuous almost everywhere.

The L^p-norm and essential supremum

Let $f : E \to \mathbb{C}$ be measurable. For $1 \le p < \infty$, define

$$\|f\|_{L^p(E)} = \left(\int_E |f|^p \right)^{1/p}.$$

Define

$$\|f\|_{L^\infty(E)} = \inf \left\{ M > 0 : m(\{x \in E : |f(x)| > M\}) = 0 \right\}.$$

This is also denoted

$$\|f\|_{L^\infty(E)} = \operatorname*{ess\,sup}_{x \in E} |f(x)|.$$

For a real-valued measurable f, one similarly writes

$$\operatorname*{ess\,inf}_{x\in E} f = \sup\{M \in \mathbb{R} : m(\{x \in E : f(x) < M\}) = 0\},$$

and

$$\operatorname*{ess\,sup}_{x\in E} f = \inf\{M \in \mathbb{R} : m(\{x \in E : f(x) > M\}) = 0\}.$$

Essential bounds

If $f : E \to \mathbb{C}$ is measurable, then

$$|f(x)| \le \|f\|_{L^\infty(E)} \quad \text{almost everywhere on } E.$$

If $E = [a, b]$ and $f \in C([a, b])$, then

$$\|f\|_{L^\infty([a,b])} = \|f\|_\infty = \sup_{x\in[a,b]} |f(x)|.$$

Proof

The first assertion follows directly from the definition of essential supremum. For the second, the Extreme Value Theorem supplies $c \in [a, b]$ such that

$$|f(c)| = \sup_{x\in[a,b]} |f(x)| = \|f\|_\infty.$$

Certainly

$$\|f\|_{L^\infty([a,b])} \le \|f\|_\infty.$$

Suppose, for contradiction, that the inequality is strict. Continuity at c gives a $\delta > 0$ such that

$$|x - c| < \delta \quad \Longrightarrow \quad ||f(x)| - |f(c)|| < \frac{|f(c)| - \|f\|_{L^\infty}}{2}.$$

Thus, on a nonempty interval about c,

$$|f(x)| > \frac{|f(c)| + \|f\|_{L^\infty}}{2}.$$

By the defining infimum, choose M with

$$\|f\|_{L^\infty} \le M < \frac{|f(c)| + \|f\|_{L^\infty}}{2}$$

and

$$m(\{x \in [a, b] : |f(x)| > M\}) = 0.$$

The interval just found is contained in this null set, a contradiction.

Hölder's and Minkowski's inequalities

Let $1 \le p \le \infty$.

1. If $p, q \in [1, \infty]$ are Hölder conjugates,
$$\frac{1}{p} + \frac{1}{q} = 1,$$
and $f, g : E \to \mathbb{C}$ are measurable, then
$$\int_E |fg| \leq \|f\|_{L^p(E)} \|g\|_{L^q(E)}.$$
2. If $f, g : E \to \mathbb{C}$ are measurable, then
$$\|f + g\|_{L^p(E)} \leq \|f\|_{L^p(E)} + \|g\|_{L^p(E)}.$$

The finite-dimensional precedents

The proofs for ℓ^p used:

1. the convexity inequality
$$A^t B^{1-t} \leq tA + (1-t)B \qquad (A, B > 0,\ t \in (0,1));$$
2. Young's inequality
$$|xy| \leq \frac{|x|^p}{p} + \frac{|y|^q}{q};$$
3. finite-dimensional Hölder's inequality; and
4. finite-dimensional Minkowski's inequality.

The same argument works here, replacing finite sums by Lebesgue integrals and using monotonicity almost everywhere.

Proof of Hölder's inequality

The endpoint cases are immediate. Suppose $1 < p, q < \infty$, and first assume that both norms are nonzero. Young's inequality gives
$$\begin{aligned}\int_E \frac{|f|}{\|f\|_p} \frac{|g|}{\|g\|_q} &\leq \int_E \left[\frac{1}{p}\frac{|f|^p}{\|f\|_p^p} + \frac{1}{q}\frac{|g|^q}{\|g\|_q^q}\right] \\ &= \frac{1}{p} + \frac{1}{q} = 1.\end{aligned}$$
Multiplying by $\|f\|_p \|g\|_q$ gives
$$\int_E |fg| \leq \|f\|_p \|g\|_q.$$
If either norm is zero, the assertion is immediate because the corresponding function vanishes almost everywhere.

Proof of Minkowski's inequality

For $1 < p < \infty$,

$$|f+g|^p \le |f+g|^{p-1}|f| + |f+g|^{p-1}|g|.$$

Integrating and applying Hölder's inequality,

$$\int_E |f+g|^p \le \big\||f+g|^{p-1}\big\|_q (\|f\|_p + \|g\|_p).$$

Since $(p-1)q = p$,

$$\big\||f+g|^{p-1}\big\|_q = \left(\int_E |f+g|^p\right)^{1/q}.$$

Therefore

$$\left(\int_E |f+g|^p\right)^{1/p} \le \|f\|_p + \|g\|_p.$$

The cases $p = 1$ and $p = \infty$ follow directly from the pointwise triangle inequality and the definition of essential supremum.

Abbreviated notation

From now on, $\|f\|_{L^p(E)}$ will usually be written $\|f\|_p$, including the case $p = \infty$. On bounded continuous functions, $\|\cdot\|_{L^\infty}$ agrees with the usual uniform norm $\|\cdot\|_\infty$.

The L^p space

For $1 \le p \le \infty$, define

$$L^p(E) = \{f : E \to \mathbb{C} : f \text{ is measurable and } \|f\|_p < \infty\}.$$

Two such functions are identified when they agree almost everywhere. Thus, strictly speaking, the elements are equivalence classes

$$[f] = \{g : E \to \mathbb{C} : g \text{ is measurable, } \|g\|_p < \infty, \text{ and } g = f \text{ almost everywhere}\}.$$

We nevertheless continue to call them functions.

$L^p(E)$ is a normed vector space

With pointwise addition and the natural scalar multiplication, $L^p(E)$ is a complex vector space, and $\|\cdot\|_p$ is a norm on it.

Proof: well-definedness

If $f = g$ almost everywhere, then $|f|^p = |g|^p$ almost everywhere, and so

$$\|f\|_p = \|g\|_p.$$

Thus the norm is well defined on equivalence classes.
Addition and scalar multiplication are also independent of the chosen representatives. Indeed,

if $f = f'$ and $g = g'$ almost everywhere, then the exceptional sets

$$E_1 = \{x : f(x) \neq f'(x)\}, \qquad E_2 = \{x : g(x) \neq g'(x)\}$$

have measure zero. Outside $E_1 \cup E_2$,

$$f + g = f' + g',$$

and $E_1 \cup E_2$ still has measure zero. Scalar multiplication is similar.

Proof: the norm axioms

Positivity is clear. If $\|f\|_p = 0$, then

$$\int_E |f|^p = 0,$$

so $f = 0$ almost everywhere by page 99; hence $[f]$ is the zero element of $L^p(E)$.
For $k \in \mathbb{C}$,

$$\|kf\|_p = \left(\int_E |k|^p |f|^p\right)^{1/p} = |k| \, \|f\|_p.$$

Finally, the triangle inequality is Minkowski's inequality.

Exhaustion by bounded intervals

Let $E \subset \mathbb{R}$ be measurable and $1 \leq p < \infty$. Then

$$f \in L^p(E) \quad \Longleftrightarrow \quad \lim_{n\to\infty} \int_{[-n,n]\cap E} |f|^p < \infty.$$

In that case, the limit equals $\int_E |f|^p$.

Proof

For each n,

$$\int_{[-n,n]\cap E} |f|^p = \int_E \chi_{[-n,n]} |f|^p.$$

The sequence $\chi_{[-n,n]}|f|^p$ increases pointwise to $|f|^p$. The Monotone Convergence Theorem gives

$$\int_E |f|^p = \lim_{n\to\infty} \int_{[-n,n]\cap E} |f|^p.$$

A decay condition for membership in every L^p

Let $f : \mathbb{R} \to \mathbb{C}$ be measurable. If there are constants $C > 0$ and $q > 1$ such that, for almost every $x \in \mathbb{R}$,

$$|f(x)| \leq C(1 + |x|)^{-q},$$

then

$$f \in L^p(\mathbb{R}) \qquad \text{for every } p \geq 1.$$

Proof

For every n,

$$\int_{[-n,n]} |f|^p \le C^p \int_{-n}^{n} (1+|x|)^{-pq}\,dx.$$

The integrand is continuous, so the Riemann and Lebesgue integrals agree. Since $pq > 1$, elementary integration provides a constant B_p, independent of n, such that

$$\int_{-n}^{n} (1+|x|)^{-pq}\,dx \le B_p.$$

The preceding exhaustion theorem now gives $f \in L^p(\mathbb{R})$.

Density of continuous functions in $L^p([a,b])$

Let $a < b$ and $1 \le p < \infty$. If $f \in L^p([a,b])$, then for every $\varepsilon > 0$ there is a $g \in C([a,b])$ such that

$$g(a) = g(b) = 0 \qquad \text{and} \qquad \|f - g\|_p < \varepsilon.$$

Meaning of the density theorem

The theorem says, more precisely, that

$$C_0([a,b]) := \{g \in C([a,b]) : g(a) = g(b) = 0\}$$

is dense in $L^p([a,b])$ for $p < \infty$. It follows almost directly from the approximation theorem on page 111.

Riesz–Fischer Theorem

For every $1 \le p \le \infty$, the space $L^p(E)$ is a Banach space.

Proof for $1 \le p < \infty$: absolutely summable series

It is enough to prove that every absolutely summable series converges. Let $(f_k)_{k=1}^{\infty} \subset L^p(E)$ satisfy

$$\sum_{k=1}^{\infty} \|f_k\|_p = M < \infty.$$

Set

$$g_n(x) = \sum_{k=1}^{n} |f_k(x)|.$$

By Minkowski's inequality,

$$\|g_n\|_p \le \sum_{k=1}^{n} \|f_k\|_p \le M. \tag{*}$$

The functions g_n^p increase pointwise to

$$\left(\sum_{k=1}^{\infty} |f_k|\right)^p.$$

Fatou's Lemma and $(*)$ give

$$\int_E \left(\sum_{k=1}^{\infty} |f_k|\right)^p \leq \liminf_{n\to\infty} \int_E g_n^p \leq M^p.$$

Hence $\sum_k |f_k| < \infty$ almost everywhere. Define

$$f(x) = \begin{cases} \sum_{k=1}^{\infty} f_k(x), & \sum_{k=1}^{\infty} |f_k(x)| < \infty, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$g(x) = \begin{cases} \sum_{k=1}^{\infty} |f_k(x)|, & \sum_{k=1}^{\infty} |f_k(x)| < \infty, \\ 0, & \text{otherwise}. \end{cases}$$

The completeness of $\mathbb{C}$ ensures the pointwise convergence defining f. Moreover,

$$\left|\sum_{k=1}^{n} f_k - f\right|^p = \left|\sum_{k=n+1}^{\infty} f_k\right|^p \leq \left(\sum_{k=n+1}^{\infty} |f_k|\right)^p \leq |g|^p$$

almost everywhere, and $\|g\|_p \leq M$.
Also,

$$|f| \leq \sum_{k=1}^{\infty} |f_k| = g$$

almost everywhere, so $f \in L^p(E)$. By the Dominated Convergence Theorem,

$$\begin{aligned} \lim_{n\to\infty} \left\|\sum_{k=1}^{n} f_k - f\right\|_p &= \lim_{n\to\infty} \left(\int_E \left|\sum_{k=1}^{n} f_k - f\right|^p\right)^{1/p} \\ &= \left(\int_E \lim_{n\to\infty} \left|\sum_{k=1}^{n} f_k - f\right|^p\right)^{1/p} = 0. \end{aligned}$$

Thus the series converges in $L^p(E)$.

Proof for $p = \infty$

Let $(f_k) \subset L^\infty(E)$ satisfy

$$\sum_{k=1}^{\infty} \|f_k\|_\infty < \infty.$$

For each k, the exceptional set

$$E_k = \{x \in E : |f_k(x)| > \|f_k\|_\infty\}$$

has measure zero, and so does $\bigcup_k E_k$. Outside this union,

$$\sum_{k=1}^{\infty} |f_k(x)| \le \sum_{k=1}^{\infty} \|f_k\|_\infty < \infty.$$

Define

$$f(x) = \begin{cases} \sum_{k=1}^{\infty} f_k(x), & x \notin \bigcup_k E_k, \\ 0, & x \in \bigcup_k E_k. \end{cases}$$

Then

$$\|f\|_\infty \le \sum_{k=1}^{\infty} \|f_k\|_\infty < \infty.$$

Since the numerical series of norms is Cauchy, given $\varepsilon > 0$ there is an M such that

$$\sum_{k=n+1}^{\infty} \|f_k\|_\infty < \varepsilon \qquad (n \ge M).$$

For $n \ge M$ and outside the null exceptional set,

$$\left| f(x) - \sum_{k=1}^{n} f_k(x) \right| = \left| \sum_{k=n+1}^{\infty} f_k(x) \right| \le \sum_{k=n+1}^{\infty} \|f_k\|_\infty < \varepsilon.$$

Therefore

$$\left\| f - \sum_{k=1}^{n} f_k \right\|_\infty < \varepsilon,$$

and the series converges in $L^\infty(E)$.

Completion of continuous functions

For $p < \infty$, $C([a,b])$ is dense in $L^p([a,b])$, so $L^p([a,b])$ may be viewed as a completion of the continuous functions. In contrast, $C([a,b])$ is not dense in $L^\infty([a,b])$.

17 Hilbert Spaces

Pre-Hilbert space

A *pre-Hilbert space* H is a complex vector space equipped with a Hermitian inner product

$$\langle \cdot, \cdot \rangle : H \times H \to \mathbb{C}$$

such that:

1. for all $\lambda_1, \lambda_2 \in \mathbb{C}$ and $v_1, v_2, w \in H$,

$$\langle \lambda_1 v_1 + \lambda_2 v_2, w \rangle = \lambda_1 \langle v_1, w \rangle + \lambda_2 \langle v_2, w \rangle;$$

2. $$\langle v, w\rangle = \overline{\langle w, v\rangle};$$

3. $$\langle v, v\rangle \geq 0, \qquad \langle v, v\rangle = 0 \iff v = 0.$$

These axioms also imply:

1. if $\langle v, w\rangle = 0$ for every $w \in H$, then $v = 0$;
2. $$\langle v, \lambda w\rangle = \overline{\lambda}\langle v, w\rangle;$$
3. $$\langle v, w_1 + w_2\rangle = \langle v, w_1\rangle + \langle v, w_2\rangle.$$

Norm induced by an inner product

If H is a pre-Hilbert space, define

$$\|v\| = \sqrt{\langle v, v\rangle}.$$

Cauchy–Schwarz inequality

For every $u, v \in H$,

$$|\langle u, v\rangle| \leq \|u\|\,\|v\|.$$

Proof

For $t \in \mathbb{R}$,

$$\begin{aligned} 0 \leq \|u + tv\|^2 &= \langle u + tv, u + tv\rangle \\ &= \|u\|^2 + t^2\|v\|^2 + 2t\,\mathrm{Re}\langle u, v\rangle. \end{aligned}$$

This quadratic is nonnegative at its minimum

$$t_{\min} = -\frac{\mathrm{Re}\langle u, v\rangle}{\|v\|^2},$$

and hence

$$|\,\mathrm{Re}\langle u, v\rangle| \leq \|u\|\,\|v\|. \tag{$*$}$$

If $\langle u, v\rangle = 0$, the result is immediate. Otherwise, let

$$\lambda = \frac{\overline{\langle u, v\rangle}}{|\langle u, v\rangle|}, \qquad |\lambda| = 1.$$

Then

$$\begin{aligned} |\langle u, v\rangle| = \frac{|\langle u, v\rangle|^2}{|\langle u, v\rangle|} &= \frac{\overline{\langle u, v\rangle}}{|\langle u, v\rangle|}\langle u, v\rangle \\ &= \lambda\langle u, v\rangle = \langle \lambda u, v\rangle = \mathrm{Re}\langle \lambda u, v\rangle \\ &\leq \|\lambda u\|\,\|v\| = \|u\|\,\|v\|, \end{aligned}$$

where the inequality is $(*)$ and $\|\lambda u\| = \|u\|$.

18 Basic Hilbert Space Theory

The inner-product norm

If H is a pre-Hilbert space, then

$$\|v\| = \sqrt{\langle v, v\rangle}$$

defines a norm on H.

Proof

Positive definiteness is inherited from the Hermitian inner product. For $\lambda \in \mathbb{C}$,

$$\|\lambda v\|^2 = \langle \lambda v, \lambda v\rangle = \lambda\overline{\lambda}\langle v, v\rangle = |\lambda|^2\|v\|^2.$$

Finally, Cauchy–Schwarz gives

$$\begin{aligned}\|u+v\|^2 &= \|u\|^2 + \|v\|^2 + 2\,\mathrm{Re}\langle u, v\rangle \\ &\le \|u\|^2 + \|v\|^2 + 2\|u\|\,\|v\| = (\|u\| + \|v\|)^2.\end{aligned}$$

Taking square roots proves the triangle inequality.

Continuity of the inner product

If $u_n \to u$ and $v_n \to v$ in a pre-Hilbert space, then

$$\langle u_n, v_n\rangle \longrightarrow \langle u, v\rangle.$$

Proof

$$\begin{aligned}|\langle u_n, v_n\rangle - \langle u, v\rangle| &= |\langle u_n - u, v_n\rangle + \langle u, v_n - v\rangle| \\ &\le \|u_n - u\|\,\|v_n\| + \|u\|\,\|v_n - v\|.\end{aligned}$$

Moreover,

$$|\|v_n\| - \|v\|| \le \|v_n - v\|,$$

so $\|v_n\| \to \|v\|$, and in particular $(\|v_n\|)$ is bounded. The displayed upper bound therefore tends to zero.

Hilbert space

A *Hilbert space* is a pre-Hilbert space which is complete with respect to the norm induced by its inner product.

Fundamental examples

1. $\mathbb{C}^n$, with

$$\langle z, w\rangle = \sum_{j=1}^{n} z_j\overline{w_j}.$$

2. ℓ^2, with
$$\langle a, b\rangle = \sum_{j=1}^{\infty} a_j \overline{b_j}.$$

3. For measurable $E \subset \mathbb{R}$, $L^2(E)$, with
$$\langle f, g\rangle = \int_E f\overline{g}.$$

Parallelogram law

If H is a pre-Hilbert space, then for every $u, v \in H$,
$$\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2). \tag{P}$$

Conversely, if the norm of a complex normed vector space satisfies (P), then the norm is induced by a Hermitian inner product.

The forward direction is a direct expansion; the converse is proved on page 136. In particular, the ℓ^p and L^p norms arise from inner products only when $p = 2$: for $p \neq 2$, one can find vectors or functions for which the parallelogram law fails.

Orthogonal and orthonormal

In a pre-Hilbert space H, vectors $u, v \in H$ are *orthogonal*, written $u \perp v$, if
$$\langle u, v\rangle = 0.$$

A subset $\{e_\lambda\}_{\lambda\in\Lambda} \subset H$ is *orthonormal* if
$$\|e_\lambda\| = 1 \quad \text{and} \quad \lambda \neq \mu \Longrightarrow e_\lambda \perp e_\mu.$$

We will mainly use finite or countably infinite orthonormal sets.

Examples of orthonormal sets

1. Any subset of the standard basis of $\mathbb{C}^n$.

2. The standard coordinate vectors
$$e_n = (0, \ldots, 0, 1, 0, \ldots)$$
form an orthonormal set in ℓ^2.

3. The functions
$$\left\{\frac{1}{\sqrt{2\pi}} e^{inx}\right\}_{n\in\mathbb{Z}}$$
form an orthonormal set in $L^2([-\pi, \pi])$.

For the last example,
$$\left\langle \frac{1}{\sqrt{2\pi}} e^{imx}, \frac{1}{\sqrt{2\pi}} e^{inx} \right\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(m-n)x}\,dx = \begin{cases} 1, & m = n, \\ 0, & m \neq n. \end{cases}$$

Separable topological space

A topological space is *separable* if it has a countable dense subset.

For example, $\mathbb{R}$ is separable because $\mathbb{Q}$ is countable and dense.

Separability of ℓ^p and $L^p([a,b])$

For $1 \le p < \infty$, both ℓ^p and $L^p([a,b])$ are separable.

Recall the Weierstrass Approximation Theorem: for every $f \in C([a,b])$, there are polynomials P_n such that

$$P_n \longrightarrow f \quad \text{uniformly on } [a,b].$$

Proof for $L^p([a,b])$

Continuous functions are dense in $L^p([a,b])$ by page 124. Given $f \in L^p([a,b])$ and $\varepsilon > 0$, choose $g \in C([a,b])$ with

$$\|f - g\|_p < \frac{\varepsilon}{4}.$$

Weierstrass gives a polynomial P_M with

$$\sup_{x\in[a,b]} |P_M(x) - g(x)| < \frac{\varepsilon}{4(b-a)^{1/p}}.$$

Then

$$\|P_M - g\|_p < \frac{\varepsilon}{4}.$$

Write

$$P_M(x) = \sum_{j=0}^{k} \alpha_j x^j, \qquad \alpha_j = a_j + ib_j.$$

Using the density of $\mathbb{Q}$ in $\mathbb{R}$, approximate every a_j, b_j by rationals p_j, q_j closely enough that

$$P(x) = \sum_{j=0}^{k} (p_j + iq_j)x^j$$

satisfies

$$\|P_M - P\|_p < \frac{\varepsilon}{4}.$$

Hence $\|f - P\|_p < \varepsilon$.
The collection

$$\mathcal{A} = \left\{ \sum_{j=0}^{k} (p_j + iq_j)x^j : k \in \mathbb{N},\ p_j, q_j \in \mathbb{Q} \right\}$$

is dense in $L^p([a,b])$. It is countable because it is a countable union of finite Cartesian products of the countable set $\mathbb{Q}$.

Proof for ℓ^p

If $a = (a_n) \in \ell^p$, then the series $\sum_n |a_n|^p$ converges. Thus a can first be approximated by a

terminating sequence

$$a' = (a_1, \dots, a_M, 0, \dots),$$

and then each of its finitely many entries can be approximated by a number in $\mathbb{Q} + i\mathbb{Q}$. The terminating sequences with entries in $\mathbb{Q} + i\mathbb{Q}$ form a countable dense subset of ℓ^p.

Bessel's inequality

Let $\{e_n\}_{n\in\Lambda}$ be an at most countable orthonormal subset of a pre-Hilbert space H. Then, for every $u \in H$,

$$\sum_{n\in\Lambda} |\langle u, e_n\rangle|^2 \le \|u\|^2.$$

Proof

First suppose the set is finite, $\{e_1, \dots, e_N\}$. Orthonormality gives

$$\left\| \sum_{n=1}^{N} \langle u, e_n\rangle e_n \right\|^2 = \sum_{n=1}^{N} |\langle u, e_n\rangle|^2$$

and

$$\left\langle u, \sum_{n=1}^{N} \langle u, e_n\rangle e_n \right\rangle = \sum_{n=1}^{N} |\langle u, e_n\rangle|^2.$$

Therefore

$$\begin{aligned} 0 &\le \left\| u - \sum_{n=1}^{N} \langle u, e_n\rangle e_n \right\|^2 \\ &= \|u\|^2 - \sum_{n=1}^{N} |\langle u, e_n\rangle|^2. \end{aligned}$$

This proves the finite assertion. For a countable set, take the limit of the finite partial sums; limits in $\mathbb{R}$ preserve inequalities.

18.1 Gram–Schmidt and maximal orthonormal sets

Maximal orthonormal subset

An orthonormal subset $\{e_\lambda\}_{\lambda\in\Lambda}$ of a pre-Hilbert space H is *maximal* if the only $u \in H$ satisfying

$$\langle u, e_\lambda\rangle = 0 \qquad (\lambda \in \Lambda)$$

is $u = 0$.

The standard basis of $\mathbb{C}^n$ is maximal, while every proper subset of it is not. The coordinate vectors $\{e_n\}$ are maximal in ℓ^2.

The scalar field matters

Both the inner product and the basis depend on whether $\mathbb{C}^n$ is viewed over $\mathbb{R}$ or over $\mathbb{C}$. Over $\mathbb{R}$, its dimension is $2n$ and its real standard basis has $2n$ vectors. Over $\mathbb{C}$, its dimension is n, and the usual n coordinate vectors are orthonormal for the Hermitian inner product

$$\langle v, w\rangle = \sum_{j=1}^{n} v_j \overline{w_j}.$$

Throughout this course, $\mathbb{C}^n$ is regarded as a vector space over $\mathbb{C}$.

Existence of maximal orthonormal subsets

Every nontrivial pre-Hilbert space has a maximal orthonormal subset.

The proof, which we omit, uses the Axiom of Choice, or equivalently Zorn's Lemma, in the same spirit as the proof that every vector space has a Hamel basis.

The separable case

Every nontrivial separable pre-Hilbert space has a countable maximal orthonormal subset.

Proof: Gram–Schmidt construction

Let $(v_j)_{j=1}^{\infty}$ be a dense sequence in H, with $v_j \neq 0$. By induction, construct integers $m(n)$ and orthonormal sets

$$\{e_1, \dots, e_{m(n)}\}$$

such that

$$\operatorname{span}\{e_1, \dots, e_{m(n)}\} = \operatorname{span}\{v_1, \dots, v_n\},$$

omitting an additional vector precisely when it is already in the preceding span.
For $n = 1$, take $e_1 = v_1/\|v_1\|$. Suppose the construction has reached v_k. If v_{k+1} is already in $\operatorname{span}\{v_1, \dots, v_k\}$, add nothing. Otherwise set

$$w_{k+1} = v_{k+1} - \sum_{j=1}^{m(k)} \langle v_{k+1}, e_j\rangle e_j$$

and define

$$e_{m(k)+1} = \frac{w_{k+1}}{\|w_{k+1}\|}.$$

For $1 \leq \ell \leq m(k)$,

$$\begin{aligned}\langle e_{m(k)+1}, e_\ell\rangle &= \frac{1}{\|w_{k+1}\|}\left\langle v_{k+1} - \sum_{j=1}^{m(k)} \langle v_{k+1}, e_j\rangle e_j, e_\ell\right\rangle \\ &= \frac{1}{\|w_{k+1}\|}\left(\langle v_{k+1}, e_\ell\rangle - \langle v_{k+1}, e_\ell\rangle\right) = 0.\end{aligned}$$

Thus the enlarged set is orthonormal.

Proof: maximality

Let

$$S = \bigcup_{n=1}^{\infty} \{e_1, \dots, e_{m(n)}\}.$$

Suppose that $u \perp e_\ell$ for every $e_\ell \in S$. Since the v_j are dense, choose a subsequence $v_{j_k} \to u$. Each v_{j_k} lies in the span of finitely many e_ℓ, so

$$v_{j_k} = \sum_{\ell=1}^{m(j_k)} \langle v_{j_k}, e_\ell \rangle e_\ell$$

and

$$\|v_{j_k}\|^2 = \sum_{\ell=1}^{m(j_k)} |\langle v_{j_k}, e_\ell \rangle|^2.$$

Because $\langle u, e_\ell \rangle = 0$, Bessel's inequality gives

$$\begin{aligned} \|v_{j_k}\|^2 &= \sum_{\ell=1}^{m(j_k)} |\langle v_{j_k} - u, e_\ell \rangle|^2 \\ &\leq \|v_{j_k} - u\|^2 \longrightarrow 0. \end{aligned}$$

But $v_{j_k} \to u$, so $\|v_{j_k}\| \to \|u\|$. Hence $\|u\| = 0$, and $u = 0$. Thus S is maximal.

Step and continuous approximation in L^p

Let $a < b$, $1 \leq p < \infty$, and $f \in L^p([a,b])$. For every $\varepsilon > 0$, there are a step function $\psi \in L^p([a,b])$ and a continuous $g \in C([a,b])$, with $g(a) = g(b) = 0$, such that

$$\|f - \psi\|_p + \|f - g\|_p < \varepsilon.$$

For $p = 1$, this is the approximation theorem on page 111. The proof for general $p < \infty$ is the same after first establishing the density of step functions in $L^p([a,b])$.

Further L^p approximation facts

Let $1 \leq p < \infty$.

1. If $1 \leq q \leq p$, then
$$L^p([a,b]) \subset L^q([a,b]).$$
2. If $f \in L^p(\mathbb{R})$ and $\varepsilon > 0$, there are $g \in C(\mathbb{R})$ and $R > 0$ such that
$$|x| > R \Longrightarrow g(x) = 0, \qquad \|f - g\|_p < \varepsilon.$$
Thus compactly supported continuous functions are dense in $L^p(\mathbb{R})$.
3. The space $L^p(\mathbb{R})$ is separable.

Proof of part 1

If $f \in L^p([a,b])$, split the interval into the sets $\{|f| \le 1\}$ and $\{|f| > 1\}$. Since $q \le p$,

$$\int_{[a,b]} |f|^q \le m([a,b]) + \int_{[a,b]} |f|^p < \infty.$$

Proof of part 2

Since $f \in L^p(\mathbb{R})$, the Monotone Convergence Theorem gives

$$\int_{[-M,M]} |f|^p \longrightarrow \int_{\mathbb{R}} |f|^p.$$

Thus, for sufficiently large M,

$$\left(\int_{\mathbb{R}\setminus[-M,M]} |f|^p \right)^{1/p} < \frac{\varepsilon}{2}.$$

Apply the continuous approximation theorem on $[-M,M]$ to obtain a $g_0 \in C([-M,M])$ with

$$g_0(-M) = g_0(M) = 0, \qquad \|f - g_0\|_{L^p([-M,M])} < \frac{\varepsilon}{2}.$$

Extend g_0 by zero outside $[-M,M]$. The endpoint conditions make the extension g continuous on $\mathbb{R}$, and

$$\|f - g\|_p \le \|f - g_0\|_{L^p([-M,M])} + \|f\|_{L^p(\mathbb{R}\setminus[-M,M])} < \varepsilon.$$

Proof of part 3

Separability follows by the same rational-polynomial construction used for $L^p([a,b])$, now using the density of compactly supported continuous functions and taking a countable union over intervals with rational endpoints.

Previously proved homework facts

- $L^\infty(E)$ is a Banach space; this is the $p = \infty$ case of the Riesz–Fischer Theorem on page 124.
- If $f \in C([a,b])$, then
$$\|f\|_{L^\infty([a,b])} = \|f\|_\infty,$$
as proved on page 119.

Continuous functions are not dense in L^∞

The subspace $C([a,b])$ is not dense in $L^\infty([a,b])$.

Proof

Fix $c \in (a,b)$ and let

$$f(x) = \begin{cases} 1, & x \in [a,c), \\ 0, & x \in [c,b]. \end{cases}$$

Suppose that a continuous g satisfied $\|f-g\|_\infty < 1/4$. Then

$$g > \frac{1}{2} \quad \text{almost everywhere on } [a,c), \qquad g < \frac{1}{2} \quad \text{almost everywhere on } [c,b].$$

By continuity and the Intermediate Value Theorem, $g(c) = 1/2$. Continuity at c then gives a $\delta > 0$ such that

$$|x-c| < \delta \quad \Longrightarrow \quad g(x) \in \left(\frac{1}{4}, \frac{3}{4}\right).$$

On a nonempty interval about c, this forces

$$|f(x) - g(x)| > \frac{1}{4},$$

a set of positive measure. Hence $\|f-g\|_\infty \geq 1/4$, a contradiction.

Multiplication operators

Let $a < b$, $1 \leq p \leq \infty$, and $g \in L^\infty([a,b])$. Define

$$Tf = fg$$

pointwise for $f \in L^p([a,b])$. Then

$$T \in \mathcal{B}(L^p([a,b]), L^p([a,b])) \qquad \text{and} \qquad \|T\| = \|g\|_\infty.$$

Proof sketch

Linearity is immediate, and

$$|fg| \leq \|g\|_\infty |f|$$

almost everywhere. Thus

$$\|Tf\|_p \leq \|g\|_\infty \|f\|_p$$

and $\|T\| \leq \|g\|_\infty$. Conversely, by the definition of essential supremum, for every $\varepsilon > 0$ the set

$$A_\varepsilon = \{x : |g(x)| > \|g\|_\infty - \varepsilon\}$$

has positive measure. Testing T on a normalized indicator of A_ε gives

$$\|T\| \geq \|g\|_\infty - \varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves equality. The argument for $p = \infty$ is the same.

Polarization identity

Let H be a complex pre-Hilbert space. Then, for every $u, v \in H$,

$$\langle u, v\rangle = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\right).$$

Proof

Expanding the four squared norms,

$$\|u+v\|^2 - \|u-v\|^2 = 4\,\mathrm{Re}\langle u, v\rangle,$$

while

$$\|u+iv\|^2 - \|u-iv\|^2 = 4\,\mathrm{Re}\langle u, iv\rangle.$$

Since

$$\langle u, iv\rangle = -i\langle u, v\rangle,$$

we have

$$i\,\mathrm{Re}\langle u, iv\rangle = i\,\mathrm{Im}\langle u, v\rangle.$$

Combining real and imaginary parts yields the stated identity.

Converse to the parallelogram law

Suppose H is a complex normed vector space whose norm satisfies the parallelogram law. Then there is a Hermitian inner product on H such that

$$\|u\| = \sqrt{\langle u, u\rangle} \qquad (u \in H).$$

Proof outline

Define

$$\langle u, v\rangle = \frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\right).$$

Positive definiteness and conjugate symmetry follow directly. For additivity, the parallelogram law gives

$$\|v+w+u\|^2 - 2\|w+u\|^2 + \|v+w-u\|^2 = 2\|v+w\|^2 - \|v+u-w\|^2 - \|v-u-w\|^2,$$

which is rearranged to prove

$$\langle v+w, u\rangle = \langle v, u\rangle + \langle w, u\rangle.$$

Additivity first yields homogeneity for integer scalars, then for rational scalars. Continuity of the norm and density of $\mathbb{Q}$ in $\mathbb{R}$ extend it to real scalars. Finally, writing $\lambda = \mathrm{Re}\,\lambda + i\,\mathrm{Im}\,\lambda$ gives complex homogeneity. Thus the defined form is a Hermitian inner product, and the polarization formula gives

$$\langle u, u\rangle = \|u\|^2.$$

19 Orthonormal Bases and Fourier Series

Orthonormal basis

Let H be a Hilbert space. An *orthonormal basis*, also called a Schauder basis in these notes, is an at most countable maximal orthonormal subset $\{e_n\}$ of H.

An orthonormal basis need not be a Hamel basis: an element of H need not be a finite linear combination of basis vectors.

Fourier–Bessel series

Let $\{e_n\}_{n\in\mathbb{N}}$, or $\{e_1,\dots,e_N\}$, be an orthonormal basis of a Hilbert space H. Then, for every $u \in H$,

$$\sum_{n=1}^{\infty} \langle u, e_n\rangle e_n \longrightarrow u.$$

Thus

$$u = \sum_{n=1}^{\infty} \langle u, e_n\rangle e_n.$$

Completeness is essential

The proof uses completeness, so the result requires a Hilbert space, not merely a pre-Hilbert space.

Proof

Bessel's inequality on page 130 shows that

$$\sum_{n=1}^{\infty} |\langle u, e_n\rangle|^2 \le \|u\|^2 < \infty.$$

Therefore, given $\varepsilon > 0$, there is an M such that

$$\sum_{n=M+1}^{\infty} |\langle u, e_n\rangle|^2 < \varepsilon^2.$$

For $m > \ell \ge M$, orthonormality gives

$$\left\| \sum_{n=1}^{m} \langle u, e_n\rangle e_n - \sum_{n=1}^{\ell} \langle u, e_n\rangle e_n \right\|^2 = \sum_{n=\ell+1}^{m} |\langle u, e_n\rangle|^2 < \varepsilon^2.$$

Thus the partial sums are Cauchy.
Completeness supplies a $\widetilde{u} \in H$ such that

$$\widetilde{u} = \sum_{n=1}^{\infty} \langle u, e_n\rangle e_n.$$

For every ℓ, continuity of the inner product gives

$$\begin{aligned} \langle u - \widetilde{u}, e_\ell\rangle &= \lim_{m\to\infty} \left\langle u - \sum_{n=1}^{m} \langle u, e_n\rangle e_n, e_\ell \right\rangle \\ &= \langle u, e_\ell\rangle - \langle u, e_\ell\rangle = 0. \end{aligned}$$

Maximality implies $u - \widetilde{u} = 0$.

An orthonormal basis implies separability

If a Hilbert space H has an orthonormal basis, then H is separable.

Proof

For a countable basis $\{e_n\}$, let

$$S = \bigcup_{N=1}^{\infty} \left\{ \sum_{n=1}^{N} q_n e_n : q_1, \ldots, q_N \in \mathbb{Q} + i\mathbb{Q} \right\}.$$

This is countable. Truncating the Fourier–Bessel series and approximating its finitely many complex coefficients by elements of $\mathbb{Q} + i\mathbb{Q}$ shows that S is dense in H.

Separability and orthonormal bases

For a Hilbert space H,

$$H \text{ is separable} \iff H \text{ has an orthonormal basis.}$$

Proof

The forward direction is the Gram–Schmidt construction on page 131, and the converse is the preceding theorem.

Parseval's identity

If H is a Hilbert space and $\{e_n\}$ is an at most countable orthonormal basis, then for every $u \in H$,

$$\sum_{n} |\langle u, e_n \rangle|^2 = \|u\|^2.$$

Proof

The finite case is immediate. In the countably infinite case, Fourier–Bessel convergence and continuity of the inner product give

$$\begin{aligned} \|u\|^2 &= \lim_{m\to\infty} \left\langle \sum_{n=1}^{m} \langle u, e_n \rangle e_n, \sum_{\ell=1}^{m} \langle u, e_\ell \rangle e_\ell \right\rangle \\ &= \lim_{m\to\infty} \sum_{n=1}^{m} |\langle u, e_n \rangle|^2 = \sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2. \end{aligned}$$

Bessel versus Parseval

Bessel's inequality holds for every at most countable orthonormal set. For an orthonormal basis, maximality strengthens the inequality to Parseval's equality.

Classification of separable Hilbert spaces

Every infinite-dimensional separable Hilbert space H is isometrically isomorphic to ℓ^2. More precisely, there is a bijective linear operator $T : H \to \ell^2$ such that

$$\|Tu\|_{\ell^2} = \|u\|_H \quad \text{and} \quad \langle Tu, Tv\rangle_{\ell^2} = \langle u, v\rangle_H.$$

Proof

Choose an orthonormal basis $\{e_n\}_{n=1}^{\infty}$ of H and define

$$Tu = (\langle u, e_n\rangle)_{n=1}^{\infty}.$$

Parseval's identity shows that $Tu \in \ell^2$ and

$$\|Tu\|_{\ell^2} = \|u\|_H.$$

Linearity follows from linearity of the inner product. The polarization identity then gives preservation of inner products. If $Tu_1 = Tu_2$, then

$$\langle u_1 - u_2, e_n\rangle = 0 \qquad (n \in \mathbb{N}),$$

and maximality gives $u_1 = u_2$. Thus T is injective.
If $c = (c_n) \in \ell^2$, the partial sums of $\sum_n c_n e_n$ are Cauchy because

$$\left\| \sum_{n=\ell+1}^{m} c_n e_n \right\|^2 = \sum_{n=\ell+1}^{m} |c_n|^2.$$

Completeness gives a $u \in H$ with

$$u = \sum_{n=1}^{\infty} c_n e_n.$$

Then $Tu = c$, so T is surjective.

If H is finite-dimensional, separable, and $\dim_{\mathbb{C}} H = n$, the same reasoning gives $H \cong \mathbb{C}^n$.

19.1 Fourier series

The exponential orthonormal system

The set

$$\left\{ \frac{e^{inx}}{\sqrt{2\pi}} \right\}_{n\in\mathbb{Z}}$$

is an orthonormal subset of $L^2([-\pi, \pi])$.

This was verified directly on page 129. Because the Lebesgue integral agrees with the Riemann integral on continuous functions, we henceforth write

$$\int_a^b f = \int_{[a,b]} f.$$

Fourier coefficients, partial sums, and series

For $f \in L^2([-\pi,\pi])$, its n-th Fourier coefficient is

$$\widehat{f}(n) = \left\langle f, \frac{e^{int}}{\sqrt{2\pi}} \right\rangle \frac{1}{\sqrt{2\pi}} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}\,dt, \qquad n \in \mathbb{Z}.$$

For $N \in \mathbb{N} \cup \{0\}$, the N-th partial Fourier sum is

$$S_N f(x) = \sum_{|n|\le N} \widehat{f}(n)e^{inx}.$$

The Fourier series of f is

$$\sum_{n\in\mathbb{Z}} \widehat{f}(n)e^{inx}.$$

The pointwise convergence question is delicate; Dirichlet's criterion answers it under additional hypotheses. The Hilbert-space question is: does

$$\|f - S_N f\|_2 = \left(\int_{-\pi}^{\pi} |f(x) - S_N f(x)|^2\,dx\right)^{1/2} \longrightarrow 0$$

for every $f \in L^2([-\pi,\pi])$?

Since $L^2([-\pi,\pi])$ is a Hilbert space, this is equivalent to asking whether

$$\left\{\frac{e^{inx}}{\sqrt{2\pi}}\right\}_{n\in\mathbb{Z}}$$

is maximal. Maximality implies that the orthonormal set spans the Hilbert space by the Fourier–Bessel theorem, and the converse is immediate.

The answer is yes. Convergence in the L^2-norm is often called *convergence in the mean*. We prove it by Fejér's method.

Dirichlet-kernel representation

For $f \in L^2([-\pi,\pi])$ and $N \in \mathbb{N} \cup \{0\}$,

$$S_N f(x) = \int_{-\pi}^{\pi} D_N(x-t)f(t)\,dt,$$

where the Dirichlet kernel is

$$D_N(x) = \frac{1}{2\pi}\sum_{|n|\le N} e^{inx} = \begin{cases} \dfrac{2N+1}{2\pi}, & x = 0, \\ \dfrac{\sin((N+\frac{1}{2})x)}{2\pi\sin(x/2)}, & x \neq 0. \end{cases}$$

Proof

The limit at zero shows that D_N is continuous; in fact it is smooth. Since the sum is finite,

$$S_N f(x) = \sum_{|n|\le N} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int}\, dt \right) e^{inx}$$
$$= \int_{-\pi}^{\pi} f(t) \left[\frac{1}{2\pi} \sum_{|n|\le N} e^{in(x-t)} \right] dt.$$

It remains to evaluate the finite geometric sum.
For $x \neq 0$,

$$\frac{1}{2\pi} \sum_{n=-N}^{N} e^{inx} = \frac{1}{2\pi} e^{-iNx} \sum_{n=0}^{2N} (e^{ix})^n$$
$$= \frac{1}{2\pi} e^{-iNx} \frac{1 - e^{i(2N+1)x}}{1 - e^{ix}}$$
$$= \frac{1}{2\pi} \frac{e^{i(N+\frac{1}{2})x} - e^{-i(N+\frac{1}{2})x}}{e^{ix/2} - e^{-ix/2}}$$
$$= \frac{\sin((N+\frac{1}{2})x)}{2\pi \sin(x/2)}.$$

At $x = 0$, the sum equals $(2N+1)/(2\pi)$. Thus the bracket above is $D_N(x-t)$.

Cesàro–Fourier mean

For $f \in L^2([-\pi, \pi])$, the N-th Cesàro–Fourier mean is

$$\sigma_N f(x) = \frac{1}{N+1} \sum_{k=0}^{N} S_k f(x),$$

the mean of the first $N+1$ Fourier partial sums.

Proving $S_N f \to f$ directly is the original problem, not a simplification. Fejér's method instead proves that the means $\sigma_N f$ converge.

Why Cesàro means help

Cesàro means often behave better than the original sequence. For a sequence (a_k), define

$$\sigma_N a = \frac{1}{N} \sum_{k=1}^{N} a_k.$$

The sequence

$$1, -1, 1, -1, \ldots$$

does not converge, while its Cesàro means converge to 0. Similarly, we will prove

$$\sigma_N f = \frac{1}{N+1} \sum_{k=0}^{N} S_k f \longrightarrow f.$$

20 Fejér's Theorem and Convergence of Fourier Series

Recall that, for $f \in L^2([-\pi,\pi])$,

$$\widehat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}\,dt, \qquad S_N f(x) = \sum_{|n|\le N} \widehat{f}(n)e^{inx},$$

and

$$\sigma_N f(x) = \frac{1}{N+1}\sum_{k=0}^{N} S_k f(x).$$

Our goal is

$$\|\sigma_N f - f\|_2 \longrightarrow 0.$$

Fejér-kernel representation

For $f \in L^2([-\pi,\pi])$,

$$\sigma_N f(x) = \int_{-\pi}^{\pi} K_N(x-t)f(t)\,dt,$$

where the *Fejér kernel* is

$$K_N(x) = \frac{1}{N+1}\sum_{k=0}^{N} D_k(x) = \begin{cases} \dfrac{N+1}{2\pi}, & x = 0, \\ \dfrac{1}{2\pi(N+1)}\left[\dfrac{\sin((N+1)x/2)}{\sin(x/2)}\right]^2, & x \neq 0. \end{cases}$$

It satisfies:

1. $K_N(x) \ge 0$;
2. $K_N(x) = K_N(-x)$, and K_N is 2π-periodic;
3. $$\int_{-\pi}^{\pi} K_N(x)\,dx = 1;$$
4. for $\delta \in (0,\pi]$ and $\delta \le |x| \le \pi$,
$$|K_N(x)| = K_N(x) \le \frac{1}{2\pi(N+1)\sin^2(\delta/2)}.$$

The kernels become concentrated at 0 as $N \to \infty$. Thus $K_N(x-t)$ is concentrated near $x = t$, where $f(t) \approx f(x)$, while its total integral is 1.

> ***Proof of the formula***
>
> By the Dirichlet-kernel representation on page 140,
>
> $$\sigma_N f(x) = \int_{-\pi}^{\pi}\left[\frac{1}{N+1}\sum_{k=0}^{N} D_k(x-t)\right] f(t)\,dt.$$

It remains to show that

$$Q_N(x) := \frac{1}{N+1}\sum_{k=0}^{N} D_k(x) = K_N(x).$$

The case $x = 0$ is immediate. For $x \neq 0$,

$$\begin{aligned} Q_N(x) &= \frac{1}{2\pi(N+1)\sin(x/2)} \sum_{k=0}^{N} \sin\big((k+\tfrac{1}{2})x\big) \\ &= \frac{1}{4\pi(N+1)\sin^2(x/2)} \sum_{k=0}^{N} \big(\cos(kx) - \cos((k+1)x)\big) \\ &= \frac{1-\cos((N+1)x)}{4\pi(N+1)\sin^2(x/2)} \\ &= \frac{1}{2\pi(N+1)} \left[\frac{\sin((N+1)x/2)}{\sin(x/2)}\right]^2. \end{aligned}$$

Proof of the four properties

Nonnegativity follows from the squared formula. Evenness and 2π-periodicity follow by direct substitution.
For the Dirichlet kernel,

$$\int_{-\pi}^{\pi} D_k(x)\,dx = \frac{1}{2\pi}\sum_{|n|\le k} \int_{-\pi}^{\pi} e^{inx}\,dx = 1,$$

because only the $n = 0$ term survives. Therefore

$$\int_{-\pi}^{\pi} K_N(x)\,dx = \frac{1}{N+1}\sum_{k=0}^{N} \int_{-\pi}^{\pi} D_k(x)\,dx = 1.$$

Finally, $\sin^2(x/2)$ is even and increasing on $[0,\pi]$. Thus, if $\delta \le |x| \le \pi$,

$$\sin^2(x/2) \ge \sin^2(\delta/2),$$

while the numerator of K_N is at most 1. This gives the fourth property.

Fejér's Theorem

If $f \in C([-\pi,\pi])$ is 2π-periodic, meaning $f(-\pi) = f(\pi)$, then

$$\sigma_N f \longrightarrow f \quad \text{uniformly on } [-\pi,\pi].$$

Proof

Extend f to a continuous 2π-periodic function on $\mathbb{R}$. It is bounded and uniformly continuous. Let $\varepsilon > 0$. Choose $\delta \in (0,\pi)$ so that

$$|y-z| < \delta \quad \Longrightarrow \quad |f(y) - f(z)| < \frac{\varepsilon}{2}.$$

For this fixed δ, choose M so large that, for $N \geq M$,

$$\frac{2\|f\|_\infty}{(N+1)\sin^2(\delta/2)} < \frac{\varepsilon}{2}. \tag{$*$}$$

Periodicity and a change of variables give

$$\sigma_N f(x) = \int_{-\pi}^{\pi} K_N(t) f(x-t)\, dt.$$

Using $\int K_N = 1$,

$$|\sigma_N f(x) - f(x)| \leq \int_{-\pi}^{\pi} K_N(t)|f(x-t) - f(x)|\, dt.$$

Split this integral into $|t| < \delta$ and $\delta \leq |t| \leq \pi$. Since $K_N \geq 0$,

$$\begin{aligned} |\sigma_N f(x) - f(x)| &\leq \frac{\varepsilon}{2}\int_{|t|<\delta} K_N(t)\, dt \\ &\quad + 2\|f\|_\infty \int_{\delta\leq|t|\leq\pi} K_N(t)\, dt \\ &\leq \frac{\varepsilon}{2} + \frac{2\|f\|_\infty}{2\pi(N+1)\sin^2(\delta/2)}\int_{-\pi}^{\pi} dt \\ &< \varepsilon \end{aligned}$$

by $(*)$. The bound is independent of x, so convergence is uniform.

Why the Dirichlet kernel cannot replace K_N

The proof would still work for kernels whose L^1-norms were uniformly bounded. The Dirichlet kernels have the analogous periodicity, mass, and concentration properties, but they change sign and

$$\int_{-\pi}^{\pi} |D_N(x)|\, dx \asymp \log N.$$

Hence their L^1-norms are unbounded. The nonnegativity of the Fejér kernel is crucial.

Conjugation and integration

If g is complex-valued and Lebesgue integrable on $[a, b]$, then

$$\overline{\int_a^b g} = \int_a^b \overline{g}.$$

Proof

Writing $g = \operatorname{Re} g + i \operatorname{Im} g$,

$$\overline{\int g} = \int \operatorname{Re} g - i \int \operatorname{Im} g = \int (\operatorname{Re} g - i \operatorname{Im} g) = \int \overline{g}.$$

Fubini's Theorem for Lebesgue integrals

If $f : [a,b] \times [c,d] \to \mathbb{C}$ is Lebesgue integrable, then

$$\iint_{[a,b]\times[c,d]} f = \int_a^b \left(\int_c^d f(x,y)\, dy \right) dx = \int_c^d \left(\int_a^b f(x,y)\, dx \right) dy.$$

Fubini's Theorem for Riemann integrals

If $f : [a,b] \times [c,d] \to \mathbb{R}$ is continuous, then

$$\iint_{[a,b]\times[c,d]} f(x,y)\, d(x,y) = \int_a^b \int_c^d f(x,y)\, dy\, dx = \int_c^d \int_a^b f(x,y)\, dx\, dy.$$

The same conclusion holds for complex-valued continuous f, by applying the real theorem to the real and imaginary parts.

The Fejér means are L^2-contractions

For every $f \in L^2([-\pi,\pi])$,

$$\|\sigma_N f\|_2 \le \|f\|_2.$$

Proof for continuous periodic functions

Suppose first that f is continuous and 2π-periodic. From

$$\sigma_N f(x) = \int_{-\pi}^{\pi} f(x-t) K_N(t)\, dt$$

and Fubini's Theorem,

$$\begin{aligned} \|\sigma_N f\|_2^2 &= \int_{-\pi}^{\pi} \left| \int_{-\pi}^{\pi} f(x-t) K_N(t)\, dt \right|^2 dx \\ &= \iint K_N(s) K_N(t) \left[\int_{-\pi}^{\pi} f(x-s)\overline{f(x-t)}\, dx \right] ds\, dt \\ &\le \iint K_N(s) K_N(t) \|f(\cdot - s)\|_2 \|f(\cdot - t)\|_2\, ds\, dt. \end{aligned}$$

Periodicity gives

$$\|f(\cdot - s)\|_2 = \|f\|_2$$

for every s. Since $K_N \ge 0$ and $\int K_N = 1$,

$$\|\sigma_N f\|_2^2 \le \|f\|_2^2 \left(\int K_N \right)^2 = \|f\|_2^2.$$

Proof for general L^2 functions

Continuous functions are dense in $L^2([-\pi,\pi])$. Choose $f_j \in C([-\pi,\pi])$ with $\|f_j - f\|_2 \to 0$, using periodic approximants. For fixed N, the kernel K_N is bounded, and

$$|\sigma_N f_j(x) - \sigma_N f(x)| \le \|K_N\|_\infty \|f_j - f\|_1.$$

On a finite-measure set, L^2-convergence implies L^1-convergence:

$$\|f_j - f\|_1 \le \|f_j - f\|_2 \sqrt{m([-\pi,\pi])} \longrightarrow 0.$$

Hence $\sigma_N f_j \to \sigma_N f$ pointwise, and the Dominated Convergence Theorem gives convergence in L^2. Therefore

$$\|\sigma_N f\|_2 = \lim_{j\to\infty} \|\sigma_N f_j\|_2 \le \lim_{j\to\infty} \|f_j\|_2 = \|f\|_2.$$

Finite-measure embedding

If $E \subset \mathbb{R}$ is measurable with $m(E) < \infty$ and $f_n \to f$ in $L^2(E)$, then $f_n \to f$ in $L^1(E)$, because

$$\|f_n - f\|_1 \le \|f_n - f\|_2 \, m(E)^{1/2}.$$

Convergence of Fejér means in L^2

For every $f \in L^2([-\pi,\pi])$,

$$\|\sigma_N f - f\|_2 \longrightarrow 0.$$

Consequently,

$$\widehat{f}(n) = 0 \text{ for every } n \in \mathbb{Z} \quad \Longrightarrow \quad f = 0.$$

Proof

Let $\varepsilon > 0$. Choose a continuous 2π-periodic g such that

$$\|f - g\|_2 < \frac{\varepsilon}{3}.$$

By Fejér's Theorem, choose M such that for $N \ge M$,

$$|\sigma_N g(x) - g(x)| < \frac{\varepsilon}{3\sqrt{2\pi}} \qquad (x \in [-\pi,\pi]).$$

Since σ_N is linear and contractive,

$$\begin{aligned} \|\sigma_N f - f\|_2 &\le \|\sigma_N(f-g)\|_2 + \|\sigma_N g - g\|_2 + \|g - f\|_2 \\ &\le \|f - g\|_2 + \|\sigma_N g - g\|_2 + \|g - f\|_2 \\ &< \varepsilon. \end{aligned}$$

Thus $\sigma_N f \to f$.
If all Fourier coefficients vanish, then every $S_k f = 0$, so every $\sigma_N f = 0$. The convergence just proved forces $f = 0$.

It follows that

$$\left\{ \frac{e^{inx}}{\sqrt{2\pi}} \right\}_{n\in\mathbb{Z}}$$

is a maximal orthonormal set, hence an orthonormal basis of $L^2([-\pi,\pi])$.

Further Fourier convergence results

- The corresponding convergence is true in $L^p([-\pi,\pi])$ for $1 < p < \infty$.
- Carleson's theorem gives pointwise almost-everywhere convergence of Fourier series.
- Dirichlet's criterion supplies hypotheses for pointwise convergence everywhere.

These require stronger tools from harmonic analysis.

Convolution

For Lebesgue integrable functions f, g, their convolution is

$$(f * g)(x) = \int_{\mathbb{R}} f(x-t)g(t)\,dt = \int_{\mathbb{R}} f(t)g(x-t)\,dt.$$

For functions originally defined on $[-\pi,\pi]$, one may extend them by zero outside that interval before convolving. Convolution is a central tool in functional analysis.

21 Minimizers, Orthogonal Complements, and Riesz Representation

Motivation from quotient norms

For a normed vector space V and a subspace W, the quotient norm was defined by

$$\|v + W\|_{V/W} = \inf_{w \in W} \|v + w\|_V.$$

It is natural to ask when such an infimum is attained.

Length minimizer in a closed convex set

Let C be a nonempty, closed, convex subset of a Hilbert space H. Then there is a unique $v \in C$ such that

$$\|v\| = \inf_{u \in C} \|u\|.$$

Convexity means that, whenever $v_1, v_2 \in C$ and $t \in [0,1]$,

$$tv_1 + (1-t)v_2 \in C.$$

Both closedness and convexity are essential: a nonclosed convex set may have no minimizer, and a closed nonconvex set may have several.

> ***Proof: existence***
>
> Let
>
> $$d = \inf_{u \in C} \|u\|.$$
>
> Choose $u_n \in C$ such that $\|u_n\| \to d$. We claim that (u_n) is Cauchy. Given $\varepsilon > 0$, choose N such that for $n \geq N$,
>
> $$2\|u_n\|^2 < 2d^2 + \frac{\varepsilon^2}{2}.$$

For $n, m \geq N$, the parallelogram law gives

$$\|u_n - u_m\|^2 = 2\|u_n\|^2 + 2\|u_m\|^2 - 4\left\|\frac{u_n + u_m}{2}\right\|^2.$$

The midpoint belongs to C, so its norm is at least d. Therefore

$$\|u_n - u_m\|^2 < 2d^2 + \frac{\varepsilon^2}{2} + 2d^2 + \frac{\varepsilon^2}{2} - 4d^2 = \varepsilon^2.$$

Completeness gives $u_n \to u \in H$, and closedness of C gives $u \in C$. By continuity of the norm, $\|u\| = d$.

Proof: uniqueness

If $\widetilde{u} \in C$ is another minimizer, then the midpoint lies in C. The parallelogram law gives

$$\|u - \widetilde{u}\|^2 = 2\|u\|^2 + 2\|\widetilde{u}\|^2 - 4\left\|\frac{u + \widetilde{u}}{2}\right\|^2 \leq 2d^2 + 2d^2 - 4d^2 = 0.$$

Hence $u = \widetilde{u}$.

21.1 Orthogonal complements

Orthogonal decomposition

Let H be a Hilbert space and $W \subset H$ a subspace.

1. The orthogonal complement
$$W^\perp = \{u \in H : \langle u, w\rangle = 0 \text{ for every } w \in W\}$$
is a closed subspace of H.

2. If W is closed, then
$$H = W \oplus W^\perp.$$
Equivalently, every $u \in H$ has a unique representation
$$u = w + w^\perp, \qquad w \in W, \quad w^\perp \in W^\perp.$$

Proof of part 1

It is immediate that $W^\perp$ is a subspace and $W \cap W^\perp = \{0\}$. To show closedness, let $u_n \in W^\perp$ and $u_n \to u$. For every $w \in W$, continuity of the inner product gives

$$\langle u, w\rangle = \lim_{n\to\infty} \langle u_n, w\rangle = 0.$$

Thus $u \in W^\perp$.

Proof of part 2

If $W = H$, the assertion is immediate. Otherwise fix $u \in H \setminus W$ and consider the affine set

$$C = u + W = \{u + w : w \in W\}.$$

It is nonempty and convex. It is closed because, if $u + w_n \to v$, then $w_n \to v - u$, and the closedness of W implies $v - u \in W$.
The minimizer theorem supplies a $v \in C$ such that

$$\|v\| = \inf_{\widetilde{v} \in C} \|\widetilde{v}\| = \inf_{w \in W} \|u + w\|.$$

Since $v \in u + W$, we have $u - v \in W$. We claim that $v \in W^{\perp}$. Fix $w \in W$. For real t,

$$v + tw \in C$$

and

$$\|v + tw\|^2 = \|v\|^2 + t^2\|w\|^2 + 2t\,\mathrm{Re}\langle v, w\rangle.$$

This polynomial has a minimum at $t = 0$, so

$$\mathrm{Re}\langle v, w\rangle = 0.$$

Applying the same argument to $v + itw$ gives

$$\mathrm{Im}\langle v, w\rangle = 0.$$

Hence $v \in W^{\perp}$, and

$$u = (u - v) + v \in W + W^{\perp}.$$

If

$$u = w_1 + w_1^{\perp} = w_2 + w_2^{\perp},$$

then

$$w_2 - w_1 = w_1^{\perp} - w_2^{\perp} \in W \cap W^{\perp} = \{0\}.$$

Thus the decomposition is unique.

Double orthogonal complement

If $W \subset H$ is a subspace, then

$$\overline{W} = (W^{\perp})^{\perp}.$$

Proof

If $w \in \overline{W}$, choose $w_n \in W$ with $w_n \to w$. For every $v \in W^{\perp}$,

$$\langle w, v\rangle = \lim_{n \to \infty} \langle w_n, v\rangle = 0,$$

so $\overline{W} \subset (W^{\perp})^{\perp}$.
The closure $\overline{W}$ is itself a closed subspace. Also

$$\overline{W}^{\perp} = W^{\perp} :$$

one inclusion is immediate from $W \subset \overline{W}$, and the other follows by approximating any $w \in \overline{W}$ with vectors in W and using continuity of the inner product. Apply the orthogonal decomposition to $\overline{W}$:

$$H = \overline{W} \oplus W^{\perp}.$$

If $v \in (W^{\perp})^{\perp}$, write

$$v = w + w^{\perp}, \qquad w \in \overline{W}, \quad w^{\perp} \in W^{\perp}.$$

Since $v \perp W^{\perp}$,

$$0 = \langle v, w^{\perp} \rangle = \langle w, w^{\perp} \rangle + \|w^{\perp}\|^2 = \|w^{\perp}\|^2.$$

Thus $w^{\perp} = 0$ and $v = w \in \overline{W}$.

Projection

A bounded linear operator $P : H \to H$ is a *projection* if

$$P^2 = P.$$

Orthogonal projection onto a closed subspace

Let W be a closed subspace of a Hilbert space H. Define $\Pi_W : H \to H$ by

$$\Pi_W(w + w^{\perp}) = w, \qquad w \in W, \quad w^{\perp} \in W^{\perp}.$$

Then Π_W is a bounded linear projection.

Proof

The direct-sum decomposition makes Π_W well defined. If

$$v_j = w_j + w_j^{\perp} \qquad (j = 1, 2),$$

then

$$\lambda_1 v_1 + \lambda_2 v_2 = (\lambda_1 w_1 + \lambda_2 w_2) + (\lambda_1 w_1^{\perp} + \lambda_2 w_2^{\perp}),$$

so Π_W is linear.
For $v = w + w^{\perp}$,

$$\|v\|^2 = \|w\|^2 + \|w^{\perp}\|^2 \geq \|w\|^2.$$

Hence

$$\|\Pi_W v\| = \|w\| \leq \|v\|,$$

so Π_W is bounded. Finally,

$$\Pi_W(\Pi_W v) = \Pi_W w = w = \Pi_W v,$$

and therefore $\Pi_W^2 = \Pi_W$.

21.2 The Riesz Representation Theorem

Riesz Representation Theorem

If H is a Hilbert space, then for every

$$f \in H' = \mathcal{B}(H, \mathbb{C})$$

there is a unique $v \in H$ such that

$$f(u) = \langle u, v \rangle \qquad (u \in H).$$

Relation to ℓ^2

This pairing appeared when identifying $(\ell^2)'$ with ℓ^2. Here it is recognized as the Hilbert-space inner product; among the ℓ^p spaces, only ℓ^2 is Hilbert.

Proof: uniqueness and the zero functional

If both $v, \widetilde{v}$ represent f, then

$$\langle u, v - \widetilde{v} \rangle = 0 \qquad (u \in H).$$

Taking $u = v - \widetilde{v}$ gives $v = \widetilde{v}$. If $f = 0$, take $v = 0$.

Proof: existence

Suppose $f \neq 0$. Choose $u_1 \in H$ with $f(u_1) \neq 0$ and set

$$u_0 = \frac{u_1}{f(u_1)}, \qquad f(u_0) = 1.$$

Let

$$C = \{u \in H : f(u) = 1\} = f^{-1}(\{1\}).$$

Since f is bounded and linear, it is continuous; hence C is closed. It is nonempty, and it is convex because for $u_1, u_2 \in C$ and $t \in [0, 1]$,

$$f(tu_1 + (1-t)u_2) = t + (1-t) = 1.$$

The minimizer theorem gives $v_0 \in C$ with

$$\|v_0\| = \inf_{u \in C} \|u\|.$$

The vector v_0 is nonzero because $f(v_0) = 1$. Let

$$N = \ker f = \{w \in H : f(w) = 0\}.$$

Then

$$C = v_0 + N.$$

The same minimum argument used in the orthogonal-decomposition theorem shows that

$$v_0 \in N^{\perp}.$$

Define

$$v = \frac{v_0}{\|v_0\|^2}.$$

For any $u \in H$,

$$f(u - f(u)v_0) = f(u) - f(u)f(v_0) = 0,$$

so $u - f(u)v_0 \in N$. Since $v_0 \perp N$,

$$\begin{aligned}\langle u, v\rangle &= \frac{1}{\|v_0\|^2}\langle u, v_0\rangle \\ &= \frac{1}{\|v_0\|^2}\langle u - f(u)v_0 + f(u)v_0, v_0\rangle \\ &= 0 + f(u) = f(u).\end{aligned}$$

22 Adjoints of Bounded Operators on Hilbert Spaces

Existence of the adjoint

Let H be a Hilbert space and $A : H \to H$ a bounded linear operator. There is a unique bounded linear operator $A^* : H \to H$ such that

$$\langle Au, v\rangle = \langle u, A^*v\rangle \qquad (u, v \in H). \tag{*}$$

The operator A^* is the *Hermitian adjoint* of A, and

$$\|A^*\| = \|A\|.$$

Proof: uniqueness and construction

If both A_1^* and A_2^* satisfy $(*)$, then

$$\langle u, (A_1^* - A_2^*)v\rangle = 0 \qquad (u, v \in H),$$

so $A_1^* = A_2^*$.
For fixed $v \in H$, define

$$f_v(u) = \langle Au, v\rangle.$$

This is linear, and

$$|f_v(u)| \le \|Au\|\,\|v\| \le \|A\|\,\|u\|\,\|v\|.$$

Thus $f_v \in H'$. By the Riesz Representation Theorem, there is a unique vector, denoted A^*v, such that

$$f_v(u) = \langle u, A^*v\rangle.$$

This defines $A^* : H \to H$.

Proof: linearity, boundedness, and equality of norms

For $v_1, v_2 \in H$ and $\lambda_1, \lambda_2 \in \mathbb{C}$,

$$\begin{aligned}\langle u, A^*(\lambda_1 v_1 + \lambda_2 v_2)\rangle &= \langle Au, \lambda_1 v_1 + \lambda_2 v_2\rangle \\ &= \overline{\lambda_1}\langle Au, v_1\rangle + \overline{\lambda_2}\langle Au, v_2\rangle \\ &= \langle u, \lambda_1 A^* v_1 + \lambda_2 A^* v_2\rangle.\end{aligned}$$

Hence A^* is linear.
If $\|v\| = 1$ and $A^*v \neq 0$, then

$$\begin{aligned}\|A^*v\|^2 = \langle A^*v, A^*v\rangle &= \langle AA^*v, v\rangle \\ &\leq \|AA^*v\|\,\|v\| \leq \|A\|\,\|A^*v\|.\end{aligned}$$

Therefore $\|A^*v\| \leq \|A\|$, and $\|A^*\| \leq \|A\|$.
Moreover,

$$\langle A^*u, v\rangle = \langle u, Av\rangle,$$

so $(A^*)^* = A$. Applying the preceding inequality to A^* gives

$$\|A\| = \|(A^*)^*\| \leq \|A^*\|.$$

Thus $\|A^*\| = \|A\|$.

Finite-dimensional matrices

Let $H = \mathbb{C}^n$ with its usual inner product, and let $A = (a_{ij}) \in M_{n\times n}(\mathbb{C})$. Then

$$(Au)_i = \sum_{j=1}^{n} a_{ij}u_j.$$

The adjoint is the conjugate transpose:

$$A^* = (\overline{a_{ji}}), \qquad (A^*v)_j = \sum_{i=1}^{n} \overline{a_{ij}}v_i.$$

Indeed,

$$\langle Au, v\rangle = \sum_{i,j} a_{ij}u_j\overline{v_i} = \sum_j u_j \overline{\sum_i \overline{a_{ij}}v_i} = \langle u, A^*v\rangle.$$

A matrix operator on ℓ^2

Let $(A_{ij})_{i,j\geq 1} \subset \mathbb{C}$ satisfy

$$\sum_{i,j=1}^{\infty} |A_{ij}|^2 < \infty.$$

Define

$$(Aa)_i = \sum_{j=1}^{\infty} A_{ij}a_j.$$

Writing $A_i = (A_{i1}, A_{i2}, \dots)$, Cauchy–Schwarz gives

$$\begin{aligned}\|Aa\|_{\ell^2}^2 &= \sum_{i=1}^{\infty} |\langle A_i, \overline{a}\rangle|^2 \\ &\le \|a\|_{\ell^2}^2 \sum_{i,j=1}^{\infty} |A_{ij}|^2.\end{aligned}$$

Hence A is bounded. For $a, b \in \ell^2$,

$$\langle Aa, b\rangle = \sum_j a_j \overline{\sum_i \overline{A_{ij}} b_i}.$$

Therefore

$$(A^* b)_j = \sum_{i=1}^{\infty} \overline{A_{ij}} b_i.$$

Integral operators

Let $K \in C([0,1] \times [0,1])$ and define

$$Af(x) = \int_0^1 K(x,y) f(y)\, dy$$

on $L^2([0,1])$. Then

$$A^* g(x) = \int_0^1 \overline{K(y,x)} g(y)\, dy.$$

The order of the kernel variables matters: in general $K(x,y) \neq K(y,x)$.

Range and null space of an adjoint

For a bounded operator $A : H \to H$,

$$\operatorname{Ran}(A)^{\perp} = \operatorname{Ker}(A^*).$$

Here $\operatorname{Ran}(A) = \operatorname{Im} A$ and $\operatorname{Ker}(A)$ is the null space of A.

Proof

For $v \in H$,

$$\begin{aligned} v \in \operatorname{Ker}(A^*) &\iff A^* v = 0 \\ &\iff \langle u, A^* v\rangle = 0 \quad \text{for every } u \in H \\ &\iff \langle Au, v\rangle = 0 \quad \text{for every } u \in H \\ &\iff v \in \operatorname{Ran}(A)^{\perp}. \end{aligned}$$

23 Sequential Compactness in Hilbert Spaces

Sequential compactness

Let X be a metric space. A subset $K \subset X$ is compact if every sequence in K has a subsequence converging to an element of K.

Every finite set is compact. In metric spaces, this sequential definition is equivalent to the open-cover definition.

Heine–Borel theorem

A subset $K \subset \mathbb{C}^n$ is compact if and only if it is closed and bounded.

Examples include closed intervals, finite sets together with their limit points, and closed balls in $\mathbb{C}^n$ or $\mathbb{R}^n$.

Failure of Heine–Borel in infinite dimensions

Let H be an infinite-dimensional Hilbert space. The closed unit ball

$$F = \{u \in H : \|u\| \leq 1\}$$

is closed and bounded but not compact.
Indeed, choose a countably infinite orthonormal set $\{e_n\}$. Then $e_n \in F$, but for $n \neq k$,

$$\|e_n - e_k\|^2 = \|e_n\|^2 + \|e_k\|^2 - 2\,\mathrm{Re}\langle e_n, e_k\rangle = 2.$$

No subsequence is Cauchy, hence none converges.

Equi-small tails

Let H be a Hilbert space and $\{e_k\}_{k=1}^{\infty}$ a countable orthonormal set. A subset $K \subset H$ has *equi-small tails* with respect to $\{e_k\}$ if, for every $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that

$$\sum_{k>N} |\langle v, e_k\rangle|^2 < \varepsilon^2 \qquad (v \in K).$$

Bessel's inequality gives small tails for each fixed v, with an index $N(v)$. Equi-small tails require one N that works uniformly for every $v \in K$, analogous to the distinction between pointwise and uniform convergence.

Finite sets have equi-small tails

Every finite $K \subset H$ has equi-small tails with respect to every countable orthonormal set. For each $v \in K$, choose N_v by Bessel's inequality and take

$$N = \max_{v \in K} N_v.$$

A convergent sequence and its limit

Let $v_n \to v$ in a Hilbert space H, and let $\{e_k\}_{k=1}^{\infty}$ be a countable orthonormal set. Then:

1. $$K = \{v_n : n \in \mathbb{N}\} \cup \{v\}$$

is compact;

2. K has equi-small tails with respect to $\{e_k\}$.

Proof of compactness

Take any sequence in K. If one of the finitely many early terms, or the limit v, repeats infinitely often, there is a constant convergent subsequence. Otherwise the indices tend to infinity along a subsequence, and that subsequence converges to v. Thus every sequence in K has a subsequence converging to an element of K.

Proof of equi-small tails

Let $\varepsilon > 0$. Choose M such that

$$n \geq M \Longrightarrow \|v_n - v\| < \frac{\varepsilon}{2}.$$

By Bessel's inequality, choose N_0 sufficiently large that

$$\sum_{k>N_0} |\langle v, e_k\rangle|^2 < \frac{\varepsilon^2}{4}$$

and, for each of the finitely many vectors $v_1, \ldots, v_{M-1}$,

$$\sum_{k>N_0} |\langle v_n, e_k\rangle|^2 < \varepsilon^2.$$

Increase N_0 if necessary so that all these finite requirements hold simultaneously.
For $n \geq M$, the triangle inequality in ℓ^2 and Bessel's inequality give

$$\begin{aligned}\left(\sum_{k>N_0} |\langle v_n, e_k\rangle|^2\right)^{1/2} &\leq \left(\sum_{k>N_0} |\langle v_n - v, e_k\rangle|^2\right)^{1/2} \\ &\quad + \left(\sum_{k>N_0} |\langle v, e_k\rangle|^2\right)^{1/2} \\ &\leq \|v_n - v\| + \frac{\varepsilon}{2} < \varepsilon.\end{aligned}$$

The same N_0 also handles v and the finitely many early terms, so K has equi-small tails.

24 Compact Subsets of a Hilbert Space and Finite-Rank Operators

24.1 Compactness

Recall that $K \subset X$ is compact if every sequence in K has a subsequence converging in K. The theorem on page 155 shows that a convergent sequence in a Hilbert space, together with its limit, is compact and has equi-small tails with respect to every countable orthonormal subset. We now characterize all compact subsets of a separable Hilbert space, beginning with a general fact about metric spaces.

Bounded subset of a metric space

Let (M, d) be a metric space. A subset $B \subset M$ is *bounded* if, for every $x \in B$, there is an $r > 0$ such that

$$B \subset B(x, r).$$

Thus B can be contained in an open ball of radius r about any of its points.

Why no origin appears

In $\mathbb{R}^n$ and $\mathbb{C}^n$, boundedness is normally written using the origin: $|x| = |x - 0| < r$ or $\|x\| = \|x - 0\| < r$. A general metric space need not have a zero element, so we use an arbitrary $x \in B$.

Compact sets are closed and bounded

If M is a metric space and $K \subset M$ is compact, then K is closed and bounded.

Proof

To prove closedness, suppose $x_n \in K$ and $x_n \to x \in M$. Compactness gives a subsequence converging to a point of K, while every subsequence of the convergent sequence x_n also converges to x. Hence $x \in K$.
For boundedness, argue by contradiction. If K is unbounded, choose $x_0 \in K$ and, for each $n \in \mathbb{N}$, choose $x_n \in K$ such that

$$d(x_0, x_n) > n.$$

We claim that x_n has no convergent subsequence.
Indeed, if $x_{n_k} \to x \in K$, then for every $\varepsilon > 0$ there is N such that $d(x, x_{n_k}) < \varepsilon$ for $k \geq N$. Therefore

$$d(x_0, x_{n_k}) < \varepsilon + d(x, x_0).$$

Choose $N_0 \in \mathbb{N}$ with $N_0 \geq \varepsilon + d(x, x_0)$. For $k \geq \max\{N, N_0\}$, the way x_n was chosen gives

$$d(x_0, x_{n_k}) > n_k \geq k \geq N_0 \geq \varepsilon + d(x, x_0),$$

a contradiction. Thus x_n has no convergent subsequence, contradicting compactness. Hence K is bounded.

Compactness by equi-small tails

Let H be a separable Hilbert space and let $\{e_k\}_{k=1}^{\infty}$ be an orthonormal basis of H. A subset $K \subset H$ is compact if and only if it is closed, bounded, and has equi-small tails with respect to $\{e_k\}_{k=1}^{\infty}$.

Recall that separability means having a countable dense subset, and an orthonormal basis is a maximal countable orthonormal set; see page 136.

Proof: compactness implies the three properties

Suppose K is compact. By the previous theorem, it is closed and bounded. If its tails were not equi-small, then there would be an $\varepsilon_0 > 0$ such that, for every $N \in \mathbb{N}$, some $u_N \in K$ satisfied

$$\sum_{k=N}^{\infty} |\langle u_N, e_k \rangle|^2 \geq \varepsilon_0^2. \tag{*}$$

Compactness gives a subsequence v_n of u_N and a vector $v \in K$ such that $v_n \to v$.
For every n, inequality $(*)$ holds for v_n with the corresponding tail index. Hence $\{v_n : n \in \mathbb{N}\} \cup \{v\}$ would fail to have equi-small tails, contradicting the result proved on page 155. Therefore K has equi-small tails.

Proof: the three properties imply compactness

Suppose K is closed, bounded, and has equi-small tails. Let $\{u_n\}_{n=1}^{\infty} \subset K$. Because K is closed, it is enough to construct a convergent subsequence; its limit will then belong to K.
Every bounded sequence in $\mathbb{C}$ has a convergent subsequence. Indeed, this follows from the real Bolzano–Weierstrass theorem and the facts that

$$a_n + b_n i \to a + bi \quad \Longleftrightarrow \quad a_n \to a \text{ and } b_n \to b,$$

and that $a_n + b_n i$ is bounded if and only if both real sequences are bounded.
Since K is bounded, there is $C > 0$ with $\|u_n\| \leq C$ for all n. Consequently,

$$|\langle u_n, e_k \rangle| \leq \|u_n\| \, \|e_k\| \leq C \qquad (n, k \in \mathbb{N}). \tag{†}$$

Thus $\{\langle u_n, e_1 \rangle\}_{n=1}^{\infty}$ has a convergent subsequence. Write its index sequence as $\{n_1(j)\}_{j=1}^{\infty}$. The sequence $\{\langle u_{n_1(j)}, e_2 \rangle\}_{j=1}^{\infty}$ is still bounded by (†), so it has a convergent subsequence indexed by $\{n_2(j)\}_{j=1}^{\infty}$, where $n_2(j)$ is itself a subsequence of $n_1(j)$. Repeat the construction.
For every $\ell \in \mathbb{N}$, this produces nested sequences of natural numbers

$$\{n_\ell(j)\}_{j=1}^{\infty} \subset \{n_{\ell-1}(j)\}_{j=1}^{\infty} \subset \cdots \subset \{n_1(j)\}_{j=1}^{\infty}$$

such that, for $k = 1, \ldots, \ell$,

$$\lim_{j \to \infty} \langle u_{n_\ell(j)}, e_k \rangle \quad \text{exists.}$$

Define the diagonal subsequence

$$v_\ell = u_{n_\ell(\ell)}, \qquad \ell \in \mathbb{N}.$$

For every fixed k, the tail $\{\langle v_\ell, e_k \rangle\}_{\ell=k}^{\infty}$ is a subsequence of the convergent sequence $\{\langle u_{n_k(\ell)}, e_k \rangle\}_{\ell=1}^{\infty}$. Hence

$$\lim_{\ell \to \infty} \langle v_\ell, e_k \rangle \quad \text{exists for every } k \in \mathbb{N}.$$

It remains to show that v_ℓ is Cauchy.
Let $\varepsilon > 0$. Equi-small tails give $N \in \mathbb{N}$ such that

$$\sum_{k=N}^{\infty} |\langle v_\ell, e_k\rangle|^2 < \frac{\varepsilon^2}{16} \qquad (\ell \in \mathbb{N}).$$

Since each of the first N coordinate sequences converges, there is $M = \max\{M_1, \dots, M_N\}$ such that, for $\ell, m \geq M$,

$$\sum_{k=1}^{N} |\langle v_\ell, e_k\rangle - \langle v_m, e_k\rangle|^2 < \frac{\varepsilon^2}{4}.$$

By the Fourier–Bessel series, the orthonormal-basis property, and the triangle inequality in ℓ^2,

$$\begin{aligned}
\|v_\ell - v_m\| &= \left(\sum_{k=1}^{\infty} |\langle v_\ell - v_m, e_k\rangle|^2\right)^{1/2} \\
&\leq \left(\sum_{k=1}^{N} |\langle v_\ell - v_m, e_k\rangle|^2\right)^{1/2} + \left(\sum_{k=N}^{\infty} |\langle v_\ell - v_m, e_k\rangle|^2\right)^{1/2} \\
&< \frac{\varepsilon}{2} + \left(\sum_{k=N}^{\infty} |\langle v_\ell, e_k\rangle|^2\right)^{1/2} + \left(\sum_{k=N}^{\infty} |\langle v_m, e_k\rangle|^2\right)^{1/2} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \varepsilon.
\end{aligned}$$

Thus v_ℓ is Cauchy, hence convergent in H; closedness of K places the limit in K.

The Hilbert cube

The set

$$K = \{(a_k)_{k=1}^{\infty} \in \ell^2 : |a_k| \leq 2^{-k}\} \subset \ell^2,$$

known as the *Hilbert cube*, is compact.

Towards another test

The criterion just proved reduces compactness to closedness, boundedness, and equi-small tails with respect to an orthonormal basis. Verifying equi-small tails is still not always easy, so we now give another approach, which does not single out a basis.

Finite-dimensional approximability

Let $K \subset H$, where H is a Hilbert space. Then K is compact if and only if K is closed and bounded and, for every $\varepsilon > 0$, there is a finite-dimensional subspace $W \subset H$ such that

$$\inf_{w \in W} \|u - w\| < \varepsilon \qquad (u \in K).$$

The intuition is that compactness gives equi-small tails relative to a countable orthonormal subset. For some N, the tail of every Fourier–Bessel series is small, so every vector of K can almost be spanned by the first N vectors of that orthonormal family. The proof is given on page 167, after some preliminaries on finite-dimensional spaces.

24.2 Finite-Rank Operators

In what follows, H is a Hilbert space and $\mathcal{B}(H, H)$ is abbreviated as $\mathcal{B}(H)$.

Finite-rank operator

An operator $T \in \mathcal{B}(H)$ is a *finite-rank operator* if $\operatorname{Ran}(T)$, a subspace of H, is finite-dimensional. We write

$$T \in \mathcal{R}(H).$$

The zero operator is finite-rank. On ℓ^2, another example is

$$T(a_1, a_2, \dots) = \left(a_1, \frac{a_2}{2}, \dots, \frac{a_n}{n}, 0, 0, \dots\right).$$

Finite-rank operators form a subspace

The set $\mathcal{R}(H)$ is a subspace of $\mathcal{B}(H)$.

Proof sketch

Certainly $0 \in \mathcal{R}(H)$, and $cT \in \mathcal{R}(H)$ whenever $T \in \mathcal{R}(H)$. If $S, T \in \mathcal{R}(H)$, then

$$\begin{aligned} \dim \operatorname{Ran}(T+S) &= \dim(\operatorname{Ran} T + \operatorname{Ran} S) \\ &= \dim \operatorname{Ran} T + \dim \operatorname{Ran} S - \dim(\operatorname{Ran} S \cap \operatorname{Ran} T) \\ &\le \dim \operatorname{Ran} T + \dim \operatorname{Ran} S < \infty. \end{aligned}$$

Thus $T + S \in \mathcal{R}(H)$.

Matrix form of a finite-rank operator

An operator $T \in \mathcal{B}(H)$ belongs to $\mathcal{R}(H)$ if and only if there are an orthonormal set $\{e_k\}_{k=1}^{L}$ and constants $\{c_{ij}\}_{i,j=1}^{L} \subset \mathbb{C}$ such that

$$Tu = \sum_{i,j=1}^{L} c_{ij} \langle u, e_j \rangle e_i.$$

Proof

The displayed formula immediately implies $\operatorname{Ran} T \subset \operatorname{span}\{e_1, \dots, e_L\}$, so T has finite rank. Conversely, suppose $T \in \mathcal{R}(H)$. Choose an orthonormal basis $\{\widetilde{e}_k\}_{k=1}^{N}$ of $\operatorname{Ran} T$. Then

$$Tu = \sum_{k=1}^{N} \langle Tu, \widetilde{e}_k \rangle \widetilde{e}_k.$$

By the adjoint theorem, with $v_k = T^* \widetilde{e}_k$,

$$Tu = \sum_{k=1}^{N} \langle u, v_k \rangle \widetilde{e}_k. \tag{*}$$

Apply Gram–Schmidt to $\{\widetilde{e}_1, \dots, \widetilde{e}_N, v_1, \dots, v_N\}$. This yields an orthonormal set $\{e_1, \dots, e_L\}$

with the same span. Write

$$\widetilde{e}_k = \sum_{i=1}^{L} a_{ki} e_i, \qquad v_k = \sum_{j=1}^{L} b_{kj} e_j.$$

Substitution in $(*)$ gives

$$\begin{aligned} Tu &= \sum_{k=1}^{N} \left\langle u, \sum_{j=1}^{L} b_{kj} e_j \right\rangle \sum_{i=1}^{L} a_{ki} e_i \\ &= \sum_{i,j=1}^{L} \left(\sum_{k=1}^{N} a_{ki} \overline{b_{kj}} \right) \langle u, e_j \rangle e_i. \end{aligned}$$

Taking $c_{ij} = \sum_{k=1}^{N} a_{ki} \overline{b_{kj}}$ proves the result. Thus finite-rank operators are essentially matrices over $\mathbb{C}$.

Adjoints and the operator ideal property

1) If $T \in \mathcal{R}(H)$, then $T^* \in \mathcal{R}(H)$.
2) If $T \in \mathcal{R}(H)$ and $A, B \in \mathcal{B}(H)$, then $ATB \in \mathcal{R}(H)$.

Proof

By the preceding theorem, write

$$Tu = \sum_{i,j=1}^{L} c_{ij} \langle u, e_j \rangle e_i.$$

For $u, v \in H$,

$$\begin{aligned} \langle u, T^* v \rangle = \langle Tu, v \rangle &= \sum_{i,j=1}^{L} c_{ij} \langle u, e_j \rangle \langle e_i, v \rangle \\ &= \left\langle u, \sum_{i,j=1}^{L} \overline{c_{ij}} \langle v, e_i \rangle e_j \right\rangle. \end{aligned}$$

Therefore

$$T^* v = \sum_{i,j=1}^{L} \overline{c_{ij}} \langle v, e_i \rangle e_j,$$

and the converse direction of the preceding theorem gives $T^* \in \mathcal{R}(H)$.
For the second assertion, $\operatorname{Ran}(TB) \subset \operatorname{Ran}(T)$, so TB is finite-rank. Choose a basis $\{e_k\}_{k=1}^{N}$ of $\operatorname{Ran}(TB)$. Then

$$A[\operatorname{Ran}(TB)] = A(\operatorname{span}\{e_1, \ldots, e_N\}) = \operatorname{span}\{Ae_1, \ldots, Ae_N\}.$$

The vectors $Ae_1, \ldots, Ae_N$ are well-defined because $A \in \mathcal{B}(H)$, and therefore

$$\dim \operatorname{Ran}(ATB) \leq N < \infty.$$

Finite-rank operators are not closed in operator norm

On ℓ^2, define

$$T_n(a_1, a_2, \ldots) = \left(a_1, \frac{a_2}{2}, \ldots, \frac{a_n}{n}, 0, 0, \ldots\right).$$

In matrix form, T_n is diagonal with entries $1, \frac{1}{2}, \ldots, \frac{1}{n}, 0, 0, \ldots$. Then $T_n \in \mathcal{R}(H)$ and

$$\|T_n - T\| \to 0, \qquad T(a_1, a_2, \ldots) = \left(a_1, \frac{a_2}{2}, \frac{a_3}{3}, \ldots\right),$$

because $\|T - T_n\| \leq 1/\sqrt{n}$. However,

$$T(e_k) = \frac{1}{k} e_k \in \mathrm{Ran}(T) \qquad (k \in \mathbb{N}),$$

so the entire standard orthonormal basis lies in $\mathrm{Ran}(T)$ after scalar rescaling. Thus $\mathrm{Ran}(T)$ is infinite-dimensional and $T \notin \mathcal{R}(H)$.

What is the closure $\overline{\mathcal{R}(H)}$ in $\mathcal{B}(H)$? The answer is given by compact operators.

Compact operator

An operator $K \in \mathcal{B}(H)$ is a *compact operator* if

$$\overline{K(\{u \in H : \|u\| \leq 1\})}$$

is compact.

Compact operators and finite-rank approximation

If H is separable, then $K \in \mathcal{B}(H)$ is compact if and only if $K \in \overline{\mathcal{R}(H)}$.

This is proved on page 185. The proof depends on the finite-dimensional approximability theorem, which we establish after some preliminaries on finite-dimensional normed spaces.

24.3 Equivalent Norms and Finite-Dimensional Spaces

Equivalent norms

Let V be a vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and let $\|\cdot\|_a, \|\cdot\|_b$ be norms. They are *equivalent* if there are constants $m, M > 0$ such that

$$m\|v\|_a \leq \|v\|_b \leq M\|v\|_a \qquad (v \in V).$$

We write $\|\cdot\|_a \simeq \|\cdot\|_b$.

Denote the set of all norms on V by $\mathcal{N}(V)$.

Equivalence really is an equivalence relation

The relation $\simeq$ is an equivalence relation on $\mathcal{N}(V)$.

Proof

It is a binary relation. Reflexivity follows by taking $m = M = 1$.
For symmetry, if $m\|v\|_a \le \|v\|_b \le M\|v\|_a$, then

$$\frac{1}{M}\|v\|_b \le \|v\|_a \le \frac{1}{m}\|v\|_b.$$

For transitivity, suppose $\|\cdot\|_a \simeq \|\cdot\|_b$ and $\|\cdot\|_b \simeq \|\cdot\|_c$. Choose $m, M, n, N > 0$ such that

$$m\|v\|_a \le \|v\|_b \le M\|v\|_a, \qquad n\|v\|_b \le \|v\|_c \le N\|v\|_b.$$

Then

$$mn\|v\|_a \le n\|v\|_b \le \|v\|_c \le N\|v\|_b \le MN\|v\|_a.$$

What equivalent norms preserve

Let $\|\cdot\|_a$ and $\|\cdot\|_b$ be equivalent norms, with induced metrics d_a, d_b. Then:

1) a sequence converges in (V, d_a) if and only if it converges in (V, d_b);
2) (V, d_a) is complete if and only if (V, d_b) is complete;
3) $S \subset V$ is open in (V, d_a) if and only if it is open in (V, d_b);
4) $S \subset V$ is closed in (V, d_a) if and only if it is closed in (V, d_b).

Proof of the first assertion

It suffices to prove one direction, since norm equivalence is symmetric. Suppose $v_n \to v$ in (V, d_a). For every $\varepsilon > 0$, eventually $\|v_n - v\|_a < \varepsilon/M$. Hence

$$\|v_n - v\|_b \le M\|v_n - v\|_a < \varepsilon,$$

so $v_n \to v$ in (V, d_b).

Proof of the remaining assertions

For completeness, it is enough to observe that the Cauchy property is preserved by equivalent norms. A sequence is Cauchy and then convergent under one norm if and only if it is Cauchy and then convergent under the other; the first assertion transfers convergence. The reverse direction is identical.
For closedness, suppose S is closed in (V, d_a). By the sequential characterization of closed sets, every convergent sequence in S converges to a point of S. If $v_n \in S$ converges in (V, d_b), the first assertion says that it converges to the same vector in (V, d_a); therefore its limit lies in S. Thus S is closed in (V, d_b). Openness follows from closedness by taking complements.

Totally bounded metric space

A metric space M is *totally bounded* if, for every $\varepsilon > 0$, there are finitely many open balls of

radius ε whose union covers M:

$$M \subset \bigcup_{i=1}^{N} B(x_i, \varepsilon).$$

The term "totally bounded" is somewhat redundant, but we keep it because it is standard terminology in topology.

Complete and totally bounded implies compact

If a metric space M is complete and totally bounded, then it is sequentially compact.

Proof

Let $\{x_n\}_{n=1}^{\infty} \subset M$. Since M is totally bounded, choose a finite set $D_0 = \{y_1, \dots, y_{k(0)}\} \subset M$ such that

$$M \subset \bigcup_{i=1}^{k(0)} B(y_i, 1).$$

Because D_0 is finite, one ball contains infinitely many x_n. Thus there is $y_0 \in D_0$ such that

$$A_0 = \{n \in \mathbb{N} : x_n \in B(y_0, 1)\}$$

is infinite.
Because M has a finite cover by balls of radius 2^{-1}, one of these balls contains infinitely many terms indexed by A_0. Hence there is a point y_1 such that

$$A_1 = \{n \in A_0 : x_n \in B(y_1, 2^{-1})\}$$

is infinite. Repeating the process, if A_k is infinite, choose a finite cover of M by balls of radius 2^{-k-1} and obtain y_{k+1} such that

$$A_{k+1} = \{n \in A_k : x_n \in B(y_{k+1}, 2^{-k-1})\}$$

is infinite. We obtain nested infinite sets

$$\cdots \subset A_{k+1} \subset A_k \subset \cdots \subset A_1 \subset A_0 \subset \mathbb{N}.$$

Choose inductively a strictly increasing sequence n_k such that $n_k \in A_k$. We claim that x_{n_k} is Cauchy. Given $\varepsilon > 0$, choose N with $2^{-N+1} < \varepsilon$. For $k, \ell \geq N$, the nested-set property implies that both n_k and n_ℓ lie in whichever of A_k, A_ℓ has the smaller index; without loss of generality suppose $A_\ell \subset A_k$. Then

$$x_{n_k}, x_{n_\ell} \in B(y_k, 2^{-k}),$$

and therefore

$$d(x_{n_k}, x_{n_\ell}) \leq d(x_{n_k}, y_k) + d(y_k, x_{n_\ell}) < 2^{-k} + 2^{-k} \leq 2^{-N+1} < \varepsilon.$$

Completeness now gives a limit in M, so every sequence in M has a convergent subsequence.

Heine–Borel

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. For every $n \in \mathbb{N}$, a subset $S \subset \mathbb{K}^n$ is compact if and only if it is closed and bounded with respect to the Euclidean norm on $\mathbb{R}^n$, or the standard norm on $\mathbb{C}^n$.

The result can be proved using sequential compactness and Bolzano–Weierstrass. This is precisely where completeness of the scalar field is needed: $\mathbb{R} \neq \mathbb{Q}$. It extends to $\mathbb{R}^n$ and $\mathbb{C}^n$, because convergence is equivalent to convergence in every component. We do not spend time on the routine proof.

Continuous images of compact sets

If $f : M \to Y$ is continuous from a compact metric space M into a metric space Y, then $f(M)$ is compact.

Proof

Let $\{V_\alpha\}_{\alpha \in A}$ be an open cover of $f(M)$. Continuity makes every $f^{-1}(V_\alpha)$ open in M, and these sets cover M. Compactness gives indices $\alpha_1, \ldots, \alpha_n$ such that

$$M \subset \bigcup_{i=1}^{n} f^{-1}(V_{\alpha_i}).$$

Applying f gives

$$f(M) \subset \bigcup_{i=1}^{n} V_{\alpha_i}.$$

Generalized Heine–Borel

Let V be a finite-dimensional normed vector space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then $A \subset V$ is compact if and only if A is closed and bounded.

Proof

Step 1: every norm on $\mathbb{K}^n$ is equivalent to the Euclidean norm. The Euclidean unit sphere

$$S = \{x \in \mathbb{K}^n : \|x\|_2 = 1\}$$

is closed and bounded, hence compact by Heine–Borel. If $\|\cdot\|$ is any norm, the function $x \mapsto \|x\|$ is continuous.
By the extreme value theorem there are $m, M > 0$, the absolute minimum and maximum on S, such that, after rescaling by homogeneity,

$$m\|x\|_2 \leq \|x\| \leq M\|x\|_2 \qquad (x \in \mathbb{K}^n).$$

Thus every norm on $\mathbb{K}^n$ is equivalent to the Euclidean norm.

Step 2: a linear isomorphism $T : \mathbb{K}^n \to X$ is a homeomorphism. Let $T : \mathbb{K}^n \to X$ be a linear isomorphism, where $\dim X = n$. Recall:

- an isomorphism is a bijective structure-preserving, here linear, map;
- a homeomorphism is a bijective continuous map with continuous inverse.

Choose any norm on X. Since T is injective, $x \mapsto \|Tx\|$ is a norm on $\mathbb{K}^n$. Equivalence of norms gives

$$m\|x\|_{\mathbb{K}^n} \le \|Tx\| \le M\|x\|_{\mathbb{K}^n}.$$

The right inequality makes T bounded and therefore continuous. By bijectivity, the left inequality can be rewritten as

$$\|T^{-1}y\|_{\mathbb{K}^n} \le \frac{1}{m}\|y\|, \qquad y \in X,$$

so T^{-1} is bounded and continuous. Thus T is a homeomorphism.

Step 3: conclusion. Homeomorphisms map closed sets to closed sets, compact sets to compact sets, and bounded sets to bounded sets. Therefore generalized Heine–Borel in X follows from Heine–Borel in $\mathbb{K}^n$, these three properties, and the existence of a linear isomorphism $T : \mathbb{K}^n \to X$.

We are now ready to prove an important fact in functional analysis: the equivalence of norms.

All norms in finite dimensions are equivalent

Let V be finite-dimensional over the complete field $\mathbb{K}$. Then all norms on V are equivalent.

Proof

It is enough to find one norm $\|\cdot\|_*$ equivalent to every norm on V, because $\simeq$ is transitive. Let $\{v_1, \dots, v_n\}$ be a basis of V, and define

$$\|x\|_* = \max_{1\le i\le n} |x_i|, \qquad x = \sum_{i=1}^{n} x_i v_i.$$

This is a norm. Let $\|\cdot\|$ be any other norm on V.
The unit sphere

$$S = \{x \in V : \|x\|_* = 1\}$$

is compact. One way to see this is that the coordinate map identifies $(V, \|\cdot\|_*)$ homeomorphically with $\mathbb{K}^n$, and S is closed and bounded there. Moreover $x \mapsto \|x\|$ is continuous with respect to $\|\cdot\|_*$: indeed,

$$\big|\|x\| - \|y\|\big| \le \|x - y\| \le \left(\sum_{i=1}^{n} \|v_i\|\right) \|x - y\|_*.$$

The extreme value theorem gives $m, M > 0$ such that

$$m \le \|x\| \le M \qquad (x \in S).$$

Rescaling yields

$$m\|x\|_* \le \|x\| \le M\|x\|_* \qquad (x \in V).$$

Hence $\|\cdot\|_* \simeq \|\cdot\|$. In particular, compactness in a finite-dimensional vector space over a complete field is independent of the chosen norm.

Finite-dimensional normed spaces are Banach

Every finite-dimensional normed vector space over a complete scalar field is a Banach space.

Proof

First consider $(V, \|\cdot\|_*)$ with the basis and maximum-coordinate norm from the preceding proof. Let x_k be Cauchy and write

$$x_k = \sum_{i=1}^{n} x_k^{(i)} v_i.$$

Then

$$\|x_k - x_m\|_* = \max_{1\le i\le n} |x_k^{(i)} - x_m^{(i)}| \to 0.$$

Each scalar sequence $x_k^{(i)}$ is therefore Cauchy, so completeness of $\mathbb{K}$ gives $x_k^{(i)} \to x^{(i)} \in \mathbb{K}$. It follows immediately that

$$x_k \longrightarrow \sum_{i=1}^{n} x^{(i)} v_i \quad \text{in } \|\cdot\|_*.$$

Thus $(V, \|\cdot\|_*)$ is Banach. Since every other norm on V is equivalent to $\|\cdot\|_*$, completeness is preserved, and V is Banach under any norm.

We now return to the finite-dimensional-approximability theorem from page 159. A finite-dimensional subspace $W \subset H$ is complete, hence closed. Therefore the orthogonal-decomposition theorems from page 147 and page 149 apply:

$$H = W \oplus W^{\perp}, \qquad u = w + w^{\perp}.$$

The length-minimizing vector in W is w, and

$$\inf_{\widetilde{w}\in W} \|u - \widetilde{w}\| = \|w^{\perp}\|, \qquad u = w + w^{\perp}.$$

If Π_W is the orthogonal projection onto W and $I : H \to H$ is the identity, then

$$\|w^{\perp}\| = \|u - w\| = \|u - \Pi_W u\| = \|(I - \Pi_W)u\|.$$

Consequently,

$$\forall u \in K, \quad \inf_{w\in W} \|u - w\| < \varepsilon \quad \Longleftrightarrow \quad \sup_{u\in K} \|(I - \Pi_W)u\| < \varepsilon$$

for some subspace $W \subset H$. We use this equivalent formulation to complete the proof.

Proof of finite-dimensional approximability: the separable case

First suppose H is separable. If K is compact, choose an orthonormal basis $\{e_k\}_{k=1}^{\infty}$ of H. By the equi-small-tail theorem, for each $\varepsilon > 0$ there is N such that

$$\sum_{k=N}^{\infty} |\langle u, e_k\rangle|^2 < \frac{\varepsilon^2}{4} \qquad (u \in K).$$

Set

$$W_N = \operatorname{span}\{e_1, \dots, e_N\}, \qquad \Pi_N = \Pi_{W_N}.$$

The Fourier–Bessel series gives

$$u = \sum_{k=1}^{\infty} \langle u, e_k \rangle e_k, \qquad \Pi_N u = \sum_{k=1}^{N} \langle u, e_k \rangle e_k.$$

Therefore

$$\|(I - \Pi_N)u\|^2 = \left\| \sum_{k=N}^{\infty} \langle u, e_k \rangle e_k \right\|^2 = \sum_{k=N}^{\infty} |\langle u, e_k \rangle|^2,$$

so

$$\sup_{u \in K} \inf_{w \in W_N} \|u - w\| = \sup_{u \in K} \|(I - \Pi_N)u\| < \frac{\varepsilon}{2} < \varepsilon.$$

Conversely, suppose $K \subset H$ is closed and bounded and, for every $\varepsilon > 0$, there is a finite-dimensional $W \subset H$ with

$$\sup_{u \in K} \|(I - \Pi_W)u\| < \varepsilon.$$

Let $\{u_n\}_{n=1}^{\infty} \subset K$. For every orthogonal projection,

$$\|\Pi_W u_n\| \leq \|u_n\|,$$

so boundedness of K implies boundedness of the projected sequence.
Since W is a finite-dimensional normed vector space over the complete field $\mathbb{C}$, Bolzano–Weierstrass gives a convergent subsequence of $\{\Pi_W u_n\}$. We now make the same kind of diagonal construction used in the proof on page 156.
For $j = 1$, take $\varepsilon = 1$. Choose a finite-dimensional $W_1 \subset H$ such that

$$\sup_{u \in K} \|(I - \Pi_1)u\| < 1, \qquad \Pi_1 = \Pi_{W_1}.$$

The sequence $\Pi_1 u_n$ is bounded, hence has a convergent subsequence $\{\Pi_1 u_{n_1(i)}\}_{i=1}^{\infty}$.
For $j = 2$, take $\varepsilon = \frac{1}{2}$. Choose a finite-dimensional $W_2 \subset H$ with

$$\sup_{u \in K} \|(I - \Pi_2)u\| < \frac{1}{2}.$$

The sequence $\{\Pi_2 u_{n_1(i)}\}$ is bounded in W_2, so it has a convergent subsequence $\{\Pi_2 u_{n_2(i)}\}$, where $\{n_2(i)\} \subset \{n_1(i)\}$.
Inductively, we obtain nested index sequences

$$\cdots \subset \{n_{j+1}(i)\}_{i=1}^{\infty} \subset \{n_j(i)\}_{i=1}^{\infty} \subset \cdots \subset \{n_1(i)\}_{i=1}^{\infty} \subset \mathbb{N}$$

such that

$$\Pi_j u_{n_j(i)} \text{ converges in } W_j, \qquad \sup_{u \in K} \|(I - \Pi_j)u\| < \frac{1}{j}.$$

Choose the diagonal subsequence

$$\{u_{n_i(i)}\}_{i=1}^{\infty} = \{u_{n_1(1)}, u_{n_2(2)}, u_{n_3(3)}, \ldots\}.$$

The indices satisfy

$$n_1(1) < n_2(2) < \cdots,$$

so this really is a subsequence of u_n. For every fixed j, its tail $\{u_{n_i(i)}\}_{i=j}^{\infty}$ is a subsequence of $\{u_{n_j(i)}\}_{i=1}^{\infty}$. We claim the diagonal sequence is Cauchy.
Given $\varepsilon > 0$, choose $j > 3/\varepsilon$. Since $\{\Pi_j u_{n_i(i)}\}_{i=j}^{\infty}$ is a subsequence of a convergent sequence in W_j, it is Cauchy. Thus, for all sufficiently large k, ℓ,

$$\|\Pi_j u_{n_k(k)} - \Pi_j u_{n_\ell(\ell)}\| < \frac{\varepsilon}{3}.$$

Also,

$$\|(I - \Pi_j)u\| < \frac{1}{j} < \frac{\varepsilon}{3} \qquad (u \in K).$$

Combining these estimates,

$$\begin{aligned}\|u_{n_k(k)} - u_{n_\ell(\ell)}\| &\le \|\Pi_j u_{n_k(k)} - \Pi_j u_{n_\ell(\ell)}\| \\ &\quad + \|(I - \Pi_j)u_{n_k(k)}\| + \|(I - \Pi_j)u_{n_\ell(\ell)}\| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.\end{aligned}$$

The diagonal subsequence is Cauchy. It converges in the Hilbert space H, and closedness of K puts its limit in K. Thus every sequence in K has a convergent subsequence, so K is compact.

To extend the forward direction to a nonseparable Hilbert space, notice that separability was used only to obtain an orthonormal basis of H; the converse did not require separability.

Proof of finite-dimensional approximability: the nonseparable case

Suppose K is compact. First, K itself is separable. For every $p \in \mathbb{N}$, the family

$$\mathcal{C}_p = \{B(x, 1/p) : x \in K\}$$

is an open cover of K. Compactness gives a finite subcover. The union of the centers from these finite subcovers, over all $p \in \mathbb{N}$, is a countable dense subset of K. Indeed, given $\varepsilon > 0$, choose $p > 1/\varepsilon$.
Let

$$H' = \left\{ \sum_{i=1}^{n} c_i x_i : n \in \mathbb{N},\ c_i \in \mathbb{K},\ x_i \in K \right\},$$

the set of all finite linear combinations of elements of K. It is a separable pre-Hilbert subspace of H: using coefficients in $\mathbb{Q} + i\mathbb{Q}$ and the countable dense set of centers above gives a countable dense subset. Its closure $\overline{H'}$ is a separable Hilbert subspace of H.
Apply the already-proved separable case inside $\overline{H'}$. It yields a finite-dimensional subspace $W \subset \overline{H'}$ satisfying the required approximation condition. Since W is also a subspace of H, the proof is complete.

24.4 Closed Spans of Orthonormal Sets

Closed span and best approximation

Let H be a Hilbert space, let $\{e_n\}_{n=1}^{\infty}$ be a countably infinite orthonormal subset of H, and let

$$W = \left\{ \sum_{i=1}^{k} c_i e_i : k \in \mathbb{N},\ c_i \in \mathbb{C} \right\}$$

be the subspace of all finite linear combinations of these vectors. Then:

1) $w \in \overline{W}$ if and only if there is $\{c_n\}_{n=1}^{\infty} \in \ell^2$ such that

$$w = \sum_{n=1}^{\infty} c_n e_n;$$

2) for every $u \in H$ and $w \in \overline{W}$,

$$\left\| u - \sum_{n=1}^{\infty} \langle u, e_n \rangle e_n \right\| \leq \|u - w\|,$$

with equality if and only if

$$w = \sum_{n=1}^{\infty} \langle u, e_n \rangle e_n.$$

Proof of the first assertion

The space $\overline{W}$ is a closed subspace of H, hence a Hilbert space. It is infinite-dimensional: otherwise, if $\dim \overline{W} = n$, any $n+1$ vectors in $\overline{W}$ would be linearly dependent, whereas $\{e_1, \ldots, e_{n+1}\} \subset \overline{W}$ is orthonormal and therefore linearly independent.
We claim that $\{e_n\}_{n=1}^{\infty}$ is maximal in $\overline{W}$. Suppose $v \in \overline{W}$ and

$$\langle e_n, v \rangle = 0 \qquad (n \in \mathbb{N}).$$

If $w = \sum_{i=1}^{k} c_i e_i \in W$, then

$$\langle w, v \rangle = \sum_{i=1}^{k} c_i \langle e_i, v \rangle = 0.$$

For any $w \in \overline{W}$, choose $w_j \in W$ with $w_j \to w$. Continuity of the inner product gives

$$\langle w, v \rangle = \lim_{j \to \infty} \langle w_j, v \rangle = 0.$$

Taking $w = v$ gives $\langle v, v \rangle = 0$, hence $v = 0$. Thus the orthonormal set is maximal and is an orthonormal basis of $\overline{W}$.
The Fourier-series isomorphism

$$T : \overline{W} \to \ell^2, \qquad Tu = \{\langle u, e_n \rangle\}_{n=1}^{\infty},$$

is an isometric isomorphism. Therefore $w \in \overline{W}$ exactly when it has an expansion $w = \sum c_n e_n$ with $(c_n) \in \ell^2$.

Proof of the best-approximation assertion

Write

$$w = \sum_{n=1}^{\infty} c_n e_n.$$

Using orthonormality and the complex inner-product convention,

$$\begin{aligned}
\|u - w\|^2 &= \langle u - w, u - w \rangle \\
&= \|u\|^2 - 2\,\mathrm{Re} \sum_{n=1}^{\infty} \overline{c_n} \langle u, e_n \rangle + \sum_{n=1}^{\infty} |c_n|^2 \\
&= \|u\|^2 - \sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2 + \sum_{n=1}^{\infty} |\langle u, e_n \rangle - c_n|^2 \\
&\geq \|u\|^2 - \sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2.
\end{aligned}$$

Equality holds exactly when

$$\sum_{n=1}^{\infty} |\langle u, e_n \rangle - c_n|^2 = 0,$$

that is, exactly when $c_n = \langle u, e_n \rangle$ for every n, or

$$w = \sum_{n=1}^{\infty} \langle u, e_n \rangle e_n.$$

Moreover,

$$\left\| u - \sum_{n=1}^{\infty} \langle u, e_n \rangle e_n \right\|^2 = \|u\|^2 - \sum_{n=1}^{\infty} |\langle u, e_n \rangle|^2.$$

Taking square roots proves the assertion.

Connection with orthogonal projection

The vector

$$\sum_{n=1}^{\infty} \langle u, e_n \rangle e_n$$

is $\Pi_{\overline{W}} u$. Thus the inequality says that the unique closest element of $\overline{W}$ to $u \in H$ is its orthogonal projection. This is the same length-minimizing property proved on page 147–page 148. Alternatively, if $H = \overline{W} \oplus \overline{W}^{\perp}$, then

$$\|u - w\|^2 = \left\| u - \sum_{n=1}^{\infty} \langle u, e_n \rangle e_n \right\|^2 + \left\| \sum_{n=1}^{\infty} \langle u, e_n \rangle e_n - w \right\|^2,$$

and equality follows from positive definiteness.

24.5 Sobolev Spaces on the Circle

The Sobolev space $H^s(\mathbb{T})$

Let $s \geq 0$. A function $f \in L^2([-\pi, \pi])$ represents an element of the Sobolev space $H^s(\mathbb{T})$ of order s if

$$\sum_{n \in \mathbb{Z}} |\widehat{f}(n)|^2 (1 + |n|^2)^s < \infty.$$

Here $\mathbb{T}$ denotes the one-dimensional torus, represented as the circle $[-\pi, \pi]$, and

$$\widehat{f}(n) = \frac{1}{\sqrt{2\pi}} \left\langle f, \frac{e^{int}}{\sqrt{2\pi}} \right\rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int}\, dt$$

is the n-th Fourier coefficient.

Sobolev norms as weighted L^2 norms

Roughly speaking, $H^s(\mathbb{T})$ is a subset of $L^2([-\pi, \pi])$ whose elements have better regularity. Parseval's identity and maximality of the trigonometric orthonormal basis give, for $f \in L^2([-\pi, \pi])$,

$$\|f\|_2^2 = \sum_{n \in \mathbb{Z}} \left| \left\langle f, \frac{e^{int}}{\sqrt{2\pi}} \right\rangle \right|^2 = 2\pi \sum_{n \in \mathbb{Z}} |\widehat{f}(n)|^2 = 2\pi \|f\|_{H^0(\mathbb{T})}^2.$$

For $s > 0$, the H^s-norm weights the n-th Fourier coefficient by $(1 + |n|^2)^s$. Since this weight is at least 1,

$$H^s(\mathbb{T}) \subset L^2([-\pi, \pi]).$$

Basic properties and Sobolev embedding

1) If $f \in C^k([-\pi, \pi])$ and

$$f^{(j)}(-\pi) = f^{(j)}(\pi), \qquad j = 0, \ldots, k-1,$$

then $f \in H^s(\mathbb{T})$ for every $0 \leq s < k$.

2) The space $H^s(\mathbb{T})$ is a vector space, and

$$\langle f, g \rangle_{H^s(\mathbb{T})} = \sum_{n \in \mathbb{Z}} \widehat{f}(n) \overline{\widehat{g}(n)} (1 + |n|^2)^s$$

is a Hermitian inner product on it.

3) $H^s(\mathbb{T})$ is a Hilbert space.

4) If $s > \frac{1}{2}$, then $H^s(\mathbb{T})$ embeds continuously into $C([-\pi, \pi])$. More precisely:

   a) there is $C(s) > 0$ such that, for every $f \in H^s(\mathbb{T})$,

   $$\sum_{n \in \mathbb{Z}} |\widehat{f}(n)| \leq C(s) \|f\|_{H^s(\mathbb{T})};$$

   b) for every $f \in H^s(\mathbb{T})$, there is $g \in C([-\pi, \pi])$ such that $f = g$ almost everywhere and

   $$\|g\|_\infty \leq C(s) \|f\|_{H^s(\mathbb{T})}.$$

Proof of part 1

For $n \neq 0$,

$$\widehat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt - \frac{i}{2\pi}\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt.$$

Because $\sin(n\pi) = 0$, $\cos(n\pi) = \cos(-n\pi)$, and the derivatives of f match at the endpoints, repeated integration by parts gives

$$\int_{-\pi}^{\pi} f(t)\cos(nt)\,dt = \begin{cases} (-1)^{(j+1)/2}\dfrac{\displaystyle\int_{-\pi}^{\pi}\sin(nt)f^{(j)}(t)\,dt}{n^j}, & j \text{ odd}, \\ (-1)^{j/2}\dfrac{\displaystyle\int_{-\pi}^{\pi}\cos(nt)f^{(j)}(t)\,dt}{n^j}, & j \text{ even}, \end{cases}$$

and

$$\int_{-\pi}^{\pi} f(t)\sin(nt)\,dt = \begin{cases} (-1)^{(j-1)/2}\dfrac{\displaystyle\int_{-\pi}^{\pi}\cos(nt)f^{(j)}(t)\,dt}{n^j}, & j \text{ odd}, \\ (-1)^{j/2}\dfrac{\displaystyle\int_{-\pi}^{\pi}\sin(nt)f^{(j)}(t)\,dt}{n^j}, & j \text{ even}. \end{cases}$$

Substitution yields

$$\widehat{f}(n) = \begin{cases} \dfrac{(-1)^{(j+1)/2}}{2\pi}\dfrac{\displaystyle\int_{-\pi}^{\pi}[\sin(nt) + i\cos(nt)]f^{(j)}(t)\,dt}{n^j}, & j \text{ odd}, \\ \dfrac{(-1)^{j/2}}{2\pi}\dfrac{\displaystyle\int_{-\pi}^{\pi}[\cos(nt) - i\sin(nt)]f^{(j)}(t)\,dt}{n^j}, & j \text{ even}. \end{cases}$$

For $j = 0, 1, 2, \ldots$, the coefficient signs cycle so that the preceding formula reduces to

$$\widehat{f}(n) = \frac{1}{(in)^j}\frac{1}{2\pi}\int_{-\pi}^{\pi} f^{(j)}(t)e^{-int}\,dt = \frac{1}{(in)^j}\widehat{f^{(j)}}(n), \qquad j = 0, \ldots, k. \tag{$*$}$$

Since $0 \leq s \leq k$,

$$\sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2(1+|n|^2)^s \leq \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2(1+|n|^2)^k.$$

Thus it suffices to treat $s = k$.

The family $\{e^{int}/\sqrt{2\pi}\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $L^2([-\pi,\pi])$. Parseval's identity therefore gives

$$\|f\|_2^2 = \sum_{n\in\mathbb{Z}} \left|\left\langle f, \frac{e^{int}}{\sqrt{2\pi}}\right\rangle\right|^2 = 2\pi\sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2. \tag{$\dagger$}$$

For $n \neq 0$, substitute $(*)$ with $j = k$:

$$\begin{aligned}\sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2(1+|n|^2)^k &= \sum_{n\neq 0} |\widehat{f^{(k)}}(n)|^2\frac{(1+|n|^2)^k}{|n|^{2k}} \\ &\leq 2^k\sum_{n\neq 0} |\widehat{f^{(k)}}(n)|^2 < \infty.\end{aligned}$$

Here $f^{(k)} \in C([-\pi,\pi]) \subset L^2([-\pi,\pi])$, so the final finiteness follows from $(\dagger)$. Hence $f \in H^s(\mathbb{T})$ for $0 \leq s \leq k$, and in particular for $0 \leq s < k$.

Proof of part 2: $H^s(\mathbb{T})$ is a vector space

The zero function belongs to $H^s(\mathbb{T})$, and scalar closure follows immediately from $\widehat{cf}(n) = c\widehat{f}(n)$. For addition, let $f, g \in H^s(\mathbb{T})$. Linearity of the integral gives

$$\widehat{f+g}(n) = \widehat{f}(n) + \widehat{g}(n).$$

For $z_1, z_2 \in \mathbb{C}$,

$$|z_1 + z_2|^2 \le 2|z_1|^2 + 2|z_2|^2,$$

because

$$2|z_1|^2 + 2|z_2|^2 - |z_1 + z_2|^2 = |z_1 - z_2|^2 \ge 0.$$

Therefore

$$\sum_{n\in\mathbb{Z}} |\widehat{f+g}(n)|^2(1+|n|^2)^s \le 2\sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2(1+|n|^2)^s + 2\sum_{n\in\mathbb{Z}} |\widehat{g}(n)|^2(1+|n|^2)^s < \infty.$$

Thus $f + g \in H^s(\mathbb{T})$.

Proof of part 2: the inner product is Hermitian

Let $f, g \in H^s(\mathbb{T})$. Rewrite the proposed inner product as an ℓ^2 inner product:

$$\langle f, g\rangle_{H^s(\mathbb{T})} = \sum_{n\in\mathbb{Z}} [\widehat{f}(n)(1+|n|^2)^{s/2}]\,\overline{[\widehat{g}(n)(1+|n|^2)^{s/2}]}.$$

Set

$$a_n = \widehat{f}(n)(1+|n|^2)^{s/2}, \qquad b_n = \widehat{g}(n)(1+|n|^2)^{s/2}.$$

Then $a, b \in \ell^2$, and Cauchy–Schwarz gives

$$|\langle f, g\rangle_{H^s(\mathbb{T})}| = |\langle a, b\rangle_{\ell^2}| \le \|a\|_{\ell^2}\|b\|_{\ell^2} < \infty.$$

The inner product is linear in its first argument because Fourier coefficients are linear:

$$\begin{aligned}\langle \lambda_1 f_1 + \lambda_2 f_2, g\rangle_{H^s} &= \sum_{n\in\mathbb{Z}} [\lambda_1 \widehat{f_1}(n) + \lambda_2 \widehat{f_2}(n)]\overline{\widehat{g}(n)}(1+|n|^2)^s \\ &= \lambda_1\langle f_1, g\rangle_{H^s} + \lambda_2\langle f_2, g\rangle_{H^s}.\end{aligned}$$

Also

$$\langle f, g\rangle_{H^s} = \overline{\langle g, f\rangle_{H^s}},$$

and

$$\langle f, f\rangle_{H^s} = \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2(1+|n|^2)^s \ge 0.$$

If this equals zero, then $\widehat{f}(n) = 0$ for every n, hence

$$\left\langle f, \frac{e^{int}}{\sqrt{2\pi}} \right\rangle = 0 \qquad (n \in \mathbb{Z}).$$

The Fourier-series theorem gives $f = 0$ in L^2. Thus the form is positive definite.

Proof of part 3: completeness

Let $\{f_k\}_{k=1}^{\infty} \subset H^s(\mathbb{T})$ be Cauchy. Define

$$a_n^{(k)} = \widehat{f}_k(n)(1+|n|^2)^{s/2}, \qquad n \in \mathbb{Z},$$

and let $a^{(k)} = \{a_n^{(k)}\}_{n\in\mathbb{Z}}$. Then $a^{(k)} \in \ell^2$, and

$$\begin{aligned}\|a^{(k)} - a^{(m)}\|_{\ell^2} &= \left(\sum_{n\in\mathbb{Z}} |\widehat{f}_k(n) - \widehat{f}_m(n)|^2(1+|n|^2)^s\right)^{1/2} \\ &= \|f_k - f_m\|_{H^s(\mathbb{T})} \longrightarrow 0.\end{aligned}$$

Completeness of ℓ^2 gives $a = \{a_n\}_{n\in\mathbb{Z}} \in \ell^2$ such that $a^{(k)} \to a$. Define

$$\widehat{f}(n) = \frac{a_n}{(1+|n|^2)^{s/2}}, \qquad n \in \mathbb{Z},$$

and define f by its Fourier series

$$f(t) = \sum_{n\in\mathbb{Z}} \widehat{f}(n)e^{int}.$$

The Fourier coefficient defined this way is indeed the n-th Fourier coefficient of f, because orthogonality gives

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)e^{-int}\,dt = \widehat{f}(n).$$

Moreover,

$$\sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2(1+|n|^2)^s = \sum_{n\in\mathbb{Z}} |a_n|^2 = \|a\|_{\ell^2}^2 < \infty,$$

so $f \in H^s(\mathbb{T})$. Finally,

$$\|f_k - f\|_{H^s(\mathbb{T})} = \|a^{(k)} - a\|_{\ell^2} \longrightarrow 0.$$

Every Cauchy sequence converges, so $H^s(\mathbb{T})$ is complete and hence is a Hilbert space.

Proof of part 4(a)

Assume $s > \frac{1}{2}$. Write

$$\sum_{n\in\mathbb{Z}} |\widehat{f}(n)| = \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|(1+|n|^2)^{s/2}(1+|n|^2)^{-s/2}.$$

Set

$$a_n = |\widehat{f}(n)|(1+|n|^2)^{s/2}, \qquad b_n = (1+|n|^2)^{-s/2}.$$

Then $a \in \ell^2$, and

$$\begin{aligned}\|b\|_{\ell^2}^2 &= \sum_{n\in\mathbb{Z}} \frac{1}{(1+|n|^2)^s} \\ &\le 1 + 2\sum_{n=1}^{\infty} \frac{1}{n^{2s}} < \infty,\end{aligned}$$

because $2s > 1$. Cauchy–Schwarz now yields

$$\sum_{n\in\mathbb{Z}} |\widehat{f}(n)| \le \|a\|_{\ell^2}\|b\|_{\ell^2} = C(s)\|f\|_{H^s(\mathbb{T})},$$

where

$$C(s) = \left(\sum_{n\in\mathbb{Z}} \frac{1}{(1+|n|^2)^s}\right)^{1/2} < \infty.$$

Proof of part 4(b)

Recall the Weierstrass M-test: if $|f_j(x)| \le M_j$ on a set S and $\sum_j M_j < \infty$, then $\sum_j f_j$ converges absolutely and uniformly on S.
For $f \in H^s(\mathbb{T})$, define

$$g(x) = \sum_{n\in\mathbb{Z}} \widehat{f}(n)e^{inx}, \qquad x \in [-\pi,\pi].$$

Each summand $g_n(x) = \widehat{f}(n)e^{inx}$ is continuous and satisfies

$$|g_n(x)| = |\widehat{f}(n)|.$$

Part (a) shows that $\sum_{n\in\mathbb{Z}} |\widehat{f}(n)| < \infty$. The Weierstrass M-test therefore makes the Fourier series uniformly convergent. Hence $g \in C([-\pi,\pi])$.
This g is the Fourier series of f in $L^2([-\pi,\pi])$. By the Fourier-series convergence theorem,

$$\|g - f\|_2 = 0,$$

so $f = g$ almost everywhere.
For every M,

$$\left|\sum_{|n|\le M} g_n(x)\right| \le \sum_{|n|\le M} |\widehat{f}(n)| \le C(s)\|f\|_{H^s(\mathbb{T})}.$$

Letting $M \to \infty$ gives, for every $x \in [-\pi,\pi]$,

$$|g(x)| \le C(s)\|f\|_{H^s(\mathbb{T})}.$$

Therefore

$$\|g\|_\infty \le C(s)\|f\|_{H^s(\mathbb{T})}.$$

24.6 Mollifiers, Convolution, and Smooth Approximation

A family of mollifiers

Let $\varphi : \mathbb{R} \to \mathbb{R}$ be infinitely differentiable and satisfy:

a) $\varphi(-x) = \varphi(x)$ and $\varphi(x) \ge 0$ for all $x \in \mathbb{R}$;

b) $|x| \ge 1$ implies $\varphi(x) = 0$;

c) $\int_{\mathbb{R}} \varphi(x)\,dx = 1.$

For $\varepsilon > 0$, define

$$\varphi_\varepsilon(x) = \frac{1}{\varepsilon}\varphi\left(\frac{x}{\varepsilon}\right).$$

The family $\{\varphi_\varepsilon\}_{\varepsilon>0}$ is a family of *mollifiers*, or bump functions, and

$$\int_{\mathbb{R}} \varphi_\varepsilon(x)\,dx = 1.$$

An almost-everywhere convergent subsequence

Let $E \subset \mathbb{R}$ be measurable and let $h_n \in L^1(E)$ satisfy

$$\int_E |h_n| \longrightarrow 0.$$

Then there is a subsequence $\{h_{n_j}\}_{j=1}^{\infty}$ such that

$$h_{n_j}(x) \longrightarrow 0 \qquad \text{for almost every } x \in E.$$

Proof

For $j \in \mathbb{N}$, let

$$A_j = \{x \in E : |h_n(x)| > 2^{-j}\}.$$

Then

$$0 \le m(A_j) = \int_{A_j} 1 \le 2^j \int_E |h_n| \longrightarrow 0.$$

Choose an increasing subsequence of indices n_j such that

$$m(\{x \in E : |h_{n_j}(x)| > 2^{-j}\}) \le 2^{-j}.$$

Write

$$E_j = \{x \in E : |h_{n_j}(x)| > 2^{-j}\}, \qquad F = \bigcap_{n=1}^{\infty} \bigcup_{j=n}^{\infty} E_j.$$

By countable subadditivity,

$$m(F) \le \lim_{n\to\infty} \sum_{j=n}^{\infty} m(E_j) \le \lim_{n\to\infty} \sum_{j=n}^{\infty} 2^{-j} = 0.$$

If $x \in E \setminus F$, then eventually $x \notin E_j$, and hence $|h_{n_j}(x)| \le 2^{-j} \to 0$. Thus the subsequence converges to zero almost everywhere.

Support and compact support

Let $f : X \to \mathbb{C}$, where X is a topological space or a subset of a metric space. The *support* of f is

$$\operatorname{supp}(f) = \overline{\{x \in X : f(x) \neq 0\}}.$$

The function f is *compactly supported* if $\operatorname{supp}(f)$ is compact in X. We use the notation

$$C_c^\infty(X) = \{f : X \to \mathbb{C} : f \text{ is infinitely differentiable and compactly supported}\},$$
$$C_c(X) = \{f : X \to \mathbb{C} : f \text{ is continuous and compactly supported}\},$$
$$C_c^k(X) = \{f : X \to \mathbb{C} : f \text{ is } k\text{-times continuously differentiable and compactly supported}\}.$$

Convolution

Let $f, g \in L^p(\mathbb{R})$, $1 \le p < \infty$. Their *convolution* is

$$(f * g)(x) = \int_{\mathbb{R}} f(x-y)g(y)\,dy = \int_{\mathbb{R}} f(y)g(x-y)\,dy.$$

Derivatives of convolutions

Let $u \in C(\mathbb{R})$ and $v \in C_c^\infty(\mathbb{R})$. Then $u * v$ is infinitely differentiable, with

$$\frac{d^n}{dx^n}(u * v) = u * \frac{d^n v}{dx^n}.$$

Moreover:

1) if $u \in C_c(\mathbb{R})$, then $u * v \in C_c^\infty(\mathbb{R})$;

2) if $u \in C_c^\infty(\mathbb{R})$, then
$$\frac{d^n}{dx^n}(u * v) = \frac{d^n u}{dx^n} * v = u * \frac{d^n v}{dx^n}.$$

Proof

Using the second convolution formula, the difference quotient is

$$\frac{(u * v)(x+t) - (u * v)(x)}{t} = \int_{\mathbb{R}} u(y)\frac{v(x+t-y) - v(x-y)}{t}\,dy.$$

Pointwise, the integrand converges to $u(y)v'(x-y)$. Since v has compact support,

$$\left|u(y)\frac{v(x+t-y) - v(x-y)}{t}\right| \le \|u\|_\infty(\|v'\|_\infty + 1)$$

on the only bounded set of y's that matters. This is integrable. Dominated convergence therefore gives

$$\frac{d}{dx}(u * v) = u * v'.$$

Repeating proves

$$\frac{d^n}{dx^n}(u * v) = u * v^{(n)}.$$

If u is smooth, the symmetric argument gives

$$(u * v)^{(n)} = u^{(n)} * v = u * v^{(n)}.$$

If $u, v \in C_c(\mathbb{R})$, choose $M, S > 0$ such that

$$\operatorname{supp}(u) \subset [-M, M], \qquad \operatorname{supp}(v) \subset [-S, S].$$

If $(u * v)(x) \neq 0$, then for some y,

$$u(y) \neq 0, \qquad v(x-y) \neq 0.$$

Thus $y \in [-M, M]$, $x - y \in [-S, S]$, and

$$x = y + (x - y) \in [-M - S, M + S].$$

Hence

$$\operatorname{supp}(u * v) \subset [-M - S, M + S],$$

which is compact. Together with smoothness, this proves $u * v \in C_c^\infty(\mathbb{R})$.

Mollification and density of smooth compactly supported functions

Let $g \in C(\mathbb{R})$ and suppose $g(x) = 0$ whenever $|x| > R$. For $\varepsilon > 0$, define

$$g_\varepsilon(x) = \int_{\mathbb{R}} \varphi_\varepsilon(x - y) g(y)\, dy.$$

Then:

1) g_ε is infinitely differentiable and $g_\varepsilon(x) = 0$ when $|x| > R + \varepsilon$;
2) $\|g_\varepsilon - g\|_\infty \to 0$ as $\varepsilon \to 0^+$;
3) if $f \in L^2(\mathbb{R})$ and $\delta > 0$, then there are $h \in C_c^\infty(\mathbb{R})$ and $S > 0$ such that

$$h(x) = 0 \quad (|x| > S), \qquad \|f - h\|_2 < \delta.$$

In particular, $C_c^\infty(\mathbb{R})$ is dense in $L^2(\mathbb{R})$.

Proof of the first two assertions

The first assertion follows from the convolution-derivative theorem. Since

$$\operatorname{supp}(\varphi_\varepsilon) \subset [-\varepsilon, \varepsilon], \qquad \operatorname{supp}(g) \subset [-R, R],$$

the support calculation gives

$$\operatorname{supp}(g_\varepsilon) \subset [-R - \varepsilon, R + \varepsilon].$$

For uniform convergence, use $\int \varphi_\varepsilon = 1$:

$$\begin{aligned} |g_\varepsilon(x) - g(x)| &= \left| \int_{\mathbb{R}} \varphi_\varepsilon(z)[g(x - z) - g(x)]\, dz \right| \\ &\leq \int_{\mathbb{R}} \varphi_\varepsilon(z) |g(x - z) - g(x)|\, dz. \end{aligned}$$

The compactly supported continuous function g is uniformly continuous on $\mathbb{R}$. Given $\eta > 0$, choose $\varepsilon_0 > 0$ such that

$$|z| < \varepsilon_0 \quad \Longrightarrow \quad |g(x - z) - g(x)| < \eta \qquad (x \in \mathbb{R}).$$

If $0 < \varepsilon \le \varepsilon_0$, then $\varphi_\varepsilon(z) = 0$ for $|z| \ge \varepsilon$, and so

$$|g_\varepsilon(x) - g(x)| \le \eta \int_{\mathbb{R}} \varphi_\varepsilon(z)\, dz = \eta.$$

This estimate is uniform in x, proving $\|g_\varepsilon - g\|_\infty \to 0$.

Proof of density

Let $f \in L^2(\mathbb{R})$ and $\delta > 0$. Density of $C_c(\mathbb{R})$ in $L^2(\mathbb{R})$ gives $g \in C(\mathbb{R})$ and $R > 0$ such that

$$g(x) = 0 \quad (|x| > R), \qquad \|f - g\|_2 < \frac{\delta}{3}.$$

By the first two assertions, $g_\varepsilon \in C_c^\infty(\mathbb{R})$, it vanishes for $|x| > R + \varepsilon$, and $\|g_\varepsilon - g\|_\infty \to 0$. Choose $0 < \varepsilon < R$ so that

$$\|g_\varepsilon - g\|_\infty < \frac{\delta}{6\sqrt{R}}.$$

Because both functions vanish outside $[-R - \varepsilon, R + \varepsilon]$,

$$\begin{aligned} \|g_\varepsilon - g\|_2 &= \left(\int_{|x| \le R+\varepsilon} |g_\varepsilon - g|^2 \right)^{1/2} \\ &\le \frac{\delta}{6\sqrt{R}} \left[2(R + \varepsilon) \right]^{1/2} \le \frac{\delta}{3}. \end{aligned}$$

Taking $h = g_\varepsilon$ and $S = R + \varepsilon$,

$$\|h - f\|_2 \le \|h - g\|_2 + \|g - f\|_2 < \frac{\delta}{3} + \frac{\delta}{3} < \delta.$$

Approximation of the identity in $L^2(\mathbb{R})$

Let $f \in L^2(\mathbb{R})$, let $\varepsilon > 0$, and define

$$f_\varepsilon(x) = \int_{\mathbb{R}} \varphi_\varepsilon(x - y) f(y)\, dy.$$

Then:

1) $\|f_\varepsilon\|_2 \le \|f\|_2$;
2) $\|f_\varepsilon - f\|_2 \to 0$ as $\varepsilon \to 0^+$.

Proof of the norm estimate

First suppose $f \in C(\mathbb{R}) \cap L^2(\mathbb{R})$. By Fubini and Cauchy–Schwarz in $L^2(\mathbb{R})$,

$$\begin{aligned} \|f_\varepsilon\|_2^2 &= \iint_{\mathbb{R}^2} \varphi_\varepsilon(y)\varphi_\varepsilon(z) \left(\int_{\mathbb{R}} \overline{f(x - y)} f(x - z)\, dx \right) dy\, dz \\ &\le \iint_{\mathbb{R}^2} \varphi_\varepsilon(y)\varphi_\varepsilon(z) \|f(\cdot - y)\|_2 \|f(\cdot - z)\|_2\, dy\, dz \end{aligned}$$

$$= \|f\|_2^2 \left(\int_{\mathbb{R}} \varphi_\varepsilon \right)^2 = \|f\|_2^2.$$

For a general $f \in L^2(\mathbb{R})$, choose $g_n \in C(\mathbb{R})$ with $\|g_n - f\|_2 \to 0$. The already-proved estimate shows

$$|(g_n)_\varepsilon(x) - f_\varepsilon(x)| \le B\|g_n - f\|_1$$

on the compact support of the mollifier, and a subsequence converges pointwise. Dominated convergence then gives

$$\|(g_n)_\varepsilon - f_\varepsilon\|_2 \to 0.$$

Passing to the limit in $\|(g_n)_\varepsilon\|_2 \le \|g_n\|_2$ gives

$$\|f_\varepsilon\|_2 \le \|f\|_2.$$

Proof of convergence to f

Let $\eta > 0$. Choose $g \in C_c(\mathbb{R})$ such that

$$\|f - g\|_2 < \frac{\eta}{3}.$$

Using the norm estimate and the uniform convergence already proved for compactly supported continuous functions,

$$\begin{aligned} \|f_\varepsilon - f\|_2 &\le \|f_\varepsilon - g_\varepsilon\|_2 + \|g_\varepsilon - g\|_2 + \|g - f\|_2 \\ &\le \|f - g\|_2 + \|g_\varepsilon - g\|_2 + \|g - f\|_2. \end{aligned}$$

For sufficiently small ε, $\|g_\varepsilon - g\|_2 < \eta/3$. Hence

$$\|f_\varepsilon - f\|_2 < \eta.$$

24.7 Orthogonal and Convex Projections

Every orthogonal projection is canonical

Let H be a Hilbert space and suppose $P \in \mathcal{B}(H)$ satisfies

$$P^2 = P, \qquad \langle Pu, v\rangle = \langle u, Pv\rangle \quad (u, v \in H).$$

Then $W = \mathrm{Ran}(P)$ is a closed subspace of H, and

$$P = \Pi_W.$$

Proof

Let $v_n = Pw_n \in \mathrm{Ran}(P)$ and suppose $v_n \to v$ in H. Since $P^2 = P$,

$$P(v_n) = P(Pw_n) = Pw_n = v_n.$$

Continuity of P gives

$$v = \lim_{n\to\infty} v_n = \lim_{n\to\infty} Pv_n = P\Big(\lim_{n\to\infty} v_n\Big) = Pv \in W.$$

Thus W is closed.
For $v \in H$, write

$$v = Pv + (v - Pv).$$

The first term lies in W. To show the second is in $W^\perp$, take $u = Pw \in W$. Then $Pu = u$, and self-adjointness gives

$$\begin{aligned}\langle u, v - Pv\rangle &= \langle u, v\rangle - \langle u, Pv\rangle \\ &= \langle u, v\rangle - \langle Pu, v\rangle = 0.\end{aligned}$$

Hence $v - Pv \in W^\perp$, so the unique decomposition $H = W \oplus W^\perp$ gives $Pv = \Pi_W v$.

Projection onto a closed convex set

Let H be a Hilbert space and let $C \subset H$ be nonempty, closed, and convex. Then:

1) for every $u \in H$, there is a unique $v \in C$ such that

$$\|u - v\| = \inf_{w\in C} \|u - w\|.$$

We denote it by $v = P_C u$;

2) if $u \in H$ and $v \in C$, then

$$v = P_C u \quad\Longleftrightarrow\quad \mathrm{Re}\langle u - v, w - v\rangle \le 0 \quad \text{for every } w \in C.$$

Proof of existence and uniqueness

Fix $u \in H$ and define

$$C' = \{u - w : w \in C\}.$$

If $u - w_1, u - w_2 \in C'$ and $t \in [0, 1]$, then

$$t(u - w_1) + (1 - t)(u - w_2) = u - [tw_1 + (1 - t)w_2] \in C',$$

because C is convex. Thus C' is convex; it is also nonempty and closed. The minimum-norm theorem for a nonempty closed convex subset of a Hilbert space gives a unique $u - v \in C'$ such that

$$\|u - v\| = \inf_{u-w\in C'} \|u - w\| = \inf_{w\in C} \|u - w\|.$$

Proof of the variational characterization

First suppose

$$\mathrm{Re}\langle u - v, w - v\rangle \le 0 \qquad (w \in C).$$

Then

$$\begin{aligned}\|u-w\|^2 &= \|u-v+v-w\|^2 \\ &= \|u-v\|^2 - 2\,\mathrm{Re}\langle u-v, w-v\rangle + \|v-w\|^2 \\ &\geq \|u-v\|^2.\end{aligned}$$

Hence $v = P_C u$.
Conversely, suppose $v = P_C u$. For $w \in C$ and $t \in [0,1]$, let

$$w(t) = (1-t)v + tw \in C.$$

Minimality gives $\|u-v\|^2 \leq \|u-w(t)\|^2$. Expanding both sides yields, for $0 < t \leq 1$,

$$2\langle v,v\rangle - t\langle v,v\rangle - 2\,\mathrm{Re}\langle v,w\rangle + 2t\,\mathrm{Re}\langle v,w\rangle - t\langle w,w\rangle - 2\,\mathrm{Re}\langle u,v\rangle + 2\,\mathrm{Re}\langle u,w\rangle \leq 0.$$

Letting $t \to 0^+$ gives

$$2\,\mathrm{Re}\{\langle v,v\rangle - \langle v,w\rangle - \langle u,v\rangle + \langle u,w\rangle\} \leq 0.$$

Therefore

$$\mathrm{Re}\langle v-u, v-w\rangle \leq 0,$$

or equivalently

$$\mathrm{Re}\langle u-v, w-v\rangle \leq 0.$$

Coordinatewise projection onto a box in ℓ^2

Let $\{h_k\}_{k=1}^{\infty} \subset \mathbb{R}_{\geq 0}$, and set

$$C = \left\{ b = (b_k)_{k=1}^{\infty} \in \ell^2 : |b_k| \leq h_k \text{ for every } k \right\}.$$

Then C is closed and convex. For every $a = (a_k)_{k=1}^{\infty} \in \ell^2$,

$$P_C a = ((P_C a)_k)_{k=1}^{\infty},$$

where

$$(P_C a)_k = \begin{cases} a_k, & |a_k| \leq h_k, \\ h_k \dfrac{a_k}{|a_k|}, & |a_k| > h_k. \end{cases}$$

Proof

If $a^{(n)} \in C$ and $a^{(n)} \to a$ in ℓ^2, then for each fixed k,

$$|a_k^{(n)} - a_k| \leq \|a^{(n)} - a\|_{\ell^2} \longrightarrow 0.$$

Since $|a_k^{(n)}| \leq h_k$, taking limits gives $|a_k| \leq h_k$. Therefore $a \in C$, so C is closed.
If $a, b \in C$ and $t \in [0,1]$, then

$$|ta_k + (1-t)b_k| \leq t|a_k| + (1-t)|b_k| \leq th_k + (1-t)h_k = h_k.$$

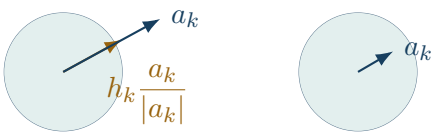
Thus $ta + (1-t)b \in C$, and C is convex.

Finally, minimizing

$$\|a-b\|_{\ell^2}^2 = \sum_{k=1}^{\infty} |a_k - b_k|^2$$

over $b \in C$ is equivalent to minimizing each coordinate distance $|a_k - b_k|$ over the closed disk $\{|b_k| \le h_k\} \subset \mathbb{C}$. If $|a_k| \le h_k$, the closest point is $b_k = a_k$. If $|a_k| > h_k$, the closest point is the radial point

$$b_k = h_k \frac{a_k}{|a_k|}.$$



$|a_k| > h_k$: radial clipping $|a_k| \le h_k$: unchanged

25 Compact Operators and the Spectrum of a Bounded Operator

25.1 Compact Operators

Let H be a Hilbert space. Recall that $K \in \mathcal{B}(H)$ is compact if

$$\overline{K(\{u \in H : \|u\| \le 1\})} = \overline{\{Ku : \|u\| \le 1\}}$$

is compact in H, and that the finite-rank operators are not closed in $\mathcal{B}(H)$; see the example on page 162.

Examples and a nonexample

1) The diagonal operator

$$K(a_1, a_2, \ldots) = \left(a_1, \frac{a_2}{2}, \frac{a_3}{3}, \ldots\right)$$

is compact on ℓ^2.

2) If $k \in C([0,1] \times [0,1])$, then

$$Tf(x) = \int_0^1 k(x,y) f(y)\, dy$$

is compact on $L^2([0,1])$. This is proved on page 203; it is a form of Carleman's theorem.

3) For

$$k(x,y) = \begin{cases} (x-1)y, & 0 \le y \le x \le 1, \\ x(y-1), & 0 \le x \le y \le 1, \end{cases}$$

the function

$$u(x) = \int_0^1 k(x,y) f(y)\, dy$$

solves the boundary-value problem

$$-u'' = f \quad \text{on } [0,1], \qquad u(0) = u(1) = 0.$$

Thus compact integral operators arise as solution operators for differential equations.

4) The identity on an infinite-dimensional Hilbert space is not compact. If e_n is an orthonormal sequence in the closed unit ball, then

$$\|e_n - e_m\|^2 = 2 \qquad (n \neq m),$$

so it has no convergent subsequence. Therefore the closed unit ball is not compact.

Compact operators are norm limits of finite-rank operators

Let H be a separable Hilbert space. An operator $T \in \mathcal{B}(H)$ is compact if and only if there are $T_n \in \mathcal{R}(H)$ such that

$$\|T - T_n\| \longrightarrow 0.$$

Consequently, the set of compact operators on H is $\overline{\mathcal{R}(H)}$, and we use this notation for it from now on.

Proof: compact implies approximable

Let $\{e_k\}_{k=1}^{\infty}$ be an orthonormal basis of H. The compact set

$$\overline{\{Tu : \|u\| \leq 1\}}$$

has equi-small tails. Thus, for every $\varepsilon > 0$, there is N such that

$$\sum_{k=N}^{\infty} |\langle Tu, e_k\rangle|^2 < \varepsilon^2 \qquad (\|u\| \leq 1). \tag{*}$$

For $n \in \mathbb{N}$, define

$$T_n u = \sum_{k=1}^{n} \langle Tu, e_k\rangle e_k.$$

Then $\mathrm{Ran}(T_n) \subset \mathrm{span}\{e_1, \ldots, e_n\}$, so T_n has finite rank. If $n \geq N$ and $\|u\| = 1$, Parseval gives

$$\begin{aligned} \|T_n u - Tu\|^2 &= \left\| \sum_{k=1}^{n} \langle Tu, e_k\rangle e_k - \sum_{k=1}^{\infty} \langle Tu, e_k\rangle e_k \right\|^2 \\ &= \sum_{k=n}^{\infty} |\langle Tu, e_k\rangle|^2 \leq \sum_{k=N}^{\infty} |\langle Tu, e_k\rangle|^2 < \varepsilon^2. \end{aligned}$$

Taking the supremum over the unit sphere yields $\|T_n - T\| \leq \varepsilon$, and hence $T_n \to T$ in operator norm.

Proof: approximable implies compact

Suppose $T_n \in \mathcal{R}(H)$ and $\|T_n - T\| \to 0$. We show that

$$\overline{\{Tu : \|u\| \leq 1\}}$$

is compact. This set is closed and bounded, since $\|Tu\| \leq \|T\|$ on the unit ball.

Given $\varepsilon > 0$, choose N so that $\|T_N - T\| < \varepsilon$, and set $W = \mathrm{Ran}(T_N)$. This is a finite-dimensional subspace. For every $\|u\| \leq 1$,

$$\inf_{w \in W} \|Tu - w\| \leq \|Tu - T_N u\| \leq \|T - T_N\| \, \|u\| < \varepsilon.$$

The finite-dimensional-approximability theorem on page 159 now shows that the closure of the image of the unit ball is compact. Thus T is compact.

Structure of the compact operators

For a separable Hilbert space H:

1) $\overline{\mathcal{R}(H)}$, the compact operators on H, is a closed subspace of $\mathcal{B}(H)$;
2) if $T \in \overline{\mathcal{R}(H)}$, then $T^* \in \overline{\mathcal{R}(H)}$;
3) if $A, B \in \mathcal{B}(H)$ and $T \in \overline{\mathcal{R}(H)}$, then $ATB \in \overline{\mathcal{R}(H)}$.

Proof

The first assertion follows from the preceding theorem: the compact operators are the closure of the subspace $\mathcal{R}(H)$.
For the adjoint assertion, choose $T_n \in \mathcal{R}(H)$ with $\|T_n - T\| \to 0$. Then $T_n^* \in \mathcal{R}(H)$, and

$$\|T_n^* - T^*\| = \|(T_n - T)^*\| = \|T_n - T\| \longrightarrow 0.$$

Hence $T^* \in \overline{\mathcal{R}(H)}$.
Finally, $AT_nB \in \mathcal{R}(H)$, and

$$\|AT_nB - ATB\| = \|A(T_n - T)B\| \leq \|A\| \, \|T_n - T\| \, \|B\| \longrightarrow 0.$$

Therefore ATB is compact.

25.2 The Spectrum

Neumann series

If $T \in \mathcal{B}(H)$ and $\|T\| < 1$, then $I - T$ is invertible and

$$(I - T)^{-1} = \sum_{n=0}^{\infty} T^n.$$

Proof

Let $S = \sum_{n=0}^{\infty} T^n$, which converges in operator norm because $\|T\| < 1$. As for a geometric series,

$$S(I - T) = I - \lim_{N \to \infty} T^N.$$

Since

$$\|T^N\| \leq \|T\|^N \longrightarrow 0,$$

positive definiteness of the operator norm gives $T^N \to 0$. Hence

$$S(I - T) = I.$$

The same computation on the other side gives $(I - T)S = I$, so $I - T$ is invertible with inverse S.

Invertible operators form an open set

The general linear group

$$GL(H) = \{T \in \mathcal{B}(H) : T \text{ is bijective}\}$$

is open in $\mathcal{B}(H)$.

Proof

Fix $T_0 \in GL(H)$. If

$$\|T - T_0\| < \|T_0^{-1}\|^{-1},$$

then

$$\|T_0^{-1}(T_0 - T)\| \leq \|T_0^{-1}\| \, \|T_0 - T\| < 1.$$

The Neumann-series theorem makes

$$I - T_0^{-1}(T_0 - T) \in GL(H).$$

Since

$$T = T_0\big[I - T_0^{-1}(T_0 - T)\big],$$

the operator T is invertible. Thus a ball about T_0 is contained in $GL(H)$.

Resolvent and spectrum

For $A \in \mathcal{B}(H)$, the *resolvent set* is

$$\mathrm{Res}(A) = \{\lambda \in \mathbb{C} : A - \lambda I \in GL(H)\}.$$

We often write $A - \lambda$ for $A - \lambda I$. The *spectrum* is

$$\mathrm{Spec}(A) = \mathbb{C} \setminus \mathrm{Res}(A).$$

For a matrix $A \in M_{n\times n}(\mathbb{C})$, λ is an eigenvalue when there is $\xi \in \mathbb{C}^n \setminus \{0\}$ with

$$A\xi = \lambda\xi.$$

Equivalently, $A - \lambda I$ has nontrivial kernel and is not invertible. Thus, in finite dimensions, $\mathrm{Spec}(A)$ is exactly the set of eigenvalues.

A diagonal 2×2 matrix

For

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix},$$

the operator $A - \lambda I$ is invertible exactly when $\lambda \neq \lambda_1, \lambda_2$. Hence

$$\mathrm{Res}(A) = \mathbb{C} \setminus \{\lambda_1, \lambda_2\}, \qquad \mathrm{Spec}(A) = \{\lambda_1, \lambda_2\}.$$

If A is diagonalizable, repeated diagonal entries are allowed; the distinct diagonal values are the eigenvalues.

Eigenvalues for bounded operators

Let $A \in \mathcal{B}(H)$. If $A - \lambda I$ is not injective, then some $u \in H \setminus \{0\}$ satisfies $Au = \lambda u$. We call $\lambda \in \mathrm{Spec}(A)$ an *eigenvalue* and u an *eigenvector*.

Spectrum can contain more than eigenvalues

On ℓ^2, let

$$T(a_1, a_2, \ldots) = \left(a_1, \frac{a_2}{2}, \frac{a_3}{3}, \ldots\right).$$

Then e_n is an eigenvector with eigenvalue $1/n$, so

$$\{1/n : n \in \mathbb{N}\} \subset \mathrm{Spec}(T).$$

In finite dimensions, injectivity is equivalent to surjectivity, but this is false in infinite dimensions.

For the preceding diagonal operator,

$$\mathrm{Spec}(T) = \{1/n : n \in \mathbb{N}\} \cup \{0\}.$$

Indeed, 0 is not an eigenvalue because T is injective, but T is not surjective: the vector

$$(1, \tfrac{1}{2}, \tfrac{1}{3}, \ldots)$$

would have to be the image of $(1, 1, 1, \ldots) \notin \ell^2$. Thus T is not invertible in $\mathcal{B}(\ell^2)$; the formal inverse

$$T^{-1}(a_1, a_2, \ldots) = (a_1, 2a_2, 3a_3, \ldots)$$

is unbounded.

In infinite dimensions, the spectrum need not consist only of eigenvalues, and a bounded operator can have no eigenvalues at all. For example, on $L^2([0,1])$,

$$Tf(x) = xf(x).$$

This multiplication operator has no eigenvalues, while

$$\mathrm{Spec}(T) = [0, 1].$$

Elementary properties of the spectrum

Let $A \in \mathcal{B}(H)$. Then:

1) $\mathrm{Res}(A)$ is open in $\mathbb{C}$;

2) $$\{\lambda \in \mathbb{C} : |\lambda| > \|A\|\} \subset \mathrm{Res}(A);$$

3) consequently, $\mathrm{Spec}(A)$ is closed and

$$\mathrm{Spec}(A) \subset \{\lambda \in \mathbb{C} : |\lambda| \leq \|A\|\},$$

so $\mathrm{Spec}(A)$ is compact.

Proof

Let $\lambda_0 \in \mathrm{Res}(A)$. Since $GL(H)$ is open, there is $\varepsilon > 0$ such that

$$\|T - (A - \lambda_0 I)\| < \varepsilon \quad \Longrightarrow \quad T \in GL(H).$$

If $|\lambda - \lambda_0| < \varepsilon$, then

$$\|(A - \lambda I) - (A - \lambda_0 I)\| = \|(\lambda - \lambda_0)I\| = |\lambda - \lambda_0| < \varepsilon.$$

Thus $A - \lambda I \in GL(H)$, and

$$B(\lambda_0, \varepsilon) \subset \mathrm{Res}(A).$$

Hence the resolvent is open.
If $|\lambda| > \|A\|$, then $\|A/\lambda\| < 1$, so

$$I - \frac{1}{\lambda} A \in GL(H)$$

by the Neumann-series theorem. Since

$$A - \lambda I = (-\lambda)\left(I - \frac{1}{\lambda} A\right)$$

is a product of invertible operators, $\lambda \in \mathrm{Res}(A)$. The assertions about the spectrum follow by taking complements.

Can $\mathrm{Spec}(A)$ be empty? No, but the proof needs complex analysis. If it were empty, then for $u, v \in H$,

$$f(\lambda) = \langle (A - \lambda I)^{-1} u, v \rangle$$

would be entire. Moreover, $\|(A - \lambda I)^{-1}\| \to 0$ as $|\lambda| \to \infty$, so $f(\lambda) \to 0$. Liouville's theorem would make f constant, hence identically zero, for every u, v. This would force $(A - \lambda I)^{-1} = 0$, impossible for an inverse.

25.3 Self-Adjoint Operators

Numerical characterization of the norm

Let $A \in \mathcal{B}(H)$ be self-adjoint, $A = A^*$. Then:

1) $\langle Au, u \rangle \in \mathbb{R}$ for every $u \in H$;

2)

$$\|A\| = \sup_{\|u\|=1} |\langle Au, u \rangle|.$$

Proof

For the first assertion,

$$\overline{\langle Au, u\rangle} = \langle u, Au\rangle = \langle u, A^*u\rangle = \langle Au, u\rangle.$$

Set

$$a = \sup_{\|u\|=1} |\langle Au, u\rangle|.$$

Cauchy–Schwarz gives

$$|\langle Au, u\rangle| \le \|Au\|\,\|u\| \le \|A\|\|u\|^2,$$

so $a \le \|A\|$.
To prove the reverse inequality, let $\|u\| = 1$, $Au \neq 0$, and set

$$v = \frac{Au}{\|Au\|}.$$

Then $\|v\| = 1$ and, because all diagonal quadratic forms are real, the polarization identity gives

$$\begin{aligned}
\|Au\| &= \langle Au, v\rangle = \mathrm{Re}\langle Au, v\rangle \\
&= \frac{1}{4}\,\mathrm{Re}\Big(\langle A(u+v), u+v\rangle - \langle A(u-v), u-v\rangle \\
&\qquad\qquad + i\langle A(u+iv), u+iv\rangle - i\langle A(u-iv), u-iv\rangle\Big) \\
&= \frac{1}{4}\,\mathrm{Re}\Big(\langle A(u+v), u+v\rangle - \langle A(u-v), u-v\rangle\Big) \\
&\le \frac{a}{4}(\|u+v\|^2 + \|u-v\|^2) \\
&= \frac{a}{4}(2\|u\|^2 + 2\|v\|^2) = a.
\end{aligned}$$

Thus $\|Au\| \le a$ for every unit vector u. Taking the supremum gives $\|A\| \le a$, and hence equality.

26 The Spectrum of Self-Adjoint Operators and Eigenspaces of Compact Operators

26.1 The Spectrum of a Self-Adjoint Operator

As in linear algebra, a self-adjoint operator is the infinite-dimensional analogue of a Hermitian matrix. We will show that its spectrum is real. Later, we will show that a compact self-adjoint operator can be diagonalized by an orthonormal basis, the analogue of orthogonal diagonalization of a Hermitian matrix.

Recall from page 188 and page 189 that

$$\mathrm{Spec}(A) \subset \{\lambda \in \mathbb{C} : |\lambda| \le \|A\|\}$$

for every $A \in \mathcal{B}(H)$, and that, for $A = A^*$,

$$\langle Au, u\rangle \in \mathbb{R}, \qquad \|A\| = \sup_{\|u\|=1} |\langle Au, u\rangle|.$$

Spectrum of a self-adjoint operator

Suppose $A = A^* \in \mathcal{B}(H)$. Then:

1) $$\mathrm{Spec}(A) \subset [-\|A\|, \|A\|];$$

2) at least one of $+\|A\|$ and $-\|A\|$ belongs to $\mathrm{Spec}(A)$.

Proof that the spectrum is real

Because the spectrum is already contained in the disk of radius $\|A\|$, it is enough to show $\mathrm{Spec}(A) \subset \mathbb{R}$.
Let $\lambda = s + it$ with $t \neq 0$. Set

$$\widetilde{A} = A - sI.$$

Then $\widetilde{A} = \widetilde{A}^*$, and

$$A - \lambda I = \widetilde{A} - itI.$$

It therefore suffices to prove that $A - itI$ is invertible whenever $A = A^*$ and $t \neq 0$.
For $u \in H$,

$$\begin{aligned} \mathrm{Im}\langle (A - itI)u, u\rangle &= \mathrm{Im}\langle Au, u\rangle - \mathrm{Im}\langle itu, u\rangle \\ &= -t\|u\|^2. \end{aligned} \tag{$*$}$$

Thus $(A - itI)u = 0$ implies $u = 0$, so $A - itI$ is injective. The same argument makes $A + itI = (A - itI)^*$ injective. By the range–kernel theorem,

$$\mathrm{Ran}(A - itI)^{\perp} = \mathrm{Ker}((A - itI)^*) = \mathrm{Ker}(A + itI) = \{0\}.$$

Hence

$$\overline{\mathrm{Ran}(A - itI)} = H.$$

It remains to show that the range is closed. If $(A - itI)u_n \to v$, then by $(*)$, Cauchy–Schwarz gives

$$|t|\,\|u_n - u_m\|^2 \leq \|(A - itI)u_n - (A - itI)u_m\|\,\|u_n - u_m\|,$$

and hence

$$\|u_n - u_m\| \leq \frac{1}{|t|}\|(A - itI)u_n - (A - itI)u_m\|.$$

The sequence u_n is Cauchy, so $u_n \to u \in H$. Continuity of $A - itI$ yields

$$(A - itI)u = v.$$

Thus $\mathrm{Ran}(A - itI)$ is closed. Being both dense and closed, it equals H, so $A - itI$ is bijective. Therefore every nonreal λ lies in the resolvent, and $\mathrm{Spec}(A) \subset \mathbb{R}$.

Proof that an endpoint belongs to the spectrum

The numerical norm formula supplies unit vectors u_n such that

$$|\langle Au_n, u_n\rangle| \longrightarrow \|A\|.$$

Passing to a subsequence, either

$$\langle Au_n, u_n\rangle \longrightarrow \|A\| \quad \text{or} \quad \langle Au_n, u_n\rangle \longrightarrow -\|A\|.$$

For the corresponding choice of sign,

$$\begin{aligned}\|(A \mp \|A\|I)u_n\|^2 &= \|Au_n\|^2 \mp 2\|A\|\langle Au_n, u_n\rangle + \|A\|^2\|u_n\|^2 \\ &\le 2\|A\|^2 \mp 2\|A\|\langle Au_n, u_n\rangle \longrightarrow 0.\end{aligned}$$

If $A \mp \|A\|I$ were invertible, then

$$1 = \|u_n\| \le \|(A \mp \|A\|I)^{-1}\|\,\|(A \mp \|A\|I)u_n\| \longrightarrow 0,$$

a contradiction. Thus one of the two endpoints is in the spectrum.

Exact spectral interval

Let $A = A^* \in \mathcal{B}(H)$, and define

$$a_- = \inf_{\|u\|=1}\langle Au, u\rangle, \qquad a_+ = \sup_{\|u\|=1}\langle Au, u\rangle.$$

Then:

1) $\mathrm{Spec}(A) \subset [a_-, a_+]$;

2) $a_-, a_+ \in \mathrm{Spec}(A)$.

Proof

Define

$$b_+ = \frac{a_+ + a_-}{2}, \qquad b_- = \frac{a_+ - a_-}{2}, \qquad B = A - b_+ I.$$

Then $B = B^*$, and

$$\begin{aligned}\sup_{\|u\|=1}\langle Bu, u\rangle &= a_+ - b_+ = b_-, \\ \inf_{\|u\|=1}\langle Bu, u\rangle &= a_- - b_+ = -b_-.\end{aligned}$$

The numerical norm formula gives

$$\|B\| = \sup_{\|u\|=1}|\langle Bu, u\rangle| = b_-.$$

The previous theorem gives

$$\mathrm{Spec}(B) \subset [-b_-, b_-].$$

Spectral translation is immediate from the definition:

$$\mathrm{Spec}(A) = \mathrm{Spec}(B) + b_+ \subset [b_+ - b_-, b_+ + b_-] = [a_-, a_+].$$

For the endpoints, choose unit-vector sequences $u_n^{\pm}$ with

$$\langle Bu_n^+, u_n^+\rangle \uparrow b_-, \qquad \langle Bu_n^-, u_n^-\rangle \downarrow -b_-.$$

Exactly as in the preceding endpoint argument,

$$\|(B - b_- I)u_n^+\| \to 0, \qquad \|(B + b_- I)u_n^-\| \to 0.$$

Neither $B - b_- I$ nor $B + b_- I$ can be invertible, since applying a bounded inverse would again imply $1 \leq 0$. Hence

$$\pm b_- \in \operatorname{Spec}(B),$$

and translating back gives

$$a_- = b_+ - b_- \in \operatorname{Spec}(A), \qquad a_+ = b_+ + b_- \in \operatorname{Spec}(A).$$

Positive operators and their spectrum

Let $A = A^* \in \mathcal{B}(H)$. Then

$$\langle Au, u\rangle \geq 0 \quad (u \in H) \quad \Longleftrightarrow \quad \operatorname{Spec}(A) \subset [0, \infty).$$

Proof

If $\langle Au, u\rangle \geq 0$ for all u, then

$$a_- = \inf_{\|u\|=1} \langle Au, u\rangle \geq 0.$$

The exact spectral interval gives

$$\operatorname{Spec}(A) \subset [a_-, a_+] \subset [0, \infty).$$

Conversely, if $\operatorname{Spec}(A) \subset [0, \infty)$, then $a_- \in \operatorname{Spec}(A)$ implies $a_- \geq 0$. Thus $\langle Au, u\rangle \geq 0$ for every unit vector u. For arbitrary $v \neq 0$,

$$\langle Av, v\rangle = \|v\|^2 \left\langle A\frac{v}{\|v\|}, \frac{v}{\|v\|}\right\rangle \geq 0.$$

An operator satisfying the equivalent conditions above is called *positive*.

26.2 Spectral Theory for Compact Self-Adjoint Operators

We will show that the spectrum of a compact self-adjoint operator consists only of eigenvalues, possibly together with 0, where 0 may be an eigenvalue or merely a limit of eigenvalues. We will also obtain an orthonormal basis of $\overline{\operatorname{Ran} T}$ consisting entirely of eigenvectors, and the spectrum will be at most countable.

This is the infinite-dimensional analogue of diagonalizing a Hermitian matrix: real eigenvalues and an orthonormal basis of eigenvectors.

Eigenspace

Let $A \in \mathcal{B}(H)$ and let λ be an eigenvalue. The corresponding *eigenspace* is

$$E_\lambda = \operatorname{Ker}(A - \lambda I) = \{u \in H : (A - \lambda I)u = 0\}.$$

It is the collection of all eigenvectors for λ, together with zero.

Eigenspaces of a compact self-adjoint operator

Suppose $A = A^* \in \mathcal{B}(H)$ is compact. Then:

1) if $\lambda \neq 0$ is an eigenvalue, then $\dim E_\lambda < \infty$;

2) if $\lambda_1 \neq \lambda_2$ are eigenvalues, then $E_{\lambda_1} \perp E_{\lambda_2}$;

3) the set of nonzero eigenvalues is at most countable. If it is enumerated as $\{\lambda_n\}_{n=1}^{\infty}$, then

$$|\lambda_n| \longrightarrow 0.$$

Proof of finite multiplicity

Suppose, for contradiction, that $\lambda \neq 0$ and $\dim E_\lambda = \infty$. Gram–Schmidt provides an orthonormal sequence $\{u_n\} \subset E_\lambda$. Since A is compact and $\|u_n\| = 1$, the sequence Au_n must have a convergent, hence Cauchy, subsequence. But for $j \neq k$,

$$\|Au_j - Au_k\|^2 = \|\lambda u_j - \lambda u_k\|^2 = |\lambda|^2 \|u_j - u_k\|^2 = 2|\lambda|^2,$$

a contradiction. Hence E_λ is finite-dimensional.

Proof of orthogonality

All eigenvalues are real. Let $u_1 \in E_{\lambda_1}$, $u_2 \in E_{\lambda_2}$, with $\lambda_1 \neq \lambda_2$. Then

$$\begin{aligned} \lambda_1 \langle u_1, u_2 \rangle &= \langle Au_1, u_2 \rangle = \langle u_1, A^* u_2 \rangle \\ &= \langle u_1, Au_2 \rangle = \lambda_2 \langle u_1, u_2 \rangle. \end{aligned}$$

Thus

$$(\lambda_1 - \lambda_2)\langle u_1, u_2 \rangle = 0,$$

and $\langle u_1, u_2 \rangle = 0$.

Proof of countability and convergence to zero

Let

$$\Lambda = \{\lambda \neq 0 : \lambda \text{ is an eigenvalue of } A\}.$$

If $\{\lambda_n\} \subset \Lambda$ are distinct, choose unit eigenvectors $u_n \in E_{\lambda_n}$. The preceding part makes u_n orthonormal, and

$$|\lambda_n| = \|\lambda_n u_n\| = \|Au_n\|.$$

We claim $\|Au_n\| \to 0$. Otherwise, some $\varepsilon_0 > 0$ and a subsequence u_{n_j} would satisfy

$$\|Au_{n_j}\| \geq \varepsilon_0. \tag{$*$}$$

Compactness supplies a further subsequence, write it e_k, such that

$$Ae_k \longrightarrow f \in H.$$

The vectors e_k remain orthonormal. Continuity of the norm and $(*)$ give

$$\|f\| = \lim_{k \to \infty} \|Ae_k\| \geq \varepsilon_0.$$

By continuity of the inner product and self-adjointness,

$$\varepsilon_0^2 \le \|f\|^2 = \lim_{k\to\infty} \langle Ae_k, f\rangle = \lim_{k\to\infty} \langle e_k, Af\rangle.$$

But Bessel's inequality for the orthonormal sequence e_k says

$$\sum_{k=1}^{\infty} |\langle e_k, Af\rangle|^2 \le \|Af\|^2 < \infty.$$

Therefore $|\langle e_k, Af\rangle| \to 0$, contradicting the preceding lower bound. Hence

$$\|Au_n\| \to 0, \qquad \text{and so} \qquad |\lambda_n| \to 0.$$

For every N,

$$\{\lambda \in \Lambda : |\lambda| > 1/N\}$$

must be finite; otherwise one could choose a sequence from it whose absolute values do not approach zero. Therefore

$$\Lambda = \bigcup_{N=1}^{\infty} \{\lambda \in \Lambda : |\lambda| > 1/N\}$$

is a countable union of finite sets, hence is at most countable.

27 The Spectral Theorem for a Compact Self-Adjoint Operator

In finite dimensions the spectrum consists exactly of eigenvalues. In infinite dimensions this fails even for compact self-adjoint operators: the diagonal operator on ℓ^2 studied on page 188 has $0 \in \mathrm{Spec}(A)$ although 0 is not an eigenvalue. The next result shows that, for a compact self-adjoint operator, 0 is the only possible exception.

Fredholm alternative

Let $A \in \mathcal{B}(H)$ be compact and let $\lambda \in \mathbb{R} \setminus \{0\}$. Then $\mathrm{Ran}(A - \lambda I)$ is closed. If, in addition, $A = A^*$, then exactly one of the following occurs:

1) $A - \lambda I$ is bijective;

2) the eigenspace

$$E_\lambda = \mathrm{Ker}(A - \lambda I)$$

is nontrivial and finite-dimensional.

Proof that the range is closed

Suppose

$$(A - \lambda I)u_n \longrightarrow f \in H.$$

Let

$$v_n = \Pi_{\mathrm{Ker}(A-\lambda I)^\perp} u_n.$$

Because the component of u_n in the kernel is annihilated,

$$(A-\lambda I)v_n = (A-\lambda I)u_n \longrightarrow f.$$

We claim v_n is bounded.
If not, choose v_{n_j} with $\|v_{n_j}\| \to \infty$. Then

$$(A-\lambda I)\frac{v_{n_j}}{\|v_{n_j}\|} = \frac{(A-\lambda I)v_{n_j}}{\|v_{n_j}\|} \longrightarrow 0. \qquad (*)$$

Compactness gives a subsequence for which

$$A\frac{v_{n_j}}{\|v_{n_j}\|}$$

converges. Since

$$\frac{v_{n_j}}{\|v_{n_j}\|} = \frac{1}{\lambda}\left[A\frac{v_{n_j}}{\|v_{n_j}\|} - (A-\lambda I)\frac{v_{n_j}}{\|v_{n_j}\|}\right],$$

the same subsequence converges to some v. Closedness of the orthogonal complement gives

$$v \in \operatorname{Ker}(A-\lambda I)^{\perp}, \qquad \|v\| = 1.$$

Passing to the limit in $(*)$ gives

$$(A-\lambda I)v = 0.$$

Hence v lies in both the kernel and its orthogonal complement, so $v = 0$, contradicting $\|v\| = 1$.
Therefore v_n is bounded.
Compactness now supplies a subsequence for which $Av_{n_j} \to w$. The identity

$$v_{n_j} = \frac{1}{\lambda}(Av_{n_j} - (A-\lambda I)v_{n_j})$$

shows that $v_{n_j} \to u$ for some $u \in H$. By continuity,

$$f = \lim_{j\to\infty}(A-\lambda I)v_{n_j} = (A-\lambda I)u \in \operatorname{Ran}(A-\lambda I).$$

Thus the range is closed.

Proof of the alternative

Now suppose $A = A^*$. Since the range is closed,

$$\begin{aligned}\operatorname{Ran}(A-\lambda I) &= \overline{\operatorname{Ran}(A-\lambda I)} = \operatorname{Ker}((A-\lambda I)^*)^{\perp} \\ &= \operatorname{Ker}(A-\lambda I)^{\perp}.\end{aligned}$$

Thus $A-\lambda I$ is either bijective or has a nontrivial finite-dimensional kernel. The finite-dimensionality was proved on page 193.

A compact self-adjoint operator attains its numerical norm

Let $A = A^*$ be a nonzero compact operator. Then A has a nonzero eigenvalue λ with a unit eigenvector u such that

$$|\lambda| = \sup_{\|v\|=1} |\langle Av, v\rangle| = |\langle Au, u\rangle| = \|A\|.$$

Proof

By the Fredholm alternative,

$$\mathrm{Spec}(A) = \{\text{eigenvalues of } A\} \cup \{0\}.$$

Since $A \neq 0$, $\|A\| \neq 0$; and the self-adjoint endpoint theorem says that at least one of $\pm\|A\|$ belongs to the spectrum. It is therefore a nonzero eigenvalue. If u is a normalized eigenvector,

$$|\langle Au, u\rangle| = |\langle \lambda u, u\rangle| = |\lambda|.$$

Maximum principle

Let $A = A^*$ be compact. Its nonzero eigenvalues, counted with geometric multiplicity, can be ordered

$$|\lambda_1| \geq |\lambda_2| \geq |\lambda_3| \geq \cdots > 0$$

with corresponding pairwise orthonormal eigenvectors u_k, so that

$$|\lambda_j| = \sup_{\substack{\|u\|=1 \\ u \in \mathrm{span}\{u_1,\ldots,u_{j-1}\}^\perp}} |\langle Au, u\rangle| = |\langle Au_j, u_j\rangle|. \qquad (*)$$

If the sequence of nonzero eigenvalues does not terminate, then

$$|\lambda_j| \longrightarrow 0.$$

The word *eigenfunctions* is synonymous with eigenvectors in function spaces. The theorem says to choose an eigenvalue of largest magnitude, then restrict A to the orthogonal complement of the eigenspace already found and repeat.

Proof by induction

We already know that the nonzero eigenvalues are at most countable, tend to zero if infinite, and have finite-dimensional eigenspaces. Only the maximum formula remains.
The base case is the preceding norm-attainment theorem. Suppose $\lambda_1, \ldots, \lambda_n$ and orthonormal eigenvectors $u_1, \ldots, u_n$ satisfying $(*)$ have been chosen.
If

$$Au = \sum_{k=1}^{n} \lambda_k \langle u, u_k\rangle u_k \qquad (u \in H),$$

then all eigenvalues, including multiplicity, have been found, and the process terminates. Otherwise define

$$A_n u = Au - \sum_{k=1}^{n} \lambda_k \langle u, u_k\rangle u_k.$$

The operator A_n is compact and self-adjoint. It vanishes on $\operatorname{span}\{u_1, \dots, u_n\}$ and agrees with A on its orthogonal complement. Moreover,

$$\operatorname{Ran}(A_n) \subset \operatorname{span}\{u_1, \dots, u_n\}^{\perp}.$$

Any nonzero eigenvalue of A_n is also an eigenvalue of A.
The norm-attainment theorem applied to A_n gives a nonzero eigenvalue λ_{n+1} with unit eigenvector u_{n+1} and

$$|\lambda_{n+1}| = \sup_{\|u\|=1} |\langle A_n u, u\rangle|.$$

Because $\operatorname{Ran}(A_n)$ lies in the orthogonal complement, the vector u_{n+1} is orthogonal to $u_1, \dots, u_n$. Also,

$$|\lambda_{n+1}| = \sup_{\substack{\|u\|=1 \\ u \perp u_1, \dots, u_n}} |\langle Au, u\rangle|.$$

Since the orthogonal complements shrink as n increases,

$$|\lambda_{n+1}| \le |\lambda_n|.$$

Finally,

$$|\lambda_{n+1}| = |\langle Au_{n+1}, u_{n+1}\rangle|.$$

Spectral theorem for a compact self-adjoint operator

Let $A = A^*$ be compact on a separable Hilbert space H. Let

$$|\lambda_1| \ge |\lambda_2| \ge \cdots > 0$$

be its nonzero eigenvalues, counted with multiplicity, and let u_k be corresponding orthonormal eigenvectors. Then:

1) $\{u_k\}$ is an orthonormal basis of both $\operatorname{Ran}(A)$ and $\overline{\operatorname{Ran}(A)}$;

2) there is an orthonormal basis $\{f_j\}$ of $\operatorname{Ker}(A)$ such that

$$\{u_k\} \cup \{f_j\}$$

is an orthonormal basis of H.

Thus H has an orthonormal basis consisting entirely of eigenvectors, and relative to that basis A is diagonal:

$$A \sim \operatorname{diag}(\lambda_1, \lambda_2, \dots, 0, 0, \dots).$$

Proof

If the maximum-principle construction terminates after n steps, then

$$Au = \sum_{k=1}^{n} \lambda_k \langle u, u_k\rangle u_k,$$

so

$$\operatorname{Ran}(A) \subset \operatorname{span}\{u_1, \dots, u_n\}.$$

The vectors $u_1, \dots, u_n$ form an orthonormal basis of the range, which is finite-dimensional and therefore closed.
Suppose the process does not terminate. We already know that $\{u_k\} \subset \mathrm{Ran}(A)$ is orthonormal. To prove maximality, take $f = Av \in \mathrm{Ran}(A)$ with

$$\langle f, u_k \rangle = 0 \qquad (k \in \mathbb{N}).$$

Then

$$0 = \langle Av, u_k \rangle = \langle v, Au_k \rangle = \lambda_k \langle v, u_k \rangle.$$

Since $\lambda_k \neq 0$, $v \perp u_k$ for every k. The maximum-principle remainder gives, for every n,

$$\|f\| = \left\| Av - \sum_{k=1}^{n} \lambda_k \langle v, u_k \rangle u_k \right\| \leq |\lambda_{n+1}| \, \|v\|.$$

As $|\lambda_{n+1}| \to 0$, we obtain $f = 0$. Hence $\{u_k\}$ is maximal in $\mathrm{Ran}(A)$.
Moreover,

$$\overline{\mathrm{Ran}(A)} \subset \overline{\mathrm{span}\{u_k\}},$$

and the Fourier–Bessel series description of the closed span shows that $\{u_k\}$ is also maximal in $\overline{\mathrm{Ran}(A)}$.
For the second assertion,

$$\overline{\mathrm{Ran}(A)} = \mathrm{Ker}(A^*)^{\perp} = \mathrm{Ker}(A)^{\perp}.$$

The kernel is closed and separable, so choose an orthonormal basis $\{f_j\}$ of it. The direct-sum decomposition

$$H = \mathrm{Ker}(A) \oplus \mathrm{Ker}(A)^{\perp} = \mathrm{Ker}(A) \oplus \overline{\mathrm{Ran}(A)}$$

shows that $\{f_j\} \cup \{u_k\}$ is an orthonormal basis of H.

27.1 Arzelà–Ascoli Theorems

Arzelà–Ascoli on an interval

Let $f_n \in C([a,b])$. Suppose:

a) there is $B > 0$ such that $\|f_n\|_\infty \leq B$ for every n;

b) the sequence is uniformly equicontinuous: for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$|x - y| < \delta \quad \Longrightarrow \quad |f_n(x) - f_n(y)| < \varepsilon \quad \text{for every } n.$$

Then some subsequence converges uniformly in $C([a,b])$.

Pointwise boundedness and equicontinuity

Let E be a set and $f_n : E \to \mathbb{C}$.

- The sequence is *pointwise bounded* on E if $\{f_n(x)\}_n$ is bounded for every $x \in E$.

- It is *uniformly bounded* if some $M > 0$ satisfies

$$|f_n(x)| \le M \qquad (x \in E,\ n \in \mathbb{N}).$$

If E is a subset of a metric space, a family $\mathcal{F}$ is *pointwise equicontinuous* if, for every $\varepsilon > 0$ and $x_0 \in E$, there is $\delta(x_0) > 0$ such that

$$d(x, x_0) < \delta(x_0) \quad \Longrightarrow \quad |f(x) - f(x_0)| < \varepsilon \qquad (f \in \mathcal{F}).$$

It is *uniformly equicontinuous* if a single $\delta > 0$ works for all $x, x_0 \in E$.

Pointwise diagonal subsequence

If f_n is a pointwise bounded sequence of complex-valued functions on a countable set E, then it has a subsequence f_{n_k} such that

$$f_{n_k}(x)$$

converges in $\mathbb{C}$ for every $x \in E$.

Proof

Enumerate $E = \{x_i\}_{i=1}^{\infty}$. Generalized Bolzano–Weierstrass gives a subsequence for which the values at x_1 converge. From it choose a subsequence for which the values at x_2 converge, and continue. Taking the diagonal subsequence gives convergence at every x_i.

Uniform limits preserve uniform equicontinuity

Let K be a compact metric space and $f_n \in C(K)$. If f_n converges uniformly on K, then the family $\{f_n\}$ is uniformly equicontinuous on K.

Proof

Uniform convergence is convergence in the supremum norm, so f_n is Cauchy. Given $\varepsilon > 0$, choose N such that

$$\|f_n - f_N\|_\infty < \varepsilon/3 \qquad (n \ge N).$$

The finite family $f_1, \dots, f_N$ is uniformly equicontinuous on compact K. Choose $\delta > 0$ so that for all these functions,

$$d(x, y) < \delta \Longrightarrow |f_i(x) - f_i(y)| < \varepsilon/3.$$

For $n \ge N$,

$$\begin{aligned} |f_n(x) - f_n(y)| &\le |f_n(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f_n(y)| \\ &< \varepsilon. \end{aligned}$$

Thus one δ works for all n.

Strong Arzelà–Ascoli theorem

Let K be a compact metric space and let $f_n \in C(K)$. If $\{f_n\}$ is pointwise bounded and uniformly equicontinuous on K, then:

1) $\{f_n\}$ is uniformly bounded on K;

2) it contains a uniformly convergent subsequence.

Proof of uniform boundedness

Given $\varepsilon > 0$, uniform equicontinuity supplies $\delta > 0$ such that

$$d(x, y) < \delta \quad \Longrightarrow \quad |f_n(x) - f_n(y)| < \varepsilon \qquad (n \in \mathbb{N}).$$

Compactness gives a finite cover

$$K \subset \bigcup_{i=1}^{r} B(p_i, \delta).$$

Pointwise boundedness supplies M_i with

$$|f_n(p_i)| \leq M_i \qquad (n \in \mathbb{N}).$$

Let $M = \max_i M_i$. For $x \in K$, choose p_i with $x \in B(p_i, \delta)$. Then

$$|f_n(x)| \leq |f_n(x) - f_n(p_i)| + |f_n(p_i)| < \varepsilon + M.$$

Hence the family is uniformly bounded.

Proof of existence of a convergent subsequence

Every compact metric space K has a countable dense subset. Indeed, for each $n \in \mathbb{N}$, compactness supplies finitely many points $p_1^{(n)}, \ldots, p_{r_n}^{(n)}$ such that

$$K \subset \bigcup_{i=1}^{r_n} B(p_i^{(n)}, 1/n).$$

The countable union of these finite sets is dense.
Let $E \subset K$ be such a countable dense set. By the pointwise diagonal subsequence theorem, choose a subsequence $g_j = f_{n_j}$ such that

$$g_j(x)$$

converges for every $x \in E$. We claim g_j converges uniformly on K.
Fix $\varepsilon > 0$. Uniform equicontinuity gives $\delta > 0$ such that

$$d(x, y) < \delta \quad \Longrightarrow \quad |g_j(x) - g_j(y)| < \varepsilon \qquad (j \in \mathbb{N}).$$

Compactness gives $y_1, \ldots, y_m \in K$ with

$$K \subset \bigcup_{i=1}^{m} B(y_i, \delta/2).$$

Density of E supplies $x_i \in E$ with $d(x_i, y_i) < \delta/2$. Hence

$$K \subset \bigcup_{i=1}^{m} B(x_i, \delta). \qquad (*)$$

Since $g_j(x_i)$ converges for each of the finitely many x_i, there is N such that

$$|g_j(x_i) - g_k(x_i)| < \varepsilon \qquad (j,k \geq N,\ i = 1,\dots,m).$$

For $x \in K$, choose x_i as in $(*)$. Then

$$\begin{aligned} |g_j(x) - g_k(x)| &\leq |g_j(x) - g_j(x_i)| + |g_j(x_i) - g_k(x_i)| + |g_k(x_i) - g_k(x)| \\ &< 3\varepsilon. \end{aligned}$$

After rescaling ε, g_j is Cauchy in $(C(K), \|\cdot\|_\infty)$. This space is complete, so g_j converges uniformly in $C(K)$.

The Arzelà–Ascoli theorem on an interval, stated at the beginning of this subsection, is the special case $K = [a,b]$; there uniform boundedness was assumed outright.

Compact families in $C(K)$

A family $\mathcal{F} \subset C(K)$, where K is a compact metric space, is compact in $(C(K), \|\cdot\|_\infty)$ if and only if it is closed, bounded, and uniformly equicontinuous on K.

Proof

If $\mathcal{F}$ is closed, bounded, and uniformly equicontinuous, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence by Arzelà–Ascoli; closedness keeps the limit in $\mathcal{F}$.
Conversely, compactness makes $\mathcal{F}$ closed and bounded. If it were not uniformly equicontinuous, there would be $\varepsilon_0 > 0$, points $x_n, y_n \in K$, and $f_n \in \mathcal{F}$ with

$$d(x_n, y_n) < 1/n, \qquad |f_n(x_n) - f_n(y_n)| \geq \varepsilon_0.$$

Compactness supplies a uniformly convergent subsequence of f_n. The previous uniform-limit theorem would make that subsequence uniformly equicontinuous, contradicting these inequalities.

A compactness criterion for $C^1([0,1])$

Let $f_n \in C^1([0,1])$ and suppose

$$B = \sup_{n \in \mathbb{N}} (\|f_n\|_2 + \|f_n'\|_2) < \infty.$$

Then:

1) $\|f_n\|_\infty \leq B$;
2) for all $x, y \in [0,1]$,
$$|f_n(x) - f_n(y)| \leq B|x - y|^{1/2};$$
3) some subsequence converges in $L^2([0,1])$.

Proof

By the fundamental theorem of calculus and Cauchy–Schwarz,

$$|f_n(x) - f_n(y)| = \left| \int_y^x f_n'(t)\, dt \right| \leq \|f_n'\|_2 |x-y|^{1/2} \leq B|x-y|^{1/2}.$$

Also,

$$|f_n(x)| \leq |f_n(y)| + \|f_n'\|_2.$$

Integrating in y and applying Cauchy–Schwarz gives

$$|f_n(x)| \leq \|f_n\|_2 + \|f_n'\|_2 \leq B.$$

Thus the sequence is uniformly bounded and uniformly equicontinuous. Arzelà–Ascoli provides a uniformly convergent subsequence. Since uniform convergence on $[0,1]$ implies L^2-convergence, the last assertion follows.

Compact integral operator with continuous kernel

Let $k \in C([a,b] \times [a,b])$, and define

$$Tf(x) = \int_a^b k(x,y) f(y)\, dy, \qquad f \in L^2([a,b]).$$

Then $T \in \overline{\mathcal{R}(L^2([a,b]))}$; equivalently, T is a compact operator.

Proof: boundedness

The continuous kernel is bounded on the compact square: choose $B > 0$ with

$$|k(x,y)| \leq B \qquad ((x,y) \in [a,b]^2).$$

If $\|f\|_2 = 1$, then

$$\begin{aligned} \|Tf\|_2^2 &= \int_a^b \left| \int_a^b k(x,y) f(y)\, dy \right|^2 dx \\ &\leq \int_a^b \left(\int_a^b |k(x,y)|\, |f(y)|\, dy \right)^2 dx \\ &\leq B^2 (b-a)^3. \end{aligned}$$

Thus

$$\|T\| \leq B(b-a)^{3/2},$$

and $T \in \mathcal{B}(L^2([a,b]))$.

Proof: compactness

Let f_n be bounded in $L^2([a,b])$, say $\|f_n\|_2 \leq M$. Uniform continuity of k gives, for every $\varepsilon > 0$,

a $\delta > 0$ such that

$$|x - x'| < \delta \quad \Longrightarrow \quad |k(x,y) - k(x',y)| < \varepsilon$$

for every $y \in [a,b]$. Hence

$$\begin{aligned} |Tf_n(x) - Tf_n(x')| &\le \int_a^b |k(x,y) - k(x',y)|\,|f_n(y)|\,dy \\ &\le \varepsilon \|f_n\|_2 \sqrt{b-a} \le \varepsilon M \sqrt{b-a}. \end{aligned}$$

Thus $\{Tf_n\}$ is uniformly equicontinuous. It is also uniformly bounded:

$$|Tf_n(x)| \le \|k\|_\infty \|f_n\|_2 \sqrt{b-a} \le M\|k\|_\infty \sqrt{b-a}.$$

Arzelà–Ascoli gives a uniformly convergent subsequence, which converges in $L^2([a,b])$. Therefore the image of the closed unit ball is relatively compact, and T is compact.

Compact balls in positive-order Sobolev spaces

Recall the Sobolev space of order $s \ge 0$,

$$H^s(\mathbb{T}) := \left\{ f \in L^2([-\pi,\pi]) : \|f\|_{H^s(\mathbb{T})} < \infty \right\},$$

where

$$\|f\|_{H^s(\mathbb{T})}^2 := \sum_{n\in\mathbb{Z}} (1+|n|^2)^s |\widehat{f}(n)|^2.$$

If $s > 0$ and $B > 0$, then

$$K := \left\{ f \in H^s(\mathbb{T}) : \|f\|_{H^s(\mathbb{T})} \le B \right\}$$

is a compact subset of $L^2([-\pi,\pi])$.

Proof

One may prove that K is closed and then combine boundedness and equi-small tails, relative to the orthonormal basis

$$\left\{ \frac{e^{inx}}{\sqrt{2\pi}} \right\}_{n\in\mathbb{Z}}$$

of $L^2([-\pi,\pi])$, with the compactness criterion on page 156–page 158. We instead prove compactness by the original diagonal argument.
Parseval's identity gives

$$\|f\|_2^2 = 2\pi \sum_{n\in\mathbb{Z}} |\widehat{f}(n)|^2.$$

Thus, for $f \in K$,

$$\|f\|_2^2 \le 2\pi \sum_{n\in\mathbb{Z}} (1+|n|^2)^s |\widehat{f}(n)|^2 = 2\pi \|f\|_{H^s(\mathbb{T})}^2 \le 2\pi B^2.$$

Therefore K is bounded in $L^2([-\pi,\pi])$.
Let $\{f_k\} \subset K$ be a bounded sequence. We first show that K has equi-small tails relative to the

displayed orthonormal basis. Given $\varepsilon > 0$, for every $f \in K$ and $N \in \mathbb{N}$,

$$\sum_{|n|>N} |\widehat{f}(n)|^2 \leq \sum_{|n|>N} \frac{(1+|n|^2)^s}{(1+N^2)^s} |\widehat{f}(n)|^2$$
$$\leq \frac{B^2}{(1+N^2)^s}.$$

Choose N so large that

$$\frac{2\pi B^2}{(1+N^2)^s} < \frac{\varepsilon^2}{16}. \tag{*}$$

With

$$e_n(x) = \frac{e^{inx}}{\sqrt{2\pi}},$$

the normalization of the Fourier coefficients gives

$$\widehat{f}(n) = \frac{1}{\sqrt{2\pi}} \langle f, e_n \rangle.$$

Consequently,

$$\sum_{|n|>N} |\langle f, e_n \rangle|^2 = 2\pi \sum_{|n|>N} |\widehat{f}(n)|^2 \leq \frac{2\pi B^2}{(1+N^2)^s} < \frac{\varepsilon^2}{16}.$$

This is the required uniform tail estimate.
For every $n \in \mathbb{Z}$ and $k \in \mathbb{N}$,

$$|\widehat{f_k}(n)| \leq \|f_k\|_{H^s(\mathbb{T})} < B.$$

Hence $\{\widehat{f_k}(n)\}_k$ is bounded in $\mathbb{C}$. We now apply a diagonal argument over $n \in \mathbb{Z}$. First choose a subsequence for which $\widehat{f_k}(0)$ converges; from it choose a subsequence for which $\widehat{f_k}(1)$ converges. Continue in the order

$$0,\ 1,\ -1,\ 2,\ -2,\ 3,\ -3, \ldots.$$

Thus, after repeated applications of generalized Bolzano–Weierstrass, the successive rows of Fourier coefficients converge to complex numbers

$$a_0,\ a_1,\ a_{-1},\ a_2, \ldots.$$

Picking the diagonal entries gives a subsequence, which we still denote by $\{f_{k(j)}\}$, such that

$$\widehat{f_{k(j)}}(n) \longrightarrow a_n \qquad \text{for every } n \in \mathbb{Z}.$$

Define

$$f(x) := \sum_{n\in\mathbb{Z}} a_n e^{inx}.$$

We claim that $f \in K$ and

$$\|f_{k(j)} - f\|_2 \longrightarrow 0.$$

For the first claim, since $\{f_{k(j)}\} \subset K$, for every j,

$$\sum_{n\in\mathbb{Z}} (1+|n|^2)^s |\widehat{f_{k(j)}}(n)|^2 \leq B^2.$$

Consequently, for every $M \in \mathbb{N}$,

$$\sum_{n=-M}^{M} (1+|n|^2)^s |\widehat{f}_{k(j)}(n)|^2 \le B^2.$$

Taking $j \to \infty$ in this finite sum gives

$$\sum_{n=-M}^{M} (1+|n|^2)^s |a_n|^2 \le B^2.$$

Now let $M \to \infty$:

$$\sum_{n\in\mathbb{Z}} (1+|n|^2)^s |a_n|^2 \le B^2.$$

By the definition of f, and because $\{e^{inx}/\sqrt{2\pi}\}_{n\in\mathbb{Z}}$ is an orthonormal basis of $L^2([-\pi,\pi])$, one has $\widehat{f}(n) = a_n$.
Therefore

$$\sum_{n\in\mathbb{Z}} (1+|n|^2)^s |a_n|^2 = \sum_{n\in\mathbb{Z}} (1+|n|^2)^s |\widehat{f}(n)|^2 = \|f\|^2_{H^s(\mathbb{T})} \le B^2,$$

so $f \in K$.
It remains to prove L^2-convergence. Fix $\varepsilon > 0$, and choose N as in $(*)$. Since $\widehat{f}_{k(j)}(n) \to a_n$ for every $n \in \mathbb{Z}$, for each $-N \le n \le N$ there is $M_n \in \mathbb{N}$ such that, whenever $j > M_n$,

$$|\widehat{f}_{k(j)}(n) - a_n|^2 < \frac{\varepsilon^2}{32\pi(2N+1)}.$$

Let

$$M := \max_{-N\le n\le N} M_n.$$

For $j > M$, split the Fourier series as

$$f_{k(j)} - f = g_1 + g_2 - g_3,$$

where

$$\widehat{g}_1(n) = \begin{cases} \widehat{f}_{k(j)}(n) - \widehat{f}(n), & |n| \le N, \\ 0, & |n| > N, \end{cases}$$

$$\widehat{g}_2(n) = \begin{cases} \widehat{f}_{k(j)}(n), & |n| > N, \\ 0, & |n| \le N, \end{cases} \qquad \widehat{g}_3(n) = \begin{cases} \widehat{f}(n) = a_n, & |n| > N, \\ 0, & |n| \le N. \end{cases}$$

Parseval's identity, the finite-coordinate estimate, and the equi-small-tail estimate yield

$$\begin{aligned} \|f_{k(j)} - f\|_2 &\le \|g_1\|_2 + \|g_2\|_2 + \|g_3\|_2 \\ &\le \left\{ 2\pi(2N+1) \frac{\varepsilon^2}{32\pi(2N+1)} \right\}^{1/2} + \left(\frac{\varepsilon^2}{16}\right)^{1/2} + \left(\frac{\varepsilon^2}{16}\right)^{1/2} \\ &= \frac{3\varepsilon}{4} < \varepsilon. \end{aligned}$$

Therefore every sequence $\{f_k\} \subset K$ has a subsequence converging in $L^2([-\pi,\pi])$ to an element $f \in K$. Hence K is compact in $L^2([-\pi,\pi])$.

Square-summable matrices define compact operators

Let $\{A_{ij}\}_{i,j=1}^{\infty}$ be a doubly indexed sequence in $\mathbb{C}$ such that

$$\sum_{i,j=1}^{\infty} |A_{ij}|^2 := \lim_{N\to\infty} \sum_{i=1}^{N}\sum_{j=1}^{N} |A_{ij}|^2 < \infty.$$

For $a = \{a_j\}_{j=1}^{\infty} \in \ell^2$, define

$$(Ta)_i := \sum_{j=1}^{\infty} A_{ij} a_j.$$

Then:

1) $T \in \mathcal{B}(\ell^2)$, with

$$\|T\| \le \left(\sum_{i,j=1}^{\infty} |A_{ij}|^2 \right)^{1/2};$$

2) if the n-th truncation T_n is defined by

$$(T_n a)_i = \begin{cases} \displaystyle\sum_{j=1}^{\infty} A_{ij} a_j, & i = 1, \dots, n, \\ 0, & i \ge n+1, \end{cases}$$

then T_n has finite rank and $\|T - T_n\| \to 0$. Consequently, T is compact.

Proof

Let $a \in \ell^2$ with $\|a\|_{\ell^2} = 1$, and write $A_i = \{A_{ij}\}_{j=1}^{\infty}$. Cauchy–Schwarz in ℓ^2 gives

$$\begin{aligned} \|Ta\|_{\ell^2}^2 &= \sum_{i=1}^{\infty} \left| \sum_{j=1}^{\infty} A_{ij} a_j \right|^2 \\ &\le \sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} |A_{ij}|^2 \right) \left(\sum_{j=1}^{\infty} |a_j|^2 \right) \\ &= \sum_{i,j=1}^{\infty} |A_{ij}|^2 < \infty. \end{aligned}$$

Thus the asserted norm bound holds and $T \in \mathcal{B}(\ell^2)$.
For fixed n,

$$T_n a = \{b_1, \dots, b_n, 0, 0, \dots\},$$

so

$$\operatorname{Ran}(T_n) \subset \operatorname{span}\{e_1, \dots, e_n\},$$

where e_i is the standard i-th unit vector. Hence $\dim \operatorname{Ran}(T_n) \le n$, and T_n has finite rank.

For every $a \in \ell^2$, Cauchy–Schwarz gives the uniform estimate

$$\|(T - T_n)a\|_{\ell^2}^2 = \sum_{i=n+1}^{\infty} \left| \sum_{j=1}^{\infty} A_{ij} a_j \right|^2 \leq \|a\|_{\ell^2}^2 \sum_{i=n+1}^{\infty} \sum_{j=1}^{\infty} |A_{ij}|^2.$$

Taking the supremum over $\|a\|_{\ell^2} = 1$ yields

$$\|T - T_n\|^2 \leq \sum_{i=n+1}^{\infty} \sum_{j=1}^{\infty} |A_{ij}|^2.$$

The right-hand side tends to zero because the nonnegative double series $\sum_{i,j} |A_{ij}|^2$ converges. Hence $\|T - T_n\| \to 0$. Since the operator-norm limit of finite-rank operators is compact, T is compact.

The left and right shift operators

For $a = \{a_n\}_{n=1}^{\infty} \in \ell^2$, define the left shift

$$La = \{a_2, a_3, a_4, \ldots\}$$

and the right shift

$$Ra = \{0, a_1, a_2, a_3, \ldots\}.$$

Then:

1) L and R are not compact;

2) R has no eigenvalues;

3)
$$\mathrm{Spec}(R) = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\};$$

4) the eigenvalues of L are precisely $\{\lambda \in \mathbb{C} : |\lambda| < 1\}$, and
$$E_\lambda = \mathrm{span}\{(1, \lambda, \lambda^2, \lambda^3, \ldots)\};$$

5)
$$\mathrm{Spec}(L) = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}.$$

Proof of part 1: noncompactness

The same example used on page 154 and page 184 shows that the bounded sets

$$\{La : \|a\|_{\ell^2} \leq 1\} \quad \text{and} \quad \{Ra : \|a\|_{\ell^2} \leq 1\}$$

have no convergent subsequence. Indeed, let e_n be the standard unit vectors. Then $e_n \in \{La : \|a\|_{\ell^2} \leq 1\}$ for every n, and the sequence $\{e_n\}$ has no convergent subsequence. Thus L is not compact. The same argument applies to R.

Proof of part 2: R has no eigenvalues

We claim that $R-\lambda I$ is injective for every $\lambda \in \mathbb{C}$. For $\lambda \neq 0$, suppose $a \in \ell^2$ and $(R-\lambda I)a = 0$. Then

$$\{0, a_1, a_2, a_3, \ldots\} - \{\lambda a_1, \lambda a_2, \lambda a_3, \ldots\} = \{0, 0, 0, \ldots\}.$$

It follows successively that

$$a_1 = 0, \qquad a_1 = \lambda a_2, \qquad a_2 = \lambda a_3, \ldots,$$

and hence $a = 0$. If $\lambda = 0$, then $\mathrm{Ker}(R) = \{0\}$, because $Ra = 0$ also forces $a = 0$. Thus $R - \lambda I$ is injective for every $\lambda \in \mathbb{C}$, so R has no eigenvalues.

Proof of part 3: the spectrum of R

Although $R^* = L$, the self-adjoint spectral results on page 189–page 191 do not apply. Since R is bounded, the general spectral theorem on page 187–page 188 gives

$$\mathrm{Spec}(R) \subset \{\lambda \in \mathbb{C} : |\lambda| \leq \|R\|\}.$$

Because $\|R\| = 1$, it remains to prove that every $|\lambda| \leq 1$ belongs to the spectrum.
We claim that $R - \lambda I$ is not surjective. If $\lambda = 0$, then $e_1 \notin \mathrm{Ran}(R)$. Suppose $0 < |\lambda| \leq 1$, and assume that $(R - \lambda I)a = e_1$. Solving recursively gives

$$a_1 = -\frac{1}{\lambda}, \qquad a_2 = -\frac{1}{\lambda^2}, \qquad a_3 = -\frac{1}{\lambda^3}, \ldots.$$

Because $|\lambda| \leq 1$,

$$\lim_{n\to\infty} |a_n|^2 = \lim_{n\to\infty} \frac{1}{|\lambda|^{2n}} \neq 0,$$

so $\sum_n |a_n|^2 = \infty$. Thus $a \notin \ell^2$, $R - \lambda I$ is not surjective, and $\lambda \in \mathrm{Spec}(R)$. Therefore

$$\mathrm{Spec}(R) = \{\lambda \in \mathbb{C} : |\lambda| \leq 1\}.$$

Proof of parts 4 and 5: the eigenvalues and spectrum of L

A complex number λ is an eigenvalue of L precisely when there is a nonzero $a \in \ell^2$ satisfying $La = \lambda a$. Equivalently,

$$a_{n+1} = \lambda a_n,$$

so

$$a_n = \lambda^{n-1} a_1.$$

If $a_1 = 0$, then $a = 0$. Thus a nonzero solution belongs to ℓ^2 exactly when

$$\sum_{n=1}^{\infty} |a_n|^2 = |a_1|^2 \sum_{n=1}^{\infty} |\lambda|^{2n-2} < \infty,$$

which holds exactly when $|\lambda| < 1$. Hence the eigenvalues of L are precisely those in the open unit disk, and

$$E_\lambda = \left\{(a_1, a_1\lambda, a_1\lambda^2, a_1\lambda^3, \ldots) : a_1 \in \mathbb{C}\right\} = \mathrm{span}\{(1, \lambda, \lambda^2, \ldots)\}.$$

Also $\|L\| = 1$. Therefore

$$\{\lambda : |\lambda| < 1\} \subset \mathrm{Spec}(L) \subset \{\lambda : |\lambda| \le 1\}.$$

The spectrum of a bounded operator is closed, so

$$\mathrm{Spec}(L) = \{\lambda \in \mathbb{C} : |\lambda| \le 1\}.$$

The Volterra operator

For $f \in L^2([0,1])$, define

$$Tf(x) := \int_0^x f(t)\,dt, \qquad x \in [0,1].$$

Then:

1) $T \in \mathcal{B}(L^2([0,1]))$, and T is compact;

2) if $Tf = 0$, then $f = 0$; thus 0 is not an eigenvalue of T;

3) if $\lambda \in \mathbb{C} \setminus \{0\}$ and $(T - \lambda I)f = 0$, then $f \in C^1([0,1])$,

$$f - \lambda f' = 0, \qquad f(0) = 0,$$

and consequently $f = 0$. Thus T has no eigenvalues.

Proof of part 1: boundedness and compactness

Suppose $f \in L^2([0,1])$ with $\|f\|_2 = 1$. Hölder's inequality gives

$$\|f\|_1 = \int_0^1 |f| \le \|f\|_2 \|1\|_2 = \|f\|_2 = 1.$$

Then

$$\begin{aligned} \|Tf\|_2^2 &= \int_0^1 \left| \int_0^x f(t)\,dt \right|^2 dx \\ &\le \int_0^1 \left(\int_0^1 |f(t)|\,dt \right)^2 dx \\ &= \int_0^1 \|f\|_1^2\,dx \le 1. \end{aligned}$$

Thus $\|Tf\|_2 \le 1$ whenever $\|f\|_2 = 1$, and $T \in \mathcal{B}(L^2([0,1]))$.
We show that every sequence in

$$\{Tf : \|f\|_2 \le 1\}$$

has a convergent subsequence by applying Arzelà–Ascoli and then Cauchy–Schwarz. First, for every $x \in [0,1]$,

$$|Tf(x)| = |\langle f, 1\rangle_{L^2([0,x])}| \le \|f\|_{L^2([0,x])} \|1\|_{L^2([0,x])} \le \|f\|_2.$$

Similarly, if $y < x$,

$$\begin{aligned} |Tf(x) - Tf(y)| &= \left| \int_y^x f(t)\,dt \right| \\ &= |\langle f, 1\rangle_{L^2([y,x])}| \\ &\le \|f\|_{L^2([y,x])} \|1\|_{L^2([y,x])} \\ &\le \|f\|_2 |x-y|^{1/2}. \end{aligned}$$

Let $\{Tf_n\}$ be a sequence with $\|f_n\|_2 \le 1$. The first estimate shows that it is uniformly bounded on $[0,1]$. Given $\varepsilon > 0$, choose $\delta = \varepsilon^2/4$. Whenever $|x-y| < \delta$,

$$|Tf_n(x) - Tf_n(y)| \le \|f_n\|_2 |x-y|^{1/2} < \frac{\varepsilon}{2} < \varepsilon.$$

Thus the sequence is uniformly equicontinuous. Arzelà–Ascoli, proved on page 199–page 201, supplies a uniformly convergent subsequence. Uniform convergence on the bounded interval $[0,1]$ implies L^2-convergence. Thus $\{Tf : \|f\|_2 \le 1\}$ is relatively compact in $L^2([0,1])$. Its closure is compact; therefore T is compact.

Proof of part 2: injectivity

Suppose $Tf = 0$. Then, for every $x \in [0,1]$,

$$\int_0^x f(t)\,dt = 0.$$

In particular, for $0 \le a < b \le 1$,

$$\int_a^b f(t)\,dt = \int_0^b f(t)\,dt - \int_0^a f(t)\,dt = 0. \tag{$**$}$$

Recall Littlewood's first principle from page 107–page 108: for every measurable $E \subset [0,1]$ and every $\varepsilon > 0$, there are finitely many open intervals $I_1, \dots, I_N$ in $[0,1]$ such that

$$m^*\left(E \triangle \bigcup_{n=1}^N I_n \right) < \frac{\varepsilon^2}{4\|f\|_2^2}.$$

We claim that

$$\int_0^1 \overline{f(t)}\, \chi_E(t)\,dt = 0$$

for every measurable $E \subset [0,1]$. By the properties of the Lebesgue integral and $(**)$,

$$\begin{aligned} \left| \int_0^1 \overline{f}\, \chi_E \right| &\le \left| \int_0^1 \overline{f} \left(\chi_E - \chi_{\bigcup_{n=1}^N I_n} \right) \right| \\ &\le \left\| \chi_E - \chi_{\bigcup_{n=1}^N I_n} \right\|_2 \|f\|_2 \\ &= m^*\left(E \triangle \bigcup_{n=1}^N I_n \right)^{1/2} \|f\|_2 \\ &< \varepsilon/2 < \varepsilon. \end{aligned}$$

As this holds for every $\varepsilon > 0$, the integral is zero.
For a simple function

$$\varphi = \sum_{i=1}^{N} a_i \chi_{A_i},$$

where the $A_i \subset [0,1]$ are measurable, linearity of the integral gives

$$\int_0^1 \overline{f}\,\varphi = \sum_{i=1}^{N} a_i \int_0^1 \overline{f}\,\chi_{A_i} = 0. \tag{†}$$

Simple functions are dense in $L^2([0,1])$: the approximation theorem on page 88 together with dominated convergence gives simple functions $\varphi_n : [0,1] \to \mathbb{C}$ such that

$$\|\varphi_n - f\|_2 \longrightarrow 0.$$

Equation (†) gives

$$\langle \varphi_n, f \rangle = 0 \qquad (n \in \mathbb{N}).$$

By continuity of the inner product,

$$\langle f, f \rangle = \Big\langle \lim_{n\to\infty} \varphi_n, f \Big\rangle = \lim_{n\to\infty} \langle \varphi_n, f \rangle = 0.$$

Hence $f = 0$ almost everywhere. Therefore $\mathrm{Ker}(T) = \{0\}$, so T is injective and 0 is not an eigenvalue.

Proof of part 3: no nonzero eigenvalues

Let $\lambda \in \mathbb{C} \setminus \{0\}$, and suppose $f \in \mathrm{Ker}(T - \lambda I)$. Then

$$Tf - \lambda f = 0,$$

so

$$f = \frac{1}{\lambda} \int_0^x f(t)\,dt.$$

The right-hand side is continuous by the fundamental theorem of calculus. Applying that theorem again shows that $f \in C^1([0,1])$, and

$$(Tf)' = f.$$

Thus

$$f - \lambda f' = 0.$$

Solving the ordinary differential equation gives

$$f(x) = Ce^{x/\lambda} \qquad (C \in \mathbb{C}).$$

But $(T - \lambda I)f = 0$ at $x = 0$ yields

$$-\lambda f(0) = 0.$$

Since $\lambda \neq 0$, $f(0) = 0$, and hence $C = 0$. Therefore $f = 0$,

$$\mathrm{Ker}(T - \lambda I) = \{0\},$$

and λ is not an eigenvalue. Together with the preceding part, this shows that T has no eigenvalues.

28 The Dirichlet Problem on an Interval

The Dirichlet problem

Let $V \in C([0,1])$ be real-valued and nonnegative. Given $f \in C([0,1])$, solve

$$\begin{cases} -u''(x) + V(x)u(x) = f(x), & x \in [0,1], \\ u(0) = u(1) = 0. \end{cases} \tag{DP}$$

We shall prove that there exists a unique solution $u \in C^2([0,1])$: uniqueness directly, and existence through the spectral theory of compact self-adjoint operators.

Uniqueness

Let $V \geq 0$ be real-valued and continuous on $[0,1]$, and let $f \in C([0,1])$. If $u_1, u_2 \in C^2([0,1])$ both solve the Dirichlet problem, then $u_1 = u_2$.

Proof

Let $u = u_1 - u_2$. Then $u \in C^2([0,1])$ and

$$\begin{cases} -u''(x) + V(x)u(x) = 0, & x \in [0,1], \\ u(0) = u(1) = 0. \end{cases}$$

Multiply the differential equation by $\overline{u(x)}$, integrate, and use integration by parts:

$$\begin{aligned} 0 &= \int_0^1 [-u''(x) + V(x)u(x)]\overline{u(x)}\,dx \\ &= -\int_0^1 u''(x)\overline{u(x)}\,dx + \int_0^1 V(x)|u(x)|^2\,dx \\ &= -u'(x)\overline{u(x)}\Big|_0^1 \int_0^1 |u'(x)|^2\,dx \int_0^1 V(x)|u(x)|^2\,dx \\ &= \int_0^1 |u'(x)|^2\,dx \int_0^1 V(x)|u(x)|^2\,dx. \end{aligned}$$

Both terms are nonnegative, so

$$\int_0^1 |u'(x)|^2\,dx = 0.$$

Since $|u'|^2$ is continuous and nonnegative, $u'(x) = 0$ for every $x \in [0,1]$. Hence $u(x) = C$, and the boundary condition $u(0) = 0$ gives $C = 0$. Thus $u_1 = u_2$.

We begin the existence proof with the case $V = 0$.

Green's operator for $V = 0$

Define the continuous Green kernel $K \in C([0,1] \times [0,1])$ by

$$K(x,y) := \begin{cases} (x-1)y, & 0 \leq y \leq x \leq 1, \\ (y-1)x, & 0 \leq x \leq y \leq 1. \end{cases}$$

Define

$$Af(x) := \int_0^1 K(x,y)f(y)\,dy.$$

Then $A \in \mathcal{B}(L^2([0,1]))$ is compact and self-adjoint. Moreover, if $f \in C([0,1])$, then $u = Af$ is the unique solution of

$$\begin{cases} -u'' = f, & \text{on } [0,1], \\ u(0) = u(1) = 0. \end{cases}$$

Proof: boundedness and compactness

Let

$$C := \sup_{(x,y)\in[0,1]^2} |K(x,y)|.$$

The extreme-value theorem gives $C < \infty$, because K is continuous. By Cauchy–Schwarz, for $f \in L^2([0,1])$,

$$\begin{aligned} |Af(x)| &= \left| \int_0^1 K(x,y)f(y)\,dy \right| \\ &\le C \int_0^1 |f(y)|\,dy \\ &= C\langle |f|, 1\rangle_{L^2([0,1])} \\ &\le C\|f\|_2\|1\|_2 = C\|f\|_2. \end{aligned} \tag{$*$}$$

Thus $A \in \mathcal{B}(L^2([0,1]))$.
Similarly,

$$\begin{aligned} |Af(x) - Af(z)| &\le \int_0^1 |K(x,y) - K(z,y)|\,|f(y)|\,dy \\ &\le \sup_{y\in[0,1]} |K(x,y) - K(z,y)|\,\|f\|_2. \end{aligned} \tag{$\dagger$}$$

The compactness proof is the same as the continuous-kernel argument on page 203–page 203. In detail, any sequence $\{Af_n\}$ with $\|f_n\|_2 \le 1$ is pointwise uniformly bounded by $(*)$, and uniformly equicontinuous by $(\dagger)$ and uniform continuity of K. Arzelà–Ascoli gives a uniformly convergent subsequence, hence an L^2-convergent subsequence. Therefore the image of the closed unit ball has compact closure, and A is compact.

Proof: self-adjointness

First let $f, g \in C([0,1])$. Fubini's theorem, stated on page 144, and the real symmetric identity

$$\overline{K(x,y)} = K(y,x)$$

give

$$\begin{aligned} \langle Af, g\rangle &= \int_0^1 \left(\int_0^1 K(x,y)f(y)\,dy \right) \overline{g(x)}\,dx \\ &= \int_0^1 f(y) \left(\int_0^1 K(x,y)\overline{g(x)}\,dx \right) dy \end{aligned}$$

$$= \int_0^1 f(y)\overline{\left(\int_0^1 K(y,x)g(x)\,dx\right)}\,dy$$
$$= \langle f, Ag\rangle.$$

Since $C([0,1])$ is dense in $L^2([0,1])$, for arbitrary $f, g \in L^2([0,1])$ choose $f_n, g_n \in C([0,1])$ with

$$f_n \to f, \qquad g_n \to g$$

in L^2. Continuity of A and of the inner product gives

$$\begin{aligned}\langle Af, g\rangle &= \left\langle A\left(\lim_{n\to\infty} f_n\right), \lim_{m\to\infty} g_m\right\rangle \\ &= \lim_{n\to\infty}\lim_{m\to\infty}\langle Af_n, g_m\rangle \\ &= \lim_{n\to\infty}\lim_{m\to\infty}\langle f_n, Ag_m\rangle \\ &= \langle f, Ag\rangle.\end{aligned}$$

Therefore A is a compact self-adjoint operator on $L^2([0,1])$.

Proof:** Af **solves the Dirichlet problem

It remains to show that $u = Af$ solves the Dirichlet problem with $V = 0$. From the definitions of A and K,

$$u(x) = Af(x) = (x-1)\int_0^x yf(y)\,dy + x\int_x^1 (y-1)f(y)\,dy.$$

The fundamental theorem of calculus verifies directly that

$$u \in C^2([0,1]), \qquad -u'' = f, \qquad u(0) = u(1) = 0.$$

Uniqueness follows from the theorem on page 213.

Operator-theoretic reformulation for $V \neq 0$

The equation

$$-u'' + Vu = f$$

can be rewritten as

$$-u'' = f - Vu.$$

The preceding result then gives

$$u = A(f - Vu), \qquad \text{or} \qquad (I + AV)u = Af.$$

We seek a compact square root $A^{1/2}$ satisfying

$$(A^{1/2})^2 = A.$$

Writing $u = A^{1/2}v$ transforms the equation into

$$(I + A^{1/2}VA^{1/2})v = A^{1/2}f.$$

The operator $A^{1/2}VA^{1/2}$ is self-adjoint and compact, so the Fredholm alternative on page 195–page 196 will apply.

The compact self-adjoint spectral theorem

The spectral theorem on page 198 provides an orthonormal basis of $L^2([0,1])$ consisting entirely of eigenvectors of A: an orthonormal basis of $\mathrm{Ran}(A)$ associated with nonzero eigenvalues, together with an orthonormal basis of $\mathrm{Ker}(A)$ associated with the eigenvalue 0.

Eigenvalues and eigenvectors of the Green operator

For the operator A above,

$$\mathrm{Ker}(A) = \{0\},$$

and its orthonormal eigenvectors and corresponding eigenvalues are

$$u_k(x) = \sqrt{2}\sin(k\pi x), \qquad \lambda_k = \frac{1}{k^2\pi^2}, \qquad k \in \mathbb{N}.$$

The family $\{\sqrt{2}\sin(k\pi x)\}_{k=1}^{\infty}$ is therefore an orthonormal basis of $L^2([0,1])$.

Proof: the range is dense and the kernel is trivial

We first claim that $\mathrm{Ran}(A)$ contains a dense subset of $L^2([0,1])$. Let u be a polynomial on $[0,1]$ with

$$u(0) = u(1) = 0,$$

and put $f = -u''$. The Green formula shows that Af is the unique solution of the zero-potential Dirichlet problem. Since u itself solves that problem,

$$u = Af.$$

Hence

$$\{u : u \text{ is a polynomial on } [0,1],\ u(0) = u(1) = 0\} \subset \mathrm{Ran}(A).$$

Set

$$C_0([0,1]) := \{g \in C([0,1]) : g(0) = g(1) = 0\}.$$

We claim that the polynomials vanishing at the endpoints are dense in this space. Indeed, let $g \in C_0([0,1])$ and $\varepsilon > 0$. By the Weierstrass approximation theorem from page 129, choose a polynomial p such that

$$\|p - g\|_\infty < \frac{\varepsilon}{2}.$$

Define

$$q(x) := p(x) - (1-x)p(0) - xp(1).$$

Then q is a polynomial with $q(0) = q(1) = 0$. Since $g(0) = g(1) = 0$,

$$|p(0)| < \frac{\varepsilon}{2}, \qquad |p(1)| < \frac{\varepsilon}{2},$$

and hence

$$\|q - g\|_\infty \le \|p - g\|_\infty \sup_{0\le x\le 1} |(1-x)p(0) + xp(1)| < \varepsilon.$$

Thus the endpoint-vanishing polynomials are dense in $C_0([0,1])$. The endpoint-vanishing density theorem on page 124 says that $C_0([0,1])$ is dense in $L^2([0,1])$. Also,

$$\|h\|_2 \le \|h\|_\infty \qquad (h \in C([0,1])).$$

Therefore the endpoint-vanishing polynomials are dense in $L^2([0,1])$, and $\mathrm{Ran}(A)$ contains a dense subset.
It follows that

$$\overline{\mathrm{Ran}(A)} = L^2([0,1]).$$

For a bounded operator,

$$\mathrm{Ran}(A)^\perp = \mathrm{Ker}(A^*).$$

Since $A = A^*$,

$$\mathrm{Ran}(A)^\perp = \mathrm{Ker}(A).$$

Also,

$$\overline{\mathrm{Ran}(A)} = (\mathrm{Ran}(A)^\perp)^\perp = \mathrm{Ker}(A)^\perp.$$

Therefore $\mathrm{Ker}(A)^\perp = L^2([0,1])$, and hence $\mathrm{Ker}(A) = \{0\}$.

Proof: the eigenvalues and eigenvectors

Suppose $Au = \lambda u$, where $\lambda \neq 0$ and $\|u\|_2 = 1$. The continuity estimate $(*)$ on page 214 gives $Au \in C([0,1])$, so

$$u = \lambda^{-1} Au \in C([0,1]).$$

The Green formula then implies $Au \in C^2([0,1])$, and consequently $u \in C^2([0,1])$. Moreover,

$$-(Au)'' = u.$$

Since $Au = \lambda u$,

$$-\lambda u'' = u,$$

or

$$u'' + \frac{1}{\lambda} u = 0.$$

The eigenvalues are real by the spectral theorem for compact self-adjoint operators. In fact, every nonzero eigenvalue is positive. Multiplying $-\lambda u'' = u$ by $\overline{u}$, integrating by parts, and using the boundary conditions gives

$$\lambda \int_0^1 |u'(x)|^2 \, dx = \int_0^1 |u(x)|^2 \, dx = 1.$$

Thus $\lambda > 0$. Solving the ordinary differential equation and imposing

$$u(0) = u(1) = 0$$

gives

$$\frac{1}{\sqrt{\lambda}} = k\pi, \qquad k \in \mathbb{N}.$$

Therefore

$$\lambda_k = \frac{1}{k^2\pi^2}, \qquad u_k(x) = C \sin(k\pi x).$$

The normalization $\|u_k\|_2 = 1$ gives $C = \sqrt{2}$, hence

$$u_k(x) = \sqrt{2} \sin(k\pi x).$$

Square roots of diagonal operators

For a diagonal matrix one has, for example,

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}.$$

The compact self-adjoint spectral theorem diagonalizes A, so its square root is obtained by taking the positive square roots of its eigenvalues.

The square root $A^{1/2}$

For $f \in L^2([0,1])$, its Fourier–Bessel series is

$$f(x) = \sum_{k=1}^{\infty} c_k \sqrt{2}\sin(k\pi x),$$

where

$$c_k = \int_0^1 f(x)\sqrt{2}\sin(k\pi x)\,dx.$$

Define

$$A^{1/2}f(x) := \sum_{k=1}^{\infty} \frac{c_k}{k\pi}\sqrt{2}\sin(k\pi x).$$

Properties of $A^{1/2}$

The operator $A^{1/2}$ is compact and self-adjoint on $L^2([0,1])$, and

$$(A^{1/2})^2 = A.$$

Proof: boundedness, self-adjointness, and the square identity

By Parseval's identity,

$$\|A^{1/2}f\|_2^2 = \sum_{k=1}^{\infty} \frac{|c_k|^2}{k^2\pi^2} \le \frac{1}{\pi^2}\sum_{k=1}^{\infty}|c_k|^2 = \frac{1}{\pi^2}\|f\|_2^2.$$

Thus $A^{1/2} \in \mathcal{B}(L^2([0,1]))$.
The operator is self-adjoint because, relative to the displayed orthonormal basis, it is diagonal with real diagonal entries $1/(k\pi)$. Explicitly, if

$$g(x) = \sum_{k=1}^{\infty} d_k \sqrt{2}\sin(k\pi x),$$

then

$$\langle A^{1/2}f, g\rangle = \left\langle \sum_{k=1}^{\infty}\frac{c_k}{k\pi}\sqrt{2}\sin(k\pi x), \sum_{k=1}^{\infty} d_k\sqrt{2}\sin(k\pi x)\right\rangle = \langle f, A^{1/2}g\rangle.$$

Finally,

$$\begin{aligned}(A^{1/2}A^{1/2})f &= A^{1/2}\left(\sum_{k=1}^{\infty}\frac{c_k}{k\pi}\sqrt{2}\sin(k\pi x)\right)\\ &= \sum_{k=1}^{\infty}\frac{c_k}{k^2\pi^2}\sqrt{2}\sin(k\pi x)\\ &= A\left(\sum_{k=1}^{\infty}c_k\sqrt{2}\sin(k\pi x)\right) = Af.\end{aligned}$$

Proof: compactness

We show that

$$\{A^{1/2}f : \|f\|_2 \le 1\}$$

has equi-small tails relative to the orthonormal basis $\{\sqrt{2}\sin(k\pi x)\}_{k=1}^{\infty}$, and that its closure has the same property.
Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ so that $1/N < \varepsilon$. If $\|f\|_2 \le 1$, then

$$\begin{aligned}\sum_{k=N}^{\infty}\left|\left\langle A^{1/2}f, \sqrt{2}\sin(k\pi x)\right\rangle\right|^2 &= \sum_{k=N}^{\infty}\frac{|c_k|^2}{k^2\pi^2}\\ &< \sum_{k=N}^{\infty}\frac{|c_k|^2}{N^2}\\ &= \frac{1}{N^2}\|f\|_2^2 < \varepsilon^2.\end{aligned}$$

If g is a limit point of this image, choose $A^{1/2}f_n \to g$. Continuity of the inner product and finite partial sums shows that the same tail estimate holds for g. The closure is therefore closed, bounded, and has equi-small tails. The compactness criterion for subsets of a separable Hilbert space gives compactness of the closure, so $A^{1/2}$ is compact.

Multiplication by a real continuous potential

Let $V \in C([0,1])$ be real-valued, and define

$$m_V f(x) := V(x)f(x), \qquad f \in L^2([0,1]).$$

Then

$$m_V \in \mathcal{B}(L^2([0,1])) \qquad \text{and} \qquad m_V^* = m_V.$$

Proof

The extreme-value theorem gives $B > 0$ with

$$|V(x)| \le B \qquad (x \in [0,1]).$$

Hence

$$\|m_V f\|_2 = \|Vf\|_2 \le B\|f\|_2,$$

so m_V is bounded. Since V is real-valued, for $f, g \in L^2([0,1])$,

$$\langle m_V f, g\rangle = \int_0^1 V f\overline{g} = \int_0^1 f\,\overline{Vg} = \langle f, m_V g\rangle.$$

The symmetrized potential operator

Let $V \in C([0,1])$ be real-valued. Then

$$T := A^{1/2} m_V A^{1/2}$$

is a compact self-adjoint operator on $L^2([0,1])$. Moreover,

$$T : L^2([0,1]) \longrightarrow C([0,1])$$

is bounded.

Proof

Compactness follows from the composition theorem on page 185. Self-adjointness follows from

$$\begin{aligned}(A^{1/2} m_V A^{1/2})^* &= (A^{1/2})^* m_V^* (A^{1/2})^* \\ &= A^{1/2} m_V A^{1/2}.\end{aligned}$$

It remains to prove that $A^{1/2}$ maps $L^2([0,1])$ boundedly into $C([0,1])$. Write

$$f(x) = \sum_{k=1}^{\infty} c_k \sqrt{2}\sin(k\pi x).$$

Then

$$A^{1/2} f(x) = \sum_{k=1}^{\infty} \frac{c_k}{k\pi}\sqrt{2}\sin(k\pi x).$$

For every $x \in [0,1]$,

$$\left|\frac{c_k}{k\pi}\sqrt{2}\sin(k\pi x)\right| \le \frac{\sqrt{2}\,|c_k|}{k\pi}.$$

Cauchy–Schwarz in ℓ^2 gives

$$\begin{aligned}\sum_{k=1}^{\infty} \frac{\sqrt{2}\,|c_k|}{k\pi} &\le \frac{\sqrt{2}}{\pi}\left(\sum_{k=1}^{\infty}\frac{1}{k^2}\right)^{1/2}\left(\sum_{k=1}^{\infty}|c_k|^2\right)^{1/2} \\ &= \frac{1}{\sqrt{3}}\|f\|_2.\end{aligned}$$

The Weierstrass M-test therefore gives uniform convergence on $[0,1]$. Since every partial sum is continuous, $A^{1/2} f \in C([0,1])$, and

$$\|A^{1/2} f\|_\infty \le \frac{1}{\sqrt{3}}\|f\|_2.$$

Thus $A^{1/2} : L^2([0,1]) \to C([0,1])$ is bounded, and consequently so is $T = A^{1/2} m_V A^{1/2}$. Finally, term-by-term evaluation gives

$$A^{1/2} f(0) = A^{1/2} f(1) = 0.$$

Solution of the Dirichlet problem on $[0,1]$

Let $V \in C([0,1])$ be real-valued and nonnegative, and let $f \in C([0,1])$. Then there is a unique $u \in C^2([0,1])$ such that

$$\begin{cases} -u'' + Vu = f, & \text{on } [0,1], \\ u(0) = u(1) = 0. \end{cases}$$

Proof: invertibility of $I + A^{1/2} m_V A^{1/2}$

The operator

$$A^{1/2} m_V A^{1/2}$$

is compact and self-adjoint. By the Fredholm alternative on page 195–page 196, it is enough to prove

$$\operatorname{Ker}(I + A^{1/2} m_V A^{1/2}) = \{0\}.$$

Suppose

$$(I + A^{1/2} m_V A^{1/2}) g = 0$$

for $g \in L^2([0,1])$. Then, using self-adjointness,

$$\begin{aligned} 0 &= \left\langle (I + A^{1/2} m_V A^{1/2}) g, g \right\rangle \\ &= \|g\|_2^2 \langle A^{1/2} m_V A^{1/2} g, g \rangle \\ &= \|g\|_2^2 \langle m_V A^{1/2} g, A^{1/2} g \rangle \\ &= \|g\|_2^2 \int_0^1 V(x) |A^{1/2} g(x)|^2 \, dx. \end{aligned}$$

Both terms are nonnegative, so $\|g\|_2 = 0$, and $g = 0$. Therefore

$$I + A^{1/2} m_V A^{1/2}$$

is invertible.

Proof: construction and verification of the solution

Define

$$v := (I + A^{1/2} m_V A^{1/2})^{-1} (A^{1/2} f), \qquad u := A^{1/2} v.$$

We claim that u is the unique solution. Uniqueness was proved on page 213. From the definition of v,

$$\begin{aligned} u + AVu &= A^{1/2} v + A^{1/2} A^{1/2} m_V A^{1/2} v \\ &= A^{1/2} (I + A^{1/2} m_V A^{1/2}) v \\ &= A^{1/2} A^{1/2} f = Af. \end{aligned}$$

Thus

$$u = Af - A(Vu) = A(f - Vu).$$

The operator $A^{1/2} : L^2([0,1]) \to C([0,1])$ shows that $u \in C([0,1])$. Hence $f - Vu \in C([0,1])$,

and the Green formula on page 213–page 215 shows that

$$u \in C^2([0,1]), \qquad -u'' = f - Vu.$$

Therefore

$$-u'' + Vu = f.$$

Finally, the endpoint identity for $A^{1/2}$ gives

$$u(0) = A^{1/2}v(0) = 0, \qquad u(1) = A^{1/2}v(1) = 0.$$

This completes the existence and uniqueness proof.